

APPENDIX Q

11,652,232

Secondary Battery And Method For Manufacturing The Same
Transaction History

Date	Transaction Description
09-27-2019	Substitute Specification Filed
09-27-2019	Preliminary Amendment
09-27-2019	Request for Foreign Priority (Priority Papers May Be Included)
09-27-2019	Information Disclosure Statement (IDS) Filed
09-27-2019	371 Completion Date
09-27-2019	Patent Term Adjustment - Ready for Examination
09-27-2019	PTO-SB-69-Authorize EPO Access to Search Results
09-27-2019	Applicants have given acceptable permission for participating foreign
09-27-2019	Information Disclosure Statement (IDS) Filed
09-27-2019	Entity Status Set To Undiscounted (Initial Default Setting or Status Change)
09-27-2019	Initial Exam Team nn
04-13-2020	Information Disclosure Statement (IDS) Filed
04-13-2020	Information Disclosure Statement (IDS) Filed
06-09-2020	Sent to Classification Contractor
06-09-2020	FITF set to YES - revise initial setting
06-10-2020	Email Notification
06-10-2020	Email Notification
06-10-2020	Notice of DO-EO Acceptance Mailed
06-10-2020	Filing Receipt
06-22-2020	Application Is Now Complete
06-22-2020	Application Dispatched from OIPE
09-17-2020	Application ready for PDX access by participating foreign offices
09-17-2020	PG-Pub Issue Notification
09-18-2020	Email Notification
10-19-2020	Information Disclosure Statement (IDS) Filed
10-19-2020	Information Disclosure Statement (IDS) Filed
05-16-2021	Case Docketed to Examiner in GAU
05-17-2021	Information Disclosure Statement considered
05-17-2021	Information Disclosure Statement considered
05-17-2021	Information Disclosure Statement considered
05-28-2021	Non-Final Rejection
06-03-2021	Electronic Review
06-03-2021	Email Notification
06-03-2021	Mail Non-Final Rejection
09-03-2021	Response after Non-Final Action
09-03-2021	Miscellaneous Incoming Letter
09-08-2021	Date Forwarded to Examiner
09-17-2021	Final Rejection
09-22-2021	Electronic Review
09-22-2021	Email Notification
09-22-2021	Mail Final Rejection (PTOL - 326)
11-02-2021	Electronic Information Disclosure Statement
11-02-2021	Information Disclosure Statement (IDS) Filed
01-18-2022	Interview Summary - Applicant Initiated - Telephonic
01-18-2022	Interview Summary Record
01-24-2022	Email Notification
01-24-2022	Mail Examiner Interview Summary (PTOL - 413)
02-08-2022	Interview Summary - Applicant Initiated - Telephonic
02-08-2022	Interview Summary Record
02-11-2022	Email Notification
02-11-2022	Mail Examiner Interview Summary (PTOL - 413)
02-18-2022	Request for Continued Examination (RCE)
02-18-2022	Request for Extension of Time - Granted

02-18-2022	Workflow - Request for RCE - Begin
02-22-2022	Date Forwarded to Examiner
02-22-2022	Disposal for a RCE - CPA - R129
03-03-2022	Non-Final Rejection
03-03-2022	Information Disclosure Statement considered
03-08-2022	Electronic Review
03-08-2022	Email Notification
03-08-2022	Mail Non-Final Rejection
07-07-2022	Response after Non-Final Action
07-07-2022	Request for Extension of Time - Granted
07-12-2022	Date Forwarded to Examiner
09-07-2022	Final Rejection
09-13-2022	Electronic Review
09-13-2022	Email Notification
09-13-2022	Mail Final Rejection (PTOL - 326)
10-19-2022	Electronic Information Disclosure Statement
10-19-2022	Information Disclosure Statement (IDS) Filed
11-15-2022	Interview Summary - Applicant Initiated - Telephonic
11-15-2022	Interview Summary Record
11-21-2022	Email Notification
11-21-2022	Mail Examiner Interview Summary (PTOL - 413)
12-01-2022	PILOT- Request for After Final Consideration Program
12-01-2022	Response after Final Action
12-07-2022	Date Forwarded to Examiner
12-16-2022	Information Disclosure Statement considered
01-04-2023	Notice of Allowance Data Verification Completed
01-09-2023	Electronic Review
01-09-2023	Email Notification
01-09-2023	Mail Notice of Allowance
04-07-2023	Issue Fee Payment Verified
04-07-2023	Issue Fee Payment Received
04-10-2023	Dispatch to FDC
04-10-2023	Application Is Considered Ready for Issue
04-26-2023	Issue Notification Mailed
04-27-2023	Electronic Review
04-27-2023	Email Notification
05-16-2023	Electronic Review
05-16-2023	Recordation of Patent Grant Mailed
05-16-2023	Email Notification
05-16-2023	Recordation of Patent eGrant
05-16-2023	Mail Patent eGrant Notification
05-16-2023	Patent eGrant Notification
05-16-2023	Patent Issue Date Used in PTA Calculation

Issue Classification 	Application/Control No.	Applicant(s)/Patent Under Reexamination
	16/498,751	Kim et al.
	Examiner	Art Unit
	ALIX E EGGERDING	1729

CPC						
Symbol					Type	Version
H01M	/	10	/	0422	I	2013-01-01
H01M	/	10	/	0431	I	2013-01-01
H01M	/	50	/	538	I	2021-01-01
H01M	/	50	/	536	A	2021-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE	Total Claims Allowed:	
(Assistant Examiner)	(Date)	15
/ALIX E EGGERDING/	16 December 2022	O.G. Print Claim(s)
Primary Examiner, Art Unit 1729		O.G. Print Figure
(Primary Examiner)	(Date)	1
		9

<i>Issue Classification</i> 	Application/Control No. 16/498,751	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H01M	/	50	/ 538
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
429	94

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	15	
/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729 (Primary Examiner)	16 December 2022 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9

<i>Issue Classification</i> 	Application/Control No. 16/498,751	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	18												
2	2	11	19												
3	4	12	20												
4	6	13	21												
5	7	14	22												
6	14	15	23												
7	15														
8	16														
9	17														

NONE	Total Claims Allowed:	
(Assistant Examiner)	(Date)	15
/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729 (Primary Examiner)	16 December 2022 (Date)	O.G. Print Claim(s) 1 O.G. Print Figure 9

Search Notes 	Application/Control No. 16/498,751	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M 50/538,531	05/18/2021	AE
H01M 10/0422	05/18/2021	AE

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor Search	05/17/2021	AE
Global Dossier	05/17/2021	AE
Searched EAST database, subclasses searched with text keywords. See Search History	05/18/2021	

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
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Search Notes 	Application/Control No. 16/498,751	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M 50/538,531	05/18/2021	AE
H01M 10/0422	05/18/2021	AE

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor Search	05/17/2021	AE
Global Dossier	05/17/2021	AE
Searched EAST database, subclasses searched with text keywords. See Search History	05/18/2021	AE
Updated above	09/17/2021	AE

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
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Search Notes 	Application/Control No. 16/498,751	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M 50/538,531	05/18/2021	AE
H01M 10/0422	05/18/2021	AE

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

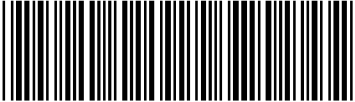
US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor Search	05/17/2021	AE
Global Dossier	05/17/2021	AE
Searched EAST database, subclasses searched with text keywords. See Search History	05/18/2021	AE
Updated above	09/17/2021	AE
Updated above	03/03/2022	AE

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
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Search Notes 	Application/Control No. 16/498,751	Applicant(s)/Patent Under Reexamination Kim et al.
	Examiner ALIX E EGGERDING	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M 50/538,531	05/18/2021	AE
H01M 10/0422	05/18/2021	AE

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor Search	05/17/2021	AE
Global Dossier	05/17/2021	AE
Searched EAST database, subclasses searched with text keywords. See Search History	05/18/2021	AE
Updated above	09/17/2021	AE
Updated above	03/03/2022	AE
Updated above	08/25/2022	AE

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L6	1	"20180145304".pn.	US-PGPUB; USPAT	OR	ON	2021/05/28 17:13
S2	14	("20010023038" "20050003264" "20150295270" "20110067227" "20120009450" "20160351966" "20110293977" "20100310926" "6569565" "20070196730" "20050287428" "20130344364" "20190379028" "20120208055").PN.	US-PGPUB; USPAT	OR	ON	2021/05/17 14:34
S3	9	("20150037638" "20110067227" "20140147714" "20120009450" "20110081574" "20160351940" "20100190056" "20160218343" "20130252054").PN.	US-PGPUB; USPAT	OR	ON	2021/05/17 14:41
S4	1	"20100310926".pn.	US-PGPUB; USPAT	OR	ON	2021/05/17 14:46
S5	21,124	(h01m10/\$.cpc. h01m50/\$.cpc.) and (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2021/05/17 14:56
S6	260	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) and (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) and (electrode cathode anode) with (tab notch) near4 (overlap\$4 plural\$4)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2021/05/17 14:57
S7	144	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) and (electrode cathode anode) with (tab notch) near4 (overlap\$4 plural\$4) and (battery mea electrode near3 assembly) with (wound jelly near3 roll cylindrical)	FPRS; EPO; JPO; DERWENT	OR	ON	2021/05/17 16:11
S8	132	S7 not S6	FPRS; EPO; JPO; DERWENT	OR	ON	2021/05/17 16:12
S9	8	("20050277019" "20090169990" "20090317707" "20030235736" "20060147793" "20020192548" "20060110658" "20070154797").PN.	US-PGPUB; USPAT	OR	ON	2021/05/17 16:15
S10	0	("2016/0351966").URPN.	USPAT	OR	ON	2021/05/17 16:31

S11	1	(15/115220).APP.	USPAT; USOCR	OR	ON	2021/05/17 16:31
S12	10	("20050287428" "20060024572" "20070196730" "20120251854" "20120328924" "20130216879" "20140045041" "7364817" "7700222" "7955736").PN. OR ("10381688").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/17 16:31
S13	6	("2007/0196730").URPN.	USPAT	OR	ON	2021/05/17 16:33
S14	10	("3490949" "4053687" "6376121" "6596434" "6818025" "7332246" "7364817").PN. OR ("7700222").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/17 16:33
S15	14	("20020164524" "20030118912" "20050225926" "5736270" "6071638" "6222720" "6328769" "7364817" "7618742" "7670715" "7678501" "7700222").PN. OR ("8310809").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/17 16:34
S16	11	("20020191370" "20030086239" "20060024572" "20060240322" "20060240323" "3237274" "5849431" "6456484" "6896993" "8310809").PN. OR ("8628877").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/17 16:35
S17	13	("3761314" "5525441" "5635312" "5736270").PN. OR ("6328769").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 10:22
S18	44	("3761314").URPN.	USPAT	OR	ON	2021/05/18 10:23
S19	5	("20050142436" "3761314").PN. OR ("7955736").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 10:24
S20	3	("2005/0142436").URPN.	USPAT	OR	ON	2021/05/18 10:34
S21	1,863	h01m50/538.cpc.	US-PGPUB; USPAT	OR	ON	2021/05/18 11:26
S22	4,637	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.)	US-PGPUB; USPAT	OR	ON	2021/05/18 13:20
S23	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) with (plural\$4 several) near4 (tab notch lead terminal)	US-PGPUB; USPAT	OR	ON	2021/05/18 13:43
S24	40	("20010038945" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683"	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 14:24

		"20050255378" "20060124973" "20060172190" "20070117011" "20090109099" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6080506" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "RE38518").PN. OR ("7927732").URPN.				
S25	49	("20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 14:29
S26	1	"20090029240".pn.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 14:29
S27	9	("2009/0029240").URPN.	USPAT	OR	ON	2021/05/18 14:30
S28	1	"20130108903".pn.	US-PGPUB; USPAT	OR	ON	2021/05/18 14:31
S29	100	("4053687" "4237199" "4283843" "4539273" "5631102").PN. OR ("5849431").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 14:37
S30	1	"20200295341".pn.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/18 15:13
S32	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead terminal)	FPRS; EPO; JPO; DERWENT	OR	ON	2021/05/27 11:35

S36	3	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	FPRS; EPO; JPO; DERWENT	OR	ON	2021/05/27 15:37
S37	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	US-PGPUB; USPAT	OR	ON	2021/05/27 15:40
S38	0	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not S37	US-PGPUB; USPAT	OR	ON	2021/05/27 16:00
S39	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not S37	US-PGPUB; USPAT	OR	ON	2021/05/27 16:00
S40	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not S37	US-PGPUB; USPAT	OR	ON	2021/05/27 16:04
S41	4	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not S37 not S40	US-PGPUB; USPAT	OR	ON	2021/05/27 16:15
S42	8	"20050287428".pn. "20090148766".pn. "20090297941".pn. "20110076547".pn. "20110076547".pn. "20160197379".pn. "7923139".pn. "7955736".pn. "8450006".pn.	US-PGPUB; USPAT	OR	ON	2021/05/27 16:18
S43	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not S37 not S40	US-PGPUB; USPAT	OR	ON	2021/05/27 16:20

S44	1	(15/573571).APP.	US-PGPUB; USOCR	OR	ON	2021/05/27 16:35
S45	2	("20130316209" "20150050535").PN. OR ("10566598").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/27 16:35
S46	1	(13/984779).APP.	USPAT; USOCR	OR	ON	2021/05/27 16:38
S47	15	("20060240319" "20070269711" "20080241646" "20090136835" "20100124694" "20100129719" "20100310910" "6019802" "6465134" "6589694" "7261792" "7642015" "7935445" "9577231" "9655393").PN. OR ("10658633").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2021/05/27 16:38
S48	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	US-PGPUB; USPAT	OR	ON	2021/05/28 09:49
S49	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not S48	US-PGPUB; USPAT	OR	ON	2021/05/28 09:49
S50	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not S48 not S49	US-PGPUB; USPAT	OR	ON	2021/05/28 09:49
S51	140	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not S48 not S49 not S50	US-PGPUB; USPAT	OR	ON	2021/05/28 09:49
S52	21	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not S48 not S49 not S50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	US-PGPUB; USPAT	OR	ON	2021/05/28 09:59
S53	274	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) not S48 not S49 not S50 and	US-PGPUB; USPAT	OR	ON	2021/05/28 10:19

		(wind\$4 wound jelly\$2roll) near4 (cell battery unit) not S52				
S54	74	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not S48 not S49 not S50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not S52	US-PGPUB; USPAT	OR	ON	2021/05/28 10:22
S55	2	(h01m50/\$.cpc.) and (mea membrane near3 electrode near3 assembly) with (terminal lead tab connect\$4) near4 (side end) not S48 not S49 not S50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not S52 not S54	US-PGPUB; USPAT	OR	ON	2021/05/28 10:40
S56	1	"20130108903".pn.	US-PGPUB; USPAT	OR	ON	2021/05/28 14:56
S57	4	"20070196730".pn. "20100081052".pn. "20090029240".pn. "20110067227".pn.	US-PGPUB; USPAT	OR	ON	2021/05/28 15:04
S58	0	"20100081052".pn. and second near3 embodiment with (roll\$4 coil\$4)	US-PGPUB; USPAT	OR	ON	2021/05/28 15:07
S59	0	"20100081052".pn. and second near3 embodiment same2 (roll\$4 coil\$4)	US-PGPUB; USPAT	OR	ON	2021/05/28 15:08

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	2	"20100081052".pn. "20130108903".pn.	US-PGPUB; USPAT	OR	ON	2021/09/17 15:17
L2	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	US-PGPUB; USPAT	OR	ON	2021/09/17 15:41
L3	28	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L2	US-PGPUB; USPAT	OR	ON	2021/09/17 15:41
L4	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L2 not L3	US-PGPUB; USPAT	OR	ON	2021/09/17 15:41
L5	22	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L2 not L3 not L4 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	US-PGPUB; USPAT	OR	ON	2021/09/17 15:41
L6	77	(h01m50/\$.cpc. h01m50/\$.ipc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L2 not L3 not L4 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L5	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2021/09/17 15:41
L8	1	"20110067227".pn.	US-PGPUB; USPAT	OR	ON	2021/09/17 15:50
L9	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	US-PGPUB; USPAT	OR	ON	2021/09/17 15:58

L10	40	(h01m50/\$.cpc. h01m\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L9	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2021/09/17 15:58
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9/17/2021 4:10:25 PM
C:\Users\aechemeyer\Documents\EAST\Workspaces\16498751.wsp

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S73	1	"20130252054".pn.	US-PGPUB; USPAT	OR	ON	2022/03/03 10:11
S74	28	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) near4 (end top bottom) with (electrode negative positive cathode anode) near4 (tab lead) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2022/03/03 11:30
S75	96	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not S74	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2022/03/03 11:33
S76	19	("4271241" "4822377").PN. OR ("5171647").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2022/03/03 11:40
S77	42	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar pad slab disc layer sheet panel) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not S74 not S75	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT	OR	ON	2022/03/03 11:43
S78	32	("3262819" "3415690" "3861960" "3986083" "3994749" "4271241" "4523376").PN. OR ("4822377").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2022/03/03 11:48
S79	3	"4822377".pn. "20110067227".pn. "20130252054".pn.	US-PGPUB; USPAT	OR	ON	2022/03/03 11:55
S80	1	(13/861455).APP.	USPAT; USOCR	OR	ON	2022/03/03 13:21
S81	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2022/03/03 13:21
S82	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797"	US-PGPUB; USPAT; USOCR	OR	ON	2022/03/03 13:22

		"20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.				
S83	3	("2013/0252054").URPN.	USPAT	OR	ON	2022/03/03 13:22
S84	17	("20040029008" "20060123609" "20060194108" "20090239133" "20100124694" "20100281685" "20130252054" "5250784" "6337758" "6534952" "6746796" "6967828" "6995971" "7875087" "8974955").PN. OR ("9373865").URPN.	US-PGPUB; USPAT; USOCR	OR	ON	2022/03/03 13:23
S85	3	"4822377".pn. "20130108903".pn. "20130252054".pn.	US-PGPUB; USPAT	OR	ON	2022/03/03 13:25
S86	3	"4822377".pn. "20110067227".pn. "20130108903".pn.	US-PGPUB; USPAT	OR	ON	2022/03/03 13:25
S87	1	"20110067227".pn. and tab near3 group with (plural\$4 number additional)	US-PGPUB; USPAT	OR	ON	2022/03/03 13:33
S88	1	"20130108903".pn. and electrode near3 tab with (across plural\$4)	US-PGPUB; USPAT	OR	ON	2022/03/03 13:37
S89	1	"20200295341".pn. and outer near4 portion with end with electrode	US-PGPUB; USPAT	OR	ON	2022/03/03 13:41
S90	1	"20130252054".pn. and number near4 (tab winding "602" "604")	US-PGPUB; USPAT	OR	ON	2022/03/03 13:47

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L91	3	"20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:44 PM
L92	48	((US-20130108903-A1 OR US-20150072204-A1 OR US- 20180026295-A1 OR US-20100081052-A1 OR US-20070117011-A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240-A1 OR US- 20200343520-A1 OR US-20110151295-A1 OR US-20210005936-A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694-A1 OR US- 20160351966-A1 OR US-20110067227-A1 OR US-20070196730-A1 OR US- 20010023038-A1 OR US-20160351940-A1 OR US-20130252054-A1 OR US- 20100310926-A1 OR US-20190027789-A1 OR US-20170125764-A1 OR US- 20110008670-A1 OR US-20190379028-A1 OR US-20050225926-A1 OR US- 20060024572-A1 OR US-20200295341-A1).did. AND PGPB.dbnm.) OR ((US- 8236441-B2 OR US- 6521374-B1 OR US- 6432574-B1 OR US- 8871373-B2 OR US- 5849431-A OR US- 9640833-B2 OR US- 6071638-A OR US- 7935445-B2 OR US- 5171647-A OR US- 4822377-A OR US- 8310809-B2 OR US-	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:45 PM

		8628877-B2 OR US-6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402-A1 OR EP-2325927-A1 OR JP-3754220-B2).did. AND FPRS.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
L93	12	("20150037638" "20110067227" "20140147714" "20070269685" "20120009450" "20050221178" "20110081574" "20110262779" "20160351940" "20100190056" "20160218343" "20130252054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:49 PM
L94	15	("2007/0269685").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:52 PM
L95	36	(Artificial Intelligence) More like doc: US-20070269685-A1	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:54 PM
L96	53	Text: (US-8084158-B2 OR EP-2181482-A1 OR WO-2009015282-A1 OR US-20090029240-A1 OR US-8236441-B2 OR EP-2181482-B1 OR KR-20100058501-A OR JP-2010534916-A OR JP-5378377-B2 OR KR-101566561-B1 OR EP-2429010-B1 OR EP-2429010-A1 OR US-7927732-B2 OR JP-2009507345-A OR KR-101304125-B1 OR US-4322484-A OR WO-2007028152-A2 OR EP-1946393-A2 OR US-20070117011-A1 OR KR-20080050452-A).did.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:54 PM
		("3823040" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237"					

		"6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "RE38518" "6780303" "6875540" "6884541" "7033697" "7927732" "20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20070117011" "20090109099").pn. OR ("8084158").urpn. AND (PGPB USPT USOC).dbnm.					
L97	33	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 04:59 PM
L98	1	"20190348703".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:01 PM
L99	1	"11152638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:02 PM
L100	11	("9299967" "20020068217" "20090208825" "20110262779" "20120009450" "20140186670" "20140199571" "20170084901" "20170092926" "20170250437" "20170346064").pn. OR ("11152638").urpn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 05:02 PM

		AND (PGPB USPT USOC).dbnm.					
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L2	14	("20010023038" "20050003264" "20150295270" "20110067227" "20120009450" "20160351966" "20110293977" "20100310926" "6569565" "20070196730" "20050287428" "20130344364" "20190379028" "20120208055").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:34 PM
L3	9	("20150037638" "20110067227" "20140147714" "20120009450" "20110081574" "20160351940" "20100190056" "20160218343" "20130252054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:41 PM
L4	1	"20100310926".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 02:46 PM
L5	21124	(h01m10/\$.cpc. h01m50/\$.cpc.) and (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 02:56 PM
L6	260	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) and (kim-j\$.inv. kim-h\$.inv. kim-s\$.inv.) and (electrode cathode anode) with (tab notch) near4 (overlap\$4 plural\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 02:57 PM
L7	144	(h01m10/\$.cpc. h01m50/\$.cpc. h01m10/\$.ipc. h01m50/\$.ipc.) and (electrode cathode anode) with (tab notch) near4 (overlap\$4 plural\$4) and (battery mea electrode near3 assembly) with (wound jelly near3 roll cylindrical)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 04:11 PM
L8	132	L7 not L6	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/17 04:12 PM
L9	8	("20050277019" "20090169990" "20090317707" "20030235736" "20060147793" "2002019254	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/17 04:15 PM

L10	0	8" "20060110658" "20070154797").PN. ("2016/0351966").URP N.	(USPAT)	OR	ON	ON	2021/05/17 04:31 PM
L11	1	(15/115220).APP.	(USPAT; USOCR)	OR	ON	ON	2021/05/17 04:31 PM
L12	10	("20050287428" "20060024572" "20070196730" "20120251854" "20120328924" "20130216879" "20140045041" "7364817" "7700222" "7955736").PN. OR ("10381688").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:31 PM
L13	6	("2007/0196730").URP N.	(USPAT)	OR	ON	ON	2021/05/17 04:33 PM
L14	10	("3490949" "4053687" "6376121" "6596434" "6818025" "7332246" "7364817").PN. OR ("7700222").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:33 PM
L15	14	("20020164524" "20030118912" "20050225926" "5736270" "6071638" "6222720" "6328769" "7364817" "7618742" "7670715" "7678501" "7700222").PN. OR ("8310809").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:34 PM
L16	11	("20020191370" "20030086239" "20060024572" "20060240322" "20060240323" "3237274" "5849431" "6456484" "6896993" "8310809").PN. OR ("8628877").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/17 04:35 PM
L17	13	("3761314" "5525441" "5635312" "5736270").PN. OR ("6328769").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 10:22 AM
L18	44	("3761314").URPN.	(USPAT)	OR	ON	ON	2021/05/18 10:23 AM
L19	5	("20050142436" "3761314").PN. OR ("7955736").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 10:24 AM
L20	3	("2005/0142436").URP N.	(USPAT)	OR	ON	ON	2021/05/18 10:34 AM
L21	1863	h01m50/538.cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 11:26 AM

L22	4637	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 01:20 PM
L23	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) with (plural\$4 several) near4 (tab notch lead terminal)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 01:43 PM
L24	40	("20010038945" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20070117011" "20090109099" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6080506" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "RE38518").PN. OR ("7927732").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:24 PM
L25	49	("20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190"	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:29 PM

		"20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.					
L26	1	"20090029240".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:29 PM
L27	9	("2009/0029240").URP N.	(USPAT)	OR	ON	ON	2021/05/18 02:30 PM
L28	1	"20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/18 02:31 PM
L29	100	("4053687" "4237199" "4283843" "4539273" "5631102").PN. OR ("5849431").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 02:37 PM
L30	1	"20200295341".pn.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/18 03:13 PM
L32	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead terminal)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/27 11:35 AM
L36	3	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/05/27 03:37 PM
L37	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 03:40 PM

L38	0	remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal (h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:00 PM
L39	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:00 PM
L40	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L37	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:04 PM
L41	4	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L37 not L40	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:15 PM
L42	8	"20050287428".pn. "20090148766".pn. "20090297941".pn. "20110076547".pn. "20110076547".pn. "20160197379".pn. "7923139".pn. "7955736".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:18 PM

L43	26	"8450006".pn. (h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L37 not L40	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/27 04:20 PM
L44	1	(15/573571).APP.	(US-PGPUB; USOCR)	OR	ON	ON	2021/05/27 04:35 PM
L45	2	("20130316209" "20150050535").PN. OR ("10566598").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/27 04:35 PM
L46	1	(13/984779).APP.	(USPAT; USOCR)	OR	ON	ON	2021/05/27 04:38 PM
L47	15	("20060240319" "20070269711" "20080241646" "20090136835" "20100124694" "20100129719" "20100310910" "6019802" "6465134" "6589694" "7261792" "7642015" "7935445" "9577231" "9655393").PN. OR ("10658633").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2021/05/27 04:38 PM
L48	66	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L49	27	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L48	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L50	26	(h01m50/\$.cpc.) and (uncoated (free no	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM

L51	140	remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:49 AM
L52	21	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L48 not L49 not L50	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 09:59 AM
L53	274	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:19 AM
L54	74	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:22 AM
L55	2	(h01m50/\$.cpc.) and (mea membrane near3 electrode near3	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 10:40 AM

		assembly) with (terminal lead tab connect\$4) near4 (side end) not L48 not L49 not L50 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L52 not L54					
L56	1	"20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 02:56 PM
L57	4	"20070196730".pn. "20100081052".pn. "20090029240".pn. "20110067227".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:04 PM
L58	0	"20100081052".pn. and second near3 embodiment with (roll\$4 coil\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:07 PM
L59	0	"20100081052".pn. and second near3 embodiment same2 (roll\$4 coil\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 03:08 PM
L60	1	"20180145304".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/05/28 05:13 PM
L61	2	"20100081052".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:17 PM
L62	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L63	28	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and (electrode anode cathode positive negative) with terminal not L62	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L64	26	(h01m50/\$.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM

L65	22	join\$4 fasten\$4) not L62 not L63 (h01m50/\$.cpc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near3 (attach\$4 add\$4 fix\$4 secur\$4 append\$4 join\$4 fasten\$4) not L62 not L63 not L64 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit)	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:41 PM
L66	77	(h01m50/\$.cpc. h01m50/\$.ipc.) and (uncoated (free no remov\$4 without)) near4 (edge top length) same (terminal lead tab connect\$4) near4 (side end) not L62 not L63 not L64 and (wind\$4 wound jelly\$2roll) near4 (cell battery unit) not L65	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/09/17 03:41 PM
L67	1	"20110067227".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:50 PM
L68	68	(h01m50/538.cpc. h01m10/0422.cpc. h01m50/531.cpc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal	(US-PGPUB; USPAT)	OR	ON	ON	2021/09/17 03:58 PM
L69	40	(h01m50/\$.cpc. h01m\$.ipc.) and (uncoated (free no remov\$4 without) near3 active near3 material) same (plural\$4 several) near4 (tab notch lead) and electrode with terminal not L68	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2021/09/17 03:58 PM
L70	3	"20100081052".pn. "20130108903".pn. "20180145304".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/01/18 12:49 PM
L71	1	"20200295341".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/08 09:51 AM
L72	59	("20010038945" "20030035993"	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/02/08 09:56 AM

		"20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20060210857" "20070117011" "20090109099" "3823040" "4049888" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "6780303" "6875540" "6884541" "7033697" "7927732" "8084158" "RE38518").PN. OR ("8236441").URPN.					
L73	1	"20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 10:11 AM
L74	28	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) near4 (end top bottom) with (electrode negative positive cathode anode) near4 (tab lead) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:30 AM
L75	96	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:33 AM

L76	19	L74 ("4271241" "4822377").PN. OR ("5171647").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 11:40 AM
L77	42	(h01m50/\$.cpc. h01m50/\$.ipc.) and insulat\$4 near4 (member plate collar pad slab disc layer sheet panel) with (electrode negative positive cathode anode) near4 (tab lead) with (cell unit mea membrane near3 assembly) and cell with (wind\$4 wound\$4 roll\$4 jelly near4 roll\$4) not L74 not L75	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/03/03 11:43 AM
L78	32	("3262819" "3415690" "3861960" "3986083" "3994749" "4271241" "4523376").PN. OR ("4822377").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 11:48 AM
L79	3	"4822377".pn. "20110067227".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 11:55 AM
L80	1	(13/861455).APP.	(USPAT; USOCR)	OR	ON	ON	2022/03/03 01:21 PM
L81	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:21 PM
L82	11	("20020192548" "20030235736" "20050277019" "20060110658" "20060147793" "20070154797" "20090169990" "20090317707" "20100124694" "5580679" "5662566").PN. OR ("9300004").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:22 PM
L83	3	("2013/0252054").URP N.	(USPAT)	OR	ON	ON	2022/03/03 01:22 PM

L84	17	("20040029008" "20060123609" "20060194108" "20090239133" "20100124694" "20100281685" "20130252054" "5250784" "6337758" "6534952" "6746796" "6967828" "6995971" "7875087" "8974955").PN. OR ("9373865").URPN.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/03/03 01:23 PM
L85	3	"4822377".pn. "20130108903".pn. "20130252054".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:25 PM
L86	3	"4822377".pn. "20110067227".pn. "20130108903".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:25 PM
L87	1	"20110067227".pn. and tab near3 group with (plural\$4 number additional)	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:33 PM
L88	1	"20130108903".pn. and electrode near3 tab with (across plural\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:37 PM
L89	1	"20200295341".pn. and outer near4 portion with end with electrode	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:41 PM
L90	1	"20130252054".pn. and number near4 (tab winding "602" "604")	(US-PGPUB; USPAT)	OR	ON	ON	2022/03/03 01:47 PM
L91	3	"20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:44 PM
L92	48	((US-20130108903-A1 OR US-20150072204- A1 OR US- 20180026295-A1 OR US-20100081052-A1 OR US-20070117011- A1 OR US- 20030211388-A1 OR US-20010038945-A1 OR US-20090029240- A1 OR US- 20200343520-A1 OR US-20110151295-A1 OR US-20210005936- A1 OR US- 20180145304-A1 OR US-20080026293-A1 OR US-20100124694- A1 OR US- 20160351966-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2022/08/25 04:45 PM

		US-20110067227-A1 OR US-20070196730-A1 OR US-20010023038-A1 OR US-20160351940-A1 OR US-20130252054-A1 OR US-20100310926-A1 OR US-20190027789-A1 OR US-20170125764-A1 OR US-20110008670-A1 OR US-20190379028-A1 OR US-20050225926-A1 OR US-20060024572-A1 OR US-20200295341-A1).did. AND PGPB.dbnm.) OR ((US-8236441-B2 OR US-6521374-B1 OR US-6432574-B1 OR US-8871373-B2 OR US-5849431-A OR US-9640833-B2 OR US-6071638-A OR US-7935445-B2 OR US-5171647-A OR US-4822377-A OR US-8310809-B2 OR US-8628877-B2 OR US-6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402-A1 OR EP-2325927-A1 OR JP-3754220-B2).did. AND FPRS.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
L93	12	("20150037638" "20110067227" "20140147714" "20070269685" "20120009450" "20050221178" "20110081574" "20110262779" "20160351940" "20100190056" "20160218343" "20130252054").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 04:49 PM
L94	15	("2007/0269685").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:52 PM
L95	36	(Artificial Intelligence)	(US-PGPUB; USPAT;	OR	ON	ON	2022/08/25

		More like doc: US-20070269685-A1 Text: (US-8084158-B2 OR EP-2181482-A1 OR WO-2009015282-A1 OR US-20090029240- A1 OR US-8236441-B2 OR EP-2181482-B1 OR KR-20100058501-A OR JP-2010534916-A OR JP-5378377-B2 OR KR- 101566561-B1 OR EP- 2429010-B1 OR EP- 2429010-A1 OR US- 7927732-B2 OR JP- 2009507345-A OR KR- 101304125-B1 OR US- 4322484-A OR WO- 2007028152-A2 OR EP-1946393-A2 OR US-20070117011-A1 OR KR-20080050452- A).did.	USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				04:54 PM
L96	53	("3823040" "4383013" "4400415" "4966822" "5464705" "5554459" "5773164" "5834133" "5866274" "5876237" "6071638" "6080506" "6083639" "6159253" "6197074" "6344292" "6432574" "6521374" "6673128" "RE38518" "6780303" "6875540" "6884541" "7033697" "7927732" "20010038945" "20030035993" "20030091893" "20030175587" "20030194601" "20030211388" "20040023108" "20040121230" "20040157115" "20040157120" "20040214076" "20040237290" "20040265683" "20050255378" "20060124973" "20060172190" "20070117011" "20090109099").pn. OR ("8084158").urpn. AND (PGPB USPT	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 04:54 PM

L97	33	USOC).dbnm. (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 04:59 PM
L98	1	"20190348703".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:01 PM
L99	1	"11152638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 05:02 PM
L100	11	("9299967" "20020068217" "20090208825" "20110262779" "20120009450" "20140186670" "20140199571" "20170084901" "20170092926" "20170250437" "20170346064").pn. OR ("11152638").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 05:02 PM
L101	1	"9012053".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:16 PM
L102	19	("2291110" "3507703" "3607417" "4816352" "7060387" "7276313" "20020061435" "20040161662" "20040202924" "20050142436" "20060240319" "20100178559").pn. OR ("9012053").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:16 PM
L103	13	("2012/0009450").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:23 PM
L104	87	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:31 PM

L105	668	central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover) (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover) NOT L97	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L106	0	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) near4 (plate cover layer) NOT L97	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L107	131	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) near4 (plate cover layer)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:32 PM
L108	0	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L97	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:32 PM
L109	231	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:37 PM

L110	174	(plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/08/25 08:37 PM
L111	1	("2014/0050956").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:42 PM
L112	15	("20020061435" "20060127751" "20090239133" "20100190056" "20110067227" "20110104541" "20110151295" "20120279053" "20140050956" "20160164133" "20170250437" "20170365839" "20180026295" "20180040918" "20190379028").pn. OR ("10700384").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	OR	ON	ON	2022/08/25 08:43 PM
L113	43	(Artificial Intelligence) More like doc: US-20140050956-A1 Text: (KR-20150030537-A OR TW-201407859-A OR JP-3187576-U OR CN-114050324-A OR KR-20180048419-A OR KR-102103900-B1 OR US-10700384-B2 OR EP-3514876-A1 OR US-20190027789-A1 OR CN-103594736-A	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT;	OR	ON	ON	2022/08/25 08:46 PM

		OR CN-108496269-B OR TW-201244225-A OR US-20040191625-A1 OR CN-102760905-A OR EP-3136494-A1 OR KR-20150122340-A OR CN-101714651-B OR CN-108496269-A OR CN-101714651-A OR CN-202308131-U).did.	IBM_TDB)				
L114	0	20150037638.pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:48 PM
L115	1	"20150037638".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/08/25 08:48 PM
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		A1).did. AND PGPB.dbnm.) OR ((US-8236441-B2 OR US-6521374-B1 OR US-6432574-B1 OR US-8871373-B2 OR US-5849431-A OR US-9640833-B2 OR US-6071638-A OR US-7935445-B2 OR US-5171647-A OR US-4822377-A OR US-8310809-B2 OR US-8628877-B2 OR US-6456484-B1).did. AND USPT.dbnm.) OR ((WO-2017047278-A1 OR WO-2012051402-A1 OR EP-2325927-A1 OR JP-3754220-B2 OR CN-114128033-A).did. AND FPRS.dbnm.) OR ((JP-2009289714-A OR JP-2013187077-A).did. AND JPAB.dbnm.) OR ((US-3761314-A).did. AND USOC.dbnm.)					
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L123	6	"20140050956".pn. "20130108903".pn. "20120009450".pn. "20110067227".pn. "20130252054".pn. "4822377".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/09/07 02:36 PM
L124	2	("20060127774" "20100 190056").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 12:17 PM
L125	183	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc.	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 12:40 PM

L126	197	h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) (h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:01 PM
L127	38	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) AND (tan lead terminal) WITH (overlap\$4 overl\$4 shingl\$4 superimpos\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:03 PM
L128	57	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4) NOT L107 AND (wound wind\$4 roll\$4) AND (tab lead terminal) WITH (overlap\$4 overl\$4 shingl\$4 superimpos\$4)	(US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:03 PM
L129	1	"20160351966".pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 01:09 PM

L130	100	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl%5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:42 PM
L131	105	(h01m50/\$.cpc. h01m10/\$.cpc. h01m50/\$.ipc. h01m10/\$.ipc.) AND (plural\$4 several multipl\$5) near4 (tab notch lead terminal) AND (center middle central inner) near4 (tab notch lead terminal) WITH (insulat\$4 plate cover)	(FPRS; EPO; JPO; DERWENT)	OR	ON	ON	2022/12/16 01:43 PM
L132	0	(h01m50/538,536.cpc. h01m10/1422,0431.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:07 PM
L133	0	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) near4 (equal\$4 axis around about)	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:07 PM
L134	0	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) SAME tab WITH ben\$4 same2 tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L135	86	(h01m50/\$.cpc. h01m10/\$.cpc.) AND	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM

L136	73423	(wound roll\$4 cylind\$5) SAME tab WITH ben\$4 .clm. (h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) tab WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L137	168	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) .clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:08 PM
L138	27	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (distribut\$4 dispers\$4 allocat\$4 allot\$4 apportion\$4 ration\$4) same3 (tab electrode) WITH (overlap\$4 laminat\$4).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:09 PM
L139	1101	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (tab electrode) WITH (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM
L140	104	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) WITH (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM
L141	56	(h01m50/\$.cpc. h01m10/\$.cpc.) AND (wound roll\$4 cylind\$5) same2 (mutipl\$4 plural\$4) near3 (tab electrode) near3 (overlap\$4 laminat\$4).clm. NOT L138	(US-PGPUB; USPAT)	OR	ON	ON	2022/12/16 02:11 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



(12) **United States Patent**
Kim et al.

(10) **Patent No.:** **US 11,652,232 B2**
(45) **Date of Patent:** **May 16, 2023**

(54) **SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME**

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Sang Uck Kim, Daejeon (KR)

(73) Assignee: **LG Energy Solution, Ltd.**

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 101 days.

(21) Appl. No.: **16/498,751**

(22) PCT Filed: **Apr. 13, 2018**

(86) PCT No.: **PCT/KR2018/004366**

§ 371 (c)(1),

(2) Date: **Sep. 27, 2019**

(87) PCT Pub. No.: **WO2018/190691**

PCT Pub. Date: **Oct. 18, 2018**

(65) **Prior Publication Data**

US 2020/0295341 A1 Sep. 17, 2020

(30) **Foreign Application Priority Data**

Apr. 14, 2017 (KR) 10-2017-0048645
Apr. 13, 2018 (KR) 10-2018-0043191

(51) **Int. Cl.**

H01M 50/538 (2021.01)
H01M 10/04 (2006.01)
H01M 50/536 (2021.01)

(52) **U.S. Cl.**

CPC ... **H01M 10/0422** (2013.01); **H01M 10/0431** (2013.01); **H01M 50/538** (2021.01); **H01M 50/536** (2021.01)

(58) **Field of Classification Search**

CPC H01M 50/538; H01M 50/531; H01M 10/0422

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

4,822,377 A * 4/1989 Wolff H01M 50/183 29/422

6,569,565 B2 5/2003 Ligeois et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 201450072 U 5/2010
CN 102024936 A 4/2011

(Continued)

OTHER PUBLICATIONS

International Search Report from Application No. PCT/KR2018/004366 dated Jul. 20, 2018, pp. 1-2.

(Continued)

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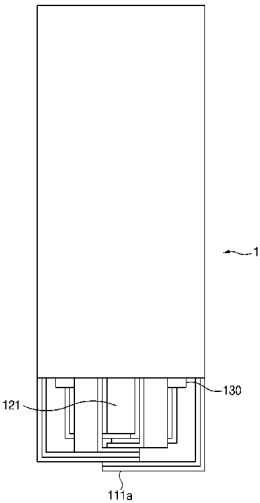
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ABSTRACT

The present invention relates to a secondary battery in which an electrode tab is improved in mechanical strength and a method for manufacturing the same.

Also, the secondary battery according to the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector and a notching tab part extending from the coating portion in a width direction of the electrode collector without coating with the active material and overlapping each other to form two or more layers when the electrode is wound.

15 Claims, 9 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2001/0023038 A1 9/2001 Ligeois et al.
2005/0003264 A1 1/2005 Oh et al.
2005/0287428 A1 12/2005 Cheon et al.
2006/0127774 A1 6/2006 Kim et al.
2007/0196730 A1 8/2007 Kozuki
2010/0081052 A1* 4/2010 Morishima H01M 4/485
429/211
2010/0190056 A1 7/2010 Turner
2010/0310926 A1 12/2010 Wang et al.
2011/0067227 A1* 3/2011 Sohn H01M 4/70
29/623.1
2011/0293977 A1 12/2011 Kim et al.
2012/0009450 A1* 1/2012 Chun H01M 50/538
429/94
2012/0208055 A1 8/2012 Ahn
2013/0108903 A1* 5/2013 Martin H01M 50/172
29/730
2013/0252054 A1* 9/2013 Barone H01M 50/538
429/94
2013/0344364 A1 12/2013 Nanaumi et al.
2014/0050956 A1* 2/2014 Huang H01M 10/052
29/623.5
2015/0295270 A1 10/2015 Chun
2016/0351966 A1* 12/2016 Batson H01M 10/052
2018/0145304 A1* 5/2018 Kogetsu H01M 4/02
2019/0379028 A1 12/2019 Lim et al.

FOREIGN PATENT DOCUMENTS

CN 203250828 U 10/2013
CN 204577523 U 8/2015

EP 1134820 A1 9/2001
JP H0935701 A 2/1997
JP H11339756 A 12/1999
JP 2003338276 A 11/2003
JP 2005276459 A 10/2005
JP 2006128106 A 5/2006
JP 2007227137 A 9/2007
JP 2008066040 A 3/2008
JP 2011065981 A 3/2011
JP 5790576 B2 10/2015
JP 2016001575 A 1/2016
JP 2020509534 A 3/2020
KR 20000051740 A 8/2000
KR 100515833 B1 9/2005
KR 20050121904 A 12/2005
KR 20060022358 A 3/2006
KR 20100071941 A 6/2010
KR 20120006389 A 1/2012
KR 20120092367 A 8/2012
KR 20130073329 A 7/2013
KR 20150117135 A 10/2015
KR 20160113612 A 9/2016
KR 20170006437 A 1/2017
WO 2012086690 A1 6/2012
WO 2014188501 A1 11/2014

OTHER PUBLICATIONS

Extended European Seach Report including Written Opinion for Application No. EP18784869.2 dated Jan. 21, 2020, 5 pages.
Chinese Search Report for Application No. 201880025247, dated Aug. 24, 2021, 2 pages.

* cited by examiner

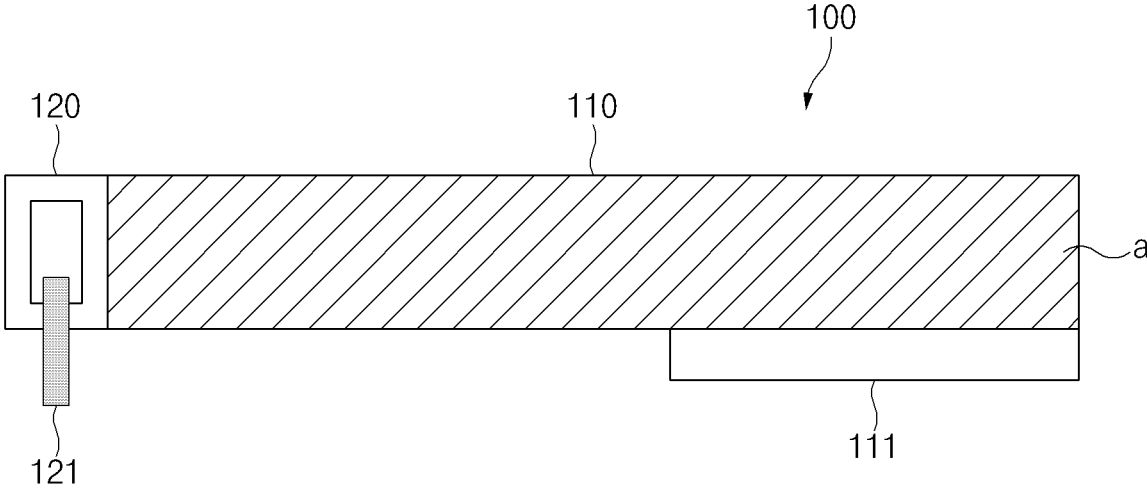


FIG.1



FIG.2

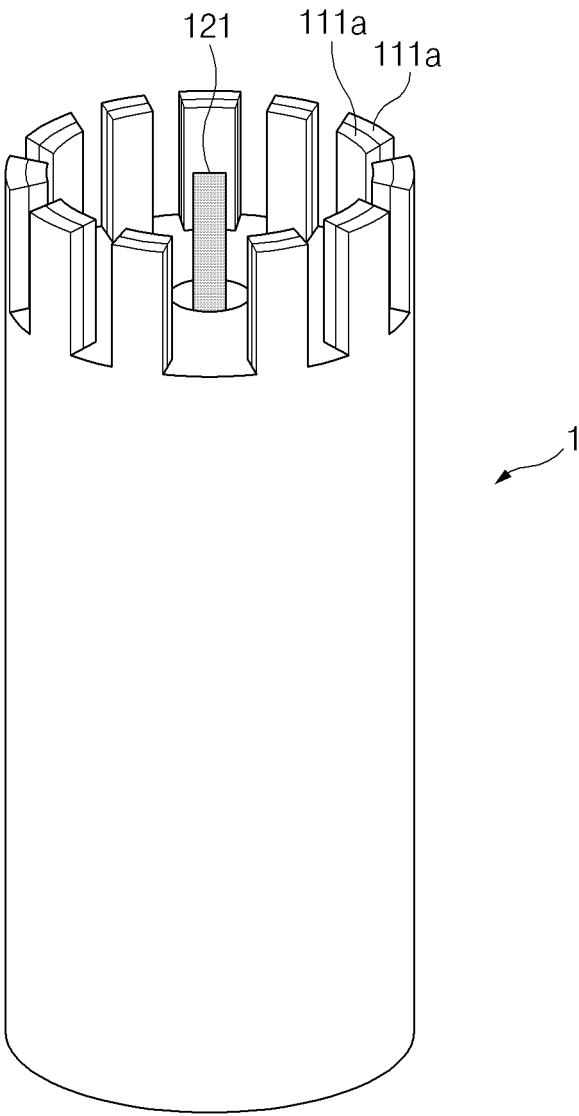


FIG. 3

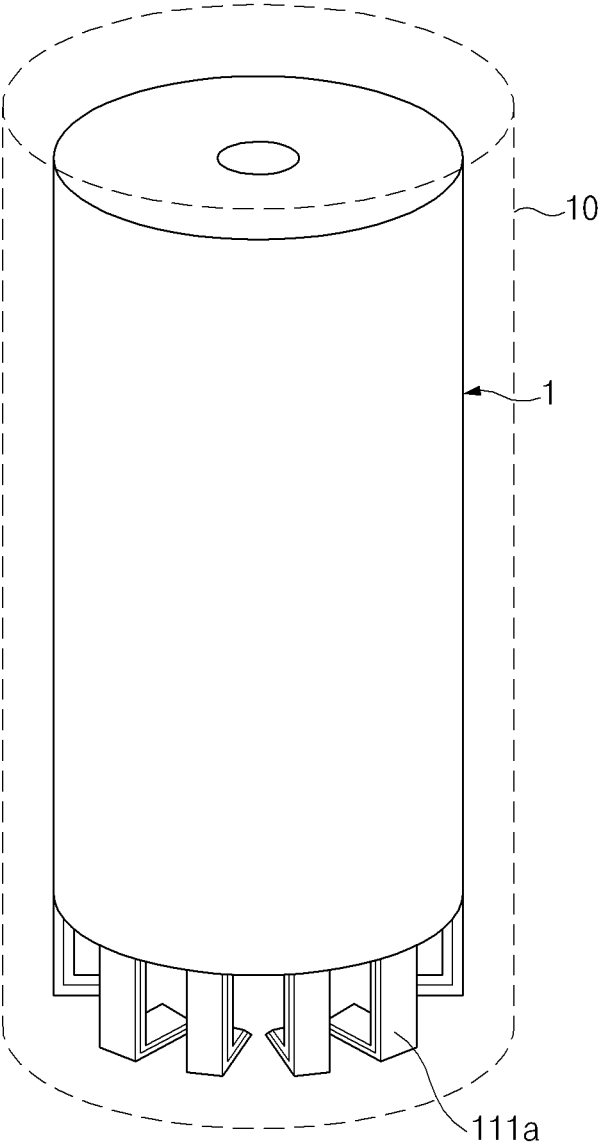


FIG.4

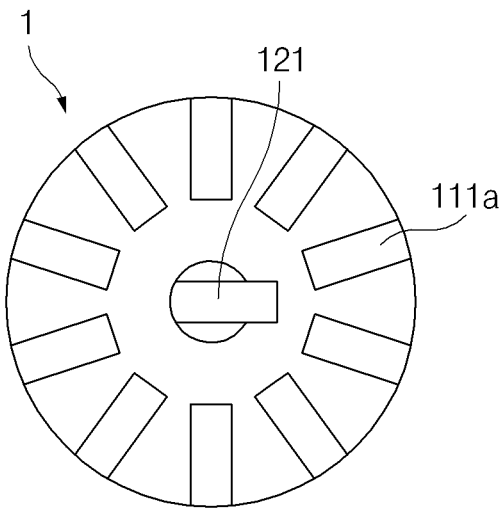


FIG. 5

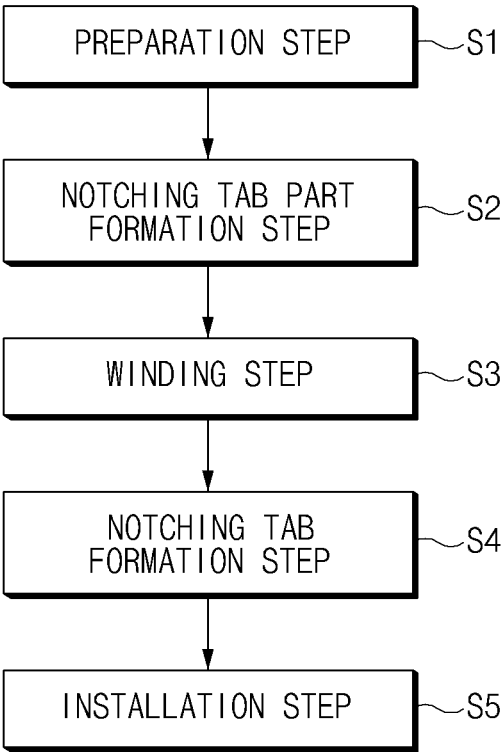


FIG.6

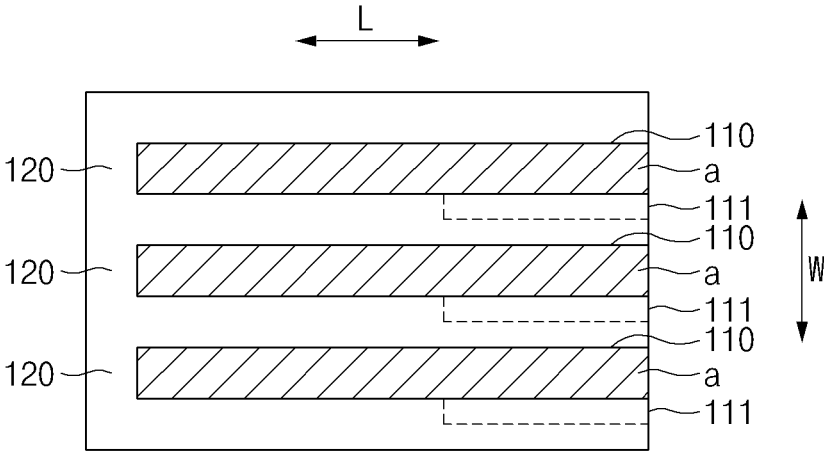


FIG.7

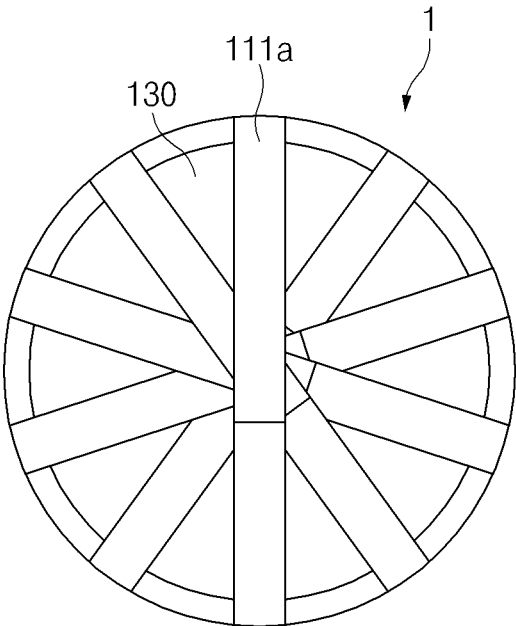


FIG. 8

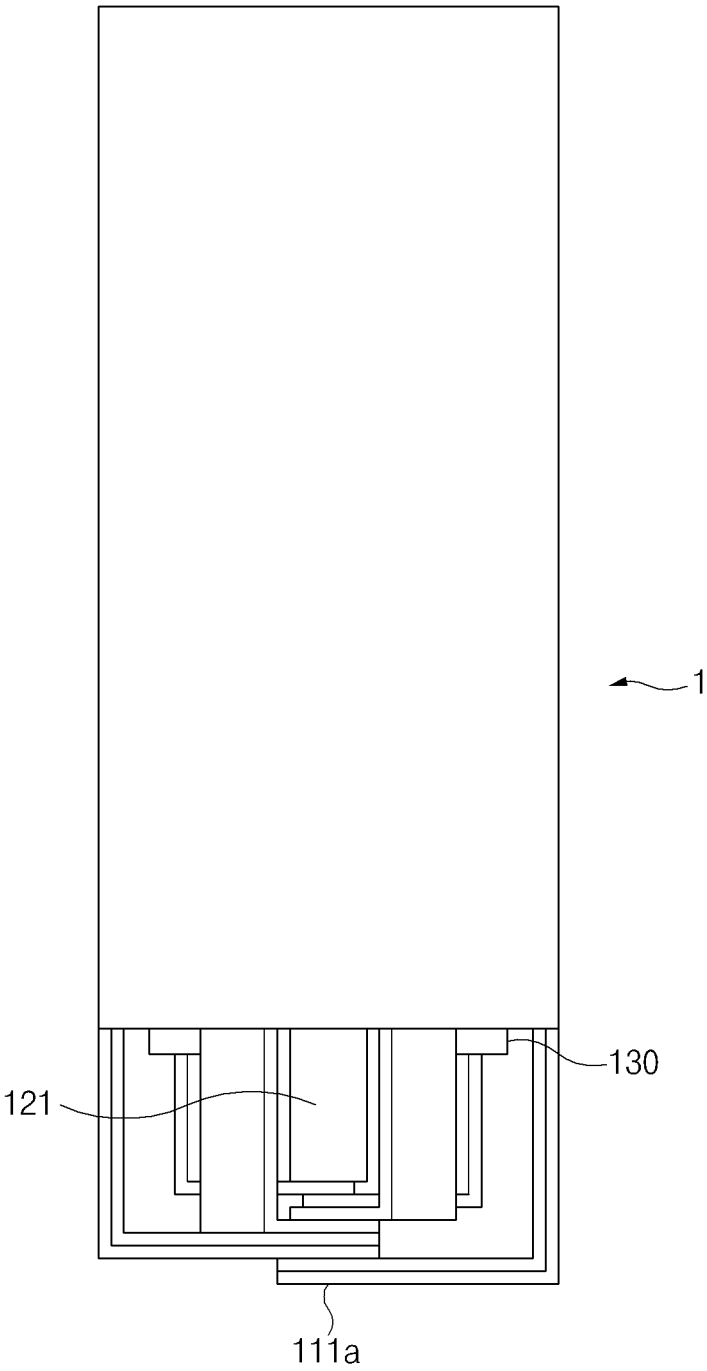


FIG. 9

SECONDARY BATTERY AND METHOD FOR
MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED
APPLICATION

The present application is a national phase entry under 35 U.S.C. § 371 of International Patent Application No. PCT/KR2018/004366, filed on Apr. 13, 2018, published in Korean, which claims priority from Korean Patent Application Nos. 10-2017-0048645, filed on Apr. 14, 2017, and 10-2018-0043191, filed on Apr. 13, 2018, the disclosures of which are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an electrode tab increases in mechanical strength and a method for manufacturing the same.

BACKGROUND ART

Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living environments surrounded by various electric and electronic devices.

Among such batteries, primary batteries and secondary batteries, which are chemical batteries using chemical reaction, are being generally used. The primary batteries are consumable batteries which are collectively referred to as dry batteries. Also, secondary batteries are rechargeable batteries that are manufactured by using a material in a redox process between current and a substance is repeatable several times. When the reduction reaction is performed on the material by the current, power is charged, and when the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

An electrode assembly and a secondary battery including the same according to the related art is disclosed in Korea Patent Publication No. 10-2012-0006389.

In the secondary battery according to the related art, it is necessary to apply a technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a three-tab structure

in which an electrode tab is applied to each of a core and an outer portion, and a middle tab is attached to an intermediate portion.

However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the electrode. Thus, there is a problem that the battery capacity is lowered due to an increase of the non-coating portion.

DISCLOSURE OF THE INVENTION

Technical Problem

Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

Technical Solution

A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

The notching tab part may continuously extend across 1/3 or more of a total length of the coating portion in the longitudinal direction.

When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs.

The foil tab may have a width of 4 mm to 120 mm.

The electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may be connected to the can member.

A bent part formed by bending each of the plurality of notching tabs may be bonded to the can member.

The secondary battery may further include an insulation member disposed between the bent parts and an end of the electrode.

The secondary battery may further include an electrode tab attached to the non-coating portion.

When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs, and the electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may overlap the electrode tab.

The electrode tab and the plurality of notching tabs may be bonded together to the can member.

A method for manufacturing a secondary battery according to an embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; a notching tab

part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction; a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly; a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

The electrode may be a negative electrode.

During the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

during the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

During the notching tab part formation step, notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

During the notching tab formation step, the notching tab part may be cut using a laser.

During the installation step, the electrode assembly may be accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

During the installation step, a bent portion of each of the plurality of notching tabs may be welded to the can member.

During the installation step, bent portions of each of the plurality of notching tabs may overlap each other and may be bonded to the can member together.

During the winding step, the notching tab part may be wound to form two or more overlapping layers.

Advantageous Effects

According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

According to the present invention, the multi-tab may be formed to minimize the electric resistance.

According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be boned to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

According to the present invention, the notching tab and the electrode tab may be bonded to the can member at the same to secure the welding processability.

According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other, thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention.

FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

Hereinafter, a secondary battery and a method for manufacturing the same according to exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

Terms or words used in the specification and claims should not be construed as limited to a lexical meaning, and should be understood as appropriate notions by the inventor based on that he/she is able to define terms to describe his/her invention in the best way to be seen by others. Therefore, embodiments and drawings described herein are simply exemplary and not exhaustive, and it will be understood that various equivalents may be made to take the place of the embodiments.

In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience in description and clarity. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120, on which the active material a is not applied on the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active material. The electrode assembly 1 may be manufactured by stacking the positive electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a jelly-roll shape.

The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

The positive electrode coating portion may be manufactured, for example, by applying the positive electrode active material to a portion of at least one surface of the aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a negative electrode non-coating portion which is not coated with the negative electrode active material.

The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

The negative electrode coating portion may be manufactured, for example, by applying the negative electrode active material to a portion of at least one surface of the copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

The separator may be manufactured by applying a polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the positive electrode and the negative electrode.

In the present invention, the negative electrode will be described as an example.

The electrode 100 has one end on which the non-coating portion 120 is formed. An electrode tab 121 may be attached to the non-coating portion 120 formed on the one end of the electrode 100.

The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode collector may be formed to extend on an area corresponding to $\frac{1}{3}$ or more of the total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

Only the active material a may be removed from the coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the coating portion 110 of the electrode 100 to form the notching tab part 111.

Thus, there may be an effect of forming the multi-tab while minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

FIG. 3 is a perspective view illustrating a state in which the notching tab part is cut to form the notching tab in FIG. 2.

As illustrated in FIG. 3, the notching tab part 111 may be cut in the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, the electrode tab may decrease in resistance.

Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a width exceeding 5 mm, deformation in the process of bonding for the electrical connection with the can member may be difficult.

The notching tab 111a that is the foil tab may increase in mechanical strength to prevent the notching tab 111a from being broken in the bent state, and thus, the notching tab 111a may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

FIG. 4 is a projective perspective view when seeing through the can member in a state in which the electrode assembly is installed in the can member according to an embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

As illustrated in FIGS. 4 and 5, in the electrode assembly 1 of the secondary battery according to the present invention, a bent part of the notching tab 111a may be welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while being accommodated in the can member 10.

The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each other while the electrical connection is smooth to minimize the electric resistance.

FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and FIG. 9 is a side view of FIG. 8.

As illustrated in FIGS. 8 and 9, in a secondary battery according to another embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

Also, the electrode assembly 1 may be accommodated in a can member 10, and the plurality of notching tabs 111a connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

In the secondary battery according to another embodiment of the present invention, an insulation member 130 may be formed between a bent part of the notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

The insulation member 130 may insulate the notching tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming into contact with the electrode assembly 1.

The plurality of notching tabs 111a may be formed on an outer portion of the end of the electrode assembly to prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

The plurality of notching tabs 111a may be formed on the outside of the electrode assembly so that the insulation member 130 is installed between the plurality of notching tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

Hereinafter, a method of manufacturing a secondary battery according an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

As illustrated in FIG. 6, a method for manufacturing a secondary battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (S3), a notching tab formation step (S4), and an installation step (S5).

The preparation step (S1) is a step of preparing an electrode 100 in which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120 on which the active material a is not applied are provided in a longitudinal direction of the electrode collector.

Here, the electrode 100 may be a negative electrode. FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120, which are formed in a longitudinal direction L of the electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

In the notching tab part formation step (S2) is a step of notching the electrode 100 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

As described above, the notching of the notching tab part 111 in the electrode 100 may a step of notching a portion of the non-coating portion on which the active material a is not applied and which is formed in the width direction W of the coating portion 11 while cutting the electrode 100 in the electrode collector to form a notch tab part 11.

Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating portion 110 and the non-coating portion 120, which are arranged in parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

In the installation step (S5), each of the notching tab 111a may be bent so that the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the notching tab 111a is bent.

Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may not be easily broken.

The bent notching tab 111a may have an end that faces a center of a bottom surface of the can member 10.

The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end of the electrode assembly 1 so as to be welded and bonded to the can member 10 together with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

Although the secondary battery and the method for manufacturing the same according to the present invention has been described above with reference to the exemplary drawings, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

The invention claimed is:

1. A secondary battery comprising:

a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector, the non-coating portion being adjacent to the coating portion in a longitudinal direction of the electrode collector, the non-coating portion being disposed at a first end of the electrode collector, the electrode being accommodated in the can member;

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part extending from part of the coating portion at a second end of the electrode collector opposite in the longitudinal direction from the first end of the electrode collector, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state;

an electrode tab attached to the non-coating portion, such that when the electrode is in the wound state, the electrode tab extends from a central portion of an end of the electrode; and

an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each

notching tab extends towards a bottom surface of the can member only from an outer portion of the end of the electrode remote from the central portion,

wherein, when the electrode is in the wound state, the plurality of notching tabs are together equally circumferentially distributed about a central axis of the electrode,

wherein, when the electrode is in the wound state, each of the plurality of notching tabs is in a bent state and overlaps the electrode tab, and the electrode tab and the plurality of notching tabs are bonded together to the can member, and

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the insulation member, and the electrode tab extending through a central opening of the insulation member.

2. The secondary battery of claim 1,

wherein the notching tab part continuously extends across $\frac{1}{3}$ or more of a total length of the coating portion in the longitudinal direction.

3. The secondary battery of claim 1, wherein each of the plurality of notching tabs has a width of 4 mm to 5 mm.

4. The secondary battery of claim 1, wherein a bent part of each of the plurality of notching tabs is bonded to the can member.

5. The secondary battery of claim 4, wherein the insulation member is disposed between the bent parts and the end of the electrode.

6. A method for manufacturing the secondary battery of claim 1, the method comprising:

a preparation step of preparing the electrode provided with the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied to the electrode collector in the longitudinal direction of the electrode collector;

a notching tab part formation step of notching the electrode to form the notching tab part not coated with the active material and extending from the coating portion in the width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across $\frac{1}{3}$ or more of a total length of the coating portion in the longitudinal direction;

a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly;

a notching tab formation step of cutting the notching tab part of the electrode assembly to form the plurality of notching tabs; and

an installation step of accommodating the electrode assembly into the can member and bonding each of the plurality of notching tabs to the can member.

7. The method of claim 6, wherein the electrode is a negative electrode.

8. The method of claim 6, wherein, during the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied are arranged parallel to each other in the width direction of the electrode collector.

9. The method of claim 8, wherein, during the notching tab part formation step, the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the coating

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portion and the non-coating portion in the longitudinal direction of the electrode collector.

10. The method of claim 6, wherein, during the notching tab part formation step, notching is performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode. 5

11. The method of claim 6, wherein, during the notching tab formation step, the notching tab part is cut using a laser.

12. The method of claim 6, wherein, during the installation step, the electrode assembly is accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member. 10

13. The method of claim 12, wherein, during the installation step a bent portion of each of the plurality of notching tabs is welded to the can member. 15

14. The method of claim 12, wherein, during the installation step bent portions of each of the plurality of notching tabs overlap each other and are bonded to the can member together. 20

15. The method of claim 6, wherein, during the winding step, the notching tab part is wound to form two or more overlapping layers.

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BIB DATA SHEET

CONFIRMATION NO. 4723

SERIAL NUMBER 16/498,751	FILING or 371(c) DATE 09/27/2019 RULE	CLASS 429	GROUP ART UNIT 1729	ATTORNEY DOCKET NO. LGCHEM 3.3F-1654		
APPLICANTS LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF						
INVENTORS Jae Woong Kim, Daejeon, KOREA, REPUBLIC OF; Hyoung Kwon Kim, Daejeon, KOREA, REPUBLIC OF; Sang Uck Kim, Daejeon, KOREA, REPUBLIC OF;						
** CONTINUING DATA ***** This application is a 371 of PCT/KR2018/004366 04/13/2018						
** FOREIGN APPLICATIONS ***** REPUBLIC OF KOREA 10-2017-0048645 04/14/2017 REPUBLIC OF KOREA 10-2018-0043191 04/13/2018						
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 06/08/2020						
Foreign Priority claimed ☒ Yes ☐ No 35 USC 119(a-d) conditions met ☒ Yes ☐ No Verified and /ALIX ECHELMAYER Acknowledged EGGERDING/ Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY KOREA, REPUBLIC OF	SHEETS DRAWINGS 9	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 2
ADDRESS LGCHEM Lerner, David, Littenberg, Krumholz & Mentlik, LLP 20 Commerce Drive Cranford, NJ 07016 UNITED STATES						
TITLE Secondary Battery And Method For Manufacturing The Same						
FILING FEE RECEIVED 1580	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

Electronic Acknowledgement Receipt	
EFS ID:	37301470
Application Number:	16498751
International Application Number:	PCT/KR2018/004366
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Leonor Dominguez
Filer Authorized By:	
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	27-SEP-2019
Filing Date:	
Time Stamp:	14:20:24
Application Type:	U.S. National Stage under 35 USC 371

Payment information:	
Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1580
RAM confirmation Number	E20199QE20563594
Deposit Account	
Authorized User	
The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:	

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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DESCRIPTION

SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

5 CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims the benefit of the priority of Korean Patent Application Nos. 10-2017-0048645, filed on April 14, 2017, and 10-2018-0043191, filed on April 13, 2018, which are hereby incorporated by reference in their
10 entirety.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an electrode
15 tab increases in mechanical strength and a method for manufacturing the same.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated
20 electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living environments surrounded by various electric and electronic devices.

[0004] Among such batteries, primary batteries and secondary
25 batteries, which are chemical batteries using chemical

reaction, are being generally used. The primary batteries are consumable batteries which are collectively referred to as dry batteries. Also, secondary batteries are rechargeable batteries that are manufactured by using a material in a redox process between current and a substance is repeatable several times. When the reduction reaction is performed on the material by the current, power is charged, and when the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

[0005] A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

[0006] An electrode assembly and a secondary battery including the same according to the related art is disclosed

in Korea Patent Publication No. 10-2012-0006389.

[0007] In the secondary battery according to the related art, it is necessary to apply a technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

[0008] A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a three-tab structure in which an electrode tab is applied to each of a core and an outer portion, and a middle tab is attached to an intermediate portion.

[0009] However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the electrode. Thus, there is a problem that the battery capacity is lowered due to an increase of the non-coating portion.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0010] Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

[0011] Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

5 **TECHNICAL SOLUTION**

[0012] A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching tab part extending from the coating portion in a width direction of the electrode collector without coating with the active material and overlapping each other to form two or more layers when the electrode is wound.

[0013] The notching tab part may continuously extend from 1/3 or more of the total length of the coating portion.

[0014] In a state in which the electrode is wound, the notching tab part may be cut to form a plurality of notching tabs.

[0015] The foil tab may have a width of 4 mm to 120 mm.

[0016] The electrode may be accommodated in a can member, and the notching tab may be bent to be connected to the can member.

25 [0017] A bent part formed by bending the notching tab may be

bonded to the can member.

[0018] The secondary battery may further include an insulation member disposed between the bent part and an end of the electrode.

5 **[0019]** The secondary battery may further include an electrode tab attached to the non-coating portion.

[0020] In the state in which the electrode is wound, the notching tab part may be notched to form a plurality of notching tabs, and the electrode may be accommodated in a can member, and the plurality of notching tabs may be bent to overlap the electrode tab.

[0021] The electrode tab and the plurality of notching tabs overlapping connected to the electrode tab may be bonded together to the can member.

15 **[0022]** Each of the plurality of notching tabs may have a notching length of 8 mm to 11 mm.

[0023] Each of the plurality of notching tabs may have a notching length of 9 mm to 10 mm.

[0024] The electrode may be a negative electrode.

20 **[0025]** A method for manufacturing a secondary battery according to an embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in

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a longitudinal direction of the electrode collector; a notching tab part formation step of notching the electrode to form a notching tab part extending without coating with the active material in a width direction of the electrode collector on an area of the electrode corresponding to 1/3 or more of the total length of the coating portion; a winding step of stacking the electrode, on which the notching tab part is formed, on a separator and the other electrode having a polarity different from that of the electrode and winding the stack to form an electrode assembly; a notching tab formation step of cutting the notching tab part in the electrode assembly to form a plurality of notching tabs; and an installation step of accommodating the electrode assembly into a can member to bond each of the notching tabs to the can member.

[0026] In the preparation step, the negative electrode may be prepared.

[0027] In the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

[0028] In the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while

cutting the coating portion and the non-coating portion, which are arranged in parallel to each other, in the longitudinal direction of the electrode collector.

5 [0029] In the notching tab part formation step, the notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion is formed while cutting the electrode.

10 [0030] In the notching tab formation step, the notching tab part may be cut by using laser.

[0031] In the installation step, the electrode assembly may be accommodated in the can member in a state in which the notching tab is bent to bond the bent portion of the notching tab to the can member.

15 [0032] In the installation step, the bent portion of the notching tab may be welded to be bonded to the can member.

[0033] in the installation step, the bent portions of the plurality of notching tabs may overlap each other to be bonded to the can member at the same time.

20 [0034] In the winding step, the notching tab part may overlap each other to form two or more layers.

ADVANTAGEOUS EFFECTS

25 [0035] According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

[0036] According to the present invention, the multi-tab may be formed to minimize the electric resistance.

[0037] According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

[0038] According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

[0039] According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0040] According to the present invention, the notching tab and the electrode tab may be bonded to the can member at the same to secure the welding processability.

[0041] According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other, thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

[0042] FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

5 [0043] FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0044] FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

10 [0045] FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

[0046] FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

15 [0047] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0048] FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed
20 in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0049] FIG. 8 is a bottom view illustrating a state in which
25 a notching tab and an electrode tab are bonded to each other

according to another embodiment of the present invention.

[0050] FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

[0051] Hereinafter, a secondary battery and a method for
5 manufacturing the same according to exemplary embodiments of
the present invention will be described in detail with
reference to the accompanying drawings.

[0052] Terms or words used in the specification and claims
should not be construed as limited to a lexical meaning, and
10 should be understood as appropriate notions by the inventor
based on that he/she is able to define terms to describe
his/her invention in the best way to be seen by others.
Therefore, embodiments and drawings described herein are
simply exemplary and not exhaustive, and it will be
15 understood that various equivalents may be made to take the
place of the embodiments.

[0053] In the drawings, the dimension of each of components
or a specific portion constituting the component is
exaggerated, omitted, or schematically illustrated for
20 convenience in description and clarity. Thus, the dimension
of each element does not entirely reflect an actual size.
Moreover, detailed descriptions related to well-known
functions or configurations will be ruled out in order not to
unnecessarily obscure subject matters of the present
25 invention.

[0054] FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

5 **[0055]** As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120, on which the active material a is not applied on
10 the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form
15 two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

[0056] The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a
20 foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

[0057] The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active
25 material. The electrode assembly 1 may be manufactured by

stacking the positive electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

[0058] Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a jelly-roll shape.

[0059] The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

[0060] The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

[0061] The positive electrode coating portion may be manufactured, for example, by applying the positive electrode active material to a portion of at least one surface of the aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

[0062] The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a

negative electrode non-coating portion which is not coated with the negative electrode active material.

[0063] The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, 5 a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

[0064] The negative electrode coating portion may be manufactured, for example, by applying the negative electrode active material to a portion of at least one surface of the 10 copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

[0065] The separator may be manufactured by applying a 15 polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

20 **[0066]** As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the 25 positive electrode and the negative electrode.

[0067] In the present invention, the negative electrode will be described as an example.

[0068] The electrode 100 has one end on which the non-coating portion 120 is formed. An electrode tab 121 may be
5 attached to the non-coating portion 120 formed on the one end of the electrode 100.

[0069] The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode
10 collector may be formed to extend on an area corresponding to 1/3 or more of the total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

[0070] Only the active material a may be removed from the
15 coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the coating portion 110 of the electrode 100 to form the notching tab part 111.

[0071] Thus, there may be an effect of forming the multi-tab
20 while minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

[0072] FIG. 3 is a perspective view illustrating a state in
25 which the notching tab part is cut to form the notching tab

in FIG. 2.

[0073] As illustrated in FIG. 3, the notching tab part 111 may be cut in the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, 5 the electrode tab may decrease in resistance.

[0074] Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a 10 width exceeding 5 mm, deformation in the process of bonding for the electrical connection with the can member may be difficult.

[0075] The notching tab 111a that is the foil tab may increase in mechanical strength to prevent the notching tab 15 111a from being broken in the bent state, and thus, the notching tab 111a may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0076] FIG. 4 is a projective perspective view when seeing 20 through the can member in a state in which the electrode assembly is installed in the can member according to an embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0077] As illustrated in FIGS. 4 and 5, in the electrode 25 assembly 1 of the secondary battery according to the present

invention, a bent part of the notching tab 111a may be welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while being
5 accommodated in the can member 10.

[0078] The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each
10 other while the electrical connection is smooth to minimize the electric resistance.

[0079] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and
15 FIG. 9 is a side view of FIG. 8.

[0080] As illustrated in FIGS. 8 and 9, in a secondary battery according to another embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a
20 plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

[0081] Also, the electrode assembly 1 may be accommodated in
25 a can member 10, and the plurality of notching tabs 111a

connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

[0082] As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

[0083] In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the

can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

[0084] In the secondary battery according to another
5 embodiment of the present invention, an insulation member 130 may be formed between a bent part of the notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

[0085] The insulation member 130 may insulate the notching
10 tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming into contact with the electrode assembly 1.

[0086] The plurality of notching tabs 111a may be formed on
an outer portion of the end of the electrode assembly to
15 prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

[0087] The plurality of notching tabs 111a may be formed on
the outside of the electrode assembly so that the insulation
member 130 is installed between the plurality of notching
20 tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

[0088] Hereinafter, a method of manufacturing a secondary
battery according an embodiment of the present invention will
25 be described in detail with reference to the accompanying

drawings.

[0089] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

5 **[0090]** As illustrated in FIG. 6, a method for manufacturing a secondary battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (s3), a notching tab formation step (S4), and an installation step (S5).

10 **[0091]** The preparation step (S1) is a step of preparing an electrode 100 in which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120 on which the active material a is not applied are provided in a longitudinal direction of the electrode
15 collector.

[0092] Here, the electrode 100 may be a negative electrode.

[0093] FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode
20 collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0094] As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120,
25 which are formed in a longitudinal direction L of the

electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

[0095] In the notching tab part formation step (S2) is a step of notching the electrode 10 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

[0096] As described above, the notching of the notching tab part 111 in the electrode 100 may a step of notching a portion of the non-coating portion on which the active material a is not applied and which is formed in the width direction W of the coating portion 11 while cutting the electrode 100 in the electrode collector to form a notch tab part 11.

[0097] Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating portion 110 and the non-coating portion 120, which are arranged in

parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

[0098] The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

[0099] While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

10 **[00100]** The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

[00101] The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

[00102] Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

[00103] The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

[00104] In the installation step (S5), each of the notching tab 111a may be bent so that the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the

notching tab 111a is bent.

[00105] Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may
5 not be easily broken.

[00106] The bent notching tab 111a may have an end that faces a center of a bottom surface of the can member 10.

[00107] The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by
10 welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

[00108] In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end
15 of the electrode assembly 1 so as to be welded and bonded to the can member 10 together with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

[00109] Although the secondary battery and the method for
20 manufacturing the same according to the present invention has been described above with reference to the exemplary drawings, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

25

CLAIMS

1. A secondary battery comprising:

an electrode (100) provided with a coating portion
5 (110) coated with an active material (a) on an electrode
collector and a non-coating portion (120) on which the active
material (a) is not applied to the electrode collector in a
longitudinal direction of the electrode collector; and

a notching tab part (111) extending from the coating
10 portion (110) in a width direction of the electrode collector
without coating with the active material and overlapping each
other to form two or more layers when the electrode (100) is
wound.

15 2. The secondary battery of claim 1, wherein the notching
tab part (111) continuously extends from 1/3 or more of the
total length of the coating portion (110).

3. The secondary battery of claim 1 or 2, wherein, in a
20 state in which the electrode (100) is wound, the notching tab
part (111) is cut to form a plurality of notching tabs (111a).

4. The secondary battery of claim 3, wherein each of the
notching tabs (111a) has a width of 4 mm to 5 mm.

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5. The secondary battery of claim 3, wherein the electrode (100) is accommodated in a can member (10), and the notching tab (111a) is bent to be connected to the can member (10).

5 6. The secondary battery of claim 5, wherein a bent part formed by bending the notching tab (111a) is bonded to the can member (10).

7. The secondary battery of claim 6, further comprising an
10 insulation member (130) disposed between the bent part and an end of the electrode (100).

8. The secondary battery of claim 1, further comprising an electrode tab (121) attached to the non-coating portion (120).
15

9. The secondary battery of claim 8, wherein, in the state in which the electrode (100) is wound, the notching tab part (111) is notched to form a plurality of notching tabs (111a), and

20 the electrode (100) is accommodated in a can member (10), and the plurality of notching tabs (111a) are bent to overlap the electrode tab (121).

10. The secondary battery of claim 9, wherein the electrode
25 tab (121) and the plurality of notching tabs (111a)

overlapping connected to the electrode tab (121) are bonded together to the can member (10).

11. The secondary battery of claim 9, wherein each of the
5 plurality of notching tabs (111a) has a notching length of 8 mm to 11 mm.

12. The secondary battery of claim 9, wherein each of the
10 plurality of notching tabs (111a) has a notching length of 9 mm to 10 mm.

13. The secondary battery of claim 1, wherein the electrode (100) is a negative electrode.

15 14. A method for manufacturing a secondary battery, the method comprising:

a preparation step (S1) of preparing an electrode (100) provided with a coating portion (110) coated with an active material (a) on an electrode collector and a non-coating
20 portion (120) on which the active material (a) is not applied to the electrode collector in a longitudinal direction of the electrode collector;

a notching tab part formation step (S2) of notching the electrode (100) to form a notching tab part (111) extending
25 without coating with the active material in a width direction

of the electrode collector on an area of the electrode (100) corresponding to 1/3 or more of the total length of the coating portion (110);

5 a winding step (S3) of stacking the electrode (100), on which the notching tab part (111) is formed, on a separator and the other electrode having a polarity different from that of the electrode (100) and winding the stack to form an electrode assembly (1);

10 a notching tab formation step (S4) of cutting the notching tab part (11) in the electrode assembly (1) to form a plurality of notching tabs (111a); and

an installation step (S5) of accommodating the electrode assembly (1) into a can member (10) to bond each of the notching tabs (111a) to the can member (10).

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15. The method of claim 14, wherein, in the preparation step (S1), the negative electrode (100) is prepared.

16. The method of claim 14, wherein, in the preparation
20 step (S1), the coating portion (110) coated with the active material (a) on the electrode collector and the non-coating portion (120) on which the active material (a) is not applied are arranged in parallel to each other in the width direction of the electrode collector.

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17. The method of claim 16, wherein, in the notching tab part formation step (S2), the notching tab part (11) on which the active material (a) is not applied is notched in the width direction of the coating portion (110) while cutting the coating portion (110) and the non-coating portion (120), which are arranged in parallel to each other, in the longitudinal direction of the electrode collector.

18. The method of claim 14, wherein, in the notching tab part formation step (S2), the notching is performed so that the notching tab part (111) on which the active material (a) is not applied is notched in the width direction of the coating portion (110) is formed while cutting the electrode (100).

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19. The method of claim 14, wherein, in the notching tab formation step (S4), the notching tab part (111) is cut by using laser.

20. The method of claim 14, wherein, in the installation step (S5), the electrode assembly (1) is accommodated in the can member (10) in a state in which the notching tab (111a) is bent to bond the bent portion of the notching tab (111a) to the can member (10).

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21. The method of claim 20, wherein, in the installation step (S5), the bent portion of the notching tab (111a) is welded to be bonded to the can member (10).

5 22. The method of claim 20, wherein, in the installation step (S5), the bent portions of the plurality of notching tabs (111a) overlap each other to be bonded to the can member (10) at the same time.

10 23. The method of claim 14, wherein, in the winding step (S3), the notching tab part (111) overlaps each other to form two or more layers.

ABSTRACT

The present invention relates to a secondary battery in which an electrode tab is improved in mechanical strength and a method for manufacturing the same.

5 Also, the secondary battery according to the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a
10 longitudinal direction of the electrode collector and a notching tab part extending from the coating portion in a width direction of the electrode collector without coating with the active material and overlapping each other to form two or more layers when the electrode is wound.

DESCRIPTION

SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

5 **CROSS-REFERENCE TO RELATED APPLICATION**

[0001] The present application is a national phase entry under 35 U.S.C. § 371 of International Patent Application No. PCT/KR2018/004366, filed on April 13, 2018, published in Korean, which claims priority from Korean Patent Application Nos. 10-2017-0048645, filed on April 14, 2017, and 10-2018-0043191, filed on April 13, 2018, the
10 disclosures of which are hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an electrode tab increases in mechanical strength and a method for manufacturing the same.

15 **BACKGROUND ART**

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living environments surrounded by various electric and electronic devices.

20 [0004] Among such batteries, primary batteries and secondary batteries, which are chemical batteries using chemical reaction, are being generally used. The primary batteries are consumable batteries which are collectively referred to as dry batteries. Also, secondary batteries are rechargeable batteries that are manufactured by using a material in a redox process between current and a substance is repeatable several times. When the
25 reduction reaction is performed on the material by the current, power is charged, and when

the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

[0005] A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode
5 conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic
10 can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

[0006] An electrode assembly and a secondary battery including the same according to the related art is disclosed in Korea Patent Publication No. 10-2012-0006389.

[0007] In the secondary battery according to the related art, it is necessary to apply a
15 technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

[0008] A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a three-tab structure in which an electrode tab is applied to each of a core and an outer
20 portion, and a middle tab is attached to an intermediate portion.

[0009] However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the electrode. Thus, there is a problem that the battery capacity is lowered due to an increase of the non-coating portion.

25 **DISCLOSURE OF THE INVENTION**

TECHNICAL PROBLEM

[0010] Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

[0011] Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

TECHNICAL SOLUTION

[0012] A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

[0013] The notching tab part may continuously extend across 1/3 or more of a total length of the coating portion in the longitudinal direction.

[0014] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs.

[0015] The foil tab may have a width of 4 mm to 120 mm.

[0016] The electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may be connected to the can member.

[0017] A bent part formed by bending each of the plurality of notching tabs may be

bonded to the can member.

[0018] The secondary battery may further include an insulation member disposed between the bent parts and an end of the electrode.

[0019] The secondary battery may further include an electrode tab attached to the non-
5 coating portion.

[0020] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs, and the electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may overlap the electrode tab.

10 **[0021]** The electrode tab and the plurality of notching tabs may be bonded together to the can member.

[0022] A method for manufacturing a secondary battery according to an embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-
15 coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or
20 more of a total length of the coating portion in the longitudinal direction; a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly; a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs;
25 and an installation step of accommodating the electrode assembly into a can member and

bonding each of the plurality of notching tabs to the can member.

[0023] The electrode may be a negative electrode.

[0024] During the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

[0025] during the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

[0026] During the notching tab part formation step, notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

[0027] During the notching tab formation step, the notching tab part may be cut using a laser.

[0028] During the installation step, the electrode assembly may be accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

[0029] During the installation step, a bent portion of each of the plurality of notching tabs may be welded to the can member.

[0030] During the installation step, bent portions of each of the plurality of notching tabs may overlap each other and may be bonded to the can member together.

[0031] During the winding step, the notching tab part may be wound to form two or more overlapping layers.

ADVANTAGEOUS EFFECTS

[0032] According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

[0033] According to the present invention, the multi-tab may be formed to minimize the electric resistance.

5 [0034] According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

[0035] According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

10 [0036] According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0037] According to the present invention, the notching tab and the electrode tab may
15 be bonded to the can member at the same to secure the welding processability.

[0038] According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other,
20 thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

[0040] FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1
25 is wound.

[0041] FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

[0042] FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

[0043] FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0044] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0045] FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0046] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention.

[0047] FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

[0048] Hereinafter, a secondary battery and a method for manufacturing the same according to exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

[0049] Terms or words used in the specification and claims should not be construed as limited to a lexical meaning, and should be understood as appropriate notions by the inventor based on that he/she is able to define terms to describe his/her invention in the best way to be seen by others. Therefore, embodiments and drawings described herein are simply exemplary and not exhaustive, and it will be understood that various equivalents

may be made to take the place of the embodiments.

5 [0050] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience in description and clarity. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

10 [0051] FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0052] As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120, on which the active material a is not applied on the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

20 [0053] The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

[0054] The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active material. The electrode assembly 1 may be manufactured by stacking the positive

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electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

[0055] Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a
5 jelly-roll shape.

[0056] The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

10 [0057] The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

[0058] The positive electrode coating portion may be manufactured, for example, by applying the positive electrode active material to a portion of at least one surface of the
15 aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

[0059] The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a
20 negative electrode non-coating portion which is not coated with the negative electrode active material.

[0060] The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

25 [0061] The negative electrode coating portion may be manufactured, for example, by

applying the negative electrode active material to a portion of at least one surface of the copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

5 **[0062]** The separator may be manufactured by applying a polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

10 **[0063]** As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the positive electrode and the negative electrode.

[0064] In the present invention, the negative electrode will be described as an example.

[0065] The electrode 100 has one end on which the non-coating portion 120 is formed.
15 An electrode tab 121 may be attached to the non-coating portion 120 formed on the one end of the electrode 100.

[0066] The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode collector may be formed to extend on an area corresponding to 1/3 or more of the
20 total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

[0067] Only the active material a may be removed from the coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the
25 coating portion 110 of the electrode 100 to form the notching tab part 111.

[0068] Thus, there may be an effect of forming the multi-tab while minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

5 [0069] FIG. 3 is a perspective view illustrating a state in which the notching tab part is cut to form the notching tab in FIG. 2.

[0070] As illustrated in FIG. 3, the notching tab part 111 may be cut in the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, the electrode tab may decrease in resistance.

10 [0071] Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a width exceeding 5 mm, deformation in the process of bonding for the electrical connection with the can member may be difficult.

[0072] The notching tab 111a that is the foil tab may increase in mechanical strength
15 to prevent the notching tab 111a from being broken in the bent state, and thus, the notching tab 111a may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0073] FIG. 4 is a projective perspective view when seeing through the can member in a state in which the electrode assembly is installed in the can member according to an
20 embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0074] As illustrated in FIGS. 4 and 5, in the electrode assembly 1 of the secondary battery according to the present invention, a bent part of the notching tab 111a may be
25 welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while

being accommodated in the can member 10.

[0075] The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each other while the electrical
5 connection is smooth to minimize the electric resistance.

[0076] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and FIG. 9 is a side view of FIG. 8.

[0077] As illustrated in FIGS. 8 and 9, in a secondary battery according to another
10 embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

[0078] Also, the electrode assembly 1 may be accommodated in a can member 10, and
15 the plurality of notching tabs 111a connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

[0079] As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and
20 the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

[0080] In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to
25 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length

of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

[0081] In the secondary battery according to another embodiment of the present invention, an insulation member 130 may be formed between a bent part of the notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

[0082] The insulation member 130 may insulate the notching tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming into contact with the electrode assembly 1.

[0083] The plurality of notching tabs 111a may be formed on an outer portion of the end of the electrode assembly to prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

[0084] The plurality of notching tabs 111a may be formed on the outside of the electrode assembly so that the insulation member 130 is installed between the plurality of notching tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

[0085] Hereinafter, a method of manufacturing a secondary battery according an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[0086] FIG. 6 is a flowchart successively illustrating a method for manufacturing a

secondary battery according to an embodiment of the present invention.

[0087] As illustrated in FIG. 6, a method for manufacturing a secondary battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (s3), a notching tab formation step (S4), and an installation step (S5).

[0088] The preparation step (S1) is a step of preparing an electrode 100 in which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120 on which the active material a is not applied are provided in a longitudinal direction of the electrode collector.

10 [0089] Here, the electrode 100 may be a negative electrode.

[0090] FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

15 [0091] As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120, which are formed in a longitudinal direction L of the electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

[0092] In the notching tab part formation step (S2) is a step of notching the electrode 20 10 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

[0093] As described above, the notching of the notching tab part 111 in the electrode 100 may a step of notching a portion of the non-coating portion on which the active 25 material a is not applied and which is formed in the width direction W of the coating

portion 11 while cutting the electrode 100 in the electrode collector to form a notch tab part 11.

[0094] Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating portion 110 and the non-coating portion 120, which are arranged in parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

[0095] The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

[0096] While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

[0097] The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

[0098] The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

[0099] Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

[00100] The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

[00101] In the installation step (S5), each of the notching tab 111a may be bent so that

the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the notching tab 111a is bent.

5 **[00102]** Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may not be easily broken.

[00103] The bent notching tab 111a may have an end that faces a center of a bottom surface of the can member 10.

10 **[00104]** The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

[00105] In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end of the electrode assembly 1 so as to be welded and bonded to the can member 10 together
15 with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

[00106] Although the secondary battery and the method for manufacturing the same according to the present invention has been described above with reference to the exemplary drawings, various changes and modifications may be made thereto by one
20 skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

DESCRIPTION

SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

5 CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is a national phase entry under 35 U.S.C. § 371 of International Patent Application No. PCT/KR2018/004366, filed on April 13, 2018, published in Korean, which claims the benefit of the priority of from Korean Patent Application Nos. 10-2017-0048645, filed on April 14, 2017, and 10-2018-0043191, filed
10 on April 13, 2018, the disclosures of which are hereby incorporated herein by reference in their-entirety entireties.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an
15 electrode tab increases in mechanical strength and a method for manufacturing the same.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living
20 environments surrounded by various electric and electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical batteries using chemical reaction, are being generally used. The primary batteries are consumable batteries which are collectively referred to as dry batteries. Also, secondary batteries are rechargeable batteries that are manufactured by using a material in
25 a redox process between current and a substance is repeatable several times. When the

reduction reaction is performed on the material by the current, power is charged, and when the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

5 **[0005]** A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several
10 times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

[0006] An electrode assembly and a secondary battery including the same according to the related art is disclosed in Korea Patent Publication No. 10-2012-0006389.

15 **[0007]** In the secondary battery according to the related art, it is necessary to apply a technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

[0008] A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a
20 three-tab structure in which an electrode tab is applied to each of a core and an outer portion, and a middle tab is attached to an intermediate portion.

[0009] However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the electrode. Thus, there is a problem that the battery capacity is lowered due to an increase
25 of the non-coating portion.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0010] Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which
5 a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

[0011] Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

TECHNICAL SOLUTION

[0012] A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching
15 tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated without coating with the active material, the notching tab part having a wound state including and overlapping each other to form two or more overlapping layers when the electrode is in a wound state.

[0013] The notching tab part may continuously extend ~~from across~~ 1/3 or more of ~~the~~
20 a total length of the coating portion in the longitudinal direction.

[0014] ~~In a state in which~~ When the electrode is in the wound state, the notching tab part may ~~be cut to~~ form a plurality of notching tabs.

[0015] The foil tab may have a width of 4 mm to 120 mm.

[0016] The electrode may be accommodated in a can member, and each of the
25 plurality of notching tab tabs may be in a bent state to and may be connected to the can

member.

[0017] A bent part formed by bending each of the plurality of notching tab ~~tabs~~ may be bonded to the can member.

[0018] The secondary battery may further include an insulation member disposed between the bent ~~part~~ parts and an end of the electrode.

[0019] The secondary battery may further include an electrode tab attached to the non-coating portion.

[0020] ~~In the state in which~~When the electrode is in the wound state, the notching tab part may ~~be notched to~~ form a plurality of notching tabs, and the electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and to may overlap the electrode tab.

[0021] The electrode tab and the plurality of notching tabs ~~overlapping connected to the electrode tab~~ may be bonded together to the can member.

~~[0022] Each of the plurality of notching tabs may have a notching length of 8 mm to 11 mm.~~

~~[0023] Each of the plurality of notching tabs may have a notching length of 9 mm to 10 mm.~~

~~[0024] The electrode may be a negative electrode.~~

~~[0025]~~[0022] A method for manufacturing a secondary battery according to an

embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; a notching tab part formation step of notching the electrode to form a notching tab part ~~extending without coating not~~ coated with the active material and extending from the coating portion in a width direction

of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across ~~on an area of the electrode corresponding to~~ 1/3 or more of ~~the~~ a total length of the coating portion in the longitudinal direction; a winding step ~~of including~~ stacking the electrode, ~~on which the notching tab part is formed, on~~ with a separator and ~~the other~~ with another electrode having a polarity different from that of the electrode and winding the ~~stack~~ electrode, the separator, and the another electrode to form an electrode assembly; a notching tab formation step of cutting the notching tab part ~~in~~ of the electrode assembly to form a plurality of notching tabs; and an installation step of accommodating the electrode assembly into a can member ~~to bond~~ and bonding each of the plurality of notching tabs to the can member.

~~{0026}~~ {0023} ~~In the preparation step, the~~ The electrode may be a negative electrode may be prepared.

~~{0027}~~ {0024} ~~In~~ During the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

~~{0028}~~ {0025} ~~In~~ during the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion, ~~which are arranged in parallel to each other,~~ in the longitudinal direction of the electrode collector.

~~{0029}~~ {0026} ~~In~~ During the notching tab part formation step, ~~the~~ notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion ~~is formed~~ while cutting the electrode.

~~{0030}~~ {0027} ~~In~~ During the notching tab formation step, the notching tab part may be cut ~~by~~ using a laser.

10 | ~~{0031}~~ [0028] ~~In~~ During the installation step, the electrode assembly may be accommodated in the can member in a state in which the plurality of notching tab is tabs are bent to bond the bent portion of the notching tab and bonded to the can member.

5 | ~~{0032}~~ [0029] ~~In~~ During the installation step, the a bent portion of each of the plurality of notching ~~tab~~ tabs may be welded to be bonded to the can member.

10 | ~~{0033}~~ [0030] ~~in~~ During the installation step, the bent portions of each of the plurality of notching tabs may overlap each other ~~to~~ and may be bonded to the can member at the same time together.

10 | ~~{0034}~~ [0031] ~~In~~ During the winding step, the notching tab part may overlap each other be wound to form two or more overlapping layers.

ADVANTAGEOUS EFFECTS

15 | ~~{0035}~~ [0032] According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

15 | ~~{0036}~~ [0033] According to the present invention, the multi-tab may be formed to minimize the electric resistance.

15 | ~~{0037}~~ [0034] According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

20 | ~~{0038}~~ [0035] According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

20 | ~~{0039}~~ [0036] According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be boned to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

25 | ~~{0040}~~ [0037] According to the present invention, the notching tab and the

electrode tab may be bonded to the can member at the same to secure the welding processability.

~~{0041}~~[0038] According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other, thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

~~{0042}~~[0039] FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

~~{0043}~~[0040] FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

~~{0044}~~[0041] FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

~~{0045}~~[0042] FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

~~{0046}~~[0043] FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

~~{0047}~~[0044] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

~~{0048}~~[0045] FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

~~{0049}~~[0046] FIG. 8 is a bottom view illustrating a state in which a notching tab

and an electrode tab are bonded to each other according to another embodiment of the present invention.

~~[0050]~~ [0047] FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

5 ~~[0051]~~ [0048] Hereinafter, a secondary battery and a method for manufacturing the same according to exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

~~[0052]~~ [0049] Terms or words used in the specification and claims should not be construed as limited to a lexical meaning, and should be understood as appropriate notions
10 by the inventor based on that he/she is able to define terms to describe his/her invention in the best way to be seen by others. Therefore, embodiments and drawings described herein are simply exemplary and not exhaustive, and it will be understood that various equivalents may be made to take the place of the embodiments.

~~[0053]~~ [0050] In the drawings, the dimension of each of components or a
15 specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience in description and clarity. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

20 ~~[0054]~~ [0051] FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

~~[0055]~~ [0052] As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion
25 110 coated with an active material a on an electrode collector and a non-coating portion

120, on which the active material a is not applied on the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

~~[0056]~~ [0053] The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

10 ~~[0057]~~ [0054] The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active material. The electrode assembly 1 may be manufactured by stacking the positive electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

15 ~~[0058]~~ [0055] Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a jelly-roll shape.

~~[0059]~~ [0056] The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

~~[0060]~~ [0057] The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

25 ~~[0061]~~ [0058] The positive electrode coating portion may be manufactured, for

example, by applying the positive electrode active material to a portion of at least one surface of the aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

5 | ~~{0062}~~[0059]_____ The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a negative electrode non-coating portion which is not coated with the negative electrode active material.

10 | ~~{0063}~~[0060]_____ The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

15 | ~~{0064}~~[0061]_____ The negative electrode coating portion may be manufactured, for example, by applying the negative electrode active material to a portion of at least one surface of the copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

20 | ~~{0065}~~[0062]_____ The separator may be manufactured by applying a polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

| ~~{0066}~~[0063]_____ As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the positive electrode and the negative electrode.

25 | ~~{0067}~~[0064]_____ In the present invention, the negative electrode will be described

as an example.

~~[0068]~~ [0065] The electrode 100 has one end on which the non-coating portion 120 is formed. An electrode tab 121 may be attached to the non-coating portion 120 formed on the one end of the electrode 100.

5 ~~[0069]~~ [0066] The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode collector may be formed to extend on an area corresponding to 1/3 or more of the total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

10 ~~[0070]~~ [0067] Only the active material a may be removed from the coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the coating portion 110 of the electrode 100 to form the notching tab part 111.

~~[0071]~~ [0068] Thus, there may be an effect of forming the multi-tab while
15 minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

~~[0072]~~ [0069] FIG. 3 is a perspective view illustrating a state in which the notching tab part is cut to form the notching tab in FIG. 2.

~~[0073]~~ [0070] As illustrated in FIG. 3, the notching tab part 111 may be cut in
20 the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, the electrode tab may decrease in resistance.

~~[0074]~~ [0071] Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a width
25 exceeding 5 mm, deformation in the process of bonding for the electrical connection with

the can member may be difficult.

~~10075~~ [0072] The notching tab 111a that is the foil tab may increase in mechanical strength to prevent the notching tab 111a from being broken in the bent state, and thus, the notching tab 111a may be ~~bonded~~ bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

~~10076~~ [0073] FIG. 4 is a projective perspective view when seeing through the can member in a state in which the electrode assembly is installed in the can member according to an embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

~~10077~~ [0074] As illustrated in FIGS. 4 and 5, in the electrode assembly 1 of the secondary battery according to the present invention, a bent part of the notching tab 111a may be welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while being accommodated in the can member 10.

~~10078~~ [0075] The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each other while the electrical connection is smooth to minimize the electric resistance.

~~10079~~ [0076] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and FIG. 9 is a side view of FIG. 8.

~~10080~~ [0077] As illustrated in FIGS. 8 and 9, in a secondary battery according to another embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to

overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

~~[0081]~~[0078] Also, the electrode assembly 1 may be accommodated in a can member 10, and the plurality of notching tabs 111a connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

~~[0082]~~[0079] As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

~~[0083]~~[0080] In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

~~[0084]~~[0081] In the secondary battery according to another embodiment of the present invention, an insulation member 130 may be formed between a bent part of the

notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

~~[0085]~~ [0082] The insulation member 130 may insulate the notching tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming
5 into contact with the electrode assembly 1.

~~[0086]~~ [0083] The plurality of notching tabs 111a may be formed on an outer portion of the end of the electrode assembly to prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

~~[0087]~~ [0084] The plurality of notching tabs 111a may be formed on the outside
10 of the electrode assembly so that the insulation member 130 is installed between the plurality of notching tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

~~[0088]~~ [0085] Hereinafter, a method of manufacturing a secondary battery
15 according an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

~~[0089]~~ [0086] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

~~[0090]~~ [0087] As illustrated in FIG. 6, a method for manufacturing a secondary
20 battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (s3), a notching tab formation step (S4), and an installation step (S5).

~~[0091]~~ [0088] The preparation step (S1) is a step of preparing an electrode 100
in which a coating portion 110 coated with an active material a on an electrode collector
25 and a non-coating portion 120 on which the active material a is not applied are provided in

a longitudinal direction of the electrode collector.

~~[0092]~~ [0089] Here, the electrode 100 may be a negative electrode.

~~[0093]~~ [0090] FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

~~[0094]~~ [0091] As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120, which are formed in a longitudinal direction L of the electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

~~[0095]~~ [0092] In the notching tab part formation step (S2) is a step of notching the electrode 100 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

~~[0096]~~ [0093] As described above, the notching of the notching tab part 111 in the electrode 100 may a step of notching a portion of the non-coating portion on which the active material a is not applied and which is formed in the width direction W of the coating portion 11 while cutting the electrode 100 in the electrode collector to form a notch tab part 11.

~~[0097]~~ [0094] Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating

portion 110 and the non-coating portion 120, which are arranged in parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

~~[0098]~~[0095] The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

~~[0099]~~[0096] While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

~~[00100]~~[0097] The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

~~[00101]~~[0098] The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

~~[00102]~~[0099] Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

~~[00103]~~[00100] The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

~~[00104]~~[00101] In the installation step (S5), each of the notching tab 111a may be bent so that the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the notching tab 111a is bent.

~~[00105]~~[00102] Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may not be easily broken.

~~[00106]~~[00103] The bent notching tab 111a may have an end that faces a center of

a bottom surface of the can member 10.

~~[00107]~~ [00104] The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

5 ~~[00108]~~ [00105] In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end of the electrode assembly 1 so as to be welded and bonded to the can member 10 together with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

10 ~~[00109]~~ [00106] Although the secondary battery and the method for manufacturing the same according to the present invention has been described above with reference to the exemplary drawings, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

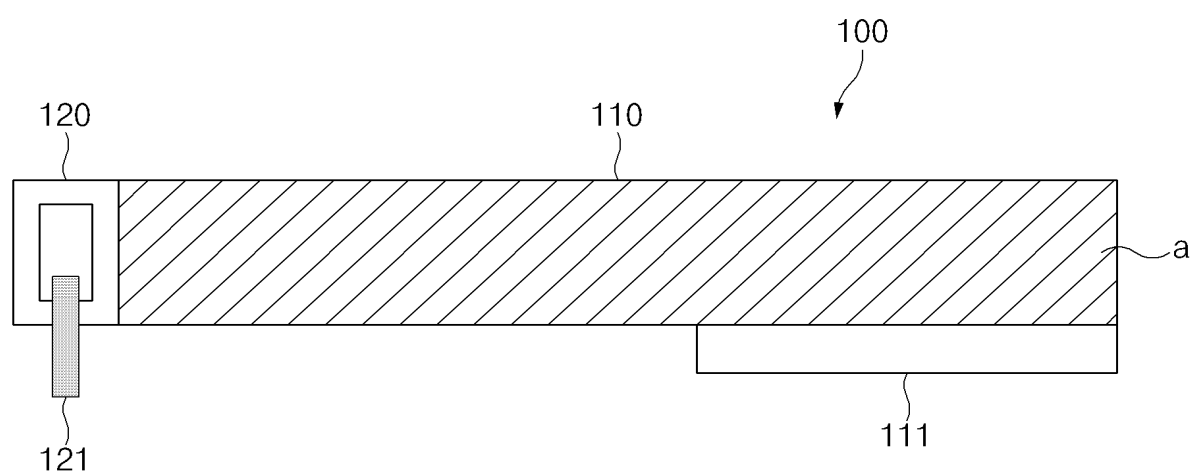


FIG.1

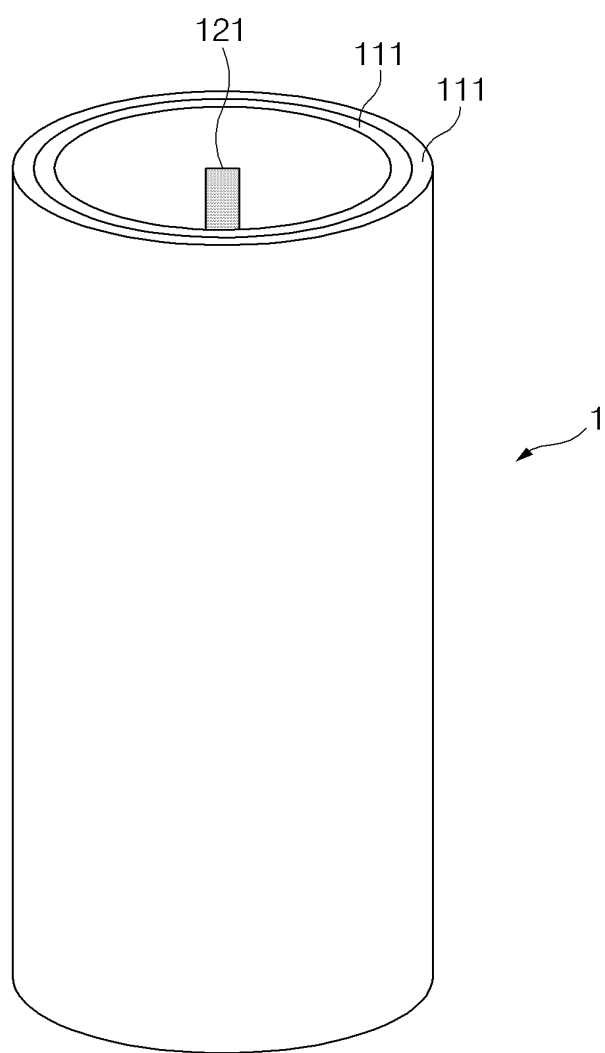


FIG.2

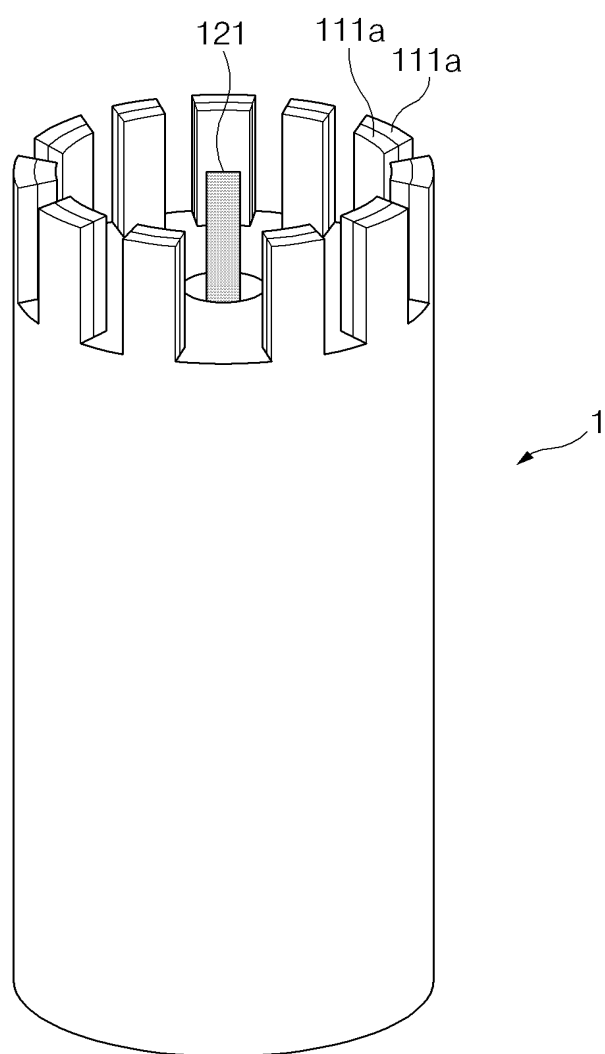


FIG.3

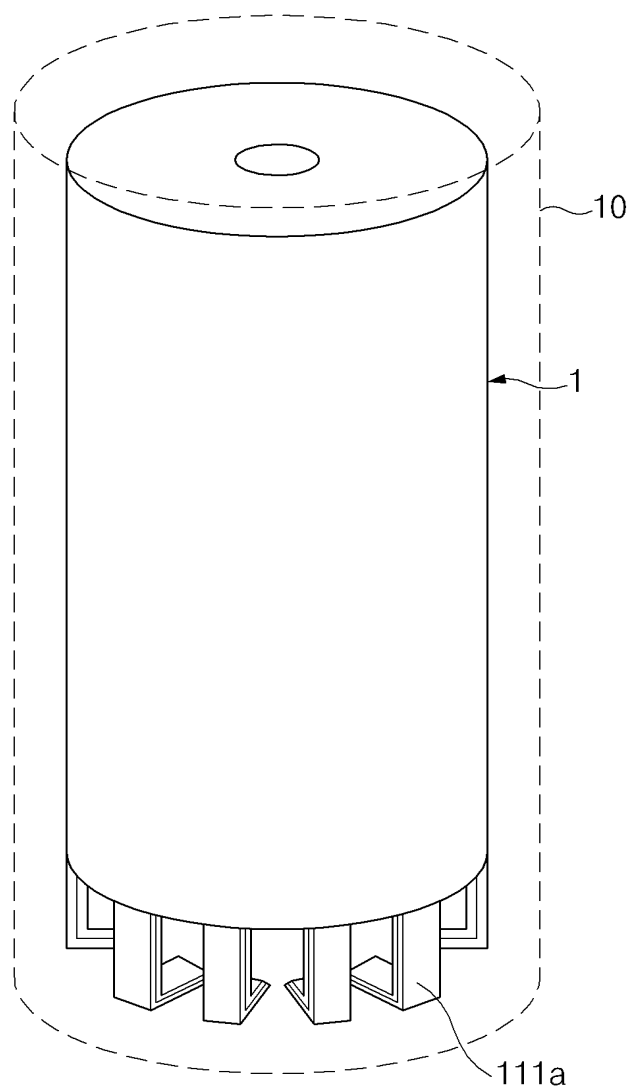


FIG.4

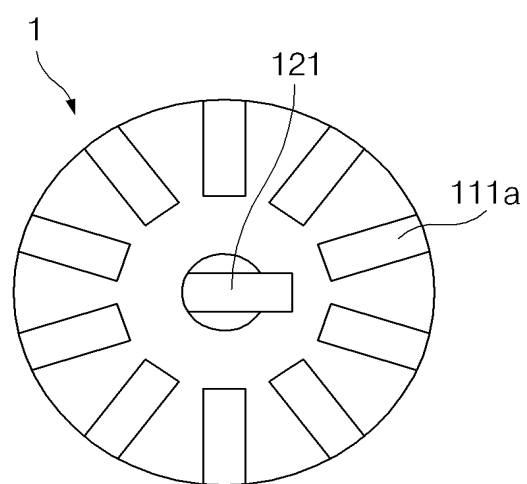


FIG. 5

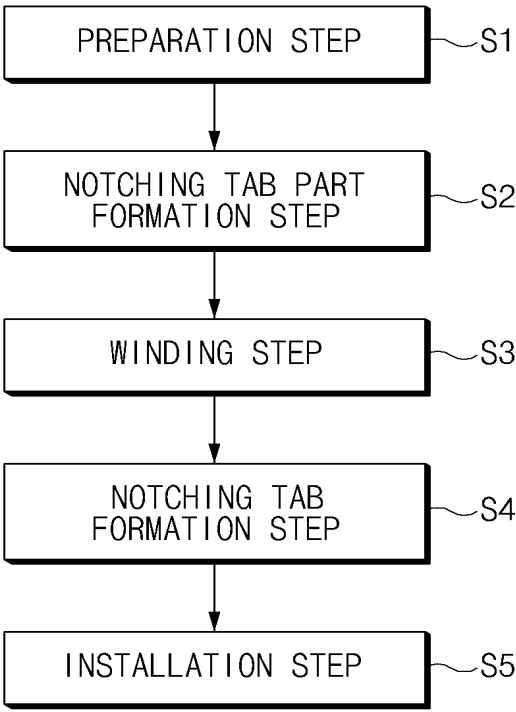


FIG.6

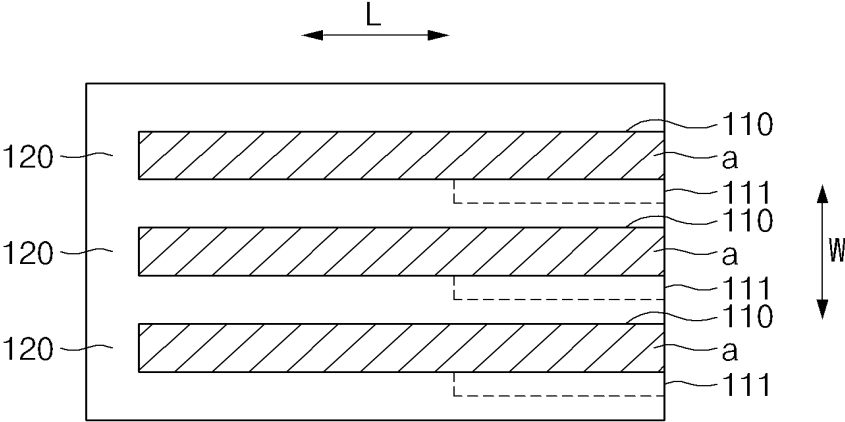


FIG.7

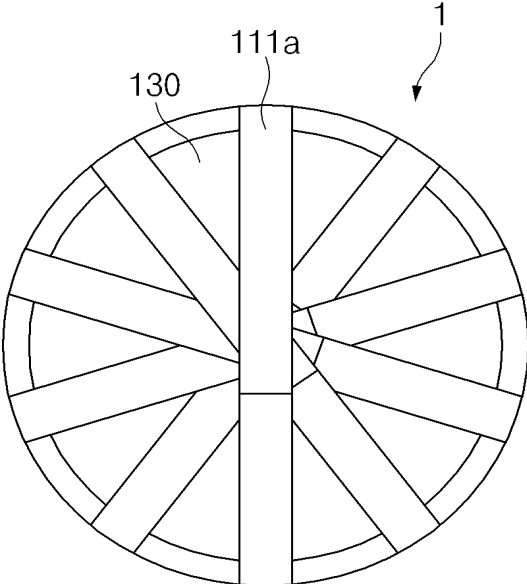


FIG.8

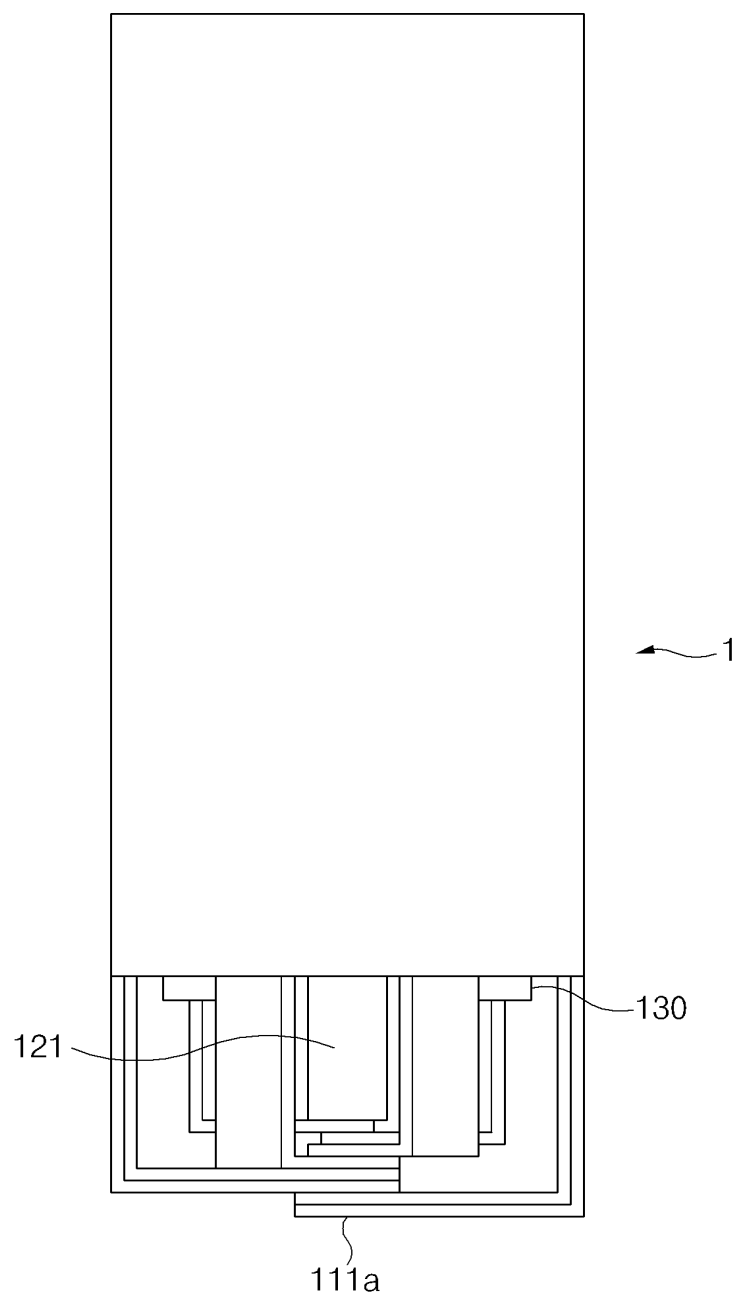


FIG.9

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-1654
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

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☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

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Inventor	1				Remove
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-1654
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

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For further information see 37 CFR 1.33(a).

☐ An Address is being provided for the correspondence Information of this application.

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Application Information:

Title of the Invention	Secondary Battery And Method For Manufacturing The Same		
Attorney Docket Number	LGCHEM 3.3F-1654	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	4

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country ⁱ

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGCHEM 3.3F-1654
	Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same	

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

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This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78. When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	a 371 of international	PCT/KR2018/004366	2018-04-13

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.

Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-1654
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2018-0043191	KR	2018-04-13	
			Remove
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2017-0048645	KR	2017-04-14	
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.
NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGCHEM 3.3F-1654
	Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. ☐ If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. ☐ If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGCHEM 3.3F-1654
	Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same	

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1		Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
		Clear	
● Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor	
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	LG Chem, Ltd.		
Mailing Address Information For Applicant:			
Address 1	128, Yeoui-daero		
Address 2	Yeongdeungpo-gu		
City	Seoul	State/Province	
Country	KR	Postal Code	07336
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button.			
Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGCHEM 3.3F-1654
	Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same	

Assignee

1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

Remove

If the Assignee or Non-Applicant Assignee is an Organization check here.

☒

Organization Name

LG Chem, Ltd.

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1

128, Yeoui-daero

Address 2

Yeongdeungpo-gu

City

Seoul

State/Province

Country i

KR

Postal Code

07336

Phone Number

Fax Number

Email Address

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

Add

Signature:

Remove

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is **not** checked in subsection 2 of the “Authorization or Opt-Out of Authorization to Permit Access” section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature

/Joseph H. Karlin/

Date (YYYY-MM-DD)

2019-09-27

First Name

Joseph

Last Name

Karlin

Registration Number

62,471

Additional Signature may be generated within this form by selecting the Add button.

Add

EFS Web 2.2.13

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGCHEM 3.3F-1654
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

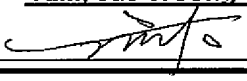
This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

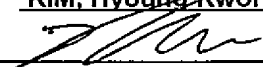
The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS		Attorney Docket Number:	
Title of Invention	SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME		
As the below named inventor(s) (hereinafter designated as the undersigned), I/We hereby declare that: This declaration is directed to: ☒ The attached application, or ☐ United States application or PCT international application number _____ filed on _____ The above-identified application was made or authorized to be made by me/us. I/We believe that I am the original inventor or an original joint inventor of a claimed invention in the application. I/We hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both. WHEREAS, the undersigned as a sole inventor (if only one name is listed below) or as a joint inventor (if plural names are listed below) has invented certain new and useful improvements in the application identified above. AND, WHEREAS LG CHEM, LTD., (hereinafter referred to as assignee), a corporation of Korea whose mailing address is 128, Yeoui-daero, Yeongdeungpo-gu, Seoul 07336, Republic of Korea is desirous of acquiring the entire right, title and interest in the same. NOW, THEREFORE, in consideration of the sum of one dollar, the receipt whereof is acknowledged, and other good and valuable consideration, the undersigned hereby sell(s), assign(s) and set(s) over to said assignee my/our entire right, title and interest for the United States of America and all other countries in and to said invention and the aforesaid utility patent application and all original, divisional, continuation, substitute or reissue applications and patents applied for or granted therefor in the United States of America and all other countries, for said invention, including without limitation all applications and patents for said invention claiming priority or benefit of the aforesaid utility application pursuant to any law or treaty, and including all claims for damages by reason of past infringement with the right to sue for and collect the same for its own use and the use of its successors, assigns and legal representatives, and including the right to claim such priority or benefit and the Commissioner of Patents and Trademarks is hereby authorized and requested to issue all patents on said improvements or resulting therefrom to said assignee herein, as assignee of the entire interest therein; and the undersigned for me/us and my/our legal representatives, heirs and assigns do hereby agree and covenant without further remuneration, to execute and deliver all original, divisional, continuation, reissue and other applications for Letters Patent on said improvements and all assignments thereof to said assignee or its assigns, to communicate to said assignee or its representatives all facts known to the undersigned respecting said improvements, whenever requested, to testify in any interferences or other legal proceedings in which any of said applications or patents may become involved, to sign all lawful papers, make all rightful oaths, and to do generally everything necessary to aid assignee, its successors, assigns and nominees to obtain patent protection for said improvements in all countries, the expenses incident to said applications to be borne and paid by said assignee. And I do hereby authorize my attorneys to insert on this deed the filing date and application number of said application when known.			
FULL NAME OF INVENTOR(S)			
Inventor one: <u>KIM, Jae Woong</u>		Date: <u>September 25, 2019</u>	
Signature: <u></u>			
☒ Additional inventors or a legal representative are being named on <u>1</u> additional form(s) attached hereto. PLEASE SEE THE WARNINGS ON THE LAST PAGE OF THIS FORM.			

SUPPLEMENTAL SHEET

DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS	ADDITIONAL INVENTOR(S) Supplemental Sheet
Inventor two: <u>KIM, Hyoung Kwon</u> Signature: <u></u>	Date: <u>September 25, 2019</u>
Inventor three: <u>KIM, Sang Uck</u> Signature: <u></u>	Date: <u>September 25, 2019</u>
Inventor four: _____ Signature: _____	Date: _____
Inventor five: _____ Signature: _____	Date: _____
Inventor six: _____ Signature: _____	Date: _____
Inventor seven: _____ Signature: _____	Date: _____
Inventor eight: _____ Signature: _____	Date: _____
Inventor nine: _____ Signature: _____	Date: _____
Inventor ten: _____ Signature: _____	Date: _____
WARNING: According to 37 C.F.R. 1.63 (c), "[a] person may not execute an oath or declaration unless the person has reviewed and understands the contents of the application, including the claims and is aware of the duty to disclose to the Office all information known to be material to patentability as defined in 1.56." Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.	

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: September 27, 2019
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

International Application No.: PCT/KR2018/004366

Confirmation No.: N/A

International Filing Date: April 13, 2018

Art Unit: N/A

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Dear Sir:

Prior to initiation of the prosecution of the above-identified pending U.S. patent application, the following amendments and remarks are respectfully submitted.

IN THE SPECIFICATION

Please amend the specification in accordance with the attached substitute specification. The substitute specification is provided in clean form without markings and also as marked-up to show all the changes relative to the immediate prior version of the English translation of the International Application. No new matter is presented by such amendments to the specification.

IN THE CLAIMS

1. (Currently Amended) A secondary battery comprising:
an electrode (100) provided with a coating portion (110) coated with an active material (a) on an electrode collector and a non-coating portion (120) on which the active material (a) is not applied to the electrode collector in a longitudinal direction of the electrode collector; and
a notching tab part (111) extending from the coating portion (110) in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated without coating with the active material, the notching tab part having a wound state including and overlapping each other to form two or more overlapping layers when the electrode (100) is in a wound state.
2. (Currently Amended) The secondary battery of claim 1, wherein the notching tab part (111) continuously extends from across 1/3 or more of ~~the a~~ total length of the coating portion in the longitudinal direction (110).
3. (Currently Amended) The secondary battery of claim 1 ~~or 2~~, wherein, ~~in a state in which when~~ the electrode (100) is in the wound state, the notching tab part (111) ~~is cut to form~~ forms a plurality of notching tabs (111a).
4. (Currently Amended) The secondary battery of claim 3, wherein each of the plurality of notching tabs (111a) has a width of 4 mm to 5 mm.
5. (Currently Amended) The secondary battery of claim 3, wherein the electrode (100) is accommodated in a can member (10), and each of the plurality of notching tabs tab (111a) is in a bent to be state and connected to the can member (10).
6. (Currently Amended) The secondary battery of claim 5, wherein a bent part ~~formed by bending of each of the plurality of notching tab (111a) tabs~~ is bonded to the can member (10).
7. (Currently Amended) The secondary battery of claim 6, further comprising an insulation member (130) disposed between the bent ~~part~~ parts and an end of the electrode (100).
8. (Currently Amended) The secondary battery of claim 1, further comprising an electrode tab (121) attached to the non-coating portion (120).
9. (Currently Amended) The secondary battery of claim 8, wherein, ~~in the state in which when~~ the electrode (100) is in the wound state, the notching tab part (111) ~~is notched to form~~

forms a plurality of notching tabs ~~(111a)~~, and

the electrode ~~(100)~~ is accommodated in a can member ~~(10)~~, and each of the plurality of notching tabs ~~(111a)~~ are is in a bent state and to overlap overlaps the electrode tab ~~(121)~~.

10. (Currently Amended) The secondary battery of claim 9, wherein the electrode tab ~~(121)~~ and the plurality of notching tabs ~~(111a)~~ overlapping connected to the electrode tab (121) are bonded together to the can member ~~(10)~~.

11-13. (Cancelled)

14. (Currently Amended) A method for manufacturing a secondary battery, the method comprising:

a preparation step ~~(S1)~~ of preparing an electrode ~~(100)~~ provided with a coating portion ~~(110)~~ coated with an active material ~~(a)~~ on an electrode collector and a non-coating portion ~~(120)~~ on which the active material ~~(a)~~ is not applied to the electrode collector in a longitudinal direction of the electrode collector;

a notching tab part formation step ~~(S2)~~ of notching the electrode ~~(100)~~ to form a notching tab part ~~(111)~~ extending without coating not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across on an area of the electrode (100) corresponding to 1/3 or more of ~~the a~~ total length of the coating portion ~~(110)~~ in the longitudinal direction;

a winding step ~~(S3)~~ of including stacking the electrode ~~(100)~~, on which the notching tab part (111) is formed, on with a separator and the other with another electrode having a polarity different from that of the electrode ~~(100)~~ and winding the ~~stack electrode, the separator, and the another electrode~~ to form an electrode assembly ~~(1)~~;

a notching tab formation step ~~(S4)~~ of cutting the notching tab part ~~(11)~~ in of the electrode assembly ~~(1)~~ to form a plurality of notching tabs ~~(111a)~~; and

an installation step ~~(S5)~~ of accommodating the electrode assembly ~~(1)~~ into a can member ~~(10)~~ to bond and bonding each of the plurality of notching tabs ~~(111a)~~ to the can member ~~(10)~~.

15. (Currently Amended) The method of claim 14, wherein, ~~in the preparation step (S1),~~ the electrode is a negative electrode (100) is prepared.

16. (Currently Amended) The method of claim 14, wherein, ~~in~~ during the preparation step ~~(S1)~~, the coating portion ~~(110)~~ coated with the active material ~~(a)~~ on the electrode collector and the non-coating portion ~~(120)~~ on which the active material ~~(a)~~ is not applied are arranged ~~in~~ parallel to each other in the width direction of the electrode collector.

17. (Currently Amended) The method of claim 16, wherein, ~~in~~ during the notching tab part formation step ~~(S2)~~, the notching tab part ~~(11)~~ on which the active material ~~(a)~~ is not applied is notched in the width direction of the coating portion ~~(110)~~ while cutting the coating portion ~~(110)~~ and the non-coating portion ~~(120)~~, ~~which are arranged in parallel to each other~~, in the longitudinal direction of the electrode collector.

18. (Currently Amended) The method of claim 14, wherein, ~~in~~ during the notching tab part formation step ~~(S2)~~, ~~the~~ notching is performed so that the notching tab part ~~(111)~~ on which the active material ~~(a)~~ is not applied is notched in the width direction of the coating portion ~~(110)~~ ~~is formed~~ while cutting the electrode ~~(100)~~.

19. (Currently Amended) The method of claim 14, wherein, ~~in~~ during the notching tab formation step ~~(S4)~~, the notching tab part ~~(111)~~ is cut ~~by~~ using a laser.

20. (Currently Amended) The method of claim 14, wherein, ~~in~~ during the installation step ~~(S5)~~, the electrode assembly ~~(1)~~ is accommodated in the can member ~~(10)~~ in a state in which the plurality of notching tab ~~(111a)~~ is tabs are bent to bond the bent portion of the notching tab ~~(111a)~~ and bonded to the can member ~~(10)~~.

21. (Currently Amended) The method of claim 20, wherein, ~~in~~ during the installation step ~~(S5)~~, ~~the~~ a bent portion of each of the plurality of notching tab ~~(111a)~~ tabs is welded ~~to be bonded~~ to the can member ~~(10)~~.

22. (Currently Amended) The method of claim 20, wherein, ~~in~~ during the installation step ~~(S5)~~, ~~the~~ bent portions of each of the plurality of notching tabs ~~(111a)~~ overlap each other to be and are bonded to the can member ~~(10)~~ at the same time together.

23. (Currently Amended) The method of claim 14, wherein, ~~in~~ during the winding step ~~(S3)~~, the notching tab part ~~(111)~~ ~~overlaps each other~~ is wound to form two or more overlapping layers.

REMARKS

The above-noted amendment of claims 1-10 and 14-23, the cancellation of claims 11-13, and the submission of revisions to the Specification are respectfully submitted prior to initiation of the prosecution of this application in the U.S. Patent and Trademark Office. The revisions to the application are submitted in order to clarify and correct the application and to conform it to all of the requirements of U.S. practice. No new matter is included in these amendments.

In view of the above, it is respectfully requested that these amendments now be entered, and that prosecution on the merits of this application now be initiated. If, however, for any reason the Examiner does not believe such action can be taken, it is respectfully requested that the Examiner telephone applicant's attorney at (908) 654-5000 in order to overcome any objections that the Examiner may have.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: September 27, 2019

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin

Registration No.: 62,471
LERNER, DAVID, LITTENBERG,
KRUMHOLZ & MENTLIK, LLP
600 South Avenue West
Westfield, New Jersey 07090
(908) 654-5000
Attorney for Applicant

WO 2018/190691 A1

(12) 특허협력조약에 의하여 공개된 국제출원
(19) 세계지식재산권기구
국제사무국
(43) 국제공개일
2018년 10월 18일 (18.10.2018) WIPO | PCT

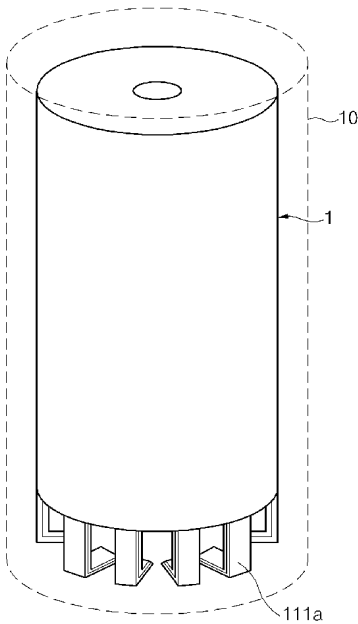


(10) 국제공개번호
WO 2018/190691 A1

- (51) 국제특허분류: *H01M 2/26* (2006.01) *H01M 10/0587* (2010.01)
H01M 10/04 (2006.01)
- (21) 국제출원번호: PCT/KR2018/004366
- (22) 국제출원일: 2018년 4월 13일 (13.04.2018)
- (25) 출원언어: 한국어
- (26) 공개언어: 한국어
- (30) 우선권정보:
10-2017-0048645 2017년 4월 14일 (14.04.2017) KR
10-2018-0043191 2018년 4월 13일 (13.04.2018) KR
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- (81) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.
- (84) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), 유라시아 (AM, AZ, BY, KG, KZ, RU, TJ, TM), 유럽 (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

(54) Title: SECONDARY BATTERY AND SECONDARY BATTERY MANUFACTURING METHOD

(54) 발명의 명칭: 이차전지 및 그 이차전지의 제조 방법



(57) Abstract: The present invention relates to a secondary battery and a secondary battery manufacturing method, wherein the mechanical strength of an electrode tab can be increased. In addition, the present invention is characterized by comprising: an electrode having a coated portion and a non-coated portion provided along the longitudinal direction of an electrode collector, the coated portion having an active material applied onto the electrode collector, and the non-coated portion having no active material applied onto the electrode collector; and a notching tab portion extending from the coated portion in the transverse direction of the electrode collector such that the active material is not applied thereto, the notching tab portion being formed to overlap in at least two tiers when the electrode is wound.

(57) 요약서: 본 발명은 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다. 또한, 본 발명은 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.

[다음 쪽 계속]

공개:

— 국제조사보고서와 함께 (조약 제21조(3))

명세서

발명의 명칭: 이차전지 및 그 이차전지의 제조 방법

기술분야

[1] 관련출원과의 상호인용

[2] 본 출원은 2017년 04월 14일자 한국 특허 출원 제10-2017-0048645호 및 2018년 04월 13일자 한국 출원 제10-2018-0043191호에 기초한 우선권의 이익을 주장하며, 해당 한국 특허 출원의 문헌에 개시된 모든 내용은 본 명세서의 일부로서 포함된다.

[3] 기술분야

[4] 본 발명은 이차전지 및 그 이차전지의 제조 방법에 관한 것으로서, 보다 자세하게는 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다.

배경기술

[5] 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 건물로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.

[6] 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정이 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게 된다.

[7] 한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.

[8] 대한민국 공개특허공보 제10-2012-0006389호에는 종래의 전극조립체 및 이를 포함하는 이차 전지가 개시되어 있다.

[9] 종래의 이차 전지는 저저항 셀 개발을 위해서 음극 탭에 저항이 낮은 금속을 적용하거나 전극 탭 수를 늘리는 기술 적용이 필요하다.

[10] 이러한 필요성에 의해 개발된 음극 멀티탭 구조는 코어와 외곽에 각각 전극 탭을 부착하는 2탭 구조 또는 코어와 외곽부에 전극 탭을 부착하고 중간측에 미들탭을 부착하는 3탭 구조였다.

- [11] 그런데 젤리롤 특성상 미들탭 부착을 위해서는 전극 중간 일정 위치에 무지부를 형성해야 하는데 무지부의 증가에 따른 전지 용량 저하의 문제점이 있었다.

발명의 상세한 설명

기술적 과제

- [12] 따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 무지부를 최소화하면서 멀티탭을 형성할 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

- [13] 또한, 본 발명의 과제는 전극 포일을 노칭하여 전극탭을 형성하면서 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

과제 해결 수단

- [14] 본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.
- [15] 상기 노칭 탭부는 상기 유지부 전체 길이의 1/3 이상으로부터 연속되게 연장될 수 있다.
- [16] 상기 전극이 권취된 상태에서 상기 노칭 탭부는 커팅되어 복수 개의 노칭탭을 형성할 수 있다.
- [17] 상기 노칭탭의 폭은 4mm ~ 5mm일 수 있다.
- [18] 상기 전극은 캔부재에 수용되고, 상기 노칭탭은 절곡되어 상기 캔부재에 연결될 수 있다.
- [19] 상기 노칭탭이 절곡되어 형성된 절곡부는 상기 캔부재에 접합될 수 있다.
- [20] 상기 절곡부와 상기 전극의 단부와의 사이에 형성된 절연부재를 포함할 수 있다.
- [21] 상기 무지부에 부착되는 전극탭을 포함할 수 있다.
- [22] 상기 전극이 권취된 상태에서 상기 노칭 탭부는 노칭되어 복수 개의 노칭탭을 형성하고, 상기 전극은 캔부재에 수용되고, 상기 복수 개의 노칭탭은 절곡되어 상기 전극탭에 겹쳐지게 연결될 수 있다.
- [23] 상기 전극탭 및 상기 전극탭과 겹쳐지게 연결된 상기 복수 개의 노칭탭은 함께 상기 캔부재에 접합될 수 있다.
- [24] 상기 복수 개의 노칭탭의 노칭 길이는 8mm ~ 11mm일 수 있다.
- [25] 상기 복수 개의 노칭탭의 노칭 길이는 9mm ~ 10mm일 수 있다.
- [26] 상기 전극은 음극일 수 있다.
- [27] 본 발명의 일 실시예에 따른 이차전지의 제조 방법은 전극집전체 상에

활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극을 준비하는 준비단계, 상기 전극에서 유지부의 길이 1/3 이상에 상기 전극집전체의 폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부가 형성되도록 상기 전극을 노칭하는 노칭 탭부 형성단계, 상기 노칭 탭부가 형성된 상기 전극을 분리막과 상기 전극과 상극인 다른 전극에 적층한 후 권취하여 전극조립체를 형성는 권취단계, 상기 전극조립체에서 상기 노칭 탭부를 커팅하여 복수 개의 노칭탭을 형성하는 노칭탭 형성단계 및 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭을 상기 캔부재에 접합하는 설치단계를 포함한다.

- [28] 상기 준비단계에서 음극의 전극을 준비할 수 있다.
- [29] 상기 준비단계는 전극집전체 상에 활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 폭방향을 기준으로 병렬로 배열할 수 있다.
- [30] 상기 노칭 탭부 형성단계는 상기 병렬로 배열된 상기 유지부와 무지부를 상기 전극집전체의 길이 방향으로 커팅하면서 상기 유지부의 폭 방향으로 활물질이 도포되지 않은 상기 노칭 탭부를 노칭할 수 있다.
- [31] 상기 노칭 탭부 형성단계는 상기 전극을 커팅하면서 상기 유지부의 폭방향으로 활물질이 도포되지 않은 상기 노칭 탭부가 형성되도록 노칭할 수 있다.
- [32] 상기 노칭탭 형성단계에서 상기 노칭 탭부를 레이저로 커팅할 수 있다.
- [33] 상기 설치단계는 상기 노칭탭을 절곡한 상태로 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭에서 절곡된 부분을 상기 캔부재에 접합할 수 있다.
- [34] 상기 설치단계에서 상기 노칭탭의 절곡된 부분을 용접에 의해 상기 캔부재에 접합할 수 있다.
- [35] 상기 설치단계에서 복수 개의 상기 노칭탭의 절곡된 부분을 서로 겹쳐지게 하여 상기 캔부재에 동시에 접합할 수 있다.
- [36] 상기 권취단계는 상기 노칭 탭부를 2겹 이상으로 겹쳐지도록 형성할 수 있다.

발명의 효과

- [37] 본 발명에 따르면, 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있다.
- [38] 본 발명에 따르면, 멀티탭을 형성하여 전기 저항을 극소화하는 효과가 있다.
- [39] 본 발명에 따르면, 무지부 형성을 극소화하여 전지 용량을 확보하는 효과가 있다.
- [40] 본 발명에 따르면, 전극 포일로부터 포일탭인 노칭탭을 형성하면서 노칭탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높이는 효과가 있다.
- [41] 본 발명에 따르면, 포일탭인 노칭탭의 기계적 강도를 강화하여 노칭탭이 절곡된 상태에서 끊어지지 않게 함에 따라 노칭탭을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높이는 효과가

있다.

[42] 본 발명에 따르면, 노칭탭과 전극탭을 동시에 캔부재에 접합할 수 있어 용접 공정성을 확보하는 효과가 있다.

[43] 본 발명에 따르면, 복수 개의 노칭탭을 전극조립체의 외곽에 형성하여 복수 개의 노칭탭과 전극조립체의 단부 사이에 절연부재를 설치할 수 있게 함에 따라 노칭탭과 전극조립체의 접촉을 방지하여 안전성을 향상하는 효과가 있다.

도면의 간단한 설명

[44] 도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이다.

[45] 도 2는 도 1을 권취한 상태로 도시한 사시도이다.

[46] 도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.

[47] 도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이다.

[48] 도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.

[49] 도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.

[50] 도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

[51] 도 8은 본 발명의 다른 실시예에 따라 노칭탭과 전극탭을 접합한 것을 도면에서 도시한 저면도이다.

[52] 도 9는 도 8을 측면에서 도시한 측면도이다.

발명의 실시를 위한 최선의 형태

[53] 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지 및 그 이차전지의 제조 방법에 대하여 상세히 설명하기로 한다.

[54] 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의 가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

[55] 도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를

불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

[56]

[57] 도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이고, 도 2는 도 1을 권취한 상태로 도시한 사시도이다.

[58]

도 1 내지 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100) 및 상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부(111)를 포함한다.(도 2에서 2겹으로 겹쳐진 노칭 탭부(111)가 도시되고 있다.)

[59]

전극집전체의 전극 포일로부터 포일탭인 노칭 탭부를 형성하면서 포일탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높일 수 있다.

[60]

전극(100)은 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 중 어느 하나일 수 있으며, 양극과 음극 사이에 분리막이 개재되도록 양극, 음극 및 분리막을 복수 회 적층하여 전극조립체(1)를 제작할 수 있다.

[61]

그리고 전극조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수 있다.

[62]

양극은 알루미늄 전극집전체일 수 있으며, 양극 활물질이 도포된 양극 유지부와, 양극 활물질이 도포되지 않은 양극 무지부를 포함할 수 있다.

[63]

양극 활물질은 LiCoO_2 , LiNiO_2 , LiMnO_2 , LiMnO_4 와 같은 리튬 함유 전이금속 산화물 또는 리튬 칼코게나이드 화합물일 수 있다.

[64]

양극 유지부는 예를 들어, 알루미늄 전극집전체의 적어도 어느 한 면의 일부에 양극 활물질을 도포하여 형성하며, 양극 활물질이 미 도포된 알루미늄 전극집전체의 나머지 부분이 양극 무지부가 될 수 있다.

[65]

음극은 구리 전극집전체일 수 있으며, 음극 활물질이 도포된 음극 유지부와 음극 활물질이 도포되지 않은 음극 무지부를 포함할 수 있다.

[66]

음극 활물질은, 결정질 탄소, 비정질 탄소, 탄소 복합체, 탄소 섬유와 같은 탄소 재료, 리튬 금속 또는 리튬 합금 등일 수 있다.

[67]

음극 유지부는 예를 들어, 구리 전극집전체의 적어도 어느 한 면의 일부에 음극 활물질을 도포하여 형성하며, 음극 활물질이 미 도포된 구리 전극집전체의 나머지 부분이 음극 무지부가 될 수 있다.

[68]

분리막은 예를 들어, 폴리에틸렌(PE), 폴리스틸렌(PS), 폴리프로필렌(PP) 및 폴리에틸렌(PE)과 폴리프로필렌(PP)의 공중합체(co-polymer)로 이루어지는 군에서 선택되는 어느 하나의 기재에 폴리비닐리덴 플로우라이드-헥사플로로프로필렌 공중합체(PVDF-HFP co-polymer)를 코팅함으로써 제조될 수 있다.

[69]

이와 같이 본 발명의 전극(100)은 전극조립체(1)를 형성하는 양극, 음극 중 어느

하나 이상일 수 있으며, 특히 양극 또는 음극 중에 저항 감소가 더 필요한 어느 하나일 수 있다.

- [70] 본 발명에서는 음극을 일 실시예로 설명한다.
- [71] 전극(100)은 일단부에 무지부(120)가 형성될 수 있고, 전극(100)의 일단부에 형성된 무지부(120)에는 전극탭(121)이 부착될 수 있다.
- [72] 노칭 탭부(111)는 전극(100)의 유지부(110)로부터 폭 방향으로 연장되도록 전극(100)을 노칭(notching)함에 따라 형성될 수 있는데, 전극(100)의 유지부(110) 전체 길이의 1/3 이상에 해당하는 영역에서 전극집전체만 연장되도록 형성되며 활물질(a)은 도포되지 않을 수 있다.
- [73] 이러한 노칭 탭부(111)는 전극(100)으로부터 연장된 유지부(110)에서 활물질(a)만을 제거하여 형성할 수 있고, 또는 전극(100)의 유지부(110) 영역에서 활물질(a)이 도포되지 않은 채 전극 포일만 연장하여 형성할 수도 있다.
- [74] 따라서 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있을 수 있고, 무지부 형성을 극소화하여 전지 용량을 확보할 수 있다.
- [75] 도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.
- [76] 도 3에 도시된 바와 같이, 노칭 탭부(111)는 권취된 상태에서 커팅되어 복수 개의 노칭탭(111a)을 형성할 수 있다. 노칭탭(111a)의 수가 많아지면 전체 전극 탭의 저항은 줄어들게 된다.
- [77] 이때, 하나의 노칭탭(111a)의 폭은 4mm ~ 5mm를 형성하는 것이 좋은데, 이러한 이유는 노칭탭(111a)의 폭이 4mm 미만이면 기계적 강도가 약할 수 있고, 노칭탭(111a)의 폭이 5mm 초과하면 캔부재와의 전기적 연결을 위한 접합의 과정에서 변형이 어려울 수 있기 때문이다.
- [78] 포일탭인 노칭탭(111a)의 기계적 강도를 강화하여 노칭탭(111a)이 절곡된 상태에서 끊어지지 않게 함에 따라 노칭탭(111a)을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높일 수 있다.
- [79] 도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이고, 도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.
- [80] 도 4 및 도 5에 도시된 바와 같이, 본 발명의 이차전지에 따른 전극조립체(1)는 캔부재(10)에 수용하면서 노칭탭(111a)을 절곡한 상태로 캔부재(10)와 전기적으로 연결되도록 캔부재(10)에 노칭탭(111a)의 절곡부를 용접 결합할 수 있다.
- [81] 노칭탭(111a)을 절곡하여 노칭탭(111a)의 끝단이 캔부재(10)의 저면 중심 측에 결합되게 하여 전기적 연결을 원활히 하면서 복수 개의 노칭탭(111a)이 전기적으로 연결되어 전기 저항을 극소화할 수 있다.
- [82] 도 8은 본 발명의 다른 실시예에 따라 노칭탭과 전극탭을 접합한 것을 도면에서 도시한 저면도이고, 도 9는 도 8을 측면에서 도시한 측면도이다.

- [83] 도 8 및 도 9에 도시된 바와 같이, 본 발명의 다른 실시예에 따른 이차전지는 전극(100)이 권취된 전극조립체(1)에서 노칭 탭부(111)는 커팅되어 복수 개의 노칭탭(111a)을 형성하고, 복수 개의 노칭탭(111a)은 절곡되어 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)에 겹쳐지게 연결될 수 있다.
- [84] 그리고 전극조립체(1)는 캔부재(10)에 수용되고, 전극탭(121)과 겹쳐지게 연결된 복수 개의 노칭탭(111a)은 전극탭(121)과 함께 캔부재(10)에 용접 등에 의해 접합될 수 있다.
- [85] 상술한 바와 같이 본 발명의 다른 실시예에 따른 이차전지는 복수 개의 노칭탭(111a)과 전극탭(121)을 서로 겹쳐지게 연결한 상태에서 캔부재(10)에 함께 용접 접합하기 때문에 용접 공정성 확보가 쉬울 수 있다.
- [86] 본 발명의 다른 실시예에 따른 이차전지는 전극조립체(1)의 단부의 지름의 길이가 16mm ~ 21mm일 때, 복수 개의 노칭탭(111a)의 노칭 길이가 8mm ~ 11mm의 범위일 수 있다. 이러한 이유는 노칭탭(111a)의 노칭 길이가 8mm미만이면 노칭탭(111a)의 길이가 짧아 노칭탭(111a)이 절곡되어 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)에 겹쳐지기 어려울 수 있다. 또한, 노칭탭(111a)의 노칭 길이가 11mm를 초과 하면 노칭탭(111a)의 길이가 필요 이상으로 길어져 재료 낭비로 인한 비용 증가와 노칭탭(111a)이 캔부재(10) 내부에서 차지하는 공간이 많아져 캔부재(10) 내부에 채워지는 전해액의 양이 줄어들어 따른 용량 저하의 원인이 될 수 있다. 바람직하게 노칭탭(111a)의 노칭 길이는 9mm ~ 10mm 인 것이 좋다.
- [87] 본 발명의 다른 실시예에 따른 이차전지는 노칭탭(111a)의 절곡부와 전극(100)이 권취된 전극조립체(1)의 단부와 사이에 절연부재(130)를 형성할 수 있다.
- [88] 절연부재(130)는 노칭탭(111a)과 전극조립체(1)의 단부와 사이를 절연하여 노칭탭(111a)과 전극조립체(1)의 사이에 접촉을 방지할 수 있다.
- [89] 복수 개의 노칭탭(111a)은 전극조립체(1)의 단부 외곽부에 형성되어 절연부재(130)를 전극조립체(1)의 단부에 설치하는 것을 방해하지 않도록 할 수 있다.
- [90] 복수 개의 노칭탭(111a)을 전극조립체의 외곽에 형성하여 복수 개의 노칭탭(111a)과 전극조립체(1)의 단부 사이에 절연부재(130)를 설치할 수 있게 함에 따라 노칭탭(111a)과 전극조립체(1)의 접촉을 방지하여 안전성을 향상할 수 있다.
- [91]
- [92] 도면을 참조하여 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 상세히 설명한다.
- [93] 도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.
- [94] 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지의 제조 방법은

준비단계(S1), 노칭 탭부 형성단계(S2), 권취단계(S3), 노칭탭 형성단계(S4) 및 설치단계(S5)를 포함한다.

- [95] 상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 전극집전체의 길이방향을 따라 구비하는 전극(100)을 준비하는 단계이다.
- [96] 이때, 전극(100)은 음극일 수 있다.
- [97] 도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.
- [98] 도 7에 도시된 바와 같이, 준비단계(S1)에서 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(110)와 무지부(120)는 전극집전체의 폭방향(W)을 기준으로 병렬로 배열할 수 있다.
- [99] 상기 노칭 탭부 형성단계(S2)는 전극(100)에서 유지부(110)의 길이 1/3 이상에 유지부(110)의 폭방향(W)으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록 전극(100)을 노칭하는 단계이다.
- [100] 이와 같이 전극(100)에서 노칭 탭부(111)를 노칭하는 것은 전극집전체에서 전극(100)을 커팅하면서 유지부(110)의 폭방향(W)으로 형성된 활물질(a)이 도포되지 않은 무지부의 일부를 노칭하여 노칭 탭부(11)를 형성하는 것일 수 있다.
- [101] 또한, 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(100)와 무지부(120)가 전극집전체의 폭방향(W)을 기준으로 병렬로 배열되었을 경우 병렬로 배열된 유지부(110)와 무지부(120)를 전극집전체의 길이 방향(L)으로 커팅하면서 하나의 유지부(110)와 폭방향(W) 기준 병렬로 배열된 다른 하나의 유지부(110a)의 사이에 형성된 활물질이 도포되지 않은 무지부 영역을 노칭하여 유지부(110)의 폭 방향(W)으로 노칭 탭부(111)가 형성되도록 할 수 있다.
- [102] 상기 권취단계(S3)는 노칭 탭부(111)가 형성된 전극(100)을 분리막과 상기 전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를 형성하는 단계일 수 있다.
- [103] 전극(100)이 권취되는 과정에서 노칭 탭부(111)는 2겹 이상으로 겹쳐지게 형성될 수 있다.
- [104] 상기 노칭탭 형성단계(S4)는 전극조립체(1)에서 노칭 탭부(111)를 커팅하여 복수 개의 노칭탭(111a)을 형성하는 단계일 수 있다.
- [105] 노칭 탭부(111)가 커팅되어 형성된 노칭탭(111a)은 노칭 탭부(111)와 동일하게 2겹 이상으로 겹쳐진 상태를 형성할 수 있다.
- [106] 이때 노칭 탭부(111)의 커팅은 레이저를 사용하여 작업 효율을 극대화할 수 있다.
- [107] 상기 설치단계(S5)는 전극조립체(1)를 캔부재(10)에 수용하여 노칭탭(111a)을 캔부재(10)에 접합하는 단계이다.

- [108] 설치단계(S5)에서 노칭탭(111a)이 캔부재(10)에 전기적으로 연결되면서 용접 접합이 용이하도록 노칭탭(111a)을 절곡하여 노칭탭(111a)이 절곡된 상태로 전극조립체(1)가 캔부재(10)에 수용되도록 할 수 있다.
- [109] 이때, 노칭탭(111a)은 2겹 이상으로 형성되었기 때문에 노칭탭(111a)이 절곡되어도 기계적 강도를 유지할 수 있어 쉽게 끊어지지 않는 효과가 있다.
- [110] 절곡된 노칭탭(111a)은 단부가 캔부재(10)의 저면 중앙을 향하도록 할 수 있다.
- [111] 캔부재(10)의 저면과 노칭탭(111a)의 결합은 노칭탭(111a)의 절곡부가 캔부재(10)의 저면에 용접 결합되게 할 수 있다.
- [112] 또한, 본 발명의 다른 실시예에 따르면 설치단계(S5)에서 복수 개의 노칭탭(111a)의 절곡된 부분을 전극조립체(1)의 단부 중앙측에서 서로 겹쳐지게 하여 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)과 함께 캔부재(10)에 동시에 용접 등에 의해 접합할 수도 있다.
- [113]
- [114] 이상과 같이 본 발명에 따른 이차전지 및 그 이차전지의 제조 방법을 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

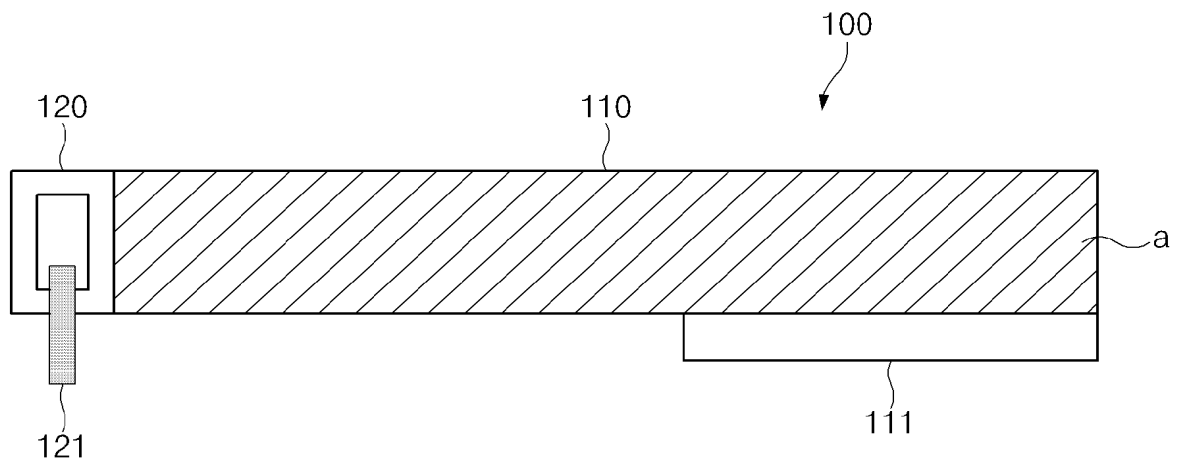
청구범위

- [청구항 1] 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100); 및
상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부(111); 를 포함하는 것을 특징으로 하는 이차전지.
- [청구항 2] 청구항 1에 있어서,
상기 노칭 탭부(111)는 상기 유지부(110) 전체 길이의 1/3 이상으로부터 연속되게 연장되는 것을 특징으로 하는 이차전지.
- [청구항 3] 청구항 1 또는 청구항 2에 있어서,
상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 커팅되어 복수 개의 노칭탭(111a)을 형성하는 것을 특징으로 하는 이차전지.
- [청구항 4] 청구항 3에 있어서,
상기 노칭탭(111a)의 폭은 4mm ~ 5mm인 것을 특징으로 하는 이차전지.
- [청구항 5] 청구항 3에 있어서,
상기 전극(100)은 캔부재(10)에 수용되고, 상기 노칭탭(111a)은 절곡되어 상기 캔부재(10)에 연결되는 것을 특징으로 하는 이차전지.
- [청구항 6] 청구항 5에 있어서,
상기 노칭탭(111a)이 절곡되어 형성된 절곡부는 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.
- [청구항 7] 청구항 6에 있어서,
상기 절곡부와 상기 전극(100)의 단부와의 사이에 형성된 절연부재(130)를 포함하는 것을 특징으로 하는 이차전지.
- [청구항 8] 청구항 1에 있어서,
상기 무지부(120)에 부착되는 전극탭(121)을 포함하는 것을 특징으로 하는 이차전지.
- [청구항 9] 청구항 8에 있어서,
상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 노칭(notching)되어 복수 개의 노칭탭(111a)을 형성하고,
상기 전극(100)은 캔부재(10)에 수용되고, 상기 복수 개의 노칭탭(111a)은 절곡되어 상기 전극탭(121)에 겹쳐지게 연결되는 것을 특징으로 하는 이차전지.
- [청구항 10] 청구항 9에 있어서,
상기 전극탭(121) 및 상기 전극탭(121)과 겹쳐지게 연결된 상기 복수 개의 노칭탭(111a)은 함께 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.

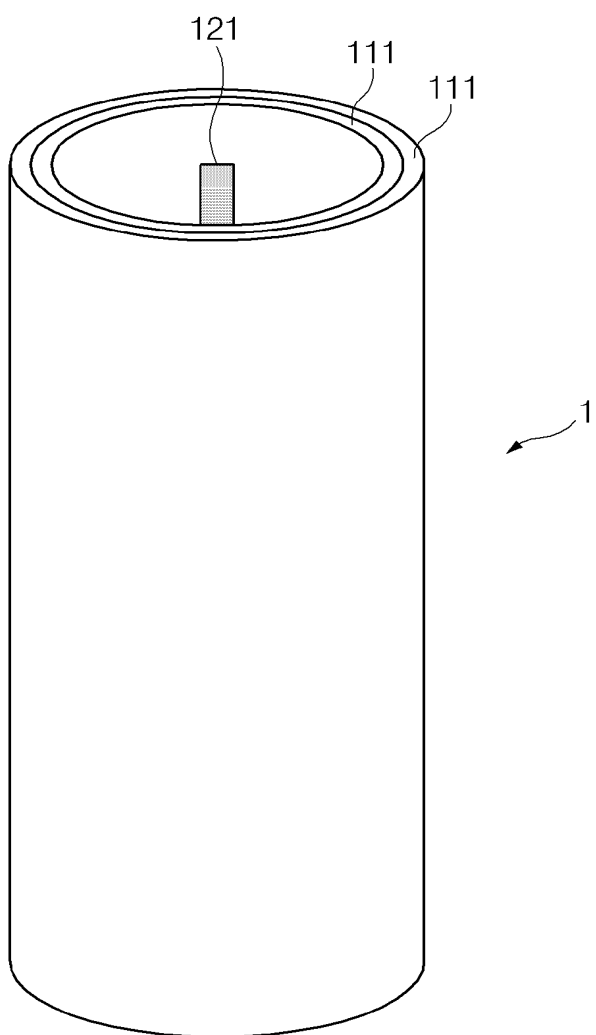
- [청구항 11] 청구항 9에 있어서,
상기 복수 개의 노칭탭(111a)의 노칭 길이는 8mm ~ 11mm인 것을
특징으로 하는 이차전지.
- [청구항 12] 청구항 9에 있어서,
상기 복수 개의 노칭탭(111a)의 노칭 길이는 9mm ~ 10mm 인 것을
특징으로 하는 이차전지.
- [청구항 13] 청구항 1에 있어서,
상기 전극(100)은 음극인 것을 특징으로 하는 이차전지.
- [청구항 14] 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지
않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는
전극(100)을 준비하는 준비단계(S1);
상기 전극(100)에서 유지부(110)의 길이 1/3 이상에 상기 전극집전체의
폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록
상기 전극(100)을 노칭하는 노칭 탭부 형성단계(S2);
상기 노칭 탭부(111)가 형성된 상기 전극(100)을 분리막과 상기
전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를
형성하는 권취단계(S3);
상기 전극조립체(1)에서 상기 노칭 탭부(111)를 커팅하여 복수 개의
노칭탭(111a)을 형성하는 노칭탭 형성단계(S4); 및
상기 전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)을 상기
캔부재(10)에 접합하는 설치단계(S5); 를 포함하는 것을 특징으로 하는
이차전지의 제조 방법.
- [청구항 15] 청구항 14에 있어서,
상기 준비단계(S1)에서 음극의 전극(100)을 준비하는 것을 특징으로 하는
이차전지의 제조 방법.
- [청구항 16] 청구항 14에 있어서,
상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와
활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 폭방향을
기준으로 병렬로 배열하는 것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 17] 청구항 16에 있어서,
상기 노칭 탭부 형성단계(S2)는 상기 병렬로 배열된 상기 유지부(110)와
무지부(120)를 상기 전극집전체의 길이 방향으로 커팅하면서 상기
유지부(110)의 폭 방향으로 활물질(a)이 도포되지 않은 상기 노칭
탭부(11)를 노칭하는 것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 18] 청구항 14에 있어서,
상기 노칭 탭부 형성단계(S2)는 상기 전극(100)을 커팅하면서 상기
유지부(110)의 폭방향으로 활물질(a)이 도포되지 않은 상기 노칭
탭부(111)가 형성되도록 노칭하는 것을 특징으로 하는 이차전지의 제조

- 방법.
- [청구항 19] 청구항 14에 있어서,
상기 노칭탭 형성단계(S4)에서 상기 노칭 탭부(111)를 레이저로 커팅하는
것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 20] 청구항 14에 있어서,
상기 설치단계(S5)는 상기 노칭탭(111a)을 절곡한 상태로 상기
전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)에서 절곡된
부분을 상기 캔부재(10)에 접합하는 것을 특징으로 하는 이차전지의 제조
방법.
- [청구항 21] 청구항 20에 있어서,
상기 설치단계(S5)에서 상기 노칭탭(111a)의 절곡된 부분을 용접에 의해
상기 캔부재(10)에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 22] 청구항 20에 있어서,
상기 설치단계(S5)에서 복수 개의 상기 노칭탭(111a)의 절곡된 부분을
서로 겹쳐지게 하여 상기 캔부재(10)에 동시에 접합하는 것을 특징으로
하는 이차전지의 제조 방법.
- [청구항 23] 청구항 14에 있어서,
상기 권취단계(S3)는 상기 노칭 탭부(111)를 2겹 이상으로 겹쳐지도록
형성하는 것을 특징으로 하는 이차전지의 제조 방법.

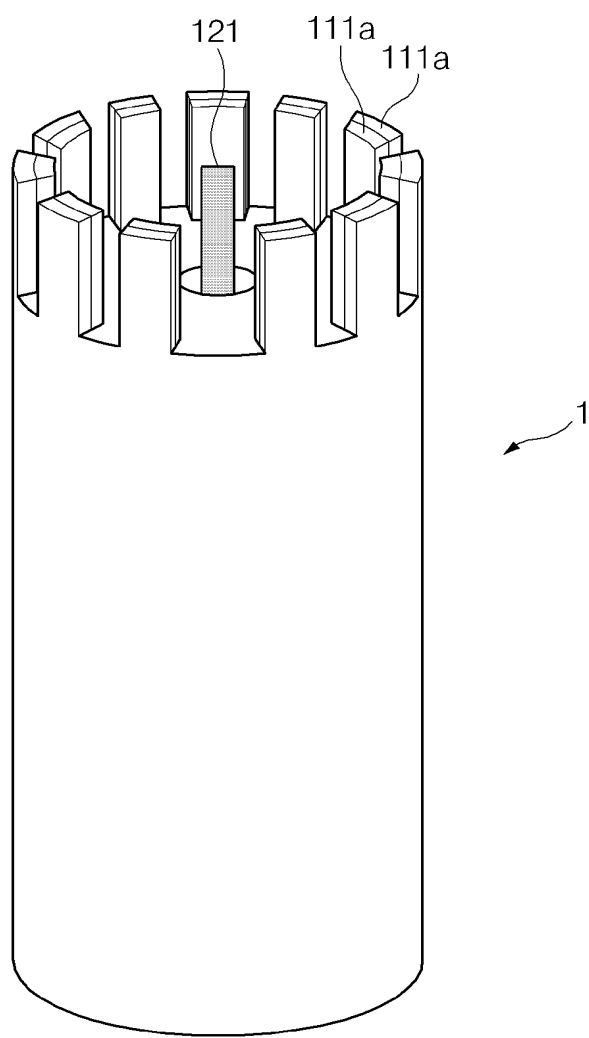
[도1]



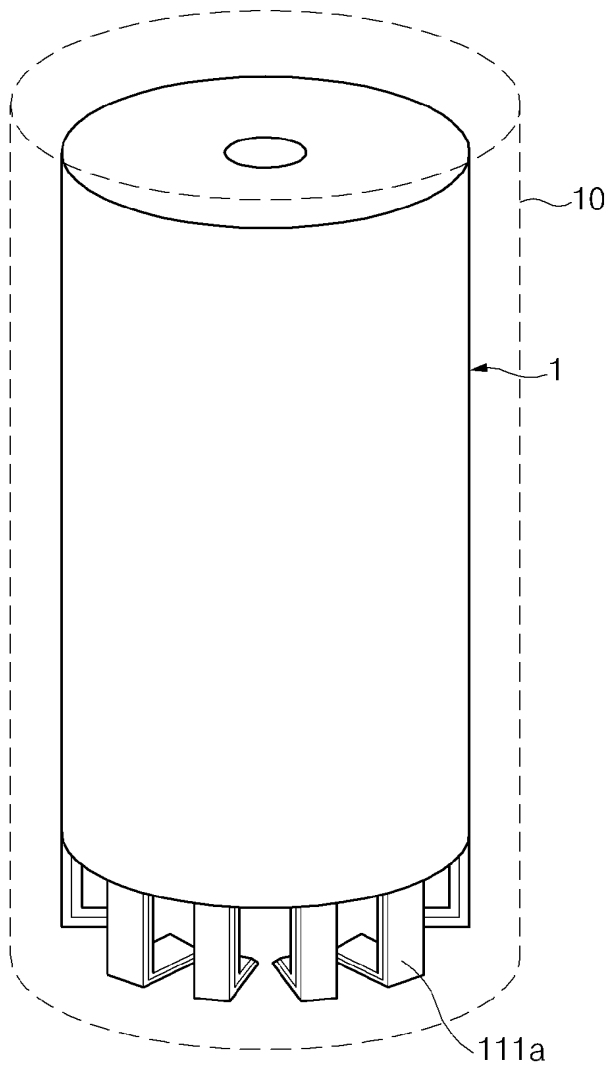
[도2]



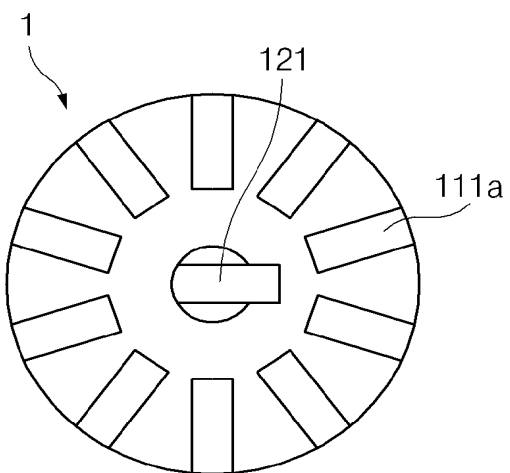
[도3]



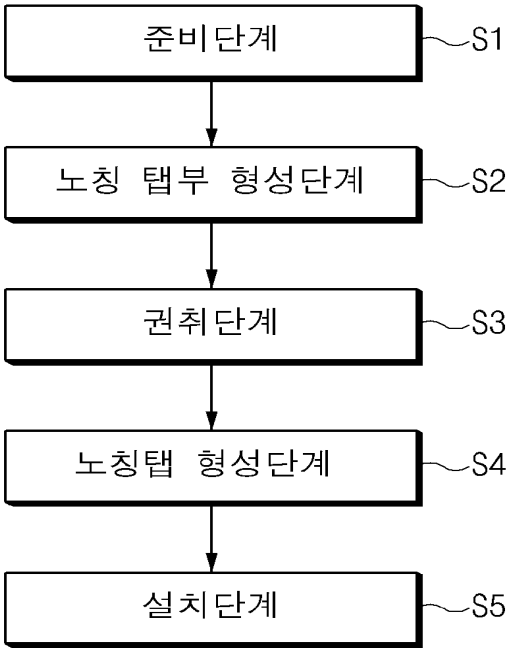
[도4]



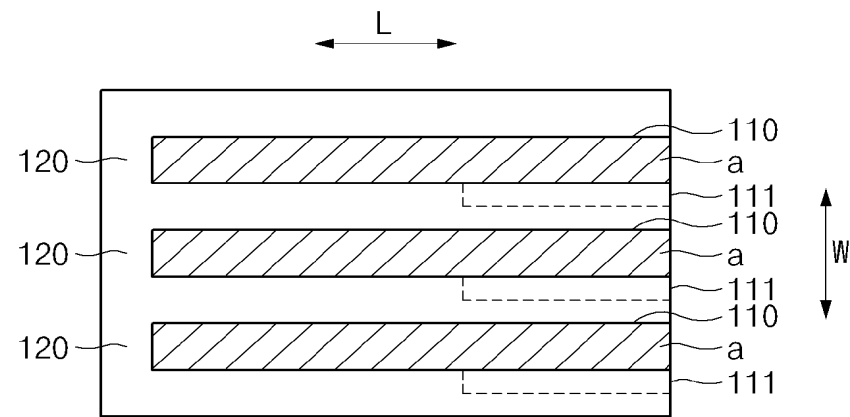
[도5]



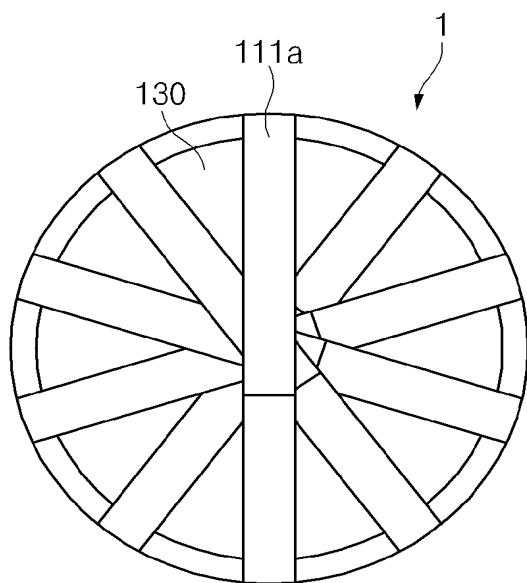
[도6]



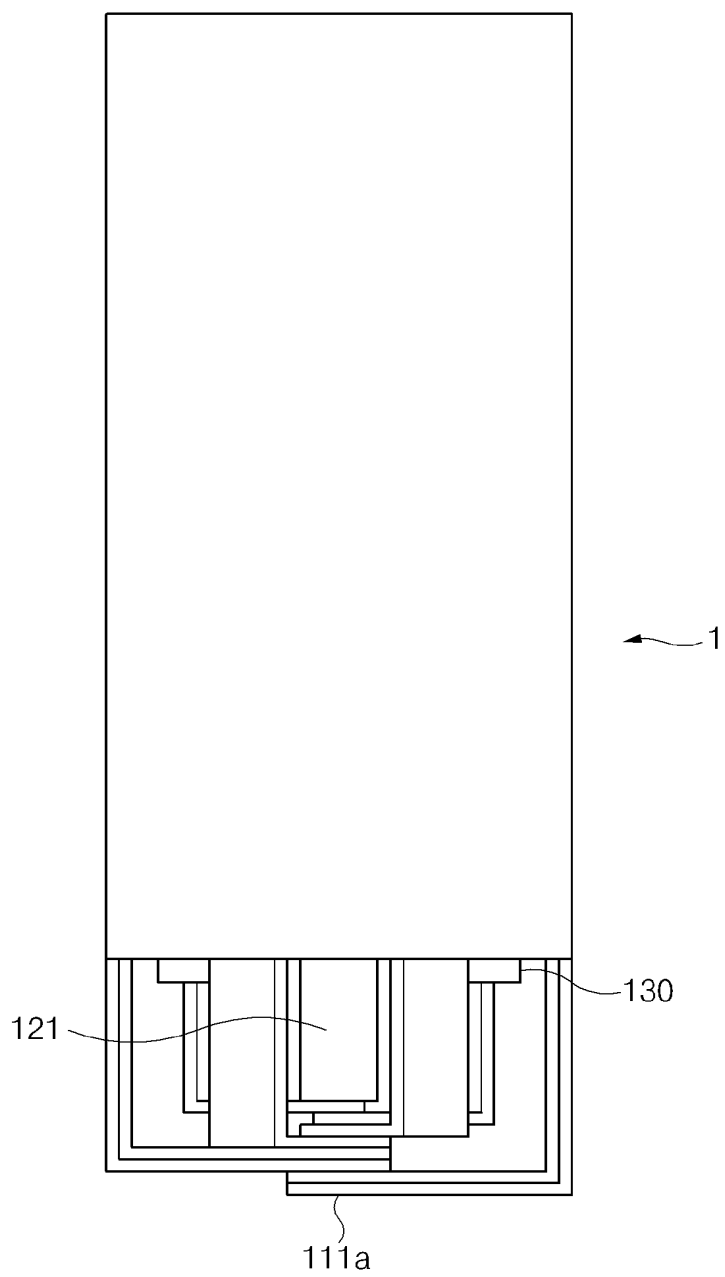
[도7]



[도8]



[도9]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2018/004366

A. CLASSIFICATION OF SUBJECT MATTER		
<i>H01M 2/26(2006.01)i, H01M 10/04(2006.01)i, H01M 10/0587(2010.01)i</i>		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED		
Minimum documentation searched (classification system followed by classification symbols) H01M 2/26; H01M 10/0525; H01M 10/05; H01M 4/02; H01M 10/0587; H01M 10/38; H01M 10/04		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched Korean Utility models and applications for Utility models: IPC as above Japanese Utility models and applications for Utility models: IPC as above		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) eKOMPASS (KIPO internal) & Keywords: coated part, non-coated part, notching tap part, multi-tap, second battery		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	KR 10-2005-0121904 A (SAMSUNG SDI CO., LTD.) 28 December 2005 See paragraphs [0026]-[0063]; claim 1; and figures 1-5.	1-4,8,13
A		5-7,9-12,14-23
Y	KR 10-0515833 B1 (SAMSUNG SDI CO., LTD.) 21 September 2005 See paragraphs [0033]-[0045]; claim 1; and figure 4.	1-4,8,13
Y	KR 10-2006-0022358 A (SAMSUNG SDI CO., LTD.) 10 March 2006 See paragraphs [0015]-[0045]; claim 1; and figures 2-4.	1-4,8,13
Y	KR 10-2012-0092367 A (SAMSUNG SDI CO., LTD.) 21 August 2012 See paragraphs [0044]-[0052]; claim 1; and figure 2.	1-4,8,13
Y	KR 10-2000-0051740 A (SAMSUNG SDI CO., LTD.) 16 August 2000 See pages 2, 3; claim 1; and figures 3, 4.	1-4,8,13
☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents: “A” document defining the general state of the art which is not considered to be of particular relevance “E” earlier application or patent but published on or after the international filing date “L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) “O” document referring to an oral disclosure, use, exhibition or other means “P” document published prior to the international filing date but later than the priority date claimed “T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention “X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone “Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art “&” document member of the same patent family		
Date of the actual completion of the international search 19 JULY 2018 (19.07.2018)		Date of mailing of the international search report 20 JULY 2018 (20.07.2018)
Name and mailing address of the ISA/KR  Korean Intellectual Property Office Government Complex-Daejeon, 189 Seonsa-ro, Daejeon 302-701, Republic of Korea Facsimile No. +82-42-481-8578		Authorized officer Telephone No.

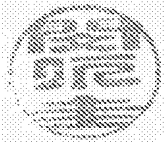
INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/KR2018/004366

Patent document cited in search report	Publication date	Patent family member	Publication date
KR 10-2005-0121904 A	28/12/2005	CN 100349321 C	14/11/2007
		CN 1713437 A	28/12/2005
		JP 2006-012827 A	12/01/2006
		KR 10-0599749 B1	12/07/2006
		US 2005-0287428 A1	29/12/2005
		US 7955736 B2	07/06/2011
KR 10-0515833 B1	21/09/2005	CN 100459233 C	04/02/2009
		CN 1574422 A	02/02/2005
		JP 2004-356085 A	16/12/2004
		JP 4418243 B2	17/02/2010
		KR 10-2004-0101652 A	03/12/2004
		US 2005-0003264 A1	06/01/2005
		US 8021781 B2	20/09/2011
KR 10-2006-0022358 A	10/03/2006	KR 10-0612236 B1	11/08/2006
KR 10-2012-0092367 A	21/08/2012	CN 102637846 A	15/08/2012
		CN 102637846 B	16/03/2016
		EP 2487738 A1	15/08/2012
		EP 2487738 B1	23/11/2016
		JP 2012-169247 A	06/09/2012
		JP 5460647 B2	02/04/2014
		KR 10-1199161 B1	12/11/2012
		US 2012-0208055 A1	16/08/2012
KR 10-2000-0051740 A	16/08/2000	US 9172109 B2	27/10/2015
		KR 10-0307452 B1	26/09/2001

국 제 조 사 보 고 서

국제출원번호
PCT/KR2018/004366

A. 발명이 속하는 기술분류(국제특허분류(IPC)) H01M 2/26(2006.01)i, H01M 10/04(2006.01)i, H01M 10/0587(2010.01)i		
B. 조사된 분야		
조사된 최소문헌(국제특허분류를 기재) H01M 2/26; H01M 10/0525; H01M 10/05; H01M 4/02; H01M 10/0587; H01M 10/38; H01M 10/04		
조사된 기술분야에 속하는 최소문헌 이외의 문헌 한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC		
국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우)) eKOMPASS(특허청 내부 검색시스템) & 키워드: 유지부, 무지부, 노칭 탭부, 멀티탭, 이차전지		
C. 관련 문헌		
카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
Y A	KR 10-2005-0121904 A (삼성에스디아이 주식회사) 2005.12.28 단락 [0026]-[0063]; 청구항 1; 및 도 1-5 참조.	1-4,8,13 5-7,9-12,14-23
Y	KR 10-0515833 B1 (삼성에스디아이 주식회사) 2005.09.21 단락 [0033]-[0045]; 청구항 1; 및 도 4 참조.	1-4,8,13
Y	KR 10-2006-0022358 A (삼성에스디아이 주식회사) 2006.03.10 단락 [0015]-[0045]; 청구항 1; 및 도 2-4 참조.	1-4,8,13
Y	KR 10-2012-0092367 A (삼성에스디아이 주식회사) 2012.08.21 단락 [0044]-[0052]; 청구항 1; 및 도 2 참조.	1-4,8,13
Y	KR 10-2000-0051740 A (삼성에스디아이 주식회사) 2000.08.16 페이지 2, 3; 청구항 1; 및 도 3, 4 참조.	1-4,8,13
☐ 추가 문헌이 C(계속)에 기재되어 있습니다. ☒ 대응특허에 관한 별지를 참조하십시오.		
* 인용된 문헌의 특별 카테고리: “A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌 “E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후에 공개된 선출원 또는 특허 문헌 “L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌 “O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌 “P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌 “T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌 “X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다. “Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다. “&” 동일한 대응특허문헌에 속하는 문헌		
국제조사의 실제 완료일 2018년 07월 19일 (19.07.2018)		국제조사보고서 발송일 2018년 07월 20일 (20.07.2018)
ISA/KR의 명칭 및 우편주소  대한민국 특허청 (35208) 대전광역시 서구 청사로 189, 4동 (둔산동, 정부대전청사) 팩스 번호 +82-42-481-8578		심사관 민인규 전화번호 +82-42-481-3326 

국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
KR 10-2005-0121904 A	2005/12/28	CN 100349321 C	2007/11/14
		CN 1713437 A	2005/12/28
		JP 2006-012827 A	2006/01/12
		KR 10-0599749 B1	2006/07/12
		US 2005-0287428 A1	2005/12/29
		US 7955736 B2	2011/06/07
KR 10-0515833 B1	2005/09/21	CN 100459233 C	2009/02/04
		CN 1574422 A	2005/02/02
		JP 2004-356085 A	2004/12/16
		JP 4418243 B2	2010/02/17
		KR 10-2004-0101652 A	2004/12/03
		US 2005-0003264 A1	2005/01/06
KR 10-2006-0022358 A	2006/03/10	US 8021781 B2	2011/09/20
		KR 10-0612236 B1	2006/08/11
KR 10-2012-0092367 A	2012/08/21	CN 102637846 A	2012/08/15
		CN 102637846 B	2016/03/16
		EP 2487738 A1	2012/08/15
		EP 2487738 B1	2016/11/23
		JP 2012-169247 A	2012/09/06
		JP 5460647 B2	2014/04/02
KR 10-2000-0051740 A	2000/08/16	KR 10-1199161 B1	2012/11/12
		US 2012-0208055 A1	2012/08/16
		US 9172109 B2	2015/10/27
		KR 10-0307452 B1	2001/09/26

WO 2018/190691 A1

(12) 특허협력조약에 의하여 공개된 국제출원
(19) 세계지식재산권기구
국제사무국
(43) 국제공개일
2018년 10월 18일 (18.10.2018) WIPO | PCT

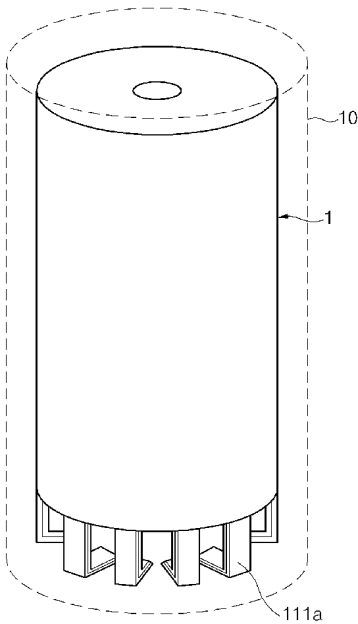


(10) 국제공개번호
WO 2018/190691 A1

- (51) 국제특허분류: *H01M 2/26* (2006.01) *H01M 10/0587* (2010.01)
H01M 10/04 (2006.01)
- (21) 국제출원번호: PCT/KR2018/004366
- (22) 국제출원일: 2018년 4월 13일 (13.04.2018)
- (25) 출원언어: 한국어
- (26) 공개언어: 한국어
- (30) 우선권정보:
10-2017-0048645 2017년 4월 14일 (14.04.2017) KR
10-2018-0043191 2018년 4월 13일 (13.04.2018) KR
- (71) 출원인: 주식회사 엘지화학 (LG CHEM, LTD.) [KR/KR]; 07336 서울시 영등포구 여의대로 128, Seoul (KR).
- (72) 발명자: 김재웅 (KIM, Jae Woong); 34122 대전시 유성구 문지로 188 LG화학 기술연구원 내, Daejeon (KR). 김형권 (KIM, Hyung Kwon); 34122 대전시 유성구 문지로 188 LG화학 기술연구원 내, Daejeon (KR). 김상욱 (KIM, Sang Uck); 34122 대전시 유성구 문지로 188 LG화학 기술연구원 내, Daejeon (KR).
- (74) 대리인: 특허법인 태평양 (BAE, KIM & LEE IP GROUP); 06626 서울시 서초구 강남대로 343, 11층, Seoul (KR).
- (81) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.
- (84) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), 유라시아 (AM, AZ, BY, KG, KZ, RU, TJ, TM), 유럽 (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

(54) Title: SECONDARY BATTERY AND SECONDARY BATTERY MANUFACTURING METHOD

(54) 발명의 명칭: 이차전지 및 그 이차전지의 제조 방법



(57) Abstract: The present invention relates to a secondary battery and a secondary battery manufacturing method, wherein the mechanical strength of an electrode tab can be increased. In addition, the present invention is characterized by comprising: an electrode having a coated portion and a non-coated portion provided along the longitudinal direction of an electrode collector, the coated portion having an active material applied onto the electrode collector, and the non-coated portion having no active material applied onto the electrode collector; and a notching tab portion extending from the coated portion in the transverse direction of the electrode collector such that the active material is not applied thereto, the notching tab portion being formed to overlap in at least two tiers when the electrode is wound.

(57) 요약서: 본 발명은 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다. 또한, 본 발명은 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.

[다음 쪽 계속]

공개:

— 국제조사보고서와 함께 (조약 제21조(3))

명세서

발명의 명칭: 이차전지 및 그 이차전지의 제조 방법

기술분야

[1] 관련출원과의 상호인용

[2] 본 출원은 2017년 04월 14일자 한국 특허 출원 제10-2017-0048645호 및 2018년 04월 13일자 한국 출원 제10-2018-0043191호에 기초한 우선권의 이익을 주장하며, 해당 한국 특허 출원의 문헌에 개시된 모든 내용은 본 명세서의 일부로서 포함된다.

[3] 기술분야

[4] 본 발명은 이차전지 및 그 이차전지의 제조 방법에 관한 것으로서, 보다 자세하게는 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다.

배경기술

[5] 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 건물로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.

[6] 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정이 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게 된다.

[7] 한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.

[8] 대한민국 공개특허공보 제10-2012-0006389호에는 종래의 전극조립체 및 이를 포함하는 이차 전지가 개시되어 있다.

[9] 종래의 이차 전지는 저저항 셀 개발을 위해서 음극 탭에 저항이 낮은 금속을 적용하거나 전극 탭 수를 늘리는 기술 적용이 필요하다.

[10] 이러한 필요성에 의해 개발된 음극 멀티탭 구조는 코어와 외곽에 각각 전극 탭을 부착하는 2탭 구조 또는 코어와 외곽부에 전극 탭을 부착하고 중간측에 미들탭을 부착하는 3탭 구조였다.

- [11] 그런데 젤리를 특성상 미들탭 부착을 위해서는 전극 중간 일정 위치에 무지부를 형성해야 하는데 무지부의 증가에 따른 전지 용량 저하의 문제점이 있었다.

발명의 상세한 설명

기술적 과제

- [12] 따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 무지부를 최소화하면서 멀티탭을 형성할 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.
- [13] 또한, 본 발명의 과제는 전극 포일을 노칭하여 전극탭을 형성하면서 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

과제 해결 수단

- [14] 본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.
- [15] 상기 노칭 탭부는 상기 유지부 전체 길이의 1/3 이상으로부터 연속되게 연장될 수 있다.
- [16] 상기 전극이 권취된 상태에서 상기 노칭 탭부는 커팅되어 복수 개의 노칭탭을 형성할 수 있다.
- [17] 상기 노칭탭의 폭은 4mm ~ 5mm일 수 있다.
- [18] 상기 전극은 캔부재에 수용되고, 상기 노칭탭은 절곡되어 상기 캔부재에 연결될 수 있다.
- [19] 상기 노칭탭이 절곡되어 형성된 절곡부는 상기 캔부재에 접합될 수 있다.
- [20] 상기 절곡부와 상기 전극의 단부와의 사이에 형성된 절연부재를 포함할 수 있다.
- [21] 상기 무지부에 부착되는 전극탭을 포함할 수 있다.
- [22] 상기 전극이 권취된 상태에서 상기 노칭 탭부는 노칭되어 복수 개의 노칭탭을 형성하고, 상기 전극은 캔부재에 수용되고, 상기 복수 개의 노칭탭은 절곡되어 상기 전극탭에 겹쳐지게 연결될 수 있다.
- [23] 상기 전극탭 및 상기 전극탭과 겹쳐지게 연결된 상기 복수 개의 노칭탭은 함께 상기 캔부재에 접합될 수 있다.
- [24] 상기 복수 개의 노칭탭의 노칭 길이는 8mm ~ 11mm일 수 있다.
- [25] 상기 복수 개의 노칭탭의 노칭 길이는 9mm ~ 10mm일 수 있다.
- [26] 상기 전극은 음극일 수 있다.
- [27] 본 발명의 일 실시예에 따른 이차전지의 제조 방법은 전극집전체 상에

활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극을 준비하는 준비단계, 상기 전극에서 유지부의 길이 1/3 이상에 상기 전극집전체의 폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부가 형성되도록 상기 전극을 노칭하는 노칭 탭부 형성단계, 상기 노칭 탭부가 형성된 상기 전극을 분리막과 상기 전극과 상극인 다른 전극에 적층한 후 권취하여 전극조립체를 형성는 권취단계, 상기 전극조립체에서 상기 노칭 탭부를 커팅하여 복수 개의 노칭탭을 형성하는 노칭탭 형성단계 및 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭을 상기 캔부재에 접합하는 설치단계를 포함한다.

- [28] 상기 준비단계에서 음극의 전극을 준비할 수 있다.
- [29] 상기 준비단계는 전극집전체 상에 활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 폭방향을 기준으로 병렬로 배열할 수 있다.
- [30] 상기 노칭 탭부 형성단계는 상기 병렬로 배열된 상기 유지부와 무지부를 상기 전극집전체의 길이 방향으로 커팅하면서 상기 유지부의 폭 방향으로 활물질이 도포되지 않은 상기 노칭 탭부를 노칭할 수 있다.
- [31] 상기 노칭 탭부 형성단계는 상기 전극을 커팅하면서 상기 유지부의 폭방향으로 활물질이 도포되지 않은 상기 노칭 탭부가 형성되도록 노칭할 수 있다.
- [32] 상기 노칭탭 형성단계에서 상기 노칭 탭부를 레이저로 커팅할 수 있다.
- [33] 상기 설치단계는 상기 노칭탭을 절곡한 상태로 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭에서 절곡된 부분을 상기 캔부재에 접합할 수 있다.
- [34] 상기 설치단계에서 상기 노칭탭의 절곡된 부분을 용접에 의해 상기 캔부재에 접합할 수 있다.
- [35] 상기 설치단계에서 복수 개의 상기 노칭탭의 절곡된 부분을 서로 겹쳐지게 하여 상기 캔부재에 동시에 접합할 수 있다.
- [36] 상기 권취단계는 상기 노칭 탭부를 2겹 이상으로 겹쳐지도록 형성할 수 있다.

발명의 효과

- [37] 본 발명에 따르면, 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있다.
- [38] 본 발명에 따르면, 멀티탭을 형성하여 전기 저항을 극소화하는 효과가 있다.
- [39] 본 발명에 따르면, 무지부 형성을 극소화하여 전지 용량을 확보하는 효과가 있다.
- [40] 본 발명에 따르면, 전극 포일로부터 포일탭인 노칭탭을 형성하면서 노칭탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높이는 효과가 있다.
- [41] 본 발명에 따르면, 포일탭인 노칭탭의 기계적 강도를 강화하여 노칭탭이 절곡된 상태에서 끊어지지 않게 함에 따라 노칭탭을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높이는 효과가 있다.

있다.

[42] 본 발명에 따르면, 노칭탭과 전극탭을 동시에 캔부재에 접합할 수 있어 용접 공정성을 확보하는 효과가 있다.

[43] 본 발명에 따르면, 복수 개의 노칭탭을 전극조립체의 외곽에 형성하여 복수 개의 노칭탭과 전극조립체의 단부 사이에 절연부재를 설치할 수 있게 함에 따라 노칭탭과 전극조립체의 접촉을 방지하여 안전성을 향상하는 효과가 있다.

도면의 간단한 설명

[44] 도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이다.

[45] 도 2는 도 1을 권취한 상태로 도시한 사시도이다.

[46] 도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.

[47] 도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이다.

[48] 도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.

[49] 도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.

[50] 도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

[51] 도 8은 본 발명의 다른 실시예에 따라 노칭탭과 전극탭을 접합한 것을 도면에서 도시한 저면도이다.

[52] 도 9는 도 8을 측면에서 도시한 측면도이다.

발명의 실시를 위한 최선의 형태

[53] 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지 및 그 이차전지의 제조 방법에 대하여 상세히 설명하기로 한다.

[54] 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의 가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

[55] 도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를

불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

[56]

[57] 도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이고, 도 2는 도 1을 권취한 상태로 도시한 사시도이다.

[58]

도 1 내지 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100) 및 상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부(111)를 포함한다.(도 2에서 2겹으로 겹쳐진 노칭 탭부(111)가 도시되고 있다.)

[59]

전극집전체의 전극 포일로부터 포일탭인 노칭 탭부를 형성하면서 포일탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높일 수 있다.

[60]

전극(100)은 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 중 어느 하나일 수 있으며, 양극과 음극 사이에 분리막이 개재되도록 양극, 음극 및 분리막을 복수 회 적층하여 전극조립체(1)를 제작할 수 있다.

[61]

그리고 전극조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수 있다.

[62]

양극은 알루미늄 전극집전체일 수 있으며, 양극 활물질이 도포된 양극 유지부와, 양극 활물질이 도포되지 않은 양극 무지부를 포함할 수 있다.

[63]

양극 활물질은 LiCoO_2 , LiNiO_2 , LiMnO_2 , LiMnO_4 와 같은 리튬 함유 전이금속 산화물 또는 리튬 칼코게나이드 화합물일 수 있다.

[64]

양극 유지부는 예를 들어, 알루미늄 전극집전체의 적어도 어느 한 면의 일부에 양극 활물질을 도포하여 형성하며, 양극 활물질이 미 도포된 알루미늄 전극집전체의 나머지 부분이 양극 무지부가 될 수 있다.

[65]

음극은 구리 전극집전체일 수 있으며, 음극 활물질이 도포된 음극 유지부와 음극 활물질이 도포되지 않은 음극 무지부를 포함할 수 있다.

[66]

음극 활물질은, 결정질 탄소, 비정질 탄소, 탄소 복합체, 탄소 섬유와 같은 탄소 재료, 리튬 금속 또는 리튬 합금 등일 수 있다.

[67]

음극 유지부는 예를 들어, 구리 전극집전체의 적어도 어느 한 면의 일부에 음극 활물질을 도포하여 형성하며, 음극 활물질이 미 도포된 구리 전극집전체의 나머지 부분이 음극 무지부가 될 수 있다.

[68]

분리막은 예를 들어, 폴리에틸렌(PE), 폴리스틸렌(PS), 폴리프로필렌(PP) 및 폴리에틸렌(PE)과 폴리프로필렌(PP)의 공중합체(co-polymer)로 이루어지는 군에서 선택되는 어느 하나의 기재에 폴리비닐리덴 플로우라이드-헥사플로로프로필렌 공중합체(PVDF-HFP co-polymer)를 코팅함으로써 제조될 수 있다.

[69]

이와 같이 본 발명의 전극(100)은 전극조립체(1)를 형성하는 양극, 음극 중 어느

하나 이상일 수 있으며, 특히 양극 또는 음극 중에 저항 감소가 더 필요한 어느 하나일 수 있다.

- [70] 본 발명에서는 음극을 일 실시예로 설명한다.
- [71] 전극(100)은 일단부에 무지부(120)가 형성될 수 있고, 전극(100)의 일단부에 형성된 무지부(120)에는 전극탭(121)이 부착될 수 있다.
- [72] 노칭 탭부(111)는 전극(100)의 유지부(110)로부터 폭 방향으로 연장되도록 전극(100)을 노칭(notching)함에 따라 형성될 수 있는데, 전극(100)의 유지부(110) 전체 길이의 1/3 이상에 해당하는 영역에서 전극집전체만 연장되도록 형성되며 활물질(a)은 도포되지 않을 수 있다.
- [73] 이러한 노칭 탭부(111)는 전극(100)으로부터 연장된 유지부(110)에서 활물질(a)만을 제거하여 형성할 수 있고, 또는 전극(100)의 유지부(110) 영역에서 활물질(a)이 도포되지 않은 채 전극 포일만 연장하여 형성할 수도 있다.
- [74] 따라서 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있을 수 있고, 무지부 형성을 극소화하여 전지 용량을 확보할 수 있다.
- [75] 도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.
- [76] 도 3에 도시된 바와 같이, 노칭 탭부(111)는 권취된 상태에서 커팅되어 복수 개의 노칭탭(111a)을 형성할 수 있다. 노칭탭(111a)의 수가 많아지면 전체 전극 탭의 저항은 줄어들게 된다.
- [77] 이때, 하나의 노칭탭(111a)의 폭은 4mm ~ 5mm를 형성하는 것이 좋은데, 이러한 이유는 노칭탭(111a)의 폭이 4mm 미만이면 기계적 강도가 약할 수 있고, 노칭탭(111a)의 폭이 5mm 초과하면 캔부재와의 전기적 연결을 위한 접합의 과정에서 변형이 어려울 수 있기 때문이다.
- [78] 포일탭인 노칭탭(111a)의 기계적 강도를 강화하여 노칭탭(111a)이 절곡된 상태에서 끊어지지 않게 함에 따라 노칭탭(111a)을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높일 수 있다.
- [79] 도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이고, 도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.
- [80] 도 4 및 도 5에 도시된 바와 같이, 본 발명의 이차전지에 따른 전극조립체(1)는 캔부재(10)에 수용하면서 노칭탭(111a)을 절곡한 상태로 캔부재(10)와 전기적으로 연결되도록 캔부재(10)에 노칭탭(111a)의 절곡부를 용접 결합할 수 있다.
- [81] 노칭탭(111a)을 절곡하여 노칭탭(111a)의 끝단이 캔부재(10)의 저면 중심 측에 결합되게 하여 전기적 연결을 원활히 하면서 복수 개의 노칭탭(111a)이 전기적으로 연결되어 전기 저항을 극소화할 수 있다.
- [82] 도 8은 본 발명의 다른 실시예에 따라 노칭탭과 전극탭을 접합한 것을 도면에서 도시한 저면도이고, 도 9는 도 8을 측면에서 도시한 측면도이다.

- [83] 도 8 및 도 9에 도시된 바와 같이, 본 발명의 다른 실시예에 따른 이차전지는 전극(100)이 권취된 전극조립체(1)에서 노칭 탭부(111)는 커팅되어 복수 개의 노칭탭(111a)을 형성하고, 복수 개의 노칭탭(111a)은 절곡되어 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)에 겹쳐지게 연결될 수 있다.
- [84] 그리고 전극조립체(1)는 캔부재(10)에 수용되고, 전극탭(121)과 겹쳐지게 연결된 복수 개의 노칭탭(111a)은 전극탭(121)과 함께 캔부재(10)에 용접 등에 의해 접합될 수 있다.
- [85] 상술한 바와 같이 본 발명의 다른 실시예에 따른 이차전지는 복수 개의 노칭탭(111a)과 전극탭(121)을 서로 겹쳐지게 연결한 상태에서 캔부재(10)에 함께 용접 접합하기 때문에 용접 공정성 확보가 쉬울 수 있다.
- [86] 본 발명의 다른 실시예에 따른 이차전지는 전극조립체(1)의 단부의 지름의 길이가 16mm ~ 21mm일 때, 복수 개의 노칭탭(111a)의 노칭 길이가 8mm ~ 11mm의 범위일 수 있다. 이러한 이유는 노칭탭(111a)의 노칭 길이가 8mm미만이면 노칭탭(111a)의 길이가 짧아 노칭탭(111a)이 절곡되어 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)에 겹쳐지기 어려울 수 있다. 또한, 노칭탭(111a)의 노칭 길이가 11mm를 초과 하면 노칭탭(111a)의 길이가 필요 이상으로 길어져 재료 낭비로 인한 비용 증가와 노칭탭(111a)이 캔부재(10) 내부에서 차지하는 공간이 많아져 캔부재(10) 내부에 채워지는 전해액의 양이 줄어들어 따른 용량 저하의 원인이 될 수 있다. 바람직하게 노칭탭(111a)의 노칭 길이는 9mm ~ 10mm 인 것이 좋다.
- [87] 본 발명의 다른 실시예에 따른 이차전지는 노칭탭(111a)의 절곡부와 전극(100)이 권취된 전극조립체(1)의 단부와 사이에 절연부재(130)를 형성할 수 있다.
- [88] 절연부재(130)는 노칭탭(111a)과 전극조립체(1)의 단부와 사이를 절연하여 노칭탭(111a)과 전극조립체(1)의 사이에 접촉을 방지할 수 있다.
- [89] 복수 개의 노칭탭(111a)은 전극조립체(1)의 단부 외곽부에 형성되어 절연부재(130)를 전극조립체(1)의 단부에 설치하는 것을 방해하지 않도록 할 수 있다.
- [90] 복수 개의 노칭탭(111a)을 전극조립체의 외곽에 형성하여 복수 개의 노칭탭(111a)과 전극조립체(1)의 단부 사이에 절연부재(130)를 설치할 수 있게 함에 따라 노칭탭(111a)과 전극조립체(1)의 접촉을 방지하여 안전성을 향상할 수 있다.
- [91]
- [92] 도면을 참조하여 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 상세히 설명한다.
- [93] 도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.
- [94] 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지의 제조 방법은

준비단계(S1), 노칭 탭부 형성단계(S2), 권취단계(S3), 노칭 탭 형성단계(S4) 및 설치단계(S5)를 포함한다.

[95] 상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 전극집전체의 길이방향을 따라 구비하는 전극(100)을 준비하는 단계이다.

[96] 이때, 전극(100)은 음극일 수 있다.

[97] 도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

[98] 도 7에 도시된 바와 같이, 준비단계(S1)에서 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(110)와 무지부(120)는 전극집전체의 폭방향(W)을 기준으로 병렬로 배열할 수 있다.

[99] 상기 노칭 탭부 형성단계(S2)는 전극(100)에서 유지부(110)의 길이 1/3 이상에 유지부(110)의 폭방향(W)으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록 전극(100)을 노칭하는 단계이다.

[100] 이와 같이 전극(100)에서 노칭 탭부(111)를 노칭하는 것은 전극집전체에서 전극(100)을 커팅하면서 유지부(110)의 폭방향(W)으로 형성된 활물질(a)이 도포되지 않은 무지부의 일부를 노칭하여 노칭 탭부(11)를 형성하는 것일 수 있다.

[101] 또한, 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(100)와 무지부(120)가 전극집전체의 폭방향(W)을 기준으로 병렬로 배열되었을 경우 병렬로 배열된 유지부(110)와 무지부(120)를 전극집전체의 길이 방향(L)으로 커팅하면서 하나의 유지부(110)와 폭방향(W) 기준 병렬로 배열된 다른 하나의 유지부(110a)의 사이에 형성된 활물질이 도포되지 않은 무지부 영역을 노칭하여 유지부(110)의 폭 방향(W)으로 노칭 탭부(111)가 형성되도록 할 수 있다.

[102] 상기 권취단계(S3)는 노칭 탭부(111)가 형성된 전극(100)을 분리막과 상기 전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를 형성하는 단계일 수 있다.

[103] 전극(100)이 권취되는 과정에서 노칭 탭부(111)는 2겹 이상으로 겹쳐지게 형성될 수 있다.

[104] 상기 노칭 탭 형성단계(S4)는 전극조립체(1)에서 노칭 탭부(111)를 커팅하여 복수 개의 노칭 탭(111a)을 형성하는 단계일 수 있다.

[105] 노칭 탭부(111)가 커팅되어 형성된 노칭 탭(111a)은 노칭 탭부(111)와 동일하게 2겹 이상으로 겹쳐진 상태를 형성할 수 있다.

[106] 이때 노칭 탭부(111)의 커팅은 레이저를 사용하여 작업 효율을 극대화할 수 있다.

[107] 상기 설치단계(S5)는 전극조립체(1)를 캔부재(10)에 수용하여 노칭 탭(111a)을 캔부재(10)에 접합하는 단계이다.

- [108] 설치단계(S5)에서 노칭탭(111a)이 캔부재(10)에 전기적으로 연결되면서 용접 접합이 용이하도록 노칭탭(111a)을 절곡하여 노칭탭(111a)이 절곡된 상태로 전극조립체(1)가 캔부재(10)에 수용되도록 할 수 있다.
- [109] 이때, 노칭탭(111a)은 2겹 이상으로 형성되었기 때문에 노칭탭(111a)이 절곡되어도 기계적 강도를 유지할 수 있어 쉽게 끊어지지 않는 효과가 있다.
- [110] 절곡된 노칭탭(111a)은 단부가 캔부재(10)의 저면 중앙을 향하도록 할 수 있다.
- [111] 캔부재(10)의 저면과 노칭탭(111a)의 결합은 노칭탭(111a)의 절곡부가 캔부재(10)의 저면에 용접 결합되게 할 수 있다.
- [112] 또한, 본 발명의 다른 실시예에 따르면 설치단계(S5)에서 복수 개의 노칭탭(111a)의 절곡된 부분을 전극조립체(1)의 단부 중앙측에서 서로 겹쳐지게 하여 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)과 함께 캔부재(10)에 동시에 용접 등에 의해 접합할 수도 있다.
- [113]
- [114] 이상과 같이 본 발명에 따른 이차전지 및 그 이차전지의 제조 방법을 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

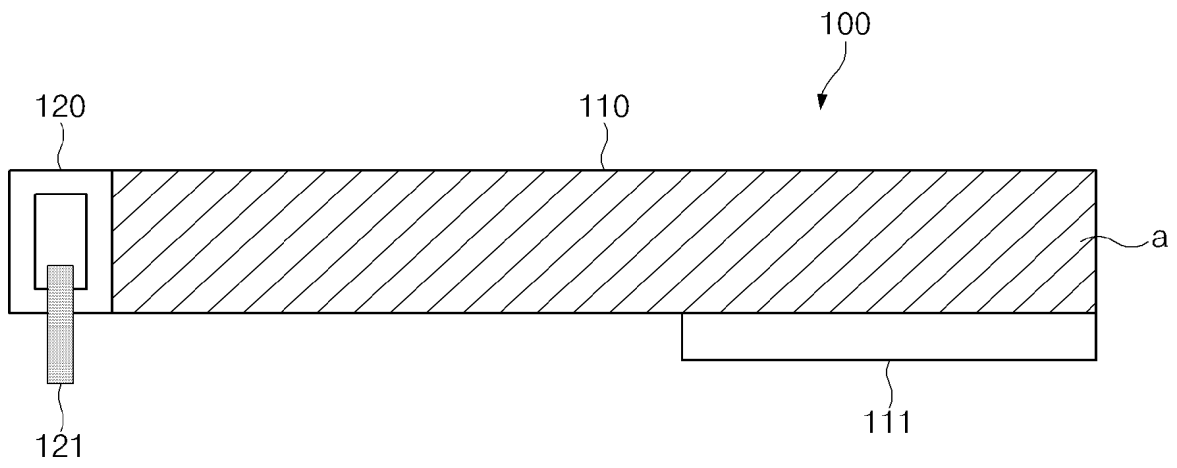
청구범위

- [청구항 1] 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100); 및
상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부(111); 를 포함하는 것을 특징으로 하는 이차전지.
- [청구항 2] 청구항 1에 있어서,
상기 노칭 탭부(111)는 상기 유지부(110) 전체 길이의 1/3 이상으로부터 연속되게 연장되는 것을 특징으로 하는 이차전지.
- [청구항 3] 청구항 1 또는 청구항 2에 있어서,
상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 커팅되어 복수 개의 노칭탭(111a)을 형성하는 것을 특징으로 하는 이차전지.
- [청구항 4] 청구항 3에 있어서,
상기 노칭탭(111a)의 폭은 4mm ~ 5mm인 것을 특징으로 하는 이차전지.
- [청구항 5] 청구항 3에 있어서,
상기 전극(100)은 캔부재(10)에 수용되고, 상기 노칭탭(111a)은 절곡되어 상기 캔부재(10)에 연결되는 것을 특징으로 하는 이차전지.
- [청구항 6] 청구항 5에 있어서,
상기 노칭탭(111a)이 절곡되어 형성된 절곡부는 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.
- [청구항 7] 청구항 6에 있어서,
상기 절곡부와 상기 전극(100)의 단부와의 사이에 형성된 절연부재(130)를 포함하는 것을 특징으로 하는 이차전지.
- [청구항 8] 청구항 1에 있어서,
상기 무지부(120)에 부착되는 전극탭(121)을 포함하는 것을 특징으로 하는 이차전지.
- [청구항 9] 청구항 8에 있어서,
상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 노칭(notching)되어 복수 개의 노칭탭(111a)을 형성하고,
상기 전극(100)은 캔부재(10)에 수용되고, 상기 복수 개의 노칭탭(111a)은 절곡되어 상기 전극탭(121)에 겹쳐지게 연결되는 것을 특징으로 하는 이차전지.
- [청구항 10] 청구항 9에 있어서,
상기 전극탭(121) 및 상기 전극탭(121)과 겹쳐지게 연결된 상기 복수 개의 노칭탭(111a)은 함께 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.

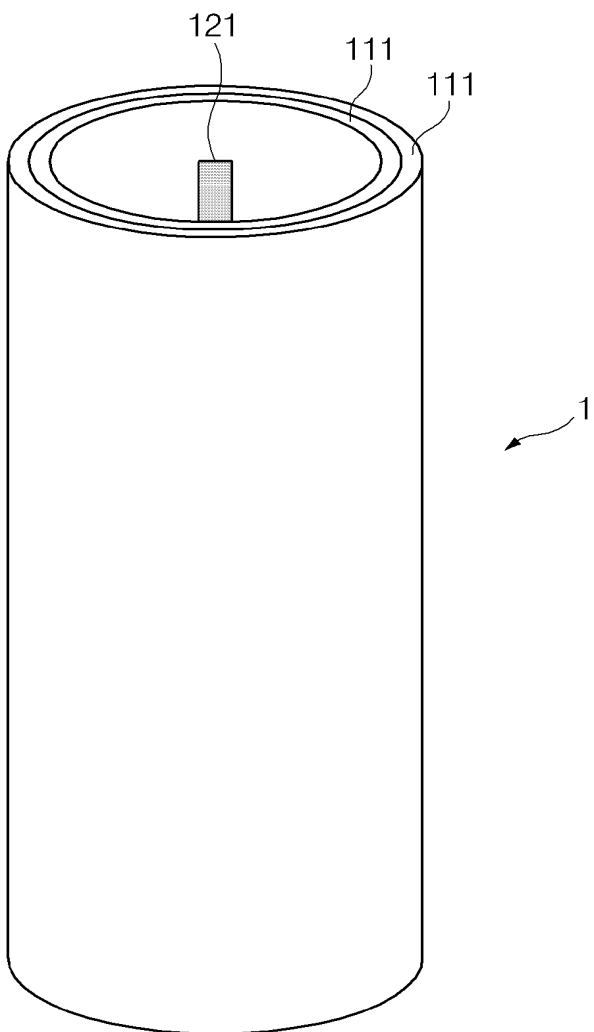
- [청구항 11] 청구항 9에 있어서,
상기 복수 개의 노칭탭(111a)의 노칭 길이는 8mm ~ 11mm인 것을
특징으로 하는 이차전지.
- [청구항 12] 청구항 9에 있어서,
상기 복수 개의 노칭탭(111a)의 노칭 길이는 9mm ~ 10mm 인 것을
특징으로 하는 이차전지.
- [청구항 13] 청구항 1에 있어서,
상기 전극(100)은 음극인 것을 특징으로 하는 이차전지.
- [청구항 14] 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지
않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는
전극(100)을 준비하는 준비단계(S1);
상기 전극(100)에서 유지부(110)의 길이 1/3 이상에 상기 전극집전체의
폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록
상기 전극(100)을 노칭하는 노칭 탭부 형성단계(S2);
상기 노칭 탭부(111)가 형성된 상기 전극(100)을 분리막과 상기
전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를
형성하는 권취단계(S3);
상기 전극조립체(1)에서 상기 노칭 탭부(111)를 커팅하여 복수 개의
노칭탭(111a)을 형성하는 노칭탭 형성단계(S4); 및
상기 전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)을 상기
캔부재(10)에 접합하는 설치단계(S5); 를 포함하는 것을 특징으로 하는
이차전지의 제조 방법.
- [청구항 15] 청구항 14에 있어서,
상기 준비단계(S1)에서 음극의 전극(100)을 준비하는 것을 특징으로 하는
이차전지의 제조 방법.
- [청구항 16] 청구항 14에 있어서,
상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와
활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 폭방향을
기준으로 병렬로 배열하는 것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 17] 청구항 16에 있어서,
상기 노칭 탭부 형성단계(S2)는 상기 병렬로 배열된 상기 유지부(110)와
무지부(120)를 상기 전극집전체의 길이 방향으로 커팅하면서 상기
유지부(110)의 폭 방향으로 활물질(a)이 도포되지 않은 상기 노칭
탭부(11)를 노칭하는 것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 18] 청구항 14에 있어서,
상기 노칭 탭부 형성단계(S2)는 상기 전극(100)을 커팅하면서 상기
유지부(110)의 폭방향으로 활물질(a)이 도포되지 않은 상기 노칭
탭부(111)가 형성되도록 노칭하는 것을 특징으로 하는 이차전지의 제조

- 방법.
- [청구항 19] 청구항 14에 있어서,
상기 노칭탭 형성단계(S4)에서 상기 노칭 탭부(111)를 레이저로 커팅하는
것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 20] 청구항 14에 있어서,
상기 설치단계(S5)는 상기 노칭탭(111a)을 절곡한 상태로 상기
전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)에서 절곡된
부분을 상기 캔부재(10)에 접합하는 것을 특징으로 하는 이차전지의 제조
방법.
- [청구항 21] 청구항 20에 있어서,
상기 설치단계(S5)에서 상기 노칭탭(111a)의 절곡된 부분을 용접에 의해
상기 캔부재(10)에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.
- [청구항 22] 청구항 20에 있어서,
상기 설치단계(S5)에서 복수 개의 상기 노칭탭(111a)의 절곡된 부분을
서로 겹쳐지게 하여 상기 캔부재(10)에 동시에 접합하는 것을 특징으로
하는 이차전지의 제조 방법.
- [청구항 23] 청구항 14에 있어서,
상기 권취단계(S3)는 상기 노칭 탭부(111)를 2겹 이상으로 겹쳐지도록
형성하는 것을 특징으로 하는 이차전지의 제조 방법.

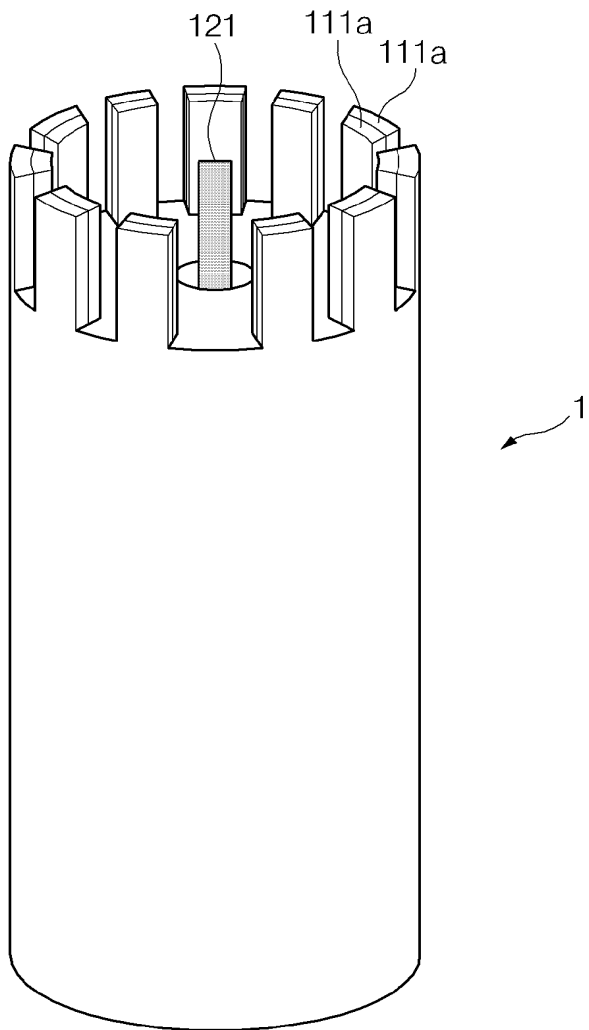
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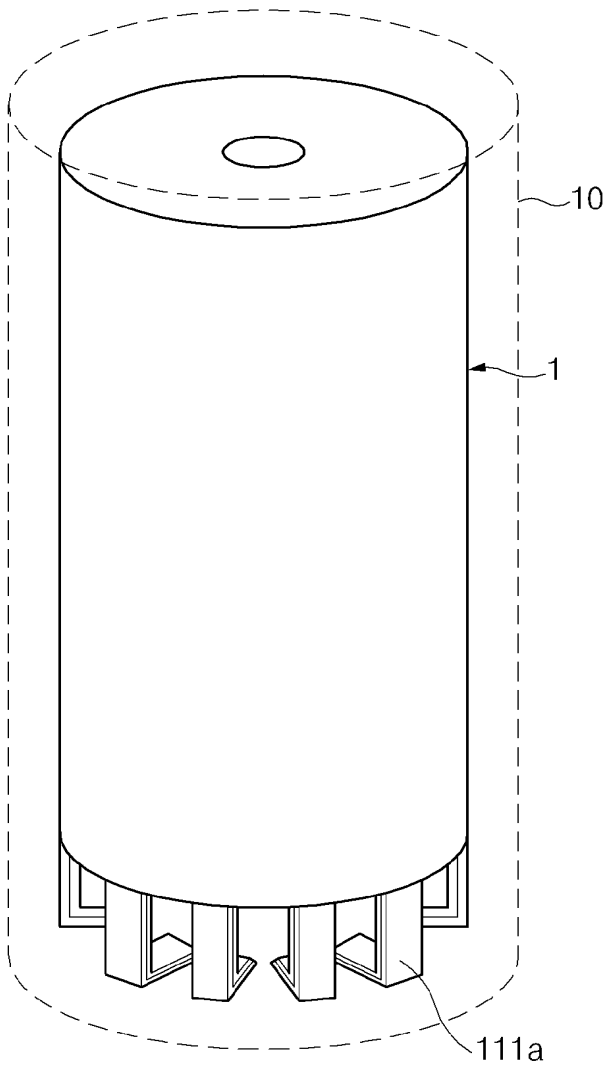
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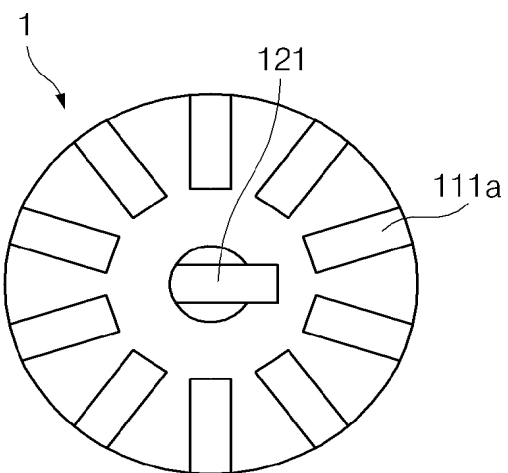
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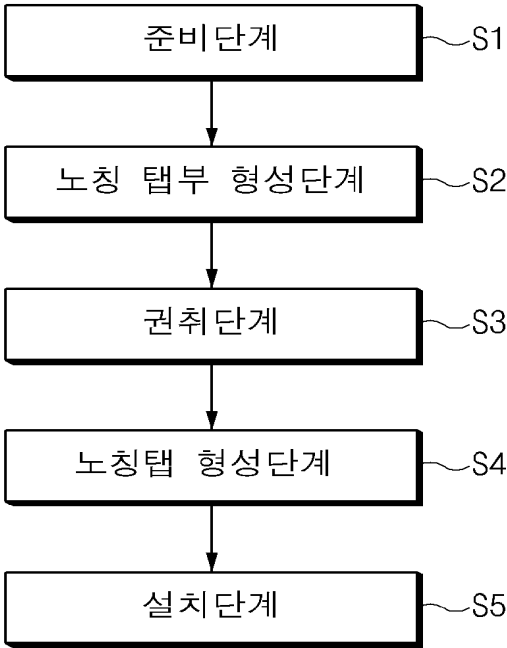
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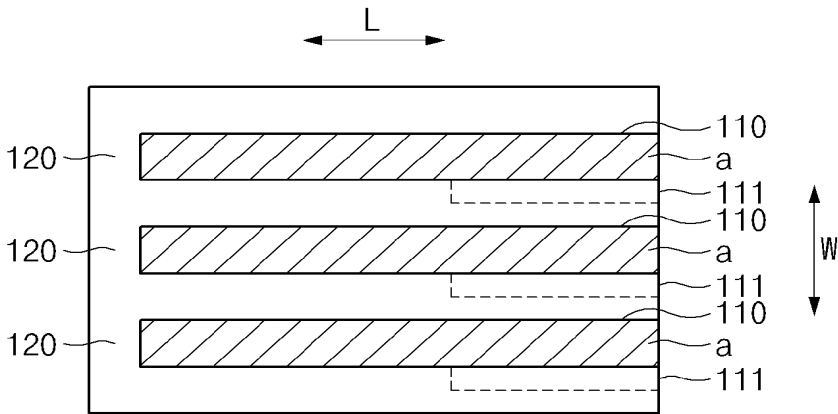
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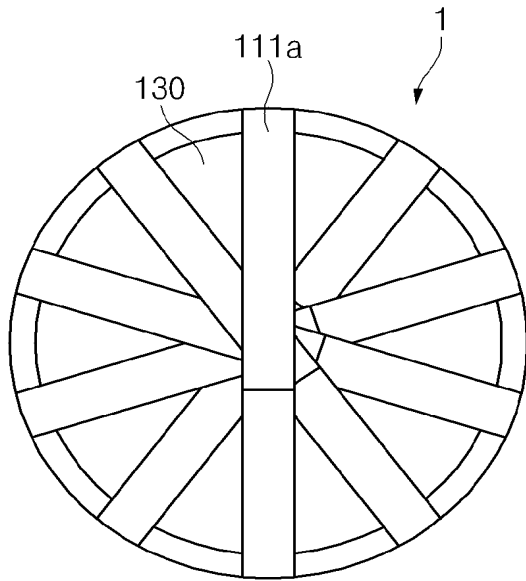
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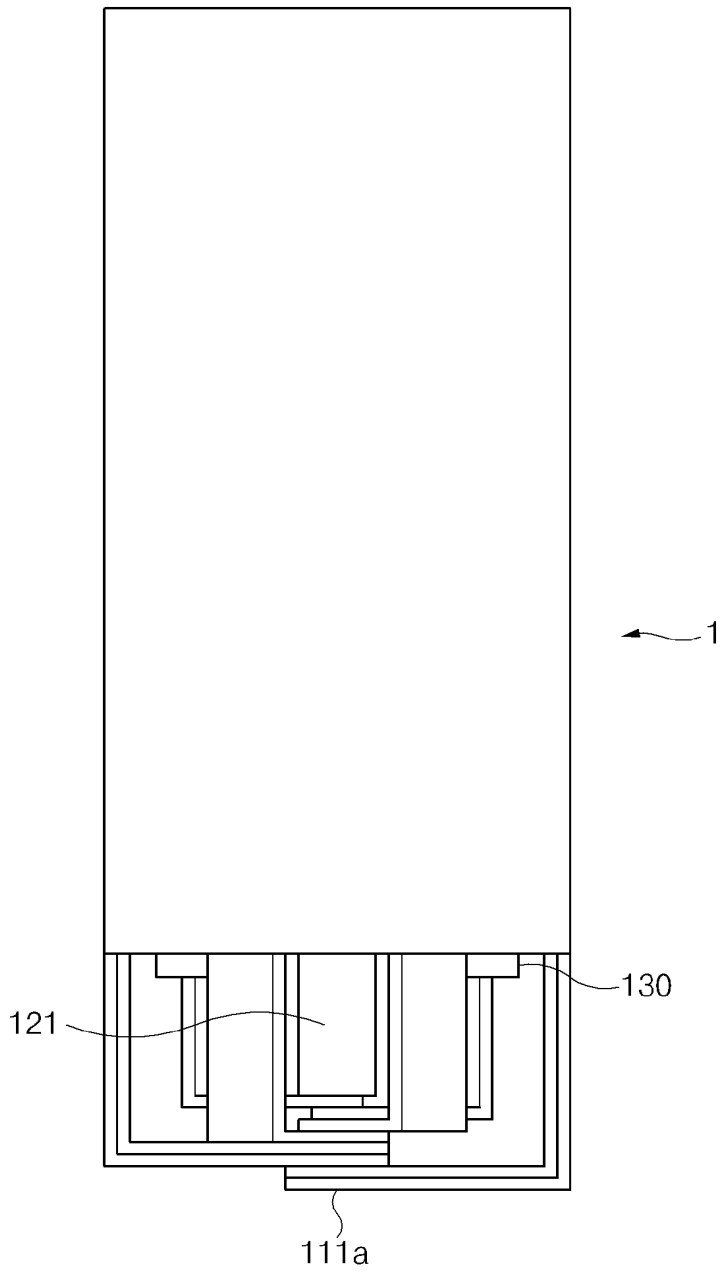
[도7]



[도8]



[도9]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2018/004366

A. CLASSIFICATION OF SUBJECT MATTER														
<i>H01M 2/26(2006.01)i, H01M 10/04(2006.01)i, H01M 10/0587(2010.01)i</i>														
According to International Patent Classification (IPC) or to both national classification and IPC														
B. FIELDS SEARCHED														
Minimum documentation searched (classification system followed by classification symbols) H01M 2/26; H01M 10/0525; H01M 10/05; H01M 4/02; H01M 10/0587; H01M 10/38; H01M 10/04														
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched Korean Utility models and applications for Utility models: IPC as above Japanese Utility models and applications for Utility models: IPC as above														
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) eKOMPASS (KIPO internal) & Keywords: coated part, non-coated part, notching tap part, multi-tap, second battery														
C. DOCUMENTS CONSIDERED TO BE RELEVANT														
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.												
Y	KR 10-2005-0121904 A (SAMSUNG SDI CO., LTD.) 28 December 2005 See paragraphs [0026]-[0063]; claim 1; and figures 1-5.	1-4, 8, 13												
A		5-7, 9-12, 14-23												
Y	KR 10-0515833 B1 (SAMSUNG SDI CO., LTD.) 21 September 2005 See paragraphs [0033]-[0045]; claim 1; and figure 4.	1-4, 8, 13												
Y	KR 10-2006-0022358 A (SAMSUNG SDI CO., LTD.) 10 March 2006 See paragraphs [0015]-[0045]; claim 1; and figures 2-4.	1-4, 8, 13												
Y	KR 10-2012-0092367 A (SAMSUNG SDI CO., LTD.) 21 August 2012 See paragraphs [0044]-[0052]; claim 1; and figure 2.	1-4, 8, 13												
Y	KR 10-2000-0051740 A (SAMSUNG SDI CO., LTD.) 16 August 2000 See pages 2, 3; claim 1; and figures 3, 4.	1-4, 8, 13												
☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.														
<table><tr><td>* Special categories of cited documents:</td><td>"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention</td></tr><tr><td>"A" document defining the general state of the art which is not considered to be of particular relevance</td><td>"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone</td></tr><tr><td>"E" earlier application or patent but published on or after the international filing date</td><td>"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art</td></tr><tr><td>"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)</td><td>"&" document member of the same patent family</td></tr><tr><td>"O" document referring to an oral disclosure, use, exhibition or other means</td><td></td></tr><tr><td>"P" document published prior to the international filing date but later than the priority date claimed</td><td></td></tr></table>			* Special categories of cited documents:	"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention	"A" document defining the general state of the art which is not considered to be of particular relevance	"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone	"E" earlier application or patent but published on or after the international filing date	"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art	"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	"&" document member of the same patent family	"O" document referring to an oral disclosure, use, exhibition or other means		"P" document published prior to the international filing date but later than the priority date claimed	
* Special categories of cited documents:	"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention													
"A" document defining the general state of the art which is not considered to be of particular relevance	"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone													
"E" earlier application or patent but published on or after the international filing date	"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art													
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	"&" document member of the same patent family													
"O" document referring to an oral disclosure, use, exhibition or other means														
"P" document published prior to the international filing date but later than the priority date claimed														
Date of the actual completion of the international search 19 JULY 2018 (19.07.2018)		Date of mailing of the international search report 20 JULY 2018 (20.07.2018)												
Name and mailing address of the ISA/KR  Korean Intellectual Property Office Government Complex-Daejeon, 189 Seonsa-ro, Daejeon 302-701, Republic of Korea Facsimile No. +82-42-481-8578		Authorized officer Telephone No.												

INTERNATIONAL SEARCH REPORT
Information on patent family members

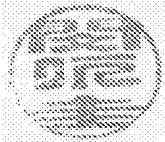
International application No.
PCT/KR2018/004366

Patent document cited in search report	Publication date	Patent family member	Publication date
KR 10-2005-0121904 A	28/12/2005	CN 100349321 C	14/11/2007
		CN 1713437 A	28/12/2005
		JP 2006-012827 A	12/01/2006
		KR 10-0599749 B1	12/07/2006
		US 2005-0287428 A1	29/12/2005
		US 7955736 B2	07/06/2011
KR 10-0515833 B1	21/09/2005	CN 100459233 C	04/02/2009
		CN 1574422 A	02/02/2005
		JP 2004-356085 A	16/12/2004
		JP 4418243 B2	17/02/2010
		KR 10-2004-0101652 A	03/12/2004
		US 2005-0003264 A1	06/01/2005
		US 8021781 B2	20/09/2011
KR 10-2006-0022358 A	10/03/2006	KR 10-0612236 B1	11/08/2006
KR 10-2012-0092367 A	21/08/2012	CN 102637846 A	15/08/2012
		CN 102637846 B	16/03/2016
		EP 2487738 A1	15/08/2012
		EP 2487738 B1	23/11/2016
		JP 2012-169247 A	06/09/2012
		JP 5460647 B2	02/04/2014
		KR 10-1199161 B1	12/11/2012
		US 2012-0208055 A1	16/08/2012
KR 10-2000-0051740 A	16/08/2000	US 9172109 B2	27/10/2015
		KR 10-0307452 B1	26/09/2001

국 제 조 사 보 고 서

국제출원번호
PCT/KR2018/004366

A. 발명이 속하는 기술분류(국제특허분류(IPC)) H01M 2/26(2006.01)i, H01M 10/04(2006.01)i, H01M 10/0587(2010.01)i		
B. 조사된 분야 조사된 최소문헌(국제특허분류를 기재) H01M 2/26; H01M 10/0525; H01M 10/05; H01M 4/02; H01M 10/0587; H01M 10/38; H01M 10/04 조사된 기술분야에 속하는 최소문헌 이외의 문헌 한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우)) eKOMPASS(특허청 내부 검색시스템) & 키워드: 유지부, 무지부, 노칭 탭부, 멀티탭, 이차전지		
C. 관련 문헌		
카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
Y A	KR 10-2005-0121904 A (삼성에스디아이 주식회사) 2005.12.28 단락 [0026]-[0063]; 청구항 1; 및 도 1-5 참조.	1-4, 8, 13 5-7, 9-12, 14-23
Y	KR 10-0515833 B1 (삼성에스디아이 주식회사) 2005.09.21 단락 [0033]-[0045]; 청구항 1; 및 도 4 참조.	1-4, 8, 13
Y	KR 10-2006-0022358 A (삼성에스디아이 주식회사) 2006.03.10 단락 [0015]-[0045]; 청구항 1; 및 도 2-4 참조.	1-4, 8, 13
Y	KR 10-2012-0092367 A (삼성에스디아이 주식회사) 2012.08.21 단락 [0044]-[0052]; 청구항 1; 및 도 2 참조.	1-4, 8, 13
Y	KR 10-2000-0051740 A (삼성에스디아이 주식회사) 2000.08.16 페이지 2, 3; 청구항 1; 및 도 3, 4 참조.	1-4, 8, 13
☐ 추가 문헌이 C(계속)에 기재되어 있습니다. ☒ 대응특허에 관한 별지를 참조하십시오.		
* 인용된 문헌의 특별 카테고리: “A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌 “T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌 “E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후에 공개된 선출원 또는 특허 문헌 “X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다. “L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌 “Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다. “O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌 “&” 동일한 대응특허문헌에 속하는 문헌 “P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌		
국제조사의 실제 완료일 2018년 07월 19일 (19.07.2018)		국제조사보고서 발송일 2018년 07월 20일 (20.07.2018)
ISA/KR의 명칭 및 우편주소 대한민국 특허청 (35208) 대전광역시 서구 청사로 189, 4동 (둔산동, 정부대전청사) 팩스 번호 +82-42-481-8578		심사관 민인규 전화번호 +82-42-481-3326



국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
KR 10-2005-0121904 A	2005/12/28	CN 100349321 C	2007/11/14
		CN 1713437 A	2005/12/28
		JP 2006-012827 A	2006/01/12
		KR 10-0599749 B1	2006/07/12
		US 2005-0287428 A1	2005/12/29
		US 7955736 B2	2011/06/07
KR 10-0515833 B1	2005/09/21	CN 100459233 C	2009/02/04
		CN 1574422 A	2005/02/02
		JP 2004-356085 A	2004/12/16
		JP 4418243 B2	2010/02/17
		KR 10-2004-0101652 A	2004/12/03
		US 2005-0003264 A1	2005/01/06
KR 10-2006-0022358 A	2006/03/10	US 8021781 B2	2011/09/20
		KR 10-0612236 B1	2006/08/11
KR 10-2012-0092367 A	2012/08/21	CN 102637846 A	2012/08/15
		CN 102637846 B	2016/03/16
		EP 2487738 A1	2012/08/15
		EP 2487738 B1	2016/11/23
		JP 2012-169247 A	2012/09/06
		JP 5460647 B2	2014/04/02
KR 10-2000-0051740 A	2000/08/16	KR 10-1199161 B1	2012/11/12
		US 2012-0208055 A1	2012/08/16
		US 9172109 B2	2015/10/27
		KR 10-0307452 B1	2001/09/26

PTO-1390 (01-18)
Approved for use through 8/31/2019. OMB 0651-0021
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

TRANSMITTAL LETTER TO THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US) CONCERNING A SUBMISSION UNDER 35 U.S.C. 371		Attorney Docket No. LGCHEM 3.3F-1654
		U.S. Application No. (if known, see 37 CFR 1.5) Not Yet Assigned
International Application No. PCT/KR2018/004366	International Filing Date 13 April 2018	Priority Date Claimed 14 April 2017
Title of Invention Secondary Battery And Method For Manufacturing The Same		
First Named Inventor Jae Woong Kim		
Applicant herewith submits to the United States Designated/Elected Office (DO/EO/US) the following items and other information.		
1. ☒ This is an express request to begin national examination procedures (35 U.S.C. 371(f)). NOTE: The express request under 35 U.S.C. 371(f) will not be effective unless the requirements under 35 U.S.C. 371(c)(1), (2), and (4) for payment of the basic national fee, copy of the International Application and English translation thereof (if required), and the oath or declaration of the inventor(s) have been received.		
2. ☒ A copy of the International Application (35 U.S.C. 371(c)(2)) is attached hereto (not required if the International Application was previously communicated by the International Bureau or was filed in the United States Receiving Office (RO/US)).		
3. An English language translation of the International Application (35 U.S.C. 371(c)(2))		
a. ☒ is attached hereto.		
b. ☐ has been previously submitted under 35 U.S.C. 154(d)(4).		
4. An oath or declaration of the inventor(s) (35 U.S.C. 371(c)(4))		
a. ☒ is attached.		
b. ☐ was previously filed in the international phase under PCT Rule 4.17(iv).		
Items 5 to 8 below concern amendments made in the international phase.		
<u>PCT Article 19 and 34 amendments</u>		
5. ☐ Amendments to the claims under PCT Article 19 are attached (not required if communicated by the International Bureau) (35 U.S.C. 371(c)(3)).		
6. ☐ English translation of the PCT Article 19 amendment is attached (35 U.S.C. 371(c)(3)).		
7. ☐ English translation of annexes (Article 19 and/or 34 amendments only) of the International Preliminary Examination Report is attached (35 U.S.C. 371(c)(5)).		
<u>Cancellation of amendments made in the international phase</u>		
8a. ☐ Do not enter the amendment made in the international phase under PCT Article 19.		
8b. ☐ Do not enter the amendment made in the international phase under PCT Article 34.		
NOTE: A proper amendment made in English under Article 19 or 34 will be entered in the U.S. national phase application absent a clear instruction from applicant not to enter the amendment(s).		
The following items 9 to 17 concern a document(s) or information included.		
9. ☒ An Information Disclosure Statement under 37 CFR 1.97 and 1.98.		
10. ☒ A preliminary amendment.		
11. ☒ An Application Data Sheet under 37 CFR 1.76.		
12. ☒ A substitute specification. NOTE: A substitute specification cannot include claims. See 37 CFR 1.125(b).		
13. ☒ A power of attorney and/or change of address letter.		
14. ☐ A computer-readable form of the sequence listing in accordance with PCT Rule 13ter.3 and 37 CFR 1.821-1.825 (not required if sequence listing in text format was indicated on the PCT Request as part of the International Application and the sequence listing was published as part of the International application).		
15. ☐ Assignment papers (<i>cover sheet and document(s)</i>). Name of Assignee: _____		
16. ☐ 37 CFR 3.73(c) Statement (<i>when there is an Assignee</i>). _____		

PTO-1390 (01-18)

Approved for use through 8/31/2019. OMB 0651-0021

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

U.S. APPLN. No. (if known – see 37 CFR 1.5) Not Yet Assigned		INTERNATIONAL APPLICATION No. PCT/KR2018/004366		ATTORNEY DOCKET No. LGCHEM 3.3F-1654	
17. ☒ Other items or information: Substitute Specification - Maked-Up Copy					
The following fees have been submitted.					CALCULATIONS
18. ☒ Basic national fee (37 CFR 1.492(a)).....					\$300
19. ☒ Examination fee (37 CFR 1.492(c))					
• If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)					\$0
• All other situations					\$760
20. ☒ Search fee (37 CFR 1.492(b))					
• If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)					\$0
• Search fee (37 CFR 1.445(a)(2)) has been paid on the international application to the USPTO as an International Searching Authority					\$140
• International Search Report prepared by an ISA other than the US and provided to the Office or previously communicated to the US by the IB					\$520
• All other situations					\$660
TOTAL OF 18, 19, and 20 =					\$ 1,580.00
☐ Additional fee for specification and drawings filed in paper over 100 sheets (excluding sequence listing in compliance with 37 CFR 1.821(c) or (e) in an electronic medium or computer program listing in an electronic medium) (37 CFR 1.492(j)).					
Fee for each additional 50 sheets of paper or fraction thereof					\$400
Total Sheets	Extra Sheets	Number of each addition 50 or fraction thereof (round up to a whole number)		RATE	
29	- 100 =			/ 50 =	x \$400
Surcharge for furnishing any of the search fee, examination fee, or the oath or declaration after the date of commencement of the national stage (37 CFR 1.492(h)).....					\$140
CLAIMS		NUMBER FILED	NUMBER EXTRA	RATE	
Total claims		20 - 20 =	0	x \$100	\$
Independent claims		2 - 3 =	0	x \$460	\$
MULTIPLE DEPENDENT CLAIM(S) (if applicable)				+ \$820	\$
Fee for submission of Sequence Listing text file of 300 MB to 800 MB (37 CFR 1.21(o)(1)).....					\$1,000
Fee for submission of Sequence Listing text file of more than 800 MB (37 CFR 1.21(o)(2)).....					\$10,000
Processing fee for furnishing the English translation later than 30 months from the earliest claimed priority date (37 CFR 1.492(i)).....					\$140 +
TOTAL OF ABOVE CALCULATIONS =					\$ 1,580.00
☐ Applicant asserts small entity status. See 37 CFR 1.27. Fees above are reduced by ½.					
☐ Applicant certifies micro entity status. See 37 CFR 1.29. Fees above are reduced by ¾. Applicant must attach form PTO/SB/15A or B or equivalent.					
TOTAL NATIONAL FEE =					\$ 1,580.00
Fee for recording the enclosed assignment (37 CFR 1.21(h)). The assignment must be accompanied by an appropriate cover sheet (37 CFR 3.28, 3.31).....					\$50.00 per property +
TOTAL FEES ENCLOSED =					\$ 1,580.00

- ADVISORY:** If filing by EFS-Web, do **NOT** attach the PTO-2038 form as a PDF along with your EFS-Web submission. Please be advised that this is **not** recommended and by doing so your **credit card information may be displayed via PAIR**. To protect your information, it is recommended to pay fees online by using the electronic payment method.

NOTE: Where an appropriate time limit under 37 CFR 1.495 has not been met, a petition to revive (37 CFR 1.137(a) or (b)) must be filed and granted to restore the International Application to pending status.

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013, and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE 1: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

NOTE 2: A U.S. national stage application may not claim priority to the international application of which it is the national phase. The filing date of a U.S. national stage application is the international filing date. See 35 U.S.C. 363.

☒ The address associated with Customer Number: 86765 **OR** ☐ Correspondence address below

Name					
Address					
City		State		Zip Code	
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Signature	/Joseph H. Karlin/	Date	September 27, 2019
Name (Print/Type)	Joseph H. Karlin	Registration No. (Attorney/Agent)	62,471

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: September 27, 2019

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

POWER OF ATTORNEY BY APPLICANT
I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.
Application Number Filing Date
(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)
[X] I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above: 86765
OR
[] I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)
Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:
[X] The address associated with the above-mentioned Customer Number
OR
[] The address associated with Customer Number:
OR
Firm or Individual Name
Address
City State Zip
Country
Telephone Email
I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):
LG Chem, Ltd.
[] Inventor or Joint Inventor (title not required below)
[] Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
[X] Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
[] Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)
SIGNATURE of Applicant for Patent
The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).
Signature Kim minh ho Date (Optional) June 28, 2016
Name Minh Kim
Title General Manager
NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.
[] Total of 1 forms are submitted.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: September 27, 2019
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

International Application No.: PCT/KR2018/004366

Confirmation No.: N/A

International Filing Date: April 13, 2018

Art Unit: N/A

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

It is respectfully requested that the reference(s) listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited reference(s) is/are legally available prior art or that the same is/are pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-1654 from which the undersigned is authorized to draw.

Dated: September 27, 2019

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/

Joseph H. Karlin

Registration No.: 62,471

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

600 South Avenue West

Westfield, New Jersey 07090

(908) 654-5000

Attorney for Applicant(s)

LD-4581

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2019-09-27
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

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	1	20050003264	A1	2005-01-06	Oh et al.	
	2	20050287428	A1	2005-12-29	Cheon et al.	
	3	20110293977	A1	2011-12-01	Kim et al.	
	4	20120009450	A1	2012-01-12	Chun	
	5	20120208055	A1	2012-08-16	Ahn	

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

	6	20150295270	A1	2015-10-15	Chun	
	7	20160351966	A1	2016-12-01	Batson et al.	

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	1	100515833	KR	B1	2005-09-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20050003264 A1 is an English language counterpart.	×
	2	20000051740	KR	A	2000-08-16	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	3	20050121904	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English language counterpart.	☒
	4	20060022358	KR	A	2006-03-10	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	5	20100071941	KR	A	2010-06-29	Lg Chemical Ltd	U.S. Publication No. 20110293977 A1 is an English language counterpart.	☒
	6	20120006389	KR	A	2012-01-18	Samsung Sdi Co., Ltd	U.S. Publication No. 20120009450 A1 is an English language counterpart.	☒
	7	20120092367	KR	A	2012-08-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20120208055 A1 is an English language counterpart.	☒

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
 (Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

	8	20130073329	KR	A	2013-07-03	Datechnology Co., Ltd	Machine generated translation is attached.	☒
	9	20150117135	KR	A	2015-10-19	Samsung Sdi Co Ltd	U.S. Publication No. 20150295270 A1 is an English language counterpart.	☒
	10	2016001575	JP	A	2016-01-07	Denso Corp	Machine generated translation is attached.	☒
	11	20160113612	KR	A	2016-09-30	A123 Systems LLC	U.S. Publication No. 20160351966 A1 is an English language counterpart.	☒
	12	20170006437	KR	A	2017-01-18	Lg Chemical Ltd	Machine generated translation is attached.	☒
	13	5790576	JP	B2	2015-10-07	Toyota Industries Corp	Machine generated translation is attached.	☒

If you wish to add additional Foreign Patent Document citation information please click the Add button

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NON-PATENT LITERATURE DOCUMENTS

Remove

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report from Application No. PCT/KR2018/004366 mailed July 20, 2018, pages 1 - 2	☒

If you wish to add additional non-patent literature document citation information please click the Add button

Add

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGCHEM 3.3F-1654

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Other Prior Art

According to the information contained in form PTO-1449 or PTO-892, there are one or more other prior art/non-patent literature documents missing from the original file history record obtained from the United States Patent and Trademark Office. Upon your request we will attempt to obtain these documents from alternative resources. Please note that additional charges will apply for this service.

PCT 출원서

출력(전자적 형태가 원본)

0	수리관청 전용	
0-1	국제출원번호	PCT/KR2018/004366
0-2	국제출원일자	2018년 04월 13일 (13.04.2018)
0-3	수리관청 명칭 및 "PCT 국제출원"	RO/KR
0-4	서식 PCT/RO/101 - PCT 출원서	
0-4-1	우측에 기재된 바와 같이 작성되었다.	PCT-SAFE 버전 3.51.080.256 MT/FOP 20180101/0.20.5.24
0-5	신청 아래 서명인은 본 국제 출원서가 특허협력조약에 의해 처리될 것을 청구합니다.	
0-6	출원인이 지정한 수리관청	대한민국 특허청 (RO/KR)
0-7	출원인 또는 대리인의 서류참조기호	2018OPLC1095
I	발명의 명칭	이차전지 및 그 이차전지의 제조 방법
II	출원인	
II-1	이 사람은	오직 출원인 (applicant only)
II-2	우측 지정국에 관한 출원인	모든 지정국 (all designated States)
II-4ko	성명	주식회사 엘지화학
II-4en	Name:	LG CHEM, LTD.
II-5ko	주소	대한민국 07336
II-5en	Address:	서울시 영등포구 여의대로 128 128, Yeoui-daero, Yeongdeungpo-gu, Seoul 07336 Republic of Korea
II-6	국적	대한민국 KR
II-7	거주국	대한민국 KR
II-11	출원인 코드	1-2001-013456-3
III-1	출원인 및/또는 발명자	
III-1-1	이 사람은	오직 발명자 (inventor only)
III-1-3	우측 지정국에 관한 발명자	모든 지정국 (all designated States)
III-1-4ko	성명	김재웅
III-1-4en	Name (LAST, First):	KIM, Jae Woong
III-1-5ko	주소	대한민국 34122
III-1-5en	Address:	대전시 유성구 문지로 188 LG화학 기술연구원 내 LG Chem Research Park, 188, Munji-ro, Yuseong-gu, Daejeon 34122 Republic of Korea

PCT 출원서

출력(전자적 형태가 원본)

III-2 III-2-1 III-2-3 III-2-4ko III-2-4en III-2-5ko III-2-5en	출원인 및/또는 발명자 이 사람은 우측 지정국에 관한 발명자 성명 Name (LAST, First): 주소 Address:	오직 발명자 (inventor only) 모든 지정국 (all designated States) 김형권 KIM, Hyoung Kwon 대한민국 34122 대전시 유성구 문지로 188 LG화학 기술연구원 내 LG Chem Research Park, 188, Munji-ro, Yuseong-gu, Daejeon 34122 Republic of Korea
III-3 III-3-1 III-3-3 III-3-4ko III-3-4en III-3-5ko III-3-5en	출원인 및/또는 발명자 이 사람은 우측 지정국에 관한 발명자 성명 Name (LAST, First): 주소 Address:	오직 발명자 (inventor only) 모든 지정국 (all designated States) 김상욱 KIM, Sang Uck 대한민국 34122 대전시 유성구 문지로 188 LG화학 기술연구원 내 LG Chem Research Park, 188, Munji-ro, Yuseong-gu, Daejeon 34122 Republic of Korea
IV-1 IV-1-1ko IV-1-1en IV-1-2ko IV-1-2en IV-1-3 IV-1-4 IV-1-5 IV-1-5(a) IV-1-6	대리인 또는 대표자 아래에 기재된 자는 관할 국제기관에 대하여 우측에 표시된 자격으로 출원인을 대리하는 것으로 선임되었다. 성명 Name: 주소 Address: 전화번호 팩스번호 이메일 주소 이메일 사용동의 수리관청, 국제조사기관, 국제사무국, 국제예비심사기관이 필요 시 이 이메일 주소를 사용하여 이 국제 출원과 관련하여 발행된 통지서를 송부할 것에 동의한다. 대리인 코드	대리인 특허법인 태평양 BAE, KIM & LEE IP GROUP 대한민국 06626 서울시 서초구 강남대로 343, 11층 11th Floor, 343, Gangnam-daero, Seocho-gu, Seoul 06626 Republic of Korea 82-2-2188-5454 82-2-2188-5455 ip@bkl.co.kr 서면 통지서에 앞서 선람용 사본 송부 9-2006-100042-1

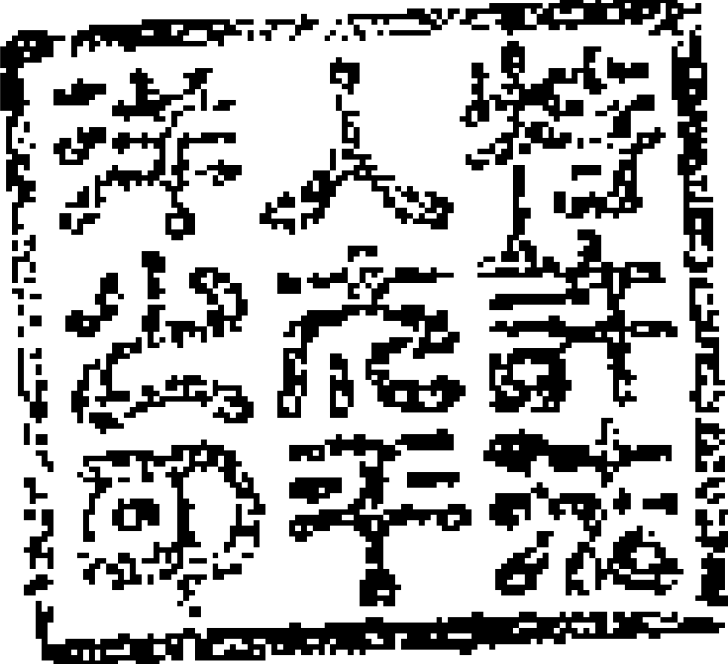
PCT 출원서

출력(전자적 형태가 원본)

V	지정국		
V-1	본 출원서의 제출로, 규칙 4.9(a)에 따라, 부여될 수 있는 모든 종류의 권리 보호를 위하여, 그리고 해당하는 경우 지역특허 및 국내특허 모두를 위하여 당해 국제출원일에 PCT에 기속되는 모든 계약국이 지정된다.		
V-2	V-2란은 출원서 제출시 또는 규칙 26의2.1에 의해 그 이후 출원서 제6기재란에 위 특정 관련 계약국의 국내 선출원에 대한 우선권주장이 포함되어 있을 경우 당해 계약국의 국내법에 의해 해당 국내 선출원의 효력이 상실되는 것을 방지하기 위한 목적으로 당해 계약국의 지정을 제외하는 데에만 사용될 수 있다 (지정 제외시 이의 취소 불가능).	KR	
VI-1	선국내출원에 대한 우선권 주장		
VI-1-1	출원일	2017년 04월 14일 (14.04.2017)	
VI-1-2	출원번호	10-2017-0048645	
VI-1-3	파리협약 당사국명 또는 WTO 회원국명	KR	
VI-2	선국내출원에 대한 우선권 주장		
VI-2-1	출원일	2018년 04월 13일 (13.04.2018)	
VI-2-2	출원번호	10-2018-0043191	
VI-2-3	파리협약 당사국명 또는 WTO 회원국명	KR	
VI-3	우선권서류 신청 수리관청에 대하여 위에 명시된 선출원의 인증증본을 준비하여 국제사무국에 송부하여 줄 것을 신청한다.	VI-1 VI-2	
VI-4	인용에 의한 보완 조약 제11조(1)(iii)(d) 또는 (e)에서 규정하는 국제출원의 요소, 또는 규칙 20.5(a)에서 규정하는 명세서, 청구 범위 또는 도면의 일부가 본 국제출원에는 포함되어 있지 않지만 조약 제11조(1)(iii) 규정의 요소 중 하나 이상이 수리관청에 최초로 접수된 날에 우선권주장의 기초가 된 선출원에 완전히 포함되어 있는 경우, 그 요소 또는 부분은 규칙 20.6 규정에 의한 확인을 조건으로, 규칙 20.6의 규정과 관련하여 본 국제출원에 있어서 인용에 의해 보완된다.		
VII-1	국제조사기관(ISA) 선택	대한민국 특허청 (ISA/KR)	
VIII	선언서	선언서 개수	
VIII-1	발명자의 신원에 관한 선언	-	
VIII-2	국제출원일에 특허출원 및 특허를 받을 수 있는 출원인의 자격에 관한 선언	-	
VIII-3	국제출원일에 선출원의 우선권을 주장할 수 있는 출원인의 자격에 관한 선언	-	
VIII-4	발명자 선언(미국에 대한 지정의 경우에 한함)	-	
VIII-5	신규성을 해치지 아니하는 개시 또는 신규성 상실의 예외에 관한 선언	-	

PCT 출원서

출력(전자적 형태가 원본)

IX	체크 리스트	용지 수	전자적 파일 첨부
IX-1	출원서(선언서 포함)	4	✓
IX-2	명세서	9	✓
IX-3	청구범위	3	✓
IX-4	요약서	1	✓
IX-5	도면	6	✓
IX-7	용지매수 소개	23	
	첨부 항목	서면 첨부	전자적 파일 첨부
IX-8	수수료 계산 용지	-	✓
IX-11	포괄위임장 사본	-	포괄위임장 번호 2018OPLC1095
IX-20	요약서에 수반되어야 할 도면 번호	4	
IX-21	국제출원의 출원 언어	한국어	
X-1	출원인, 대리인 또는 대표자의 서명 또는 날인		
X-1-1	성명	특허법인 태평양	
X-1-2	서명인의 성명	특허법인 태평양	
X-1-3	권한 (출원서를 통해 서명자의 자격이 명백하지 않은 경우에는 그 자격도 표시)	대리인	

PCT 출원서

출력(전자적 형태가 원본)

수리판청 전용

10-1	국제출원으로 제출된 서류의 실제 접수일	2018년 04월 13일 (13.04.2018)
10-2	도면	
10-2-1	접수	
10-2-2	미접수	
10-3	국제출원으로 제출된 서류를 완성하는 서류 또는 도면의 추후 기간내 제출에 따른 정정된 실제 접수일	
10-4	PCT 제11조(2)에 따라 제출이 요구된 보완서로서 기간내 제출된 보완서의 접수일	
10-5	국제조사기관(ISA)	ISA/KR
10-6	조사료 납부시까지 지연된 조사용 사본의 송부	

국제 사무국 전용

11-1	국제 사무국의 기록원본 접수일	
------	------------------	--

Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		Secondary Battery And Method For Manufacturing The Same		
First Named Inventor/Applicant Name:		Jae Woong Kim		
Filer:		Leonor Dominguez		
Attorney Docket Number:		LGCHEM 3.3F-1654		
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
NATIONAL STAGE FEE	1631	1	300	300
NATL STAGE SEARCH FEE - REPORT PROVIDED	1642	1	520	520
NATIONAL STAGE EXAM - ALL OTHER CASES	1633	1	760	760
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1580



DOCUMENT MADE AVAILABLE UNDER THE PATENT COOPERATION TREATY (PCT)

International application number:	PCT/KR2018/004366
International filing date:	13 April 2018 (13.04.2018)
Document type:	Certified copy of priority document
Document details:	Country/Office: KR
	Number: 10-2018-0043191
	Filing date: 13 April 2018 (13.04.2018)
Date of receipt at the International Bureau:	24 April 2018 (24.04.2018)
Remark:	Priority document submitted or transmitted to the International Bureau in compliance with Rule 17.1(a),(b) or (b-bis)



별첨 사본은 아래 출원의 원본과 동일함을 증명함.

This is to certify that the following application annexed hereto
is a true copy from the records of the Korean Intellectual
Property Office.

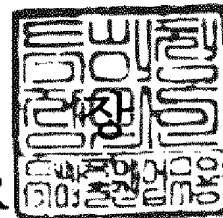
출 원 번 호 : 10-2018-0043191
Application Number

출 원 년 월 일 : 2018년 04월 13일
Filing Date APR 13, 2018

출 원 인 : 주식회사 엘지화학
Applicant(s) LG CHEM, LTD.

2018 년 04 월 23 일

특 허 청
COMMISSIONER



【서지사항】

【서류명】	특허출원서
【참조번호】	4969
【출원구분】	특허출원
【출원인】	
【명칭】	주식회사 엘지화학
【특허고객번호】	1-2001-013456-3
【대리인】	
【명칭】	특허법인 태평양
【대리인번호】	9-2006-100042-1
【지정된변리사】	이은경, 이종욱, 양성욱, 남정우
【포괄위임등록번호】	2011-015093-1
【발명의 국문명칭】	이차전지 및 그 이차전지의 제조 방법
【발명의 영문명칭】	SECONDRY BATTERY AND MANUFACTURING METHOD FOR THE SAME
【발명자】	
【성명】	김재웅
【성명의 영문표기】	KIM, Jae Woong
【주민등록번호】	790215-1XXXXXX
【우편번호】	34122
【주소】	대전광역시 유성구 문지로 188 LG화학 기술연구원 내
【발명자】	
【성명】	김형권

【성명의 영문표기】 KIM, Hyoung Kwon
【주민등록번호】 820116-1XXXXXX
【우편번호】 34122
【주소】 대전광역시 유성구 문지로 188 LG화학 기술연구원 내

【발명자】

【성명】 김상욱
【성명의 영문표기】 KIM, Sang Uck
【주민등록번호】 731025-1XXXXXX
【우편번호】 34122
【주소】 대전광역시 유성구 문지로 188 LG화학 기술연구원 내

【출원언어】 국어

【우선권 주장】

【출원국명】 KR
【출원번호】 10-2017-0048645
【출원일자】 2017.04.14
【증명서류】 미첨부

【취지】 위와 같이 특허청장에게 제출합니다.

대리인 특허법인 태평양 (서명 또는 인)

【수수료】

【출원료】 0 면 46,000 원
【가산출원료】 30 면 0 원

2018-04-13

【우선권 주장료】	1 건	18,000 원
【심사청구료】	0 항	0 원
【합계】	64,000 원	

【발명의 설명】

【발명의 명칭】

이차전지 및 그 이차전지의 제조 방법 {SECONDARY BATTERY AND MANUFACTURING METHOD FOR THE SAME}

【기술분야】

【0001】 본 발명은 이차전지 및 그 이차전지의 제조 방법에 관한 것으로서, 보다 자세하게는 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다.

【발명의 배경이 되는 기술】

【0002】 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 건물로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.

【0003】 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정이 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게

된다.

【0004】한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.

【0005】대한민국 공개특허공보 제10-2012-0006389호에는 종래의 전극조립체 및 이를 포함하는 이차 전지가 개시되어 있다.

【0006】종래의 이차 전지는 저저항 셀 개발을 위해서 음극 탭에 저항이 낮은 금속을 적용하거나 전극 탭 수를 늘리는 기술 적용이 필요하다.

【0007】이러한 필요성에 의해 개발된 음극 멀티탭 구조는 코어와 외곽에 각각 전극 탭을 부착하는 2탭 구조 또는 코어와 외곽부에 전극 탭을 부착하고 중간층에 미들탭을 부착하는 3탭 구조였다.

【0008】그런데 젤리롤 특성상 미들탭 부착을 위해서는 전극 중간 일정 위치에 무지부를 형성해야 하는데 무지부의 증가에 따른 전지 용량 저하의 문제점이 있었다.

【발명의 내용】

【해결하고자 하는 과제】

【0009】따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 무지부를 최소화하면서 멀티탭을 형성할 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

【0010】또한, 본 발명의 과제는 전극 포일을 노칭하여 전극탭을 형성하면서 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

【과제의 해결 수단】

【0011】본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.

【0012】상기 노칭 탭부는 상기 유지부 전체 길이의 1/3 이상으로부터 연속되게 연장될 수 있다.

【0013】상기 전극이 권취된 상태에서 상기 노칭 탭부는 커팅되어 복수 개의 노칭탭을 형성할 수 있다.

【0014】상기 노칭탭의 폭은 4mm ~ 5mm일 수 있다.

【0015】상기 전극은 캔부재에 수용되고, 상기 노칭탭은 절곡되어 상기 캔부재에 연결될 수 있다.

【0016】상기 노칭탭이 절곡되어 형성된 절곡부는 상기 캔부재에 접합될 수 있다.

【0017】상기 절곡부와 상기 전극의 단부와의 사이에 형성된 절연부재를 포함할 수 있다.

【0018】상기 무지부에 부착되는 전극탭을 포함할 수 있다.

【0019】상기 전극이 권취된 상태에서 상기 노칭 탭부는 노칭되어 복수 개의 노칭탭을 형성하고, 상기 전극은 캔부재에 수용되고, 상기 복수 개의 노칭탭은 절곡되어 상기 전극탭에 겹쳐지게 연결될 수 있다.

【0020】상기 전극탭 및 상기 전극탭과 겹쳐지게 연결된 상기 복수 개의 노칭탭은 함께 상기 캔부재에 접합될 수 있다.

【0021】상기 복수 개의 노칭탭의 노칭 길이는 8mm ~ 11mm일 수 있다.

【0022】상기 복수 개의 노칭탭의 노칭 길이는 9mm ~ 10mm일 수 있다.

【0023】상기 전극은 음극일 수 있다.

【0024】본 발명의 일 실시예에 따른 이차전지의 제조 방법은 전극집전체 상에 활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극을 준비하는 준비단계, 상기 전극에서 유지부의 길이 1/3 이상에 상기 전극집전체의 폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부가 형성되도록 상기 전극을 노칭하는 노칭 탭부 형성단계, 상기 노칭 탭부가 형성된 상기 전극을 분리막과 상기 전극과 상극인 다른 전극에 적층한 후 권취하여

전극조립체를 형성는 권취단계, 상기 전극조립체에서 상기 노칭 탭부를 커팅하여 복수 개의 노칭탭을 형성하는 노칭탭 형성단계 및 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭을 상기 캔부재에 접합하는 설치단계를 포함한다.

【0025】 상기 준비단계에서 음극의 전극을 준비할 수 있다.

【0026】 상기 준비단계는 전극집전체 상에 활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 폭방향을 기준으로 병렬로 배열할 수 있다.

【0027】 상기 노칭 탭부 형성단계는 상기 병렬로 배열된 상기 유지부와 무지부를 상기 전극집전체의 길이 방향으로 커팅하면서 상기 유지부의 폭 방향으로 활물질이 도포되지 않은 상기 노칭 탭부를 노칭할 수 있다.

【0028】 상기 노칭 탭부 형성단계는 상기 전극을 커팅하면서 상기 유지부의 폭방향을 활물질이 도포되지 않은 상기 노칭 탭부가 형성되도록 노칭할 수 있다.

【0029】 상기 노칭탭 형성단계에서 상기 노칭 탭부를 레이저로 커팅할 수 있다.

【0030】 상기 설치단계는 상기 노칭탭을 절곡한 상태로 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭에서 절곡된 부분을 상기 캔부재에 접합할 수 있다.

【0031】 상기 설치단계에서 상기 노칭탭의 절곡된 부분을 용접에 의해 상기 캔부재에 접합할 수 있다.

【0032】 상기 설치단계에서 복수 개의 상기 노칭탭의 절곡된 부분을 서로 겹쳐지게 하여 상기 캔부재에 동시에 접합할 수 있다.

【0033】 상기 권취단계는 상기 노칭 탭부를 2겹 이상으로 겹쳐지도록 형성할 수 있다.

【발명의 효과】

【0034】 본 발명에 따르면, 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있다.

【0035】 본 발명에 따르면, 멀티탭을 형성하여 전기 저항을 극소화하는 효과가 있다.

【0036】 본 발명에 따르면, 무지부 형성을 극소화하여 전지 용량을 확보하는 효과가 있다.

【0037】 본 발명에 따르면, 전극 포일로부터 포일탭인 노칭탭을 형성하면서 노칭탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높이는 효과가 있다.

【0038】 본 발명에 따르면, 포일탭인 노칭탭의 기계적 강도를 강화하여 노칭탭이 절곡된 상태에서 끊어지지 않게 함에 따라 노칭탭을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높이는 효과가 있다.

【0039】 본 발명에 따르면, 노칭탭과 전극탭을 동시에 캔부재에 접합할 수 있어 용접 공정성을 확보하는 효과가 있다.

【0040】본 발명에 따르면, 복수 개의 노칭탭을 전극조립체의 외곽에 형성하여 복수 개의 노칭탭과 전극조립체의 단부 사이에 절연부재를 설치할 수 있게 함에 따라 노칭탭과 전극조립체의 접촉을 방지하여 안전성을 향상하는 효과가 있다.

【도면의 간단한 설명】

【0041】도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이다.

도 2는 도 1을 권취한 상태로 도시한 사시도이다.

도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.

도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이다.

도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.

도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.

도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

도 8은 본 발명의 다른 실시예에 따라 노칭탭과 전극탭을 접합한 것을 도면에서 도시한 저면도이다.

도 9는 도 8을 측면에서 도시한 측면도이다.

【발명을 실시하기 위한 구체적인 내용】

【0042】 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지 및 그 이차전지의 제조 방법에 대하여 상세히 설명하기로 한다.

【0043】 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의 가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

【0044】 도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를 불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

【0046】도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이고, 도 2는 도 1을 권취한 상태로 도시한 사시도이다.

【0047】도 1 내지 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100) 및 상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부(111)를 포함한다.(도 2에서 2겹으로 겹쳐진 노칭 탭부(111)가 도시되고 있다.)

【0048】전극집전체의 전극 포일로부터 포일탭인 노칭 탭부를 형성하면서 노칭탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높일 수 있다.

【0049】전극(100)은 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 중 어느 하나일 수 있으며, 양극과 음극 사이에 분리막이 개재되도록 양극, 음극 및 분리막을 복수 회 적층하여 전극조립체(1)를 제작할 수 있다.

【0050】그리고 전극조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수 있다.

【0051】양극은 알루미늄 전극집전체일 수 있으며, 양극 활물질이 도포된 양극 유지부와, 양극 활물질이 도포되지 않은 양극 무지부를 포함할 수 있다.

【0052】 양극 활물질은 LiCoO_2 , LiNiO_2 , LiMnO_2 , LiMnO_4 와 같은 리튬 함유 전이금속 산화물 또는 리튬 칼코게나이드 화합물일 수 있다.

【0053】 양극 유지부는 예를 들어, 알루미늄 전극집전체의 적어도 어느 한 면의 일부에 양극 활물질을 도포하여 형성하며, 양극 활물질이 미 도포된 알루미늄 전극집전체의 나머지 부분이 양극 무지부가 될 수 있다.

【0054】 음극은 구리 전극집전체일 수 있으며, 음극 활물질이 도포된 음극 유지부와 음극 활물질이 도포되지 않은 음극 무지부를 포함할 수 있다.

【0055】 음극 활물질은, 결정질 탄소, 비정질 탄소, 탄소 복합체, 탄소 섬유와 같은 탄소 재료, 리튬 금속 또는 리튬 합금 등일 수 있다.

【0056】 음극 유지부는 예를 들어, 구리 전극집전체의 적어도 어느 한 면의 일부에 음극 활물질을 도포하여 형성하며, 음극 활물질이 미 도포된 구리 전극집전체의 나머지 부분이 음극 무지부가 될 수 있다.

【0057】 분리막은 예를 들어, 폴리에틸렌(PE), 폴리스틸렌(PS), 폴리프로필렌(PP) 및 폴리에틸렌(PE)과 폴리프로필렌(PP)의 공중합체(co-polymer)로 이루어지는 군에서 선택되는 어느 하나의 기재에 폴리비닐리덴 플로우라이드-헥사플로로프로필렌 공중합체(PVDF-HFP co-polymer)를 코팅함으로써 제조될 수 있다.

【0058】 이와 같이 본 발명의 전극(100)은 전극조립체(1)를 형성하는 양극, 음극 중 어느 하나 이상일 수 있으며, 특히 양극 또는 음극 중에 저항 감소가 더 필요한 어느 하나일 수 있다.

【0059】 본 발명에서는 음극을 일 실시예로 설명한다.

【0060】 전극(100)은 일단부에 무지부(120)가 형성될 수 있고, 전극(100)의 일단부에 형성된 무지부(120)에는 전극탭(121)이 부착될 수 있다.

【0061】 노칭 탭부(111)는 전극(100)의 유지부(110)로부터 폭 방향으로 연장되도록 전극(100)을 노칭(notching)함에 따라 형성될 수 있는데, 전극(100)의 유지부(110) 전체 길이의 1/3 이상에 해당하는 영역에서 전극집 전체만 연장되도록 형성되며 활물질(a)은 도포되지 않을 수 있다.

【0062】 이러한 노칭 탭부(111)는 전극(100)으로부터 연장된 유지부(110)에서 활물질(a) 만을 제거하여 형성할 수 있고, 또는 전극(100)의 유지부(110) 영역에서 활물질(a)이 도포되지 않은 채 전극 포일만 연장하여 형성할 수도 있다.

【0063】 따라서 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있을 수 있고, 무지부 형성을 극소화하여 전지 용량을 확보할 수 있다.

【0064】 도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.

【0065】 도 3에 도시된 바와 같이, 노칭 탭부(111)는 권취된 상태에서 커팅되어 복수 개의 노칭탭(111a)을 형성할 수 있다. 노칭탭(111a)의 수가 많아지면 전체 전극탭의 저항은 줄어들게 된다.

【0066】 이때, 하나의 노칭탭(111a)의 폭은 4mm ~ 5mm를 형성하는 것이 좋은데, 이러한 이유는 노칭탭(111a)의 폭이 4mm 미만이면 기계적 강도가 약할 수

있고, 노칭탭(111a)의 폭이 5mm 초과하면 캔부재와의 전기적 연결을 위한 접합의 과정에서 변형이 어려울 수 있기 때문이다.

【0067】 포일탭인 노칭탭(111a)의 기계적 강도를 강화하여 노칭탭(111a)이 절곡된 상태에서 끊어지지 않게 함에 따라 노칭탭(111a)을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높일 수 있다.

【0068】 도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이고, 도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.

【0069】 도 4 및 도 5에 도시된 바와 같이, 본 발명의 이차전지에 따른 전극조립체(1)는 캔부재(10)에 수용하면서 노칭탭(111a)을 절곡한 상태로 캔부재(10)와 전기적으로 연결되도록 캔부재(10)에 노칭탭(111a)의 절곡부를 용접 결합할 수 있다.

【0070】 노칭탭(111a)을 절곡하여 노칭탭(111a)의 끝단이 캔부재(10)의 저면 중심 측에 결합되게 하여 전기적 연결을 원활히 하면서 복수 개의 노칭탭(111a)이 전기적으로 연결되어 전기 저항을 극소화할 수 있다.

【0071】 도 8은 본 발명의 다른 실시예에 따라 노칭탭과 전극탭을 접합한 것을 도면에서 도시한 저면도이고, 도 9는 도 8을 측면에서 도시한 측면도이다.

【0072】 도 8 및 도 9에 도시된 바와 같이, 본 발명의 다른 실시예에 따른 이차전지는 전극(100)이 권취된 전극조립체(1)에서 노칭 탭부(111)는 커팅되어 복

수 개의 노칭탭(111a)을 형성하고, 복수 개의 노칭탭(111a)은 절곡되어 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)에 겹쳐지게 연결될 수 있다.

【0073】 그리고 전극조립체(1)는 캔부재(10)에 수용되고, 전극탭(121)과 겹쳐지게 연결된 복수 개의 노칭탭(111a)은 전극탭(121)과 함께 캔부재(10)에 용접 등에 의해 접합될 수 있다.

【0074】 상술한 바와 같이 본 발명의 다른 실시예에 따른 이차전지는 복수 개의 노칭탭(111a)과 전극탭(121)을 서로 겹쳐지게 연결한 상태에서 캔부재(10)에 함께 용접 접합하기 때문에 용접 공정성 확보가 쉬울 수 있다.

【0075】 본 발명의 다른 실시예에 따른 이차전지는 전극조립체(1)의 단부의 지름의 길이가 16mm ~ 21mm일 때, 복수 개의 노칭탭(111a)의 노칭 길이가 8mm ~ 11mm의 범위일 수 있다. 이러한 이유는 노칭탭(111a)의 노칭 길이가 8mm미만이면 노칭탭(111a)의 길이가 짧아 노칭탭(111a)이 절곡되어 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)에 겹쳐지기 어려울 수 있다. 또한, 노칭탭(111a)의 노칭 길이가 11mm를 초과 하면 노칭탭(111a)의 길이가 필요 이상으로 길어져 재료 낭비로 인한 비용 증가와 노칭탭(111a)이 캔부재(10) 내부에서 차지하는 공간이 많아져 캔부재(10) 내부에 채워지는 전해액의 양이 줄어들어 따른 용량 저하의 원인이 될 수 있다. 바람직하게 노칭탭(111a)의 노칭 길이는 9mm ~ 10mm 인 것이 좋다.

【0076】 본 발명의 다른 실시예에 따른 이차전지는 노칭탭(111a)의 절곡부와 전극(100)이 권취된 전극조립체(1)의 단부와 사이에 절연부재(130)를 형성할 수 있다.

【0077】 절연부재(130)는 노칭탭(111a)과 전극조립체(1)의 단부와 사이를 절연하여 노칭탭(111a)과 전극조립체(1)의 사이에 접촉을 방지할 수 있다.

【0078】 복수 개의 노칭탭(111a)은 전극조립체(1)의 단부 외곽부에 형성되어 절연부재(130)를 전극조립체(1)의 단부에 설치하는 것을 방해하지 않도록 할 수 있다.

【0079】 복수 개의 노칭탭(111a)을 전극조립체의 외곽에 형성하여 복수 개의 노칭탭(111a)과 전극조립체(1)의 단부 사이에 절연부재(130)를 설치할 수 있게 함에 따라 노칭탭(111a)과 전극조립체(1)의 접촉을 방지하여 안전성을 향상할 수 있다.

【0081】 도면을 참조하여 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 상세히 설명한다.

【0082】 도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.

【0083】 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지의 제조 방법은 준비단계(S1), 노칭 탭부 형성단계(S2), 권취단계(S3), 노칭탭 형성단계(S4) 및 설치단계(S5)를 포함한다.

【0084】 상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 전극집전체의 길이방향을 따라

구비하는 전극(100)을 준비하는 단계이다.

【0085】 이때, 전극(100)은 음극일 수 있다.

【0086】 도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

【0087】 도 7에 도시된 바와 같이, 준비단계(S1)에서 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(110)와 무지부(120)는 전극집전체의 폭방향(W)을 기준으로 병렬로 배열할 수 있다.

【0088】 상기 노칭 탭부 형성단계(S2)는 전극(100)에서 유지부(110)의 길이 1/3 이상에 유지부(110)의 폭방향(W)으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록 전극(100)을 노칭하는 단계이다.

【0089】 이와 같이 전극(100)에서 노칭 탭부(111)를 노칭하는 것은 전극집전체에서 전극(100)을 커팅하면서 유지부(110)의 폭방향(W)으로 형성된 활물질(a)이 도포되지 않은 무지부의 일부를 노칭하여 노칭 탭부(11)를 형성하는 것일 수 있다.

【0090】 또한, 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(100)와 무지부(120)가 전극집전체의 폭방향(W)을 기준으로 병렬로 배열되었을 경우 병렬로 배열된 유지부(110)와 무지부(120)를 전극집전체의 길이 방향(L)으로 커팅하면서 하나의 유지부(110)와 폭방향(W) 기준 병렬로 배열된 다른 하나의 유지부(110a)의 사이에 형성된 활물질이 도포되지 않은 무지부 영역을 노칭하여 유지부(110)의 폭

방향(W)으로 노칭 탭부(111)가 형성되도록 할 수 있다.

【0091】 상기 권취단계(S3)는 노칭 탭부(111)가 형성된 전극(100)을 분리막과 상기 전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를 형성하는 단계일 수 있다.

【0092】 전극(100)이 권취되는 과정에서 노칭 탭부(111)는 2겹 이상으로 겹쳐지게 형성될 수 있다.

【0093】 상기 노칭탭 형성단계(S4)는 전극조립체(1)에서 노칭 탭부(111)를 커팅하여 복수 개의 노칭탭(111a)을 형성하는 단계일 수 있다.

【0094】 노칭 탭부(111)가 커팅되어 형성된 노칭탭(111a)은 노칭 탭부(111)와 동일하게 2겹 이상으로 겹쳐진 상태를 형성할 수 있다.

【0095】 이때 노칭 탭부(111)의 커팅은 레이저를 사용하여 작업 효율을 극대화할 수 있다.

【0096】 상기 설치단계(S5)는 전극조립체(1)를 캔부재(10)에 수용하여 노칭탭(111a)을 캔부재(10)에 접합하는 단계이다.

【0097】 설치단계(S5)에서 노칭탭(111a)이 캔부재(10)에 전기적으로 연결되면서 용접 접합이 용이하도록 노칭탭(111a)을 절곡하여 노칭탭(111a)이 절곡된 상태로 전극조립체(1)가 캔부재(10)에 수용되도록 할 수 있다.

【0098】 이때, 노칭탭(111a)은 2겹 이상으로 형성되었기 때문에 노칭탭(111a)이 절곡되어도 기계적 강도를 유지할 수 있어 쉽게 끊어지지 않는 효과가 있

다.

【0099】 절곡된 노칭탭(111a)은 단부가 캔부재(10)의 저면 중앙을 향하도록 할 수 있다.

【0100】 캔부재(10)의 저면과 노칭탭(111a)의 결합은 노칭탭(111a)의 절곡부가 캔부재(10)의 저면에 용접 결합되게 할 수 있다.

【0101】 또한, 본 발명의 다른 실시예에 따르면 설치단계(S5)에서 복수 개의 노칭탭(111a)의 절곡된 부분을 전극조립체(1)의 단부 중앙측에서 서로 겹쳐지게 하여 전극조립체(1)의 단부 중앙부에 형성된 전극탭(121)과 함께 캔부재(10)에 동시에 용접 등에 의해 접합할 수도 있다.

【0103】 이상과 같이 본 발명에 따른 이차전지 및 그 이차전지의 제조 방법을 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

【부호의 설명】

【0104】 1: 전극조립체

10: 캔부재

100: 전극

110: 유지부

- 111: 노칭 탭부
- 111a: 노칭 탭
- 120: 무지부
- 121: 전극 탭
- a: 활물 질

【청구범위】

【청구항 1】

전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100); 및

상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭탭부(111); 를 포함하는 것을 특징으로 하는 이차전지.

【청구항 2】

청구항 1에 있어서,

상기 노칭 탭부(111)는 상기 유지부(110) 전체 길이의 1/3 이상으로부터 연속되게 연장되는 것을 특징으로 하는 이차전지.

【청구항 3】

청구항 1 또는 청구항 2에 있어서,

상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 커팅되어 복수 개의 노칭탭(111a)을 형성하는 것을 특징으로 하는 이차전지.

【청구항 4】

청구항 3에 있어서,

상기 노칭탭(111a)의 폭은 4mm ~ 5mm인 것을 특징으로 하는 이차전지.

【청구항 5】

청구항 3에 있어서,

상기 전극(100)은 캔부재(10)에 수용되고, 상기 노칭탭(111a)은 절곡되어 상기 캔부재(10)에 연결되는 것을 특징으로 하는 이차전지.

【청구항 6】

청구항 5에 있어서,

상기 노칭탭(111a)이 절곡되어 형성된 절곡부는 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.

【청구항 7】

청구항 6에 있어서,

상기 절곡부와 상기 전극(100)의 단부와의 사이에 형성된 절연부재(130)를 포함하는 것을 특징으로 하는 이차전지.

【청구항 8】

청구항 1에 있어서,

상기 무지부(120)에 부착되는 전극탭(121)을 포함하는 것을 특징으로 하는 이차전지.

【청구항 9】

청구항 8에 있어서,

상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 노칭(notching)되

어 복수 개의 노칭탭(111a)을 형성하고,

상기 전극(100)은 캔부재(10)에 수용되고, 상기 복수 개의 노칭탭(111a)은 절곡되어 상기 전극탭(121)에 겹쳐지게 연결되는 것을 특징으로 하는 이차전지.

【청구항 10】

청구항 9에 있어서,

상기 전극탭(121) 및 상기 전극탭(121)과 겹쳐지게 연결된 상기 복수 개의 노칭탭(111a)은 함께 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.

【청구항 11】

청구항 9에 있어서,

상기 복수 개의 노칭탭(111a)의 노칭 길이는 8mm ~ 11mm인 것을 특징으로 하는 이차전지.

【청구항 12】

청구항 9에 있어서,

상기 복수 개의 노칭탭(111a)의 노칭 길이는 9mm ~ 10mm 인 것을 특징으로 하는 이차전지.

【청구항 13】

청구항 1에 있어서,

상기 전극(100)은 음극인 것을 특징으로 하는 이차전지.

【청구항 14】

전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100)을 준비하는 준비단계(S1);

상기 전극(100)에서 유지부(110)의 길이 1/3 이상에 상기 전극집전체의 폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록 상기 전극(100)을 노칭하는 노칭 탭부 형성단계(S2);

상기 노칭 탭부(111)가 형성된 상기 전극(100)을 분리막과 상기 전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를 형성는 권취단계(S3);

상기 전극조립체(1)에서 상기 노칭 탭부(111)를 커팅하여 복수 개의 노칭탭(111a)을 형성하는 노칭탭 형성단계(S4); 및

상기 전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)을 상기 캔부재(10)에 접합하는 설치단계(S5); 를 포함하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 15】

청구항 14에 있어서,

상기 준비단계(S1)에서 음극의 전극(100)을 준비하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 16】

청구항 14에 있어서,

상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 폭방향을 기준으로 병렬로 배열하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 17】

청구항 16에 있어서,

상기 노칭 탭부 형성단계(S2)는 상기 병렬로 배열된 상기 유지부(110)와 무지부(120)를 상기 전극집전체의 길이 방향으로 커팅하면서 상기 유지부(110)의 폭 방향으로 활물질(a)이 도포되지 않은 상기 노칭 탭부(11)를 노칭하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 18】

청구항 14에 있어서,

상기 노칭 탭부 형성단계(S2)는 상기 전극(100)을 커팅하면서 상기 유지부(110)의 폭방향으로 활물질(a)이 도포되지 않은 상기 노칭 탭부(111)가 형성되도록 노칭하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 19】

청구항 14에 있어서,

상기 노칭 탭 형성단계(S4)에서 상기 노칭 탭부(111)를 레이저로 커팅하는 것

을 특징으로 하는 이차전지의 제조 방법.

【청구항 20】

청구항 14에 있어서,

상기 설치단계(S5)는 상기 노칭탭(111a)을 절곡한 상태로 상기 전극조립체 (1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)에서 절곡된 부분을 상기 캔부재 (10)에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 21】

청구항 20에 있어서,

상기 설치단계(S5)에서 상기 노칭탭(111a)의 절곡된 부분을 용접에 의해 상 기 캔부재(10)에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 22】

청구항 20에 있어서,

상기 설치단계(S5)에서 복수 개의 상기 노칭탭(111a)의 절곡된 부분을 서로 겹쳐지게 하여 상기 캔부재(10)에 동시에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 23】

청구항 14에 있어서,

상기 권취단계(S3)는 상기 노칭 탭부(111)를 2겹 이상으로 겹쳐지도록 형성 하는 것을 특징으로 하는 이차전지의 제조 방법.

【요약서】

【요약】

본 발명은 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다.

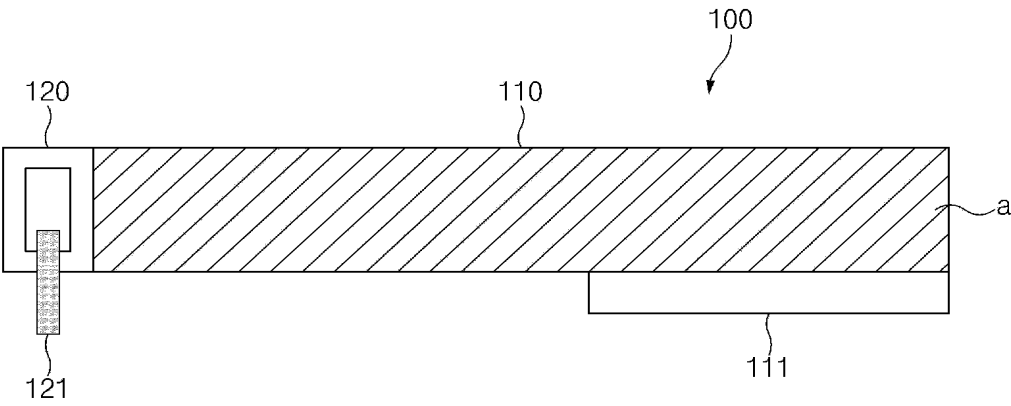
또한, 본 발명은 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.

【대표도】

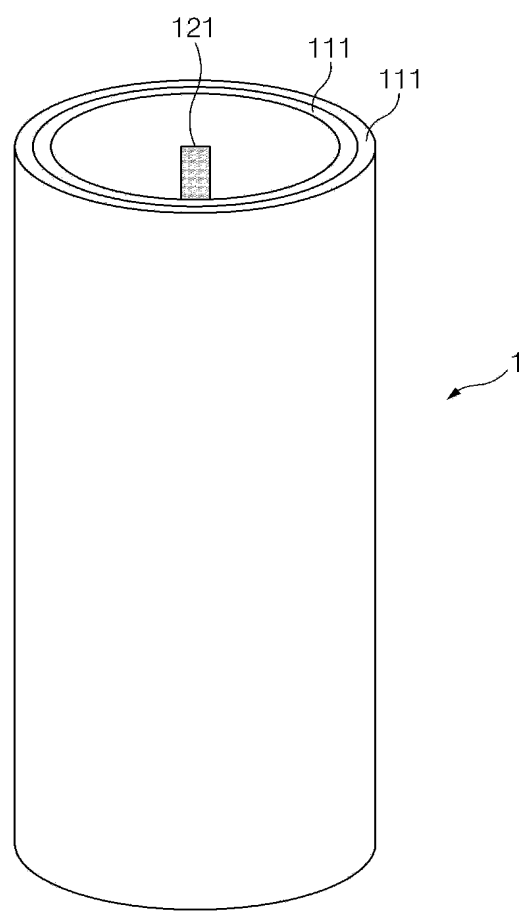
도 4

【도면】

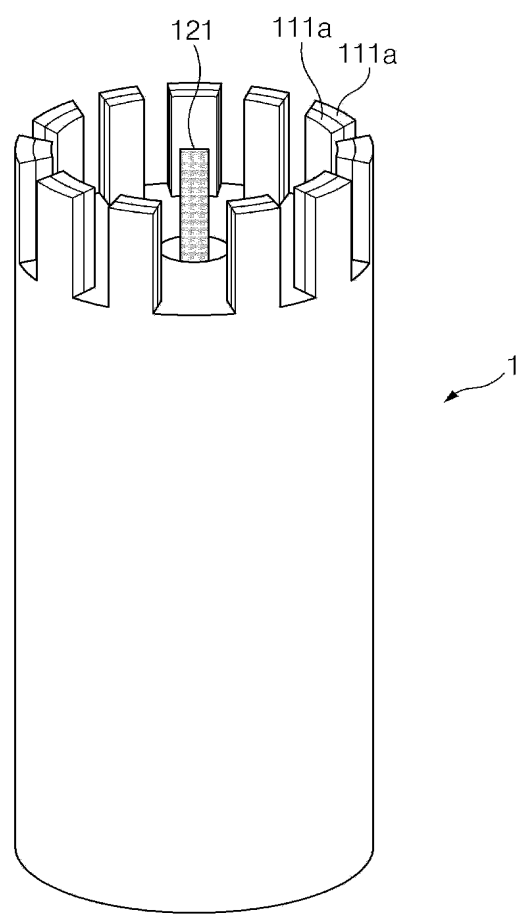
【도 1】



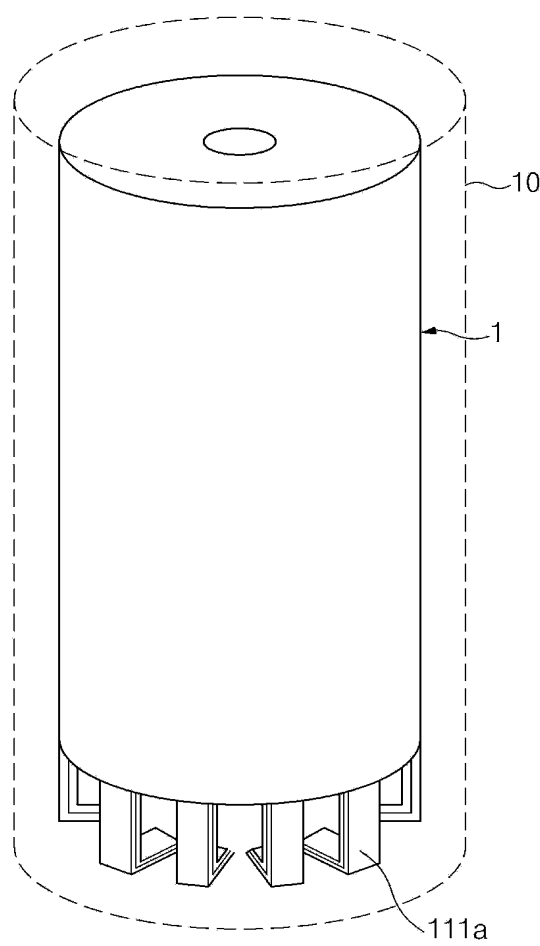
【도 2】



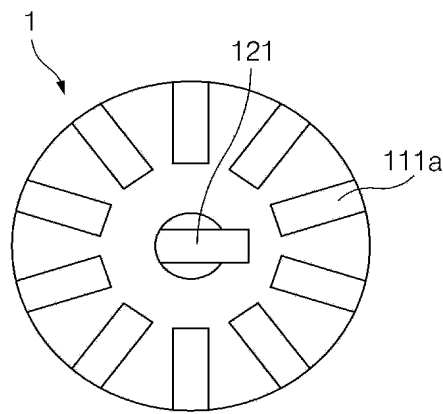
【도 3】



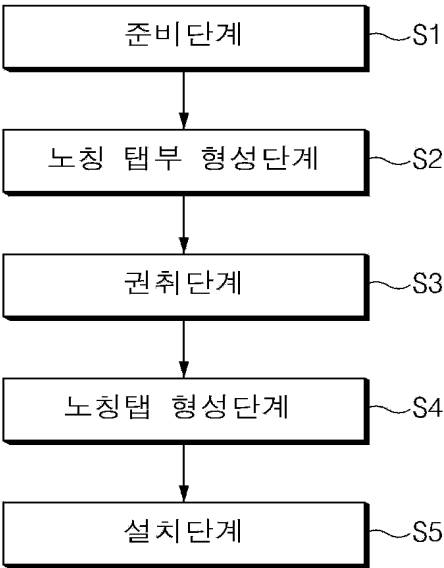
【도 4】



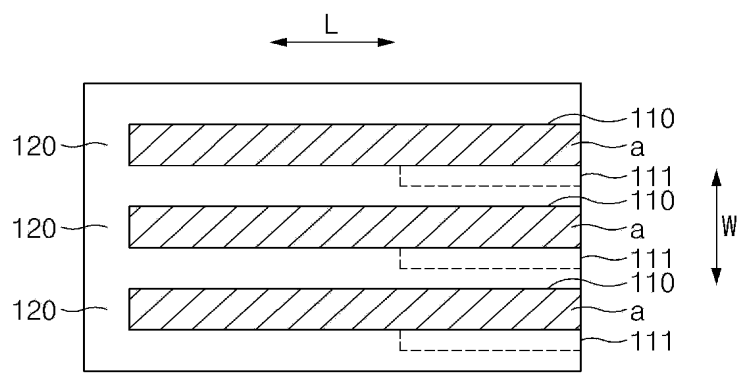
【도 5】



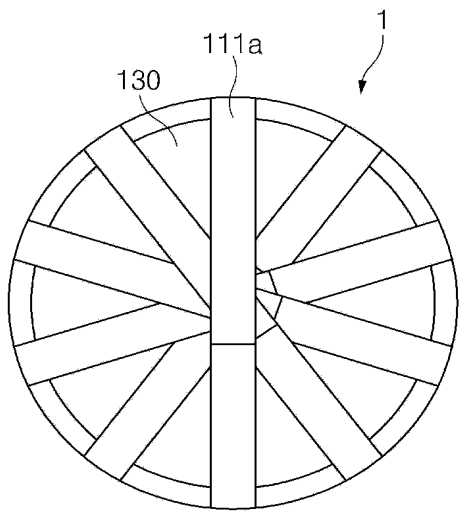
【도 6】



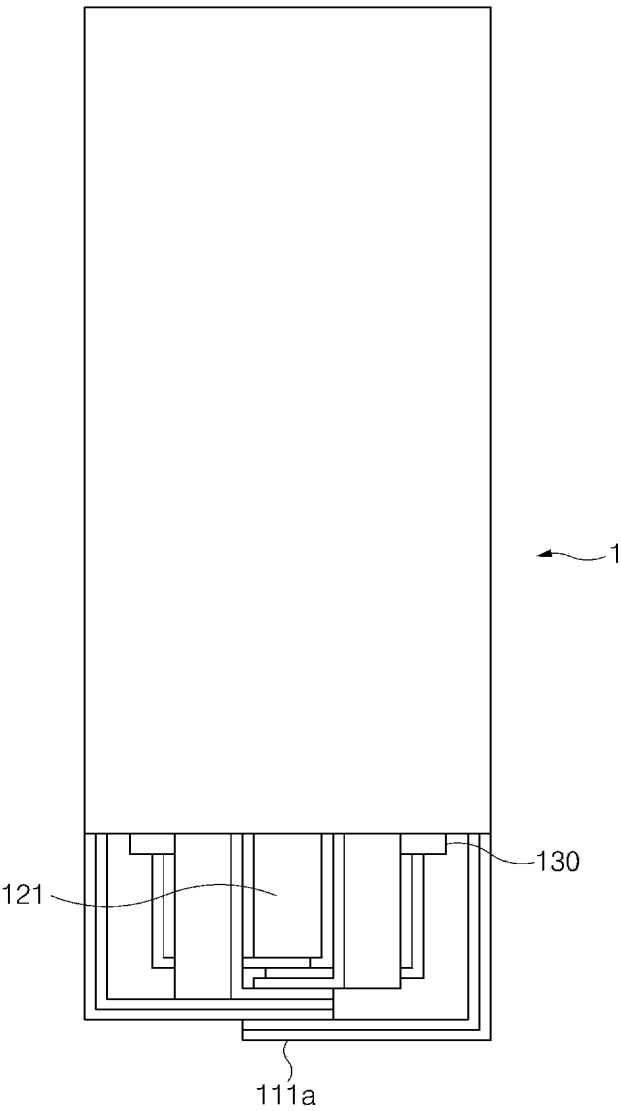
【도 7】



【도 8】



【도 9】





DOCUMENT MADE AVAILABLE UNDER THE PATENT COOPERATION TREATY (PCT)

International application number:		PCT/KR2018/004366
International filing date:		13 April 2018 (13.04.2018)
Document type:		Certified copy of priority document
Document details:	Country/Office:	KR
	Number:	10-2017-0048645
	Filing date:	14 April 2017 (14.04.2017)
Date of receipt at the International Bureau:		24 April 2018 (24.04.2018)
Remark: Priority document submitted or transmitted to the International Bureau in compliance with Rule 17.1(a),(b) or (b-bis)		



별첨 사본은 아래 출원의 원본과 동일함을 증명함.

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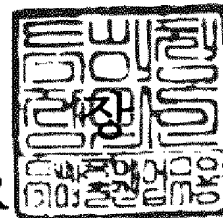
출 원 번 호 : 10-2017-0048645
Application Number

출 원 년 월 일 : 2017년 04월 14일
Filing Date APR 14, 2017

출 원 인 : 주식회사 엘지화학
Applicant(s) LG CHEM, LTD.

2018 년 04 월 23 일

특 허 청
COMMISSIONER



【서지사항】

【서류명】	특허출원서
【참조번호】	3639
【출원구분】	특허출원
【출원인】	
【명칭】	주식회사 엘지화학
【특허고객번호】	1-2001-013456-3
【대리인】	
【명칭】	특허법인 태평양
【대리인번호】	9-2006-100042-1
【지정된변리사】	이은경, 이종욱, 양성욱, 남정우
【포괄위임등록번호】	2011-015093-1
【발명의 국문명칭】	이차전지 및 그 이차전지의 제조 방법
【발명의 영문명칭】	SECONDRY BATTERY AND MANUFACTURING METHOD FOR THE SAME
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【국적】 KR

【출원언어】 국어

【취지】 위와 같이 특허청장에게 제출합니다.

대리인 특허법인 태평양 (서명 또는 인)

【수수료】

【출원료】	0	면	46,000	원
【가산출원료】	26	면	0	원
【우선권주장료】	0	건	0	원
【심사청구료】	0	항	0	원

2017-04-14

【합 계】 46,000 원

【발명의 설명】

【발명의 명칭】

이차전지 및 그 이차전지의 제조 방법 {SECONDARY BATTERY AND MANUFACTURING METHOD FOR THE SAME}

【기술분야】

【0001】 본 발명은 이차전지 및 그 이차전지의 제조 방법에 관한 것으로서, 보다 자세하게는 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다.

【발명의 배경이 되는 기술】

【0002】 물질의 물리적 반응이나 화학적 반응을 통해 전기에너지를 생성시켜 외부로 전원을 공급하게 되는 전지(cell, battery)는 각종 전기전자 기기로 둘러싸여 있는 생활환경에 따라, 건물로 공급되는 교류 전원을 획득하지 못할 경우나 직류전원이 필요할 경우 사용하게 된다.

【0003】 이와 같은 전지 중에서 화학적 반응을 이용하는 화학전지인 일차전지와 이차전지가 일반적으로 많이 사용되고 있는데, 일차전지는 건전지로 통칭되는 것으로 소모성 전지이다. 또한, 이차전지는 전류와 물질 사이의 산화환원과정이 다수 반복 가능한 소재를 사용하여 제조되는 재충전식 전지로서, 전류에 의해 소재에 대한 환원반응이 수행되면 전원이 충전되고, 소재에 대한 산화반응이 수행되면 전원이 방전되는데, 이와 같은 충전-방전이 반복적으로 수행되면서 전기가 생성되게

된다.

【0004】한편, 이차 전지 중 리튬 이온 전지는 양극 도전 포일과 음극 도전 포일에 각각 활물질을 일정한 두께로 코팅하고, 상기 양 도전 포일 사이에는 분리막이 개재되도록 하여 대략 젤리 롤(jelly roll) 내지는 원통 형태로 다수 회 권취하여 제작한 전극 조립체를 원통형 또는 각형 캔, 파우치 등에 수납하고 이를 밀봉 처리하여 제작된다.

【0005】대한민국 공개특허공보 제10-2012-0006389호에는 종래의 전극조립체 및 이를 포함하는 이차 전지가 개시되어 있다.

【0006】종래의 이차 전지는 저저항 셀 개발을 위해서 음극 탭에 저항이 낮은 금속을 적용하거나 전극 탭 수를 늘리는 기술 적용이 필요하다.

【0007】이러한 필요성에 의해 개발된 음극 멀티탭 구조는 코어와 외곽에 각각 전극 탭을 부착하는 2탭 구조 또는 코어와 외곽부에 전극 탭을 부착하고 중간층에 미들탭을 부착하는 3탭 구조였다.

【0008】그런데 젤리롤 특성상 미들탭 부착을 위해서는 전극 중간 일정 위치에 무지부를 형성해야 하는데 무지부의 증가에 따른 전지 용량 저하의 문제점이 있었다.

【발명의 내용】

【해결하고자 하는 과제】

【0009】따라서 본 발명은 위와 같은 문제점을 해결하기 위해 안출된 것으로서, 본 발명의 과제는 무지부를 최소화하면서 멀티탭을 형성할 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

【0010】또한, 본 발명의 과제는 전극 포일을 노칭하여 전극 탭을 형성하면서 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법을 제공하는 것이다.

【과제의 해결 수단】

【0011】본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.

【0012】상기 노칭 탭부는 상기 유지부 전체 길이의 1/3 이상으로부터 연속되게 연장될 수 있다.

【0013】상기 전극이 권취된 상태에서 상기 노칭 탭부는 커팅되어 복수 개의 노칭탭을 형성할 수 있다.

【0014】상기 노칭탭의 폭은 4mm ~ 5mm일 수 있다.

【0015】상기 전극은 캔부재에 수용되고, 상기 노칭탭은 절곡되어 상기 캔부재에 연결될 수 있다.

【0016】 상기 노칭탭이 절곡되어 형성된 절곡부는 상기 캔부재에 접합될 수 있다.

【0017】 상기 무지부에 부착되는 전극탭을 포함할 수 있다.

【0018】 상기 전극은 음극일 수 있다.

【0020】 본 발명의 일 실시예에 따른 이차전지의 제조 방법은 전극집전체 상에 활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극을 준비하는 준비단계, 상기 전극에서 유지부의 길이 1/3 이상에 상기 전극집전체의 폭방향으로 활물질이 없는 상태로 연장된 노칭탭부가 형성되도록 상기 전극을 노칭하는 노칭 탭부 형성단계, 상기 노칭 탭부가 형성된 상기 전극을 분리막과 상기 전극과 상극인 다른 전극에 적층한 후 권취하여 전극조립체를 형성는 권취단계, 상기 전극조립체에서 상기 노칭 탭부를 커팅하여 복수 개의 노칭탭을 형성하는 노칭탭 형성단계 및 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭을 상기 캔부재에 접합하는 설치단계를 포함한다.

【0021】 상기 준비단계에서 음극의 전극을 준비할 수 있다.

【0022】 상기 준비단계는 전극집전체 상에 활물질이 도포된 유지부와 활물질이 도포되지 않은 무지부를 상기 전극집전체의 폭방향을 기준으로 병렬로 배열할 수 있다.

【0023】상기 노칭 탭부 형성단계는 상기 병렬로 배열된 상기 유지부와 무지부를 상기 전극집전체의 길이 방향으로 커팅하면서 상기 유지부의 폭 방향으로 활물질이 도포되지 않은 상기 노칭 탭부를 노칭할 수 있다.

【0024】상기 노칭 탭부 형성단계는 상기 전극을 커팅하면서 상기 유지부의 폭방향으로 활물질이 도포되지 않은 상기 노칭 탭부가 형성되도록 노칭할 수 있다.

【0025】상기 노칭탭 형성단계에서 상기 노칭 탭부를 레이저로 커팅할 수 있다.

【0026】상기 설치단계는 상기 노칭탭을 절곡한 상태로 상기 전극조립체를 캔부재에 수용하여 상기 노칭탭에서 절곡된 부분을 상기 캔부재에 접합할 수 있다.

【0027】상기 설치단계에서 상기 노칭탭의 절곡된 부분을 용접에 의해 상기 캔부재에 접합할 수 있다.

【0028】상기 권취단계는 상기 노칭 탭부를 2겹 이상으로 겹쳐지도록 형성할 수 있다.

【발명의 효과】

【0029】본 발명에 따르면, 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있다.

【0030】본 발명에 따르면, 멀티탭을 형성하여 전기 저항을 극소화하는 효과가 있다.

【0031】 본 발명에 따르면, 무지부 형성을 극소화하여 전지 용량을 확보하는 효과가 있다.

【0032】 본 발명에 따르면, 전극 포일로부터 포일 탭을 형성하면서 포일 탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높이는 효과가 있다.

【0033】 본 발명에 따르면, 포일 탭의 기계적 강도를 강화하여 포일 탭이 절곡된 상태에서 끊어지지 않게 함에 따라 포일 탭을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높이는 효과가 있다.

【도면의 간단한 설명】

【0034】 도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이다.

도 2는 도 1을 권취한 상태로 도시한 사시도이다.

도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.

도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이다.

도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.

도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.

도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성

된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

【발명을 실시하기 위한 구체적인 내용】

【0035】 이하, 첨부된 도면을 참조하여 본 발명의 바람직한 실시예에 따른 이차전지 및 그 이차전지의 제조 방법에 대하여 상세히 설명하기로 한다.

【0036】 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다. 따라서, 본 명세서에 기재된 실시예와 도면에 도시된 구성은 본 발명의 가장 바람직한 일 실시예에 불과하고 본 발명의 기술적 사상을 모두 대변하는 것은 아니므로, 본 출원시점에 있어서 이들을 대체할 수 있는 다양한 균등물이 있을 수 있음을 이해하여야 한다.

【0037】 도면에서 각 구성요소 또는 그 구성요소를 이루는 특정 부분의 크기는 설명의 편의 및 명확성을 위하여 과장되거나 생략되거나 또는 개략적으로 도시되었다. 따라서, 각 구성요소의 크기는 실제크기를 전적으로 반영하는 것은 아니다. 관련된 공지기능 혹은 구성에 대한 구체적인 설명이 본 발명의 요지를 불필요하게 흐릴 수 있다고 판단되는 경우, 그러한 설명은 생략하도록 한다.

【0039】 도 1은 본 발명의 일 실시예에 따른 전극을 전개한 상태로 도시한 전개도이고, 도 2는 도 1을 권취한 상태로 도시한 사시도이다.

【0040】 도 1 내지 도 2에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100) 및 상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부(111)를 포함한다.(도 2에서 2겹으로 겹쳐진 노칭 탭부(111)가 도시되고 있다.)

【0041】 전극(100)은 양극 활물질이 도포된 양극과 음극 활물질이 도포된 음극 중 어느 하나일 수 있으며, 양극과 음극 사이에 분리막이 개재되도록 양극, 음극 및 분리막을 복수 회 적층하여 전극조립체(1)를 제작할 수 있다.

【0042】 그리고 전극조립체(1)는 양극과 분리막 및 음극을 적층한 적층체를 젤리 롤 형태로 권취하여 제작할 수 있다.

【0043】 양극은 알루미늄 전극집전체일 수 있으며, 양극 활물질이 도포된 양극 유지부와, 양극 활물질이 도포되지 않은 양극 무지부를 포함할 수 있다.

【0044】 양극 활물질은 LiCoO_2 , LiNiO_2 , LiMnO_2 , LiMnO_4 와 같은 리튬 함유 전이금속 산화물 또는 리튬 칼코게나이드 화합물일 수 있다.

【0045】 양극 유지부는 예를 들어, 알루미늄 전극집전체의 적어도 어느 한 면의 일부에 양극 활물질을 도포하여 형성하며, 양극 활물질이 미 도포된 알루미늄 전극집전체의 나머지 부분이 양극 무지부가 될 수 있다.

【0046】 음극은 구리 전극집전체일 수 있으며, 음극 활물질이 도포된 음극 유지부와 음극 활물질이 도포되지 않은 음극 무지부를 포함할 수 있다.

【0047】 음극 활물질은, 결정질 탄소, 비정질 탄소, 탄소 복합체, 탄소 섬유와 같은 탄소 재료, 리튬 금속 또는 리튬 합금 등일 수 있다.

【0048】 음극 유지부는 예를 들어, 구리 전극집전체의 적어도 어느 한 면의 일부에 음극 활물질을 도포하여 형성하며, 음극 활물질이 미 도포된 구리 전극집전체의 나머지 부분이 음극 무지부가 될 수 있다.

【0049】 분리막은 예를 들어, 폴리에틸렌(PE), 폴리스틸렌(PS), 폴리프로필렌(PP) 및 폴리에틸렌(PE)과 폴리프로필렌(PP)의 공중합체(co-polymer)로 이루어지는 군에서 선택되는 어느 하나의 기재에 폴리비닐리덴 플로우라이드-헥사플로로프로필렌 공중합체(PVDF-HFP co-polymer)를 코팅함으로써 제조될 수 있다.

【0050】 이와 같이 본 발명의 전극(100)은 전극조립체(1)를 형성하는 양극, 음극 중 어느 하나 이상일 수 있으며, 특히 양극 또는 음극 중에 저항 감소가 더 필요한 어느 하나일 수 있다.

【0051】 본 발명에서는 음극을 일 실시예로 설명한다.

【0052】전극(100)은 일단부에 무지부(120)가 형성될 수 있고, 전극(100)의 일단부에 형성된 무지부(120)에는 전극탭(121)이 부착될 수 있다.

【0053】노칭 탭부(111)는 전극(100)의 유지부(110)로부터 폭 방향으로 연장되도록 전극(100)을 노칭함에 따라 형성될 수 있는데, 전극(100)의 유지부(110) 전체 길이의 1/3 이상에 해당하는 영역에서 전극집전체만 연장되도록 형성되며 활물질(a)은 도포되지 않을 수 있다.

【0054】이러한 노칭 탭부(111)는 전극(100)으로부터 연장된 유지부(110)에서 활물질(a) 만을 제거하여 형성할 수 있고, 또는 전극(100)의 유지부(110) 영역에서 활물질(a)이 도포되지 않은 채 전극집전체만 연장하여 형성할 수도 있다.

【0055】도 3은 도 2에서 노칭 탭부를 커팅하여 노칭탭을 형성한 것을 도시한 사시도이다.

【0056】도 3에 도시된 바와 같이, 노칭 탭부(111)는 권취된 상태에서 커팅되어 복수 개의 노칭탭(111a)을 형성할 수 있다. 노칭탭(111a)의 수가 많아지면 전체 전극 탭의 저항은 줄어들게 된다.

【0057】이때, 하나의 노칭탭(111a)의 폭은 4mm ~ 5mm를 형성하는 것이 좋은데, 이러한 이유는 노칭탭(111a)의 폭이 4mm 미만이면 기계적 강도가 약할 수 있고, 노칭탭(111a)의 폭이 5mm 초과하면 캔부재와의 전기적 연결을 위한 접합의 과정에서 변형이 어려울 수 있기 때문이다.

【0058】 도 4는 본 발명의 일 실시예에 따른 전극조립체를 캔부재에 설치한 상태에서 캔부재를 투시하여 도시한 투시 사시도이고, 도 5는 도 4에서 전극조립체만을 저면에서 도시한 저면도이다.

【0059】 도 4 내지 도 5에 도시된 바와 같이, 본 발명의 이차전지에 따른 전극조립체(1)는 캔부재(10)에 수용하면서 노칭탭(111a)을 절곡한 상태로 캔부재(10)와 전기적으로 연결되도록 캔부재(10)에 노칭탭(111a)의 절곡부를 용접 결합할 수 있다.

【0060】 노칭탭(111a)을 절곡하여 노칭탭(111a)의 끝단이 캔부재(10)의 저면 중심 측에 결합되게 하여 전기적 연결을 원활히 하면서 복수 개의 노칭탭(111a)이 전기적으로 연결되어 전기 저항을 극소화할 수 있다.

【0062】 도면을 참조하여 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 상세히 설명한다.

【0063】 도 6은 본 발명의 일 실시예에 따른 이차전지의 제조 방법을 순차적으로 도시한 흐름도이다.

【0064】 도 6에 도시된 바와 같이, 본 발명의 일 실시예에 따른 이차전지의 제조 방법은 준비단계(S1), 노칭 탭부 형성단계(S2), 권취단계(S3), 노칭탭 형성단계(S4) 및 설치단계(S5)를 포함한다.

【0065】 상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 전극집전체의 길이방향을 따라 구비하는 전극(100)을 준비하는 단계이다.

【0066】 이때, 전극(100)은 음극일 수 있다.

【0067】 도 7은 본 발명의 일 실시예에 따라 전극집전체의 길이방향을 기준으로 형성된 유지부와 무지부가 전극집전체의 폭방향을 기준으로 병렬로 배열된 것을 도시한 평면도이다.

【0068】 도 7에 도시된 바와 같이, 준비단계(S1)에서 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(110)와 무지부(120)는 전극집전체의 폭방향(W)을 기준으로 병렬로 배열할 수 있다.

【0069】 상기 노칭 탭부 형성단계(S2)는 전극(100)에서 유지부(110)의 길이 1/3 이상에 유지부(110)의 폭방향(W)으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록 전극(100)을 노칭하는 단계이다.

【0070】 이와 같이 전극(100)에서 노칭 탭부(111)를 노칭하는 것은 전극집전체에서 전극(100)을 커팅하면서 유지부(110)의 폭방향(W)으로 형성된 활물질(a)이 도포되지 않은 무지부의 일부를 노칭하여 노칭 탭부(11)를 형성하는 것일 수 있다.

【0071】 또한, 전극집전체의 길이방향(L)을 기준으로 형성된 유지부(100)와 무지부(120)가 전극집전체의 폭방향(W)을 기준으로 병렬로 배열되었을 경우 병렬로 배열된 유지부(110)와 무지부(120)를 전극집전체의 길이 방향(L)으로 커팅하면서

하나의 유지부(110)와 폭방향(W) 기준 병렬로 배열된 다른 하나의 유지부(110a)의 사이에 형성된 활물질이 도포되지 않은 무지부 영역을 노칭하여 유지부(110)의 폭방향(W)으로 노칭 탭부(111)가 형성되도록 할 수 있다.

【0072】 상기 권취단계(S3)는 노칭 탭부(111)가 형성된 전극(100)을 분리막과 상기 전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를 형성하는 단계일 수 있다.

【0073】 전극(100)이 권취되는 과정에서 노칭 탭부(111)는 2겹 이상으로 겹쳐지게 형성될 수 있다.

【0074】 상기 노칭탭 형성단계(S4)는 전극조립체(1)에서 노칭 탭부(111)를 커팅하여 복수 개의 노칭탭(111a)을 형성하는 단계일 수 있다.

【0075】 노칭 탭부(111)가 커팅되어 형성된 노칭탭(111a)은 노칭 탭부(111)와 동일하게 2겹 이상으로 겹쳐진 상태를 형성할 수 있다.

【0076】 이때 노칭 탭부(111)의 커팅은 레이저를 사용하여 작업 효율을 극대화할 수 있다.

【0077】 상기 설치단계(S5)는 전극조립체(1)를 캔부재(10)에 수용하여 노칭 탭(111a)을 캔부재(10)에 접합하는 단계이다.

【0078】 설치단계(S5)에서 노칭탭(111a)이 캔부재(10)에 전기적으로 연결되면서 용접 접합이 용이하도록 노칭탭(111a)을 절곡하여 노칭탭(111a)이 절곡된 상태로 전극조립체(1)가 캔부재(10)에 수용되도록 할 수 있다.

【0079】 이때, 노칭탭(111a)은 2겹 이상으로 형성되었기 때문에 노칭탭(111a)이 절곡되어도 기계적 강도를 유지할 수 있어 쉽게 끊어지지 않는 효과가 있다.

【0080】 절곡된 노칭탭(111a)은 단부가 캔부재(10)의 저면 중앙을 향하도록 할 수 있다.

【0081】 캔부재(10)의 저면과 노칭탭(111a)의 결합은 노칭탭(111a)의 절곡부가 캔부재(10)의 저면에 용접 결합되게 할 수 있다.

【0083】 상술한 바와 같이 본 발명에 따르면, 무지부 형성을 극소화하면서 멀티탭을 형성하는 효과가 있고, 멀티탭을 형성하여 전기 저항을 극소화하는 효과가 있고, 무지부 형성을 극소화하여 전지 용량을 확보하는 효과가 있고, 전극 포일로부터 포일 탭을 형성하면서 포일 탭이 2겹 이상 겹쳐지게 하여 기계적 강도를 높이는 효과가 있고, 포일 탭의 기계적 강도를 강화하여 포일 탭이 절곡된 상태에서 끊어지지 않게 함에 따라 포일 탭을 절곡한 상태에서 캔부재에 접합할 수 있어 전기적 연결의 신뢰성을 확보하면서 접합력을 높이는 효과가 있다.

【0085】 이상과 같이 본 발명에 따른 이차전지 및 그 이차전지의 제조 방법을 예시된 도면을 참고하여 설명하였으나, 본 발명은 이상에서 설명된 실시예와 도면에 의해 한정되지 않으며, 특허청구범위 내에서 본 발명이 속하는 기술분야에서

통상의 지식을 가진 자들에 의해 다양한 실시가 가능하다.

【부호의 설명】

【0086】 1: 전극조립체

10: 캔부재

100: 전극

110: 유지부

111: 노칭 탭부

111a: 노칭 탭

120: 무지부

121: 전극탭

a: 활물질

【청구범위】

【청구항 1】

전극집전체 상에 활물질(a)이 도포된 유지부(110)와 상기 전극집전체 상에 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100); 및

상기 전극집전체의 폭방향으로 상기 유지부(110)로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극(100)이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭탭부(111); 를 포함하는 것을 특징으로 하는 이차전지.

【청구항 2】

청구항 1에 있어서,

상기 노칭 탭부(111)는 상기 유지부(110) 전체 길이의 1/3 이상으로부터 연속되게 연장되는 것을 특징으로 하는 이차전지.

【청구항 3】

청구항 1 또는 청구항 2에 있어서,

상기 전극(100)이 권취된 상태에서 상기 노칭 탭부(111)는 커팅되어 복수 개의 노칭탭(111a)을 형성하는 것을 특징으로 하는 이차전지.

【청구항 4】

청구항 3에 있어서,

상기 노칭탭(111a)의 폭은 4mm ~ 5mm인 것을 특징으로 하는 이차전지.

【청구항 5】

청구항 3에 있어서,

상기 전극(100)은 캔부재(10)에 수용되고, 상기 노칭탭(111a)은 절곡되어 상기 캔부재(10)에 연결되는 것을 특징으로 하는 이차전지.

【청구항 6】

청구항 5에 있어서,

상기 노칭탭(111a)이 절곡되어 형성된 절곡부(111a)는 상기 캔부재(10)에 접합되는 것을 특징으로 하는 이차전지.

【청구항 7】

청구항 1에 있어서,

상기 무지부(120)에 부착되는 전극탭(121)을 포함하는 것을 특징으로 하는 이차전지.

【청구항 8】

청구항 1에 있어서,

상기 전극(100)은 음극인 것을 특징으로 하는 이차전지.

【청구항 9】

전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 길이방향을 따라 구비하는 전극(100)을 준비하는 준비단계(S1);

상기 전극(100)에서 유지부(110)의 길이 1/3 이상에 상기 전극집전체의 폭방향으로 활물질이 없는 상태로 연장된 노칭 탭부(111)가 형성되도록 상기 전극(100)을 노칭하는 노칭 탭부 형성단계(S2);

상기 노칭 탭부(111)가 형성된 상기 전극(100)을 분리막과 상기 전극(100)과 상극인 다른 전극에 적층한 후 권취하여 전극조립체(1)를 형성는 권취단계(S3);

상기 전극조립체(1)에서 상기 노칭 탭부(111)를 커팅하여 복수 개의 노칭탭(111a)을 형성하는 노칭탭 형성단계(S4); 및

상기 전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)을 상기 캔부재(10)에 접합하는 설치단계(S5); 를 포함하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 10】

청구항 9에 있어서,

상기 준비단계(S1)에서 음극의 전극(100)을 준비하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 11】

청구항 9에 있어서,

상기 준비단계(S1)는 전극집전체 상에 활물질(a)이 도포된 유지부(110)와 활물질(a)이 도포되지 않은 무지부(120)를 상기 전극집전체의 폭방향을 기준으로 병렬로 배열하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 12】

청구항 11에 있어서,

상기 노칭 탭부 형성단계(S2)는 상기 병렬로 배열된 상기 유지부(110)와 무지부(120)를 상기 전극집전체의 길이 방향으로 커팅하면서 상기 유지부(110)의 폭 방향으로 활물질(a)이 도포되지 않은 상기 노칭 탭부(11)를 노칭하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 13】

청구항 9에 있어서,

상기 노칭 탭부 형성단계(S2)는 상기 전극(100)을 커팅하면서 상기 유지부(110)의 폭방향으로 활물질(a)이 도포되지 않은 상기 노칭 탭부(111)가 형성되도록 노칭하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 14】

청구항 9에 있어서,

상기 노칭탭 형성단계(S4)에서 상기 노칭 탭부(111)를 레이저로 커팅하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 15】

청구항 9에 있어서,

상기 설치단계(S5)는 상기 노칭탭(111a)을 절곡한 상태로 상기 전극조립체(1)를 캔부재(10)에 수용하여 상기 노칭탭(111a)에서 절곡된 부분을 상기 캔부재

(10)에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 16】

청구항 15에 있어서,

상기 설치단계(S5)에서 상기 노칭탭(111a)의 절곡된 부분을 용접에 의해 상기 캔부재(10)에 접합하는 것을 특징으로 하는 이차전지의 제조 방법.

【청구항 17】

청구항 9에 있어서,

상기 권취단계(S3)는 상기 노칭 탭부(111)를 2겹 이상으로 겹쳐지도록 형성하는 것을 특징으로 하는 이차전지의 제조 방법.

【요약서】

【요약】

본 발명은 전극 탭의 기계적 강도를 높일 수 있는 이차전지 및 그 이차전지의 제조 방법에 관한 것이다.

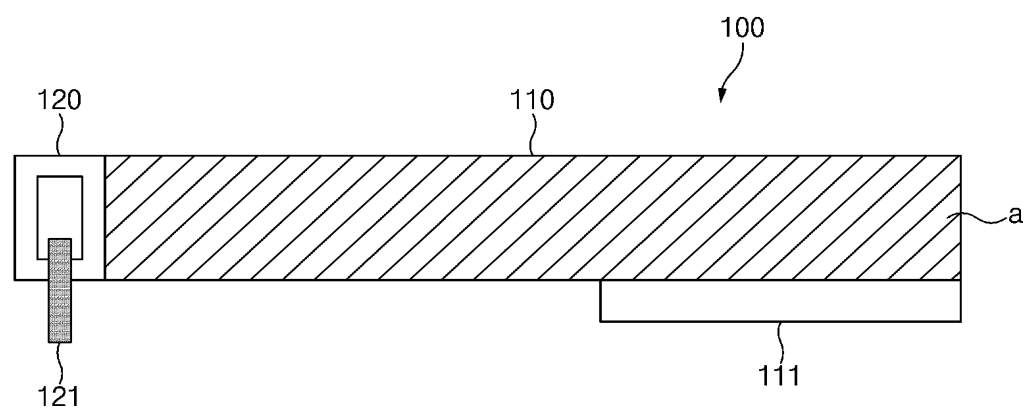
또한, 본 발명은 전극집전체 상에 활물질이 도포된 유지부와 상기 전극집전체 상에 활물질이 도포되지 않은 무지부를 상기 전극집전체의 길이방향을 따라 구비하는 전극 및 기 전극집전체의 폭방향으로 상기 유지부로부터 상기 활물질이 도포되지 않게 연장되고, 상기 전극이 권취되면 2겹 이상 겹쳐지게 형성되는 노칭 탭부를 포함하는 것을 특징으로 한다.

【대표도】

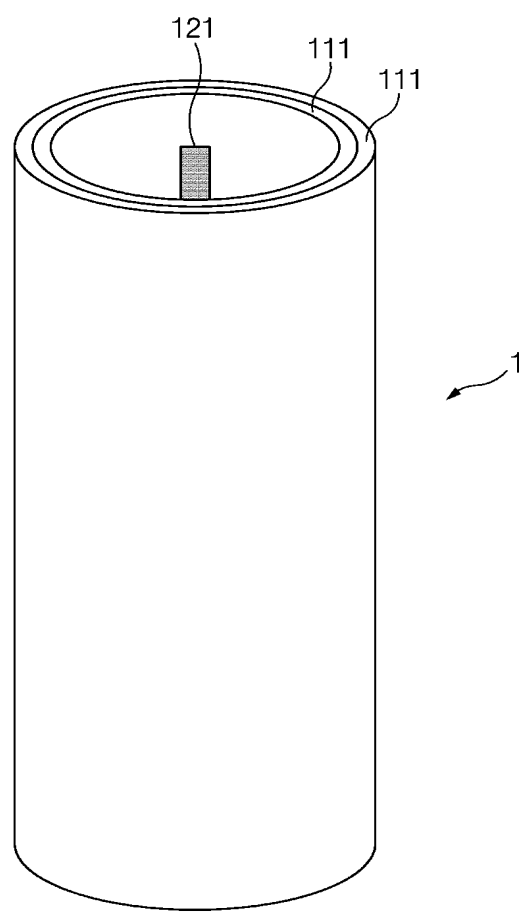
도 4

【도면】

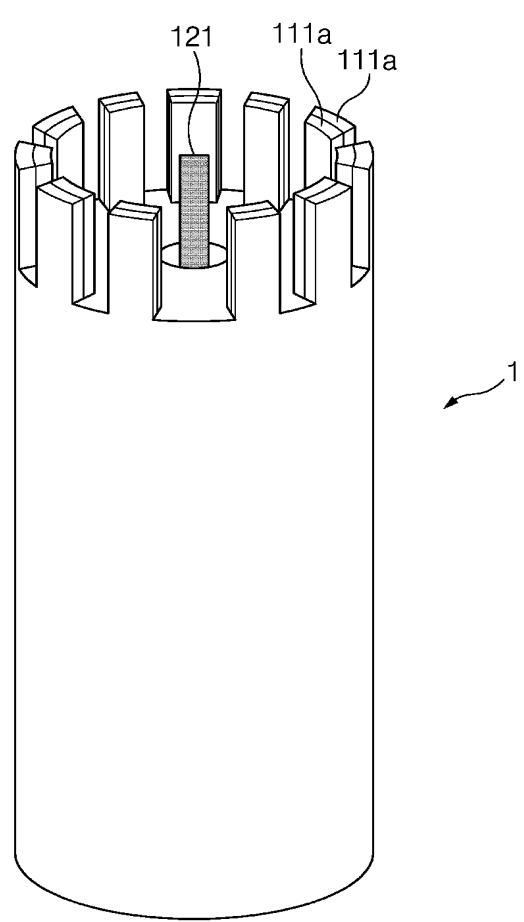
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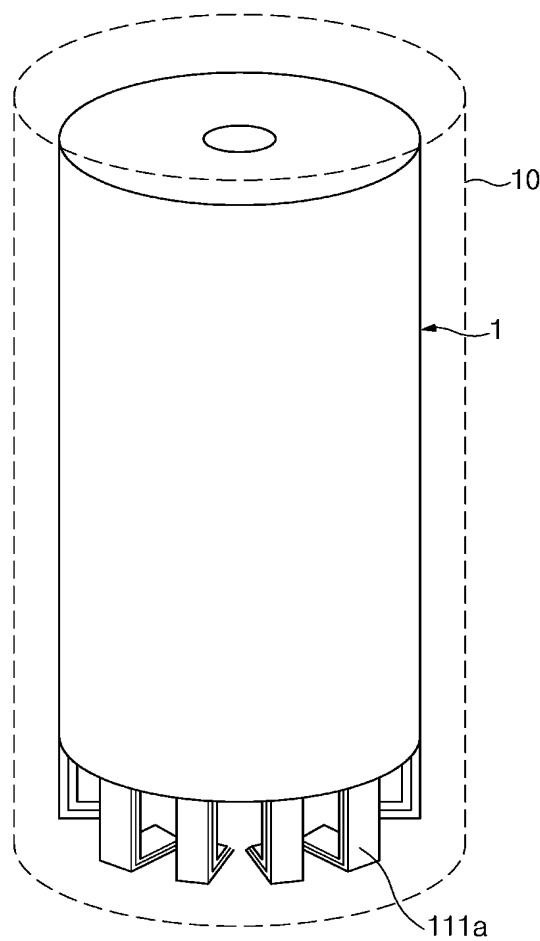
【도 2】



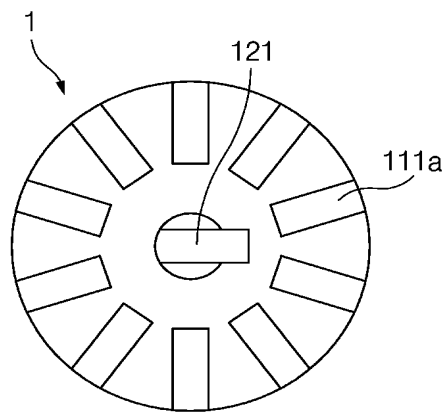
【도 3】



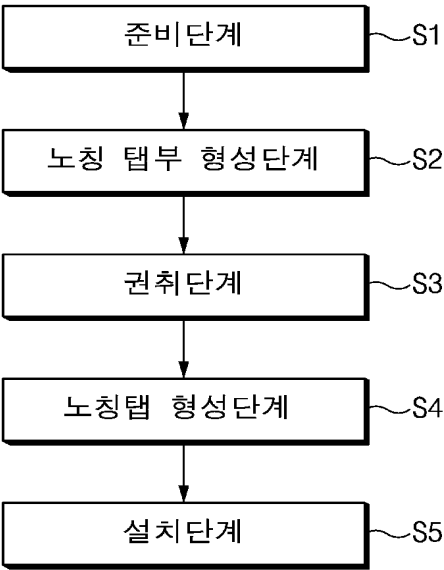
【도 4】



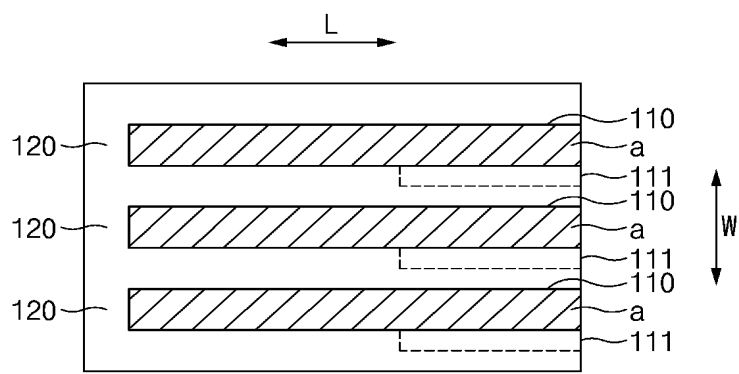
【도 5】



【도 6】



【도 7】



File History Content Report

The following content is missing from the original file history record obtained from the United States Patent and Trademark Office. No additional information is available.

Document Date - 2019-09-27

Document Title - Applicant Communication Re: Entity Status Set to Undiscounted
(Initial Default Setting or Status Change)

Electronic Acknowledgement Receipt	
EFS ID:	39144003
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Charles H. Humkey/Ashley Rowe
Filer Authorized By:	Charles H. Humkey
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	13-APR-2020
Filing Date:	
Time Stamp:	17:07:25
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment		no			
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM1957_-_EP1134820.pdf	954395	no	18
			98a65d93cf3d0fb37bb4ce35410a6346414ab2ad		
Warnings:					

Information:					
2	Non Patent Literature	LGCHEM1957_-_EESR.pdf	126077	no	5
			c21019ba9a88a5bcb114379502fd36beb3ccf4d4		
Warnings:					
Information:					
3	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM1957_-_Information_Disclosure_Statement_Fillable_PDF.pdf	1034069	no	4
			87d551c49310a7a525d1abfa1c3d0c3654679d6d		
Warnings:					
Information:					
4	Transmittal Letter	IDS_-_Information_Disclosure_Statement_-_Reference_Enclosed_LD-458.pdf	22405	no	1
			95ac946ca978ab8535119e76e342f2ea697d5ff7		
Warnings:					
Information:					
Total Files Size (in bytes):			2136946		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2020-04-13
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16498751	
	Filing Date		2019-09-27	
	First Named Inventor		Jae Woong Kim	
	Art Unit		N/A	
	Examiner Name		Not Yet Assigned	
	Attorney Docket Number		LGCHEM 3.3F-1654	

U.S.PATENTS							Remove	
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1	6569565		2003-05-27	Ligeois et al.			
If you wish to add additional U.S. Patent citation information please click the Add button.							Add	
U.S.PATENT APPLICATION PUBLICATIONS							Remove	
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
	1	20010023038	A1	2001-09-20	Ligeois et al.			
	2	20100310926	A1	2010-12-09	WANG et al.			
If you wish to add additional U.S. Published Application citation information please click the Add button.							Add	
FOREIGN PATENT DOCUMENTS							Remove	
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	1134820	EP	A1	2001-09-19	Cit Alcatel	U.S. Publication No. 20010023038 A1 and U.S. Patent No. 6,569,565 are English language counterparts.	×
If you wish to add additional Foreign Patent Document citation information please click the Add button							Add	
NON-PATENT LITERATURE DOCUMENTS							Remove	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
 (Not for submission under 37 CFR 1.99)

	Application Number		16498751
	Filing Date		2019-09-27
	First Named Inventor	Jae Woong Kim	
	Art Unit		N/A
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGCHEM 3.3F-1654	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Extended European Seach Report including Written Opinion for Application No. EP18784869.2 dated January 21, 2020, 5 pages.	

If you wish to add additional non-patent literature document citation information please click the Add button

Add

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Other Prior Art

According to the information contained in form PTO-1449 or PTO-892, there are one or more other prior art/non-patent literature documents missing from the original file history record obtained from the United States Patent and Trademark Office. Upon your request we will attempt to obtain these documents from alternative resources. Please note that additional charges will apply for this service.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: April 13, 2020
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: N/A

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

It is respectfully requested that the reference(s) listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited reference(s) is/are legally available prior art or that the same is/are pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-1654 from which the undersigned is authorized to draw.

Dated: April 13, 2020

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant(s)

LD-4528

6316429



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
16/498,751	Jae Woong Kim	LGCHEM 3.3F-1654
86765		
LGCHEM		
Lerner, David, Littenberg, Krumholz & Mentlik, LLP		
20 Commerce Drive		
Cranford, NJ 07016		
INTERNATIONAL APPLICATION NO.		
PCT/KR2018/004366		
I.A. FILING DATE		PRIORITY DATE
04/13/2018		04/14/2017

CONFIRMATION NO. 4723
371 ACCEPTANCE LETTER



Date Mailed: 06/10/2020

NOTICE OF ACCEPTANCE OF APPLICATION UNDER 35 U.S.C 371 AND 37 CFR 1.495

The applicant is hereby advised that the United States Patent and Trademark Office, in its capacity as a Designated / Elected Office (37 CFR 1.495), has ACCEPTED the above identified international application for national patentability examination in the United States Patent and Trademark Office.

The United States Application Number assigned to the application is shown above. A Filing Receipt will be issued for the present application in due course. **THE DATE APPEARING ON THE FILING RECEIPT AS THE "FILING DATE or 371(c) DATE" IS THE DATE ON WHICH THE LAST OF THE 35 U.S.C. 371 (c)(1) and (c)(2) REQUIREMENTS HAS BEEN RECEIVED IN THE OFFICE. THIS DATE IS SHOWN BELOW.** The filing date of the above identified application is the international filing date of the international application (Article 11(3) and 35 U.S.C. 363)

09/27/2019
DATE OF RECEIPT OF 35 U.S.C.
371(c)(1) and (c)(2) REQUIREMENTS

The following items have been received:

- Copy of the International Application filed on 09/27/2019
- English Translation of the IA filed on 09/27/2019
- Copy of the International Search Report filed on 09/27/2019
- Preliminary Amendments filed on 09/27/2019
- Information Disclosure Statements filed on 09/27/2019
- Inventor's Oath or Declaration filed on 09/27/2019
- Request for Immediate Examination filed on 09/27/2019
- U.S. Basic National Fees filed on 09/27/2019
- Substitute Specification filed on 09/27/2019
- Authorize Access to Search Results filed on 09/27/2019
- Priority Documents filed on 09/27/2019
- Power of Attorney filed on 09/27/2019
- Authorization to Permit Access filed on 09/27/2019
- Application Data Sheet (37 CFR 1.76) filed on 09/27/2019

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

JUELETHIA A PALMER

Telephone: (571) 272-9050



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/498,751	09/27/2019		1580	LGCHEM 3.3F-1654	20	2

CONFIRMATION NO. 4723

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

FILING RECEIPT



Date Mailed: 06/10/2020

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Jae Woong Kim, Daejeon, KOREA, REPUBLIC OF;
Hyoung Kwon Kim, Daejeon, KOREA, REPUBLIC OF;
Sang Uck Kim, Daejeon, KOREA, REPUBLIC OF;

Applicant(s)

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

Assignment For Published Patent Application

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

Power of Attorney: The patent practitioners associated with Customer Number 86765

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/KR2018/004366 04/13/2018

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2017-0048645 04/14/2017
REPUBLIC OF KOREA 10-2018-0043191 04/13/2018

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 06/08/2020

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/498,751**

Projected Publication Date: 09/17/2020

Non-Publication Request: No

Early Publication Request: No

Title

Secondary Battery And Method For Manufacturing The Same

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

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NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654

CONFIRMATION NO. 4723

PUBLICATION NOTICE

86765
LGCHEM
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016



Title:Secondary Battery And Method For Manufacturing The Same

Publication No.US-2020-0295341-A1

Publication Date:09/17/2020

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

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Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Managment, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Electronic Acknowledgement Receipt	
EFS ID:	40882811
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Charles H. Humkey/Ashley Rowe
Filer Authorized By:	Charles H. Humkey
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	19-OCT-2020
Filing Date:	27-SEP-2019
Time Stamp:	16:46:26
Application Type:	U.S. National Stage under 35 USC 371

Payment information:					
Submitted with Payment			no		
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM1957_- _JP2003338276A.pdf	378460	no	7
			d4f5dade9740a468f85a495ddcc68d3504bead0f9		
Warnings:					

Information:					
2	Foreign Reference	LGCHEM1957_- _Translation_of_JP2003338276 A.pdf	291789	no	9
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Warnings:					
Information:					
3	Foreign Reference	LGCHEM1957_- _JP2007227137A.pdf	568045	no	16
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Warnings:					
Information:					
4	Foreign Reference	LGCHEM1957_- _JP2008066040A.pdf	1634615	no	36
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Information:					
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Warnings:					
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Warnings:					
Information:					
7	Foreign Reference	LGCHEM1957_- _JP2020509534A.pdf	557970	no	14
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Warnings:					
Information:					
8	Foreign Reference	LGCHEM1957_- _WO2012086690A1.pdf	1882837	no	51
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Warnings:					
Information:					

9	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM1957_-_Information_Disclosure_Statement_Fillable_PDF.pdf	1034973	no	5
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Warnings:					
Information:					
10	Transmittal Letter	IDS_-_Information_Disclosure_Statement_-_Reference_Enclosed_LD-458.pdf	22590	no	1
			e529c076cb480e663a5aa402084fb2873c53ae69		
Warnings:					
Information:					
Total Files Size (in bytes):			7940577		
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	1725	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2020-10-19
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16498751	
	Filing Date		2019-09-27	
	First Named Inventor	Jae Woong Kim		
	Art Unit		1725	
	Examiner Name	Not Yet Assigned		
	Attorney Docket Number		LGCHEM 3.3F-1654	

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	1	20070196730	A1	2007-08-23	Kozuki			
	2	20110067227	A1	2011-03-24	Sohn			
	3	20190379028	A1	2019-12-12	Lim et al.			
	4	20130344364	A1	2013-12-26	Nanaumi et al.			
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
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Application Number		16498751
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First Named Inventor	Jae Woong Kim	
Art Unit		1725
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGCHEM 3.3F-1654

1	2003338276	JP	A	2003-11-28	Matsushita Electric Ind Co Ltd	Machine Generated Translation attached.	×
2	2007227137	JP	A	2007-09-06	Matsushita Electric Ind Co Ltd	US Publication 2007196730 is an English Language counterpart.	☒
3	2008066040	JP	A	2008-03-21	Sony Corp	Machine Generated Translation attached.	☒
4	2011065981	JP	A	2011-03-31	Samsung Sdi Co Ltd	US Publication 2011067227 is an English Language counterpart.	☒
5	2020509534	JP	A	2020-03-26	LG CHEM Ltd.	US Publication 2019379028 is an English Language counterpart.	☒
6	2012086690	WO	A1	2012-06-28	Hitachi Vehicle Energy, Ltd	US Publication 2013344364 is an English Language counterpart.	☒

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit		1725
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGCHEM 3.3F-1654

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: October 19, 2020
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1725

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

It is respectfully requested that the reference(s) listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited reference(s) is/are legally available prior art or that the same is/are pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-1654 from which the undersigned is authorized to draw.

Dated: October 19, 2020

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant(s)

LD-4523

6597718



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	06/03/2021		
LGCHEM			EXAMINER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP			EGGERDING, ALIX ECHELMEYER	
20 Commerce Drive				
Cranford, NJ 07016			ART UNIT	PAPER NUMBER
			1729	
			NOTIFICATION DATE	DELIVERY MODE
			06/03/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):
eOfficeAction@lernerdavid.com

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

2. The information disclosure statements filed 9/27/19, 4/13/20, and 10/19/20 have been considered by the examiner.

REQUIREMENT FOR UNITY OF INVENTION

As provided in 37 CFR 1.475(a), a national stage application shall relate to one invention only or to a group of inventions so linked as to form a single general inventive concept ("requirement of unity of invention"). Where a group of inventions is claimed in a national stage application, the requirement of unity of invention shall be fulfilled only when there is a technical relationship among those inventions involving one or more of the same or corresponding special technical features. The expression "special technical features" shall mean those technical features that define a contribution which each of the claimed inventions, considered as a whole, makes over the prior art.

The determination whether a group of inventions is so linked as to form a single general inventive concept shall be made without regard to whether the inventions are claimed in separate claims or as alternatives within a single claim. See 37 CFR 1.475(e).

When Claims Are Directed to Multiple Categories of Inventions:

As provided in 37 CFR 1.475 (b), a national stage application containing claims to different categories of invention will be considered to have unity of invention if the claims are drawn only to one of the following combinations of categories:

- (1) A product and a process specially adapted for the manufacture of said product; or
- (2) A product and a process of use of said product; or
- (3) A product, a process specially adapted for the manufacture of the said product, and a use of the said product; or
- (4) A process and an apparatus or means specifically designed for carrying out the said process; or
- (5) A product, a process specially adapted for the manufacture of the said product, and an apparatus or means specifically designed for carrying out the said process.

Otherwise, unity of invention might not be present. See 37 CFR 1.475 (c).

3. Restriction is required under 35 U.S.C. 121 and 372.

This application contains the following inventions or groups of inventions which are not so linked as to form a single general inventive concept under PCT Rule 13.1.

In accordance with 37 CFR 1.499, applicant is required, in reply to this action, to elect a single invention to which the claims must be restricted.

Group I, claims 1-10, drawn to a secondary battery.

Group II, claims 14-23, drawn to a method of making a secondary battery.

4. The groups of inventions listed above do not relate to a single general inventive concept under PCT Rule 13.1 because, under PCT Rule 13.2, they lack the same or corresponding special technical features for the following reasons:

Group I and II lack unity of invention because even though the inventions of these groups require the technical feature of a secondary battery comprising an electrode with a non-coating portion, having a notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state, this technical feature is not a special technical feature as it does not make

a contribution over the prior art in view of Martin et al. (US 2013/0108903). Martin teaches a secondary battery (50) including an electrode provided with non-coating portion including notching tab parts, or electrode tabs (300), forming two or more overlapping layers when the electrode is in a wound state (Figures 1, 3, and 5, [0016], [0043]).

5. During a telephone conversation with Joseph Karlin on 5/26/21 a provisional election was made without traverse to prosecute the invention of Group I, claims 1-10. Affirmation of this election must be made by applicant in replying to this Office action. Claims 14-23 are withdrawn from further consideration by the examiner, 37 CFR 1.142(b), as being drawn to a non-elected invention.

6. Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be corrected in compliance with 37 CFR 1.48(a) if one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. A request to correct inventorship under 37 CFR 1.48(a) must be accompanied by an application data sheet in accordance with 37 CFR 1.76 that identifies each inventor by his or her legal name and by the processing fee required under 37 CFR 1.17(i).

7. The examiner has required restriction between product or apparatus claims and process claims. Where applicant elects claims directed to the product/apparatus, and all product/apparatus claims are subsequently found allowable, withdrawn process claims that include all the limitations of the allowable product/apparatus claims should be considered for rejoinder. All claims directed to a nonelected process invention must include all the limitations of an allowable product/apparatus claim for that process invention to be rejoined.

In the event of rejoinder, the requirement for restriction between the product/apparatus claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully

examined for patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product/apparatus are found allowable, an otherwise proper restriction requirement between product/apparatus claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product/apparatus claim will not be rejoined. See MPEP § 821.04. Additionally, in order for rejoinder to occur, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product/apparatus claims. **Failure to do so may result in no rejoinder.** Further, note that the prohibition against double patenting rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

Claim Rejections - 35 USC § 103

8. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.
9. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.
10. Claims 1-7 are rejected under 35 U.S.C. 103 as being unpatentable over Morishima et al. (US 2010/0081052) in view of Martin et al. (US 2013/0108903).

Regarding claims 1 and 2, Morishima teaches a secondary battery comprising an electrode (7) provided with an active material coating portion (2a, 2b) coated with active material on an electrode collector, or metallic foil (1), and a non-coating portion, or uncoated parts (3a, 3b), on which active material is not applied to the electrode collector, where the non-coating portions extends continuously more than 1/3 of a total length of the coating portion in the longitudinal direction (Figure 11, [0136]).

Further regarding claim 1 and with regard to claim 3, Morishima teaches a notching tab part, or plurality of notching tabs, or tabs (5), extending from the coating portion (2a) in a width direction perpendicular to the longitudinal direction of the electrode collector (1) when the electrode is in a wound state (Figure 12, [0135], [0137]).

With further regard to claim 1, and with regard to claims 5-7, Morishima teaches that the notching tab part, or tabs, have a wound state, i.e. when the electrode group is rolled up ([0135]), but fails to teach specifically that the notching tab part forms two or more overlapping layers when the electrode is in a wound state.

Martin teaches a secondary battery comprising a notching tab part, or electrode tabs (300), which extend from the active material part of an electrode, and which are bent to form overlapping layers when the electrode is in a wound state (Figures 1, 3, and 5, [0016], [0043]). The bent tabs (300) of Martin are bonded to the can member by weld and crimp joints ([0044]).

Martin further teaches that it is desirable to provide the notching tab parts, or electrode tabs (300), in two or more overlapping layers that are well aligned in order to facilitate installation of an insulation member, or insulator (24a/24b, 34a/34b, 50), provided between the bent parts of the tabs (5) and the end of the electrode (33a, 33b), and which is used to prevent short circuits (Figure 5, [0003], [0006], [0007]).

It would have been obvious to the skilled artisan at the time of the invention to form the notching tabs (5) of Morishima to be used in the insulator arrangement such as taught by Martin in order to properly fit the tabs in the insulator and therefore prevent short circuits.

Regarding claim 4, Morishima and Martin are silent on the width of the notching tabs. The examiner finds that it would have been an obvious matter of design choice to determine the proper width of the notching tabs for the size and function of the battery. A change in size is generally recognized as being within the level of ordinary skill in the art. MPEP 2144.04 IV A

11. Claims 8-10 are rejected under 35 U.S.C. 103 as being unpatentable over Morishima in view of Martin as applied to claim 1 above, and further in view of Kogetsu et al. (US 2018/0145304).

The teachings of Morishima and Martin as discussed above are incorporated herein.

Regarding claim 8, Morishima in view of Martin teaches the secondary battery of claim 1 including the electrode and notching tab parts, but fails to teach an electrode tab attached to the non-coating portion.

Martin teaches that bending of the tabs can introduce stress that may cause eventual failure ([0003]).

Kogetsu teaches a secondary battery comprising an electrode provided with a coating portion (41) and non-coating portion (40a), where an electrode tab, or current collecting lead (24), attached to the non-coating portion ([0045]).

Kogetsu teaches that the electrode tab is attached to the non-coating portion by insulating tape to suppress motion of the electrode tab, or current collecting lead ([0026]).

It would have been obvious to the skilled artisan at the time of the invention to include, in addition to the notching tabs of Morishima in view of Martin, an electrode tab connected to the non-

coating portion by insulating tape, such as suggested by Kogetsu, in order to reduce stress on the electrode tab and ensure a connection between the electrode and the can in the instance of failure such as warned by Martin.

As for claim 9, as is discussed above, Morishima teaches a notching tab part, or plurality of notching tabs, or tabs (5), extending from the coating portion (2a) in a width direction perpendicular to the longitudinal direction of the electrode collector (1) when the electrode is in a wound state (Figure 12, [0135], [0137]).

Regarding claim 10, as is discussed above, Martin teaches a secondary battery comprising a notching tab part, or electrode tabs (300), which extend from the active material part of an electrode, and which are bent to form overlapping layers when the electrode is in a wound state (Figures 1, 3, and 5, [0016], [0043]). The bent tabs (300) of Martin are bonded to the can member by weld and crimp joints ([0044]).

Conclusion

12. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMEYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached on 8:00am - 5:00pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

<i>Notice of References Cited</i>	Application/Control No. 16/498,751		Applicant(s)/Patent Under Reexamination Kim et al.	
	Examiner ALIX E EGGERDING		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20100081052-A1	04-2010	MORISHIMA; Hideaki	H01M4/485	429/211
*	B	US-20130108903-A1	05-2013	Martin; Charles E.	H01M10/0404	429/82
*	C	US-20180145304-A1	05-2018	KOGETSU; YASUTAKA	H01M4/02	1/1
	D					
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2019-09-27
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Receipt date: 09/27/2019

16/498,751 -- GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (02-18)
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

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	1	20050003264	A1	2005-01-06	Oh et al.	
	2	20050287428	A1	2005-12-29	Cheon et al.	
	3	20110293977	A1	2011-12-01	Kim et al.	
	4	20120009450	A1	2012-01-12	Chun	
	5	20120208055	A1	2012-08-16	Ahn	

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Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

	6	20150295270	A1	2015-10-15	Chun	
	7	20160351966	A1	2016-12-01	Batson et al.	

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	1	100515833	KR	B1	2005-09-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20050003264 A1 is an English language counterpart.	×
	2	20000051740	KR	A	2000-08-16	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	3	20050121904	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English language counterpart.	☒
	4	20060022358	KR	A	2006-03-10	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	5	20100071941	KR	A	2010-06-29	Lg Chemical Ltd	U.S. Publication No. 20110293977 A1 is an English language counterpart.	☒
	6	20120006389	KR	A	2012-01-18	Samsung Sdi Co., Ltd	U.S. Publication No. 20120009450 A1 is an English language counterpart.	☒
	7	20120092367	KR	A	2012-08-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20120208055 A1 is an English language counterpart.	☒

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Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

8	20130073329	KR	A	2013-07-03	Datechnology Co., Ltd	Machine generated translation is attached.	☒
9	20150117135	KR	A	2015-10-19	Samsung Sdi Co Ltd	U.S. Publication No. 20150295270 A1 is an English language counterpart.	☒
10	2016001575	JP	A	2016-01-07	Denso Corp	Machine generated translation is attached.	☒
11	20160113612	KR	A	2016-09-30	A123 Systems LLC	U.S. Publication No. 20160351966 A1 is an English language counterpart.	☒
12	20170006437	KR	A	2017-01-18	Lg Chemical Ltd	Machine generated translation is attached.	☒
13	5790576	JP	B2	2015-10-07	Toyota Industries Corp	Machine generated translation is attached.	☒

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	1	International Search Report from Application No. PCT/KR2018/004366 mailed July 20, 2018, pages 1 - 2	☒

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Examiner Signature	/ALIX E EGGERDING/	Date Considered	05/17/2021
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Receipt date: 09/27/2019

16/498,751 - GAU: 1729

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGCHEM 3.3F-1654

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

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See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2020-04-13
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Receipt date: 04/13/2020

16/498,751 -- GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (02-18)
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number	16498751
Filing Date	2019-09-27
First Named Inventor	Jae Woong Kim
Art Unit	N/A
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-1654

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	1	6569565		2003-05-27	Ligeois et al.	

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	1	20010023038	A1	2001-09-20	Ligeois et al.	
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	1	1134820	EP	A1	2001-09-19	Cit Alcatel	U.S. Publication No. 20010023038 A1 and U.S. Patent No. 6,569,565 are English language counterparts.	×

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-1654	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Extended European Seach Report including Written Opinion for Application No. EP18784869.2 dated January 21, 2020, 5 pages.	

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Examiner Signature	/ALIX E EGGERDING/	Date Considered	05/17/2021
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit		1725
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGCHEM 3.3F-1654

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2020-10-19
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Receipt date: 10/19/2020

16/498,751 -- GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (02-18)
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
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Application Number	16498751
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First Named Inventor	Jae Woong Kim
Art Unit	1725
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-1654

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	1	20070196730	A1	2007-08-23	Kozuki	
	2	20110067227	A1	2011-03-24	Sohn	
	3	20190379028	A1	2019-12-12	Lim et al.	
	4	20130344364	A1	2013-12-26	Nanaumi et al.	

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INFORMATION DISCLOSURE
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First Named Inventor	Jae Woong Kim	
Art Unit		1725
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGCHEM 3.3F-1654

1	2003338276	JP	A	2003-11-28	Matsushita Electric Ind Co Ltd	Machine Generated Translation attached.	×
2	2007227137	JP	A	2007-09-06	Matsushita Electric Ind Co Ltd	US Publication 2007196730 is an English Language counterpart.	☒
3	2008066040	JP	A	2008-03-21	Sony Corp	Machine Generated Translation attached.	☒
4	2011065981	JP	A	2011-03-31	Samsung Sdi Co Ltd	US Publication 2011067227 is an English Language counterpart.	☒
5	2020509534	JP	A	2020-03-26	LG CHEM Ltd.	US Publication 2019379028 is an English Language counterpart.	☒
6	2012086690	WO	A1	2012-06-28	Hitachi Vehicle Energy, Ltd	US Publication 2013344364 is an English Language counterpart.	☒

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Examiner Signature	/ALIX E EGGERDING/	Date Considered	05/17/2021
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16/498,751 - GAU: 1729

**INFORMATION DISCLOSURE
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Electronic Acknowledgement Receipt	
EFS ID:	43683327
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Robert B. Cohen/Sharon McGee
Filer Authorized By:	Robert B. Cohen
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	03-SEP-2021
Filing Date:	27-SEP-2019
Time Stamp:	14:21:19
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment		no			
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Arguments/Remarks Made in an Amendment	Amendment.pdf	231588	no	7
			e2dd0cedbb09a6f41573678ac9c0ee9ac6a8fe0d		
Warnings:					

Information:					
2	Miscellaneous Incoming Letter	Amendment_Transmittal.pdf	22137	no	1
			7587c37141450cffd2b67f6d702737b1107ab7b7		
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I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: September 3, 2021

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Dear Sir:

In response to the official action mailed June 3, 2021, applicant submits the following amendments and remarks.

IN THE CLAIMS

1. (Currently Amended) A secondary battery comprising:

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each tab has a respective plurality of layer portions each extending over a same circumferentially extending angle about a center of the secondary battery.

2. (Currently Amended) ~~The~~ A secondary battery ~~of claim 1, comprising:~~

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

wherein the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.

3. (Canceled)

4. (Currently Amended) The secondary battery of claim-~~3~~ 1, wherein each of the plurality of notching tabs has a width of 4 mm to 5 mm.

5. (Currently Amended) The secondary battery of claim-~~3~~ 1, wherein the electrode is accommodated in a can member, and each of the plurality of notching tabs is in a bent state and connected to the can member.

6. (Previously Presented) The secondary battery of claim 5, wherein a bent part of each of the plurality of notching tabs is bonded to the can member.

7. (Previously Presented) The secondary battery of claim 6, further comprising an insulation member disposed between the bent parts and an end of the electrode.

8. (Previously Presented) The secondary battery of claim 1, further comprising an electrode tab attached to the non-coating portion.

9. (Previously Presented) The secondary battery of claim 8, wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs, and

the electrode is accommodated in a can member, and each of the plurality of notching tabs is in a bent state and overlaps the electrode tab.

10. (Previously Presented) The secondary battery of claim 9, wherein the electrode tab and the plurality of notching tabs are bonded together to the can member.

11-13. (Cancelled)

14. (Withdrawn) A method for manufacturing a secondary battery, the method comprising:
a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector;

a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction;

a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly;

a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and

an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

15. (Withdrawn) The method of claim 14, wherein the electrode is a negative electrode.
16. (Withdrawn) The method of claim 14, wherein, during the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied are arranged parallel to each other in the width direction of the electrode collector.
17. (Withdrawn) The method of claim 16, wherein, during the notching tab part formation step, the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.
18. (Withdrawn) The method of claim 14, wherein, during the notching tab part formation step, notching is performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.
19. (Withdrawn) The method of claim 14, wherein, during the notching tab formation step, the notching tab part is cut using a laser.
20. (Withdrawn) The method of claim 14, wherein, during the installation step, the electrode assembly is accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.
21. (Withdrawn) The method of claim 20, wherein, during the installation step a bent portion of each of the plurality of notching tabs is welded to the can member.
22. (Withdrawn) The method of claim 20, wherein, during the installation step bent portions of each of the plurality of notching tabs overlap each other and are bonded to the can member together.
23. (Withdrawn) The method of claim 14, wherein, during the winding step, the notching tab part is wound to form two or more overlapping layers.

REMARKS

The present amendments and remarks are being filed in response to the non-final office action dated June 3, 2021. After entry of the foregoing amendments, claims 1, 2, 4-10, and 14-23 will be pending. Claims 1, 2, 4, and 5 are amended. Claims 14-23 have been withdrawn as directed to non-elected species or invention. Claims 3 and 11-13 are canceled. No new matter is added by the present amendments. Applicant respectfully requests that the Examiner reconsider the claims in view of the amendments and remarks herein.

Unity Rejection

In response to the Unity Rejection that is requiring Restriction under 35 U.S.C. §§ 121 and 372, Applicant hereby affirms the election of the invention of Group I, claims 1-10 that was previously made orally on May 26, 2021.

Prior Art Rejections

Claims 1-7 were rejected under 35 U.S.C. § 103 as allegedly unpatentable over U.S. Patent Application Pub. No. 2010/0081052 to (Morishima et al. ("*Morishima*")) in view of U.S. Patent Application Pub. No. 2013/0108903 to Martin et al. ("*Martin*"). Claims 8-10 were rejected under 35 U.S.C. § 103 as allegedly unpatentable over *Morishima* in view of *Martin*, and further in view of U.S. Patent Application Pub. No. 2018/0145304 to Kogetsu et al. ("*Kogetsu*"). Applicant respectfully submits that the rejections in the Office Action are traversed for the reasons set forth below.

Applicant submits that each and every limitation of claim 1, as amended, is not taught or suggested by any of the references of record, alone or in combination. Claim 1, as amended, recites:

A secondary battery comprising:

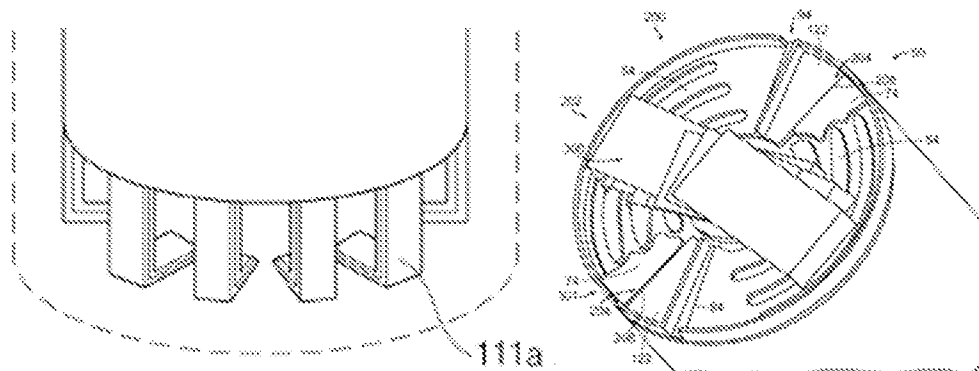
an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each tab has a respective plurality of layer portions each extending over a same circumferentially extending angle about a center of the secondary battery. (emphasis added)

The amendments to claim 1 can find support throughout the specification, and at least in as-published paragraphs [0014] and [0070], original claim 3, and figure 4.

Claim 1 is amended herein to recite the limitations dependent claim 3, which recites that when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs, and also an additional feature, reciting that “each tab has a respective plurality of layer portions each extending over a same circumferentially extending angle.” This feature is the result of cutting the tabs after the electrode is wound, which is not done in any of the cited references. This amendment recites the difference that can be seen comparing the circumferential alignment of tab portions in the claimed invention (figure 4 below, left) compared to the less-aligned tabs in *Martin* (figure 5 below, right).



Accordingly, Applicant respectfully submits that each and every limitation of amended claim 1 is not taught or suggested by any of the references of record, alone or in combination. Therefore, Applicant submits that the pending rejection of claim 1 should be withdrawn. For at least the same reasons, the rejections of dependent claims 4-10 should be also withdrawn, each of which ultimately depends from independent claim 1.

Furthermore, Applicant submits that each and every limitation of claim 2, as amended, is not taught or suggested by any of the references of record, alone or in combination. Original claim 2 has been amended merely to recite the limitations of original claim 1, thereby placing original claim 2 in independent form. Claim 2, as amended, recites:

- A secondary battery, comprising:
 - an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and
 - a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with

the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

wherein the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction. (emphasis added)

Original claim 2 recites that “the notching tab part continuously extends across 1/3 or more of a total length of the coating portion,” but original claim 1 (now part of claim 2) recites that the notching tab has a wound state. In contrast, in *Morishima*, the embodiment alleged to read on original claim 2 is not taught as being wound, so Applicant respectfully submits that the obvious rejection of claim 2 is improper.

Accordingly, Applicant respectfully submits that each and every limitation of claim 2 is not taught or suggested by any of the references of record, alone or in combination. Therefore, Applicant submits that the pending rejection of claim 2 should be withdrawn.

Furthermore, since original claim 2 is not amended herein other than to place it in independent form, if another rejection is issued rejecting claim 2 over newly-cited art, Applicant respectfully requests that the rejection be **non-final**, in accordance with MPEP § 706.07(a).

Conclusion

As it is believed that all of the rejections set forth in the Official Action have been fully met, favorable reconsideration and allowance are earnestly solicited. If, however, for any reason the Examiner does not believe that such action can be taken at this time, it is respectfully requested that the Examiner telephone applicant’s attorney at (908) 654-5000 in order to overcome any additional objections which the Examiner might have.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: September 3, 2021

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG,
KRUMHOLZ & MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

AMENDMENT TRANSMITTAL LETTER			Docket No. LGCHEM 3.3F-1654																																														
Application No. 16/498,751-Conf. #4723		Filing Date September 27, 2019		Examiner A. E. Eggerding		Art Unit 1729																																											
Applicant(s): LG Chem, Ltd.																																																	
Invention: Secondary Battery And Method For Manufacturing The Same																																																	
TO THE COMMISSIONER FOR PATENTS																																																	
Transmitted herewith is an amendment in the above-identified application.																																																	
The fee has been calculated and is transmitted as shown below.																																																	
<table><tr><th colspan="6">CLAIMS AS AMENDED</th></tr><tr><th></th><th>Claims Remaining After Amendment</th><th>Highest Number Previously Paid</th><th>Number Extra Claims Present</th><th>Rate</th><th></th></tr><tr><td>Total Claims</td><td>19</td><td>- 20 =</td><td>0</td><td>x</td><td></td></tr><tr><td>Independent Claims</td><td>3</td><td>- 3 =</td><td>0</td><td>x</td><td></td></tr><tr><td colspan="5">Multiple Dependent Claims (check if applicable)</td><td>☐</td></tr><tr><td colspan="5">Other fee (please specify):</td><td></td></tr><tr><td colspan="5">TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:</td><td>0.00</td></tr></table>								CLAIMS AS AMENDED							Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate		Total Claims	19	- 20 =	0	x		Independent Claims	3	- 3 =	0	x		Multiple Dependent Claims (check if applicable)					☐	Other fee (please specify):						TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00
CLAIMS AS AMENDED																																																	
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate																																													
Total Claims	19	- 20 =	0	x																																													
Independent Claims	3	- 3 =	0	x																																													
Multiple Dependent Claims (check if applicable)					☐																																												
Other fee (please specify):																																																	
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00																																												
☒ Large Entity ☐ Small Entity																																																	
☒ No additional fee is required for this amendment.																																																	
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☒ Credit any overpayment.																																																	
☒ Charge any additional filing or application processing fees required under 37 C.F.R. § 1.16 and 1.17.																																																	
/Joseph H. Karlin/ Joseph H. Karlin (Atty/Agent Reg. No.: 62,471) LERNER, DAVID, LITTENBERG, KRUMHOLZ & MENTLIK, LLP 20 Commerce Dr. Cranford, New Jersey 07016 (908) 654-5000				Dated: <u>September 3, 2021</u>																																													

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).	
Dated: <u>September 3, 2021</u>	Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	09/22/2021	EXAMINER	
LGCHEM			EGGERDING, ALIX ECHELMEYER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP				
20 Commerce Drive			ART UNIT	
Cranford, NJ 07016			PAPER NUMBER	
			1729	
			NOTIFICATION DATE	
			DELIVERY MODE	
			09/22/2021	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

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eOfficeAction@lernerdavid.com

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. This Office Action is in response to the amendment filed 9/3/21. Claims 1, 2, 4, and 5 are amended. Claim 3 is canceled. Claims 11-13 were previously canceled; claims 14-23 were previously withdrawn. Claims 1, 2, and 4-10 are rejected finally for the reasons provided below.

Claim Rejections - 35 USC § 103

3. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.
4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 1, 2, and 4-7 are rejected under 35 U.S.C. 103 as being unpatentable over Morishima et al. (US 2010/0081052) in view of Martin et al. (US 2013/0108903).

Regarding claims 1 and 2, Morishima teaches a secondary battery comprising an electrode (7) provided with an active material coating portion (2a, 2b) coated with active material on an electrode

collector, or metallic foil (1), and a non-coating portion, or uncoated parts (3a, 3b), on which active material is not applied to the electrode collector.

Further regarding claim 1, Morishima teaches a notching tab part, or plurality of notching tabs, or tabs (5), extending from the coating portion (2a) in a width direction perpendicular to the longitudinal direction of the electrode collector (1) when the electrode is in a wound state, forming a plurality of layer portions extending over a same circumferentially extending angle about a center of the secondary battery (Figure 12, [0135], [0137]).

Regarding claim 2, Morishima teaches that the non-coating portions extends continuously more than 1/3 of a total length of the coating portion in the longitudinal direction (Figure 11, [0136]).

With further regard to claims 1 and 2, and with regard to claims 5-7, Morishima teaches that the notching tab part, or tabs, have a wound state, i.e. when the electrode group is rolled up ([0135]), but fails to teach specifically that the notching tab part forms two or more overlapping layers when the electrode is in a wound state.

Martin teaches a secondary battery comprising a notching tab part, or electrode tabs (300), which extend from the active material part of an electrode, and which are bent to form overlapping layers when the electrode is in a wound state (Figures 1, 3, and 5, [0016], [0043]). The bent tabs (300) of Martin are bonded to the can member by weld and crimp joints ([0044]).

Martin further teaches that it is desirable to provide the notching tab parts, or electrode tabs (300), in two or more overlapping layers that are well aligned in order to facilitate installation of an insulation member, or insulator (24a/24b, 34a/34b, 50), provided between the bent parts of the tabs (5) and the end of the electrode (33a, 33b), and which is used to prevent short circuits (Figure 5, [0003], [0006], [0007]).

It would have been obvious to the skilled artisan at the time of the invention to form the notching tabs (5) of Morishima to be used in the insulator arrangement such as taught by Martin in order to properly fit the tabs in the insulator and therefore prevent short circuits.

Regarding claim 4, Morishima and Martin are silent on the width of the notching tabs. The examiner finds that it would have been an obvious matter of design choice to determine the proper width of the notching tabs for the size and function of the battery. A change in size is generally recognized as being within the level of ordinary skill in the art. MPEP 2144.04 IV A

6. Claims 8-10 are rejected under 35 U.S.C. 103 as being unpatentable over Morishima in view of Martin as applied to claim 1 above, and further in view of Kogetsu et al. (US 2018/0145304).

The teachings of Morishima and Martin as discussed above are incorporated herein.

Regarding claim 8, Morishima in view of Martin teaches the secondary battery of claim 1 including the electrode and notching tab parts, but fails to teach an electrode tab attached to the non-coating portion.

Martin teaches that bending of the tabs can introduce stress that may cause eventual failure ([0003]).

Kogetsu teaches a secondary battery comprising an electrode provided with a coating portion (41) and non-coating portion (40a), where an electrode tab, or current collecting lead (24), attached to the non-coating portion ([0045]).

Kogetsu teaches that the electrode tab is attached to the non-coating portion by insulating tape to suppress motion of the electrode tab, or current collecting lead ([0026]).

It would have been obvious to the skilled artisan at the time of the invention to include, in addition to the notching tabs of Morishima in view of Martin, an electrode tab connected to the non-

coating portion by insulating tape, such as suggested by Kogetsu, in order to reduce stress on the electrode tab and ensure a connection between the electrode and the can in the instance of failure such as warned by Martin.

As for claim 9, as is discussed above, Morishima teaches a notching tab part, or plurality of notching tabs, or tabs (5), extending from the coating portion (2a) in a width direction perpendicular to the longitudinal direction of the electrode collector (1) when the electrode is in a wound state (Figure 12, [0135], [0137]).

Regarding claim 10, as is discussed above, Martin teaches a secondary battery comprising a notching tab part, or electrode tabs (300), which extend from the active material part of an electrode, and which are bent to form overlapping layers when the electrode is in a wound state (Figures 1, 3, and 5, [0016], [0043]). The bent tabs (300) of Martin are bonded to the can member by weld and crimp joints ([0044]).

Response to Arguments

7. Applicant's arguments filed 9/3/21 have been fully considered but they are not persuasive.

First, regarding amended claim 1, Applicant argues on page 6 of the Remarks that the limitation to "each tab has a respective plurality of layer portions each extending over a same circumferentially extending angle," the examiner is not convinced. The examiner maintains that this limitation is depicted in Figures 5-7 of Martin. As to Applicant's statements concerning this feature being "the result of cutting the tabs after the electrode is wound," the examiner notes that the claims are to product claims, and further that the claims do not contain limitations to the method of making the tabs. Although the claims are interpreted in light of the specification, limitations from the specification are not read into the claims. See *In re Van Geuns*, 988 F.2d 1181, 26 USPQ2d 1057 (Fed. Cir. 1993).

Regarding amended claim 2, the arguments presented on page 7 are not convincing. Applicant argues that Morishima does not teach that the electrode is wound; however, Morishima does teach this feature (see [0135], and see the second paragraph of page 6 of the Non-Final Rejection mailed 6/3/21), and Applicant has failed to point to any specific teaching in Morishima that the electrode is not wound.

Additionally, the rejection is made over Morishima in view of Martin, and Martin teaches a wound electrode. Applicant is reminded that one cannot show nonobviousness by attacking references individually where the rejections are based on combinations of references. See *In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981); *In re Merck & Co.*, 800 F.2d 1091, 231 USPQ 375 (Fed. Cir. 1986).

Conclusion

8. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. Sohn (US 2011/0067227) teaches cutting the tabs after the electrode is wound (Figures 6A-6D and 8A-8D).

9. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

10. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMAYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached on 8:00am - 5:00pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

<i>Notice of References Cited</i>	Application/Control No. 16/498,751		Applicant(s)/Patent Under Reexamination Kim et al.	
	Examiner ALIX E EGGERDING		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20110067227-A1	03-2011	Sohn; Young-Bae	H01M4/70	29/623.1
	B					
	C					
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Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Electronic Acknowledgement Receipt	
EFS ID:	44184844
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Robert B. Cohen/Liz Semon
Filer Authorized By:	Robert B. Cohen
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	02-NOV-2021
Filing Date:	27-SEP-2019
Time Stamp:	15:42:54
Application Type:	U.S. National Stage under 35 USC 371

Payment information:	
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Payment Type	CARD
Payment was successfully received in RAM	\$ 260
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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM1957_- _CN102024936A_with_EN_abs tractpt1.pdf	19468717	no	11
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Warnings:					
Information:					
2	Foreign Reference	LGCHEM1957_- _CN102024936A_with_EN_abs tractpt2.pdf	16818010	no	11
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Information:					
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4	Foreign Reference	LGCHEM1957_- _CN201450072U_with_EN_abs tract.pdf	10351811	no	9
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Warnings:					
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5	Foreign Reference	LGCHEM1957_- _CN203250828U_with_EN_abs tractpt1.pdf	18005033	no	12
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6	Foreign Reference	LGCHEM1957_- _CN203250828U_with_EN_abs tractpt2.pdf	8368929	no	13
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7	Foreign Reference	LGCHEM1957_- _CN204577523U_with_EN_abs tract.pdf	10968714	no	11
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8	Foreign Reference	LGCHEM1957_- _Translation_of_CN204577523 U.pdf	301400	no	11
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9	Non Patent Literature	LGCHEM1957_- _CN_SR.pdf	885367	no	2
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10	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM1957_- _Information_Disclosure_State ment_Fillable_PDF.pdf	1034710	no	4
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11	Transmittal Letter	IDS_- _Form_B_- _After_Final_Rejection_LD-454. pdf	22590	no	1
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12	Fee Worksheet (SB06)	fee-info.pdf	38561	no	2
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit		1729
Examiner Name	A. E. Eggerding	
Attorney Docket Number		LGCHEM 3.3F-1654

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2021-11-02
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16498751	
	Filing Date		2019-09-27	
	First Named Inventor	Jae Woong Kim		
	Art Unit		1729	
	Examiner Name	A. E. Eggerding		
	Attorney Docket Number		LGCHEM 3.3F-1654	

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	1	102024936	CN	A	2011-04-20	Samsung Sdi Co., Ltd	U.S. Publication No. 20110067227 is the English language counterpart which was previously cited.	
	2	201450072	CN	U	2010-05-05	Dongguan Ampere Electronics Technology Co., Ltd	U.S. Publication No. 20100310926 is the English language counterpart which was previously cited.	
	3	203250828	CN	U	2013-10-23	A123 Systems, Inc	U.S. Publication No. 20130108903 is the English lanuage counterpart which was previously cited.	

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	1729	
Examiner Name	A. E. Eggerding	
Attorney Docket Number	LGCHEM 3.3F-1654	

	4	204577523	CN	U	2015-08-19	Contemporary AmpereX Technology Ltd	Machine generated translation attached.	×
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	1	Chiinese Search Report for Application No. 201880025247, dated August 24, 2021, 2 pages.	

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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Other Prior Art

According to the information contained in form PTO-1449 or PTO-892, there are one or more other prior art/non-patent literature documents missing from the original file history record obtained from the United States Patent and Trademark Office. Upon your request we will attempt to obtain these documents from alternative resources. Please note that additional charges will apply for this service.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: November 2, 2021
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

**SUBMISSION OF INFORMATION
DISCLOSURE STATEMENT UNDER 37 C.F.R. § 1.97(D)**

Dear Sir:

It is respectfully requested that the accompanying Information Disclosure Statement, filed after issuance of a final rejection, be considered pursuant to 37 C.F.R. § 1.97(d). The present submission is accompanied by a certification as required by 37 C.F.R. § 1.97(e). Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material.

The required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: November 2, 2021

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

LD-45A

Electronic Patent Application Fee Transmittal					
Application Number:		16498751			
Filing Date:		27-Sep-2019			
Title of Invention:		Secondary Battery And Method For Manufacturing The Same			
First Named Inventor/Applicant Name:		Jae Woong Kim			
Filer:		Robert B. Cohen/Liz Semon			
Attorney Docket Number:		LGCHEM 3.3F-1654			
Filed as Large Entity					
Filing Fees for U.S. National Stage under 35 USC 371					
Description		Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:					
Pages:					
Claims:					
Miscellaneous-Filing:					
Petition:					
Patent-Appeals-and-Interference:					
Post-Allowance-and-Post-Issuance:					
Extension-of-Time:					

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	1806	1	260	260
Total in USD (\$)				260



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	01/24/2022		
LGCHEM			EXAMINER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP			EGGERDING, ALIX ECHELMEYER	
20 Commerce Drive				
Cranford, NJ 07016			ART UNIT	PAPER NUMBER
			1729	
			NOTIFICATION DATE	DELIVERY MODE
			01/24/2022	ELECTRONIC

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eOfficeAction@lernerdavid.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 16/498,751	Applicant(s) Kim et al.		
	Examiner ALIX E EGGERDING	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
ALIX E EGGERDING	Primary Examiner	Telephonic
Joseph Karlin	Attorney of Record	

Date of Interview: 18 January 2022

Issues Discussed:

35 U.S.C. 103

Discussed the limitations of claims 1 and 2 as rejected in the Final Rejection of 9/22/21. The examiner maintained that the claims as filed are obvious over Morishima in view of Martin for the reasons provided .

Proposed Amendment(s)

Discussed proposed claim amendments. The examiner stated that the limitations would require further search and consideration.

☒ Attachment

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Eggerding, Alix (AU1729)

From: Karlin, Joseph H. <jkarlin@lernerdavid.com>
Sent: Tuesday, January 18, 2022 1:59 AM
To: Eggerding, Alix (AU1729)
Subject: Agenda for Interview for 16/498,751 (LGCHEM 3.3F-1654)

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Dear Examiner Eggerding,

For the Examiner Interview at 1pm today for 16/498,751, I would like to discuss the pending rejections, and also proposed amendments to independent claims 1 and 2.

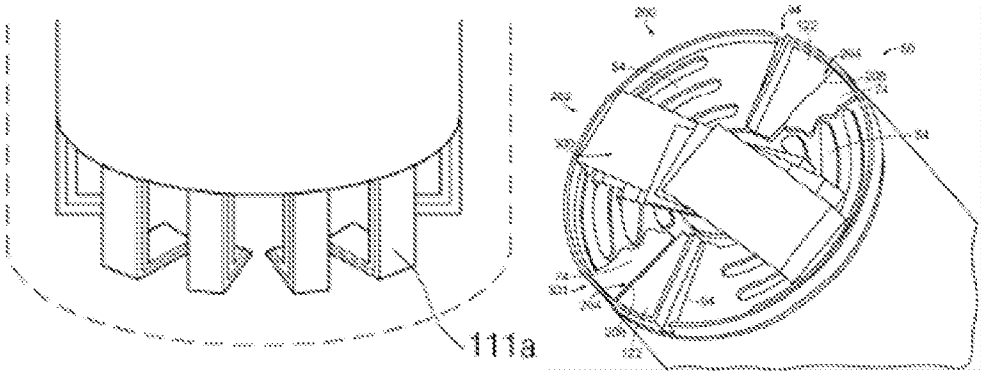
Regarding the pending rejections, Applicant does not think that the cited references actually teach the feature of claim 1 reciting that “each tab has a respective plurality of layer portions each extending over a same circumferentially extending angle about a center of the secondary battery” or the feature of claim 2 reciting that “the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.” I hope to better understand the rejections of these features so that I can explain the rejections to the Applicant.

Regarding the proposed claim amendments, we would like to discuss the following proposed amendments to independent claims 1 and 2:

1. (Currently Amended) A secondary battery comprising:
an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and
a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state,
wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery and each being in full surface contact with one another.
2. (Currently Amended) A secondary battery, comprising:
an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and
a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers each being in full surface contact with one another when the electrode is in a wound state.,
wherein the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.

The above amendments are intended to recite the difference that can be seen comparing the full surface contact of the tab portions in the claimed invention (see figure 4 below, left) compared to the less-contacting tabs in *Martin* (see figure

5 below, right). Also, the amendment to claim 1 adds “a respective plurality of layer portions each extending only over a same circumferentially extending angle,” which is also not taught in the *Martin* reference.



We hope to receive your opinion regarding whether or not the above proposed amendment are likely to overcome the pending rejections. Also we hope to receive any helpful comments, suggestions, or other amendment ideas that you may have to move the application closer to allowance.

If possible, please confirm safe receipt of this agenda.

Best regards,
Joe Karlin
Reg#: 62,471

Joseph H. Karlin
Registered Patent Attorney
jkarlin@lernerdavid.com
908-654-5000



Lerner David Littenberg Krumholz & Mentlik LLP
20 Commerce Drive, Cranford, NJ 07016
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	02/11/2022		
LGCHEM			EXAMINER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP			EGGERDING, ALIX ECHELMEYER	
20 Commerce Drive				
Cranford, NJ 07016			ART UNIT	PAPER NUMBER
			1729	
			NOTIFICATION DATE	DELIVERY MODE
			02/11/2022	ELECTRONIC

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The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 16/498,751	Applicant(s) Kim et al.		
	Examiner ALIX E EGGERDING	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
ALIX E EGGERDING	Primary Examiner	Telephonic
Joseph Karlin	Attorney of Record	

Date of Interview: 08 February 2022

Issues Discussed:

Proposed Amendment(s)

Discussed proposed amendments. The examiner stated that the amendments appear to overcome the current rejection, but would necessitate further search and consideration.

☒ Attachment

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

Eggerding, Alix (AU1729)

From: Karlin, Joseph H. <jkarlin@lernerdavid.com>
Sent: Friday, February 4, 2022 2:38 PM
To: Eggerding, Alix (AU1729)
Subject: Agenda for Second Interview for 16/498,751 (LGCHEM 3.3F-1654)

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Dear Examiner Eggerding,

For the Examiner Interview at 11am on Tuesday, February 8, 2022 for 16/498,751, I would like to discuss a proposed amendment to independent claim 1.

The proposed amendment adds a can member, an insulation member between the electrode and the can member, and having the notching tabs extend around an outer peripheral edge of the insulation member. The amended claim 1 would be as shown below:

1. (Currently Amended) A secondary battery comprising:

~~a can member;~~

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector, ~~the electrode being accommodated in the can member; and~~

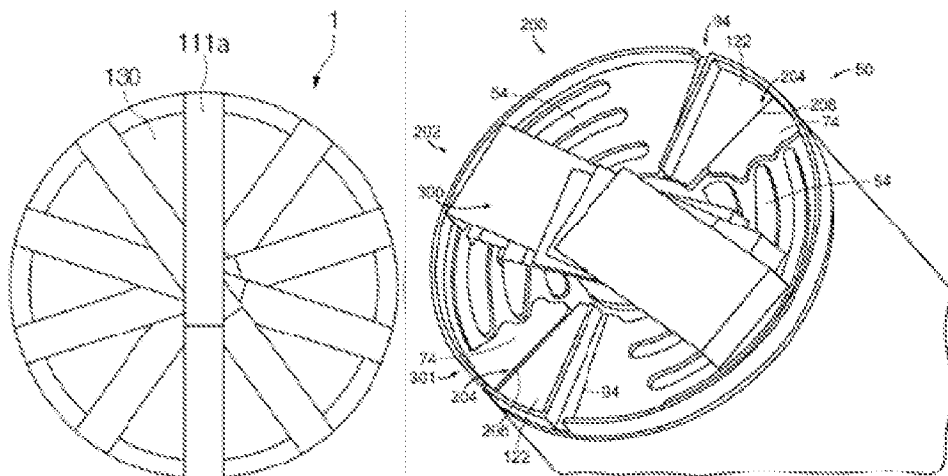
a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state; ~~and~~

~~an insulation member.~~

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each ~~notching~~ tab has a respective plurality of layer portions each extending ~~only~~ over a same circumferentially extending angle about a center of the secondary battery, ~~and each notching tab extends towards a bottom surface of the can member from an outer portion of an end of the electrode.~~

~~wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the insulation member.~~

This amendment is intended to distance claim 1 from the teachings of the *Martin* reference. To see this difference, compare figure 8 of the present invention (lower left) with figure 5 of the *Martin* reference (lower right), which teaches notching tabs extending through openings formed in a lower insulation member.



We hope to receive your opinion regarding whether or not the above proposed amendment is likely to overcome the pending rejections. Also we hope to receive any helpful comments, suggestions, or other amendment ideas that you may have to move the application closer to allowance.

If possible, please confirm safe receipt of this agenda.

Best regards,
Joe Karlin
Reg#: 62,471

Joseph H. Karlin
Registered Patent Attorney
ikarlin@lerneridavid.com
908-654-5000



Lerner David Littenberg Krumholz & Mentlik LLP
20 Commerce Drive, Cranford, NJ 07016
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Electronic Acknowledgement Receipt	
EFS ID:	45037040
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Keith McWha/Melinda Cormier
Filer Authorized By:	Keith McWha
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	18-FEB-2022
Filing Date:	27-SEP-2019
Time Stamp:	16:06:35
Application Type:	U.S. National Stage under 35 USC 371

Payment information:	
Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 2000
RAM confirmation Number	E20222HG06544720
Deposit Account	
Authorized User	
The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:	

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Arguments/Remarks Made in an Amendment	Amendment_After_Final_Action_37_CFR_Section_1.pdf	289631	no	8
			5c8195c176db5380b87dfe1c57d3559f3c117b49		
Warnings:					
Information:					
2	Miscellaneous Incoming Letter	Amendment_Transmittal.pdf	25062	no	1
			16db5d0f3891ad5de5806b9d97cb904f5b6c6e9d		
Warnings:					
Information:					
3	Request for Continued Examination (RCE)	LGCHEM1957_-_Request_for_Continued_Examination_Fillable_PDF.pdf	1349789	no	3
			07a6020c6cc03106d38b3580b0ec8bc45b1bc920		
Warnings:					
Information:					
4	Miscellaneous Incoming Letter	Certificate_of_Mailing_or_Transmission_PTO_SB-92.pdf	23399	no	1
			36d5d5a4c5d2bb6192f8aaf1c6df0cb1b8364a40		
Warnings:					
Information:					
5	Extension of Time	Two_Month_Request_for_Extension_of_Time_Under_37_CFR_1136a.pdf	23705	no	1
			5bd0d54bc7b51026c8b117068f856ba81660f433		
Warnings:					
Information:					
6	Fee Worksheet (SB06)	fee-info.pdf	40665	no	2
			05145eaea957c73ffb705d2ac54130252bd783d4		
Warnings:					
Information:					

Total Files Size (in bytes):	1752251
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I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: February 18, 2022

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

EXPEDITED PROCEDURE

Group Art Unit: 1729

Docket No. LGCHEM 3.3F-1654

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

MS AF

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

AMENDMENT UNDER 37 C.F.R. § 1.116

Dear Sir:

In response to the Office Action dated September 22, 2021, finally rejecting claims 1-2 and 4-10, please amend the above-identified U.S. patent application as follows:

IN THE CLAIMS

1. (Currently Amended) A secondary battery comprising:

a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector, the electrode being accommodated in the can member; and

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state; and

an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each notching tab extends towards a bottom surface of the can member from an outer portion of an end of the electrode,

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the insulation member.

2. (Currently Amended) ~~A~~ The secondary battery of claim 1, comprising:

~~an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and~~

~~a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.~~

wherein the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.

3. (Canceled)

4. (Previously Presented) The secondary battery of claim 1, wherein each of the plurality of notching tabs has a width of 4 mm to 5 mm.

5. (Canceled)

6. (Currently Amended) The secondary battery of claim ~~5~~ 1, wherein a bent part of each of the plurality of notching tabs is bonded to the can member.

7. (Currently Amended) The secondary battery of claim 6, ~~further comprising an~~
wherein the insulation member is disposed between the bent parts and an~~the~~ end of the electrode.

8. (Previously Presented) The secondary battery of claim 1, further comprising an electrode tab attached to the non-coating portion.

9. (Currently Amended) The secondary battery of claim 8, wherein, when the electrode is in the wound state, ~~the notching tab part forms a plurality of notching tabs, and the electrode is accommodated in a can member, and~~ each of the plurality of notching tabs is in a bent state and overlaps the electrode tab.

10. (Previously Presented) The secondary battery of claim 9, wherein the electrode tab and the plurality of notching tabs are bonded together to the can member.

11-13. (Cancelled)

14. (Withdrawn) A method for manufacturing a secondary battery, the method comprising:

a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector;

a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction;

a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly;

a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and

an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

15. (Withdrawn) The method of claim 14, wherein the electrode is a negative electrode.

16. (Withdrawn) The method of claim 14, wherein, during the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied are arranged parallel to each other in the width direction of the electrode collector.

17. (Withdrawn) The method of claim 16, wherein, during the notching tab part formation step, the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

18. (Withdrawn) The method of claim 14, wherein, during the notching tab part formation step, notching is performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

19. (Withdrawn) The method of claim 14, wherein, during the notching tab formation step, the notching tab part is cut using a laser.

20. (Withdrawn) The method of claim 14, wherein, during the installation step, the electrode assembly is accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

21. (Withdrawn) The method of claim 20, wherein, during the installation step a bent portion of each of the plurality of notching tabs is welded to the can member.

22. (Withdrawn) The method of claim 20, wherein, during the installation step bent portions of each of the plurality of notching tabs overlap each other and are bonded to the can member together.

23. (Withdrawn) The method of claim 14, wherein, during the winding step, the notching tab part is wound to form two or more overlapping layers.

REMARKS

The present amendment is responsive to the final office action dated September 22, 2021. This response is enterable after final rejection because it is accompanied by a Request for Continued Examination, filed herewith. After entry of the foregoing amendments, claims 1-2, 4, 6-10, and 14-23 will be pending. Claims 1, 2, 6, 7, and 9 are amended. Claims 14-23 are withdrawn as being directed to a non-elected species or invention. Claims 3, 5, and 11-13 are canceled. No new matter is added by the present amendments. Applicant respectfully requests that the Examiner reconsider the claims in view of the amendments and remarks herein.

Interview Summary

Applicants' representative thanks Examiner Eggerding for her time during the telephonic interviews conducted on January 18, 2022 and February 8, 2022 with the undersigned. Proposed amendments to claim 1 were discussed, and it was agreed that the amended claim 1 presented herein appears to overcome the pending rejections. No agreement was reached regarding patentability.

Prior Art Rejections

Claims 1, 2, and 4-7 were rejected under 35 U.S.C. § 103 as allegedly unpatentable over U.S. Patent Application Pub. No. 2010/0081052 to (Morishima et al. ("*Morishima*") in view of U.S. Patent Application Pub. No. 2013/0108903 to Martin et al. ("*Martin*"). Claims 8-10 were rejected under 35 U.S.C. § 103 as allegedly unpatentable over *Morishima* in view of *Martin*, and further in view of U.S. Patent Application Pub. No. 2018/0145304 to Kogetsu et al. ("*Kogetsu*"). Applicant submits that the rejections in the Office Action are traversed for the reasons set forth below.

Applicant submits that each and every limitation of claim 1, as amended, is not taught or suggested by any of the references of record, alone or in combination. Claim 1, as amended, recites:

A secondary battery comprising:

a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector, the electrode being accommodated in the can member;

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state; and

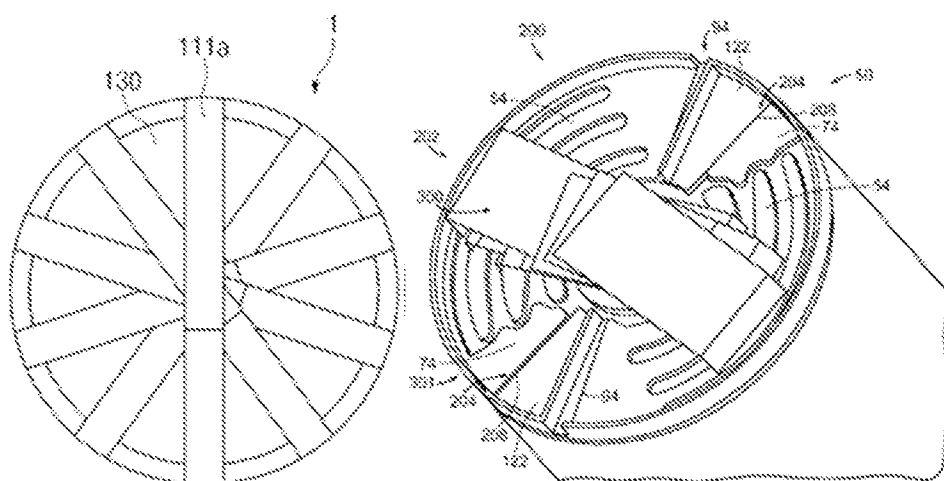
an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each notching tab extends towards a bottom surface of the can member from an outer portion of an end of the electrode,

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the insulation member. (emphasis added)

The amendments to claim 1 can find support throughout the specification, and at least in original claims 5, 7, and 8, and figures 3, 4, and 8.

The proposed amendment adds a can member, an insulation member between the electrode and the can member, and having the notching tabs extend around an outer peripheral edge of the insulation member. As amended, the configuration of the insulation member and the notching tabs of claim 1 is different from the teachings of the *Martin* reference. To see this difference, compare figure 8 of the present invention, in which the notching tabs extend around an outer peripheral edge of the insulation member (lower left), with figure 5 of the *Martin* reference (lower right), which teaches notching tabs extending through openings formed in a lower insulation member.



It is respectfully submitted that none of the references of record are alleged to teach or suggest the secondary battery of claim 1, alone or in combination. Furthermore, it was agreed during the Examiner Interview that the amendment to claim 1 presented herein appears to overcome the rejections of record.

Accordingly, Applicant respectfully submits that each and every limitation of amended claim 1 is not taught or suggested by any of the references of record, alone or in combination. Therefore, Applicant submits that the pending rejection of independent claim 1 under 35 U.S.C. § 103 should be withdrawn. Also, for at least the same reasons, the rejections of dependent claims 2, 4, and 6-10 should be withdrawn, each of which ultimately depends from independent claim 1.

Conclusion

In view of the above, each of the presently pending claims in this application is believed to be in immediate condition for allowance. Accordingly, the Examiner is respectfully requested to withdraw the outstanding rejection of the claims and to pass this application to issue.

If, however, for any reason the Examiner does not believe that such action can be taken at this time, it is respectfully requested that the Examiner telephone Applicant's attorney at (908) 654-5000 in order to overcome any additional objections which the Examiner might have.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: February 18, 2022

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG,
KRUMHOLZ & MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

AMENDMENT TRANSMITTAL LETTER			After Final Rejection	
Application No. 16/498,751	Filing Date September 27, 2019	Examiner A. E. Eggerding	Group Art Unit 1729	
Applicant(s): Jae Woong Kim, Hyoung Kwon Kim, and Sang Uck Kim			Docket No. LGCHEM 3.3F-1654	

Invention: Secondary Battery And Method For Manufacturing The Same

TO THE COMMISSIONER FOR PATENTS

Transmitted herewith is an amendment in the above-identified application.
The fee has been calculated and is transmitted as shown below.

CLAIMS AS AMENDED					
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate	
Total Claims	18	- 20 =	0	x	
Independent Claims	2	- 3 =	0	x	
Multiple Dependent Claims (check if applicable) ☐					
Other fee (please specify):					
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00

- ☒ Large Entity
- ☐ Small Entity
- ☒ No additional fee is required for this amendment.
- ☐ Please charge Deposit Account _____ in the amount _____ .
A duplicate copy of this sheet is enclosed.
- ☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☒ The Commissioner is hereby authorized to charge and credit Deposit 12-1095 as described below.

☒ Credit any overpayment.

☒ Charge any additional filing or application processing fees required under 37 CFR 1.16 and 1.17.

/Joseph H. Karlin/
Joseph H. Karlin
Attorney Reg. 62,471

Dated: February 18, 2022

LERNER, DAVID, LITTENBERG, KRUMHOLZ & MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).
Dated: February 18, 2022
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)							
Application Number	16498751	Filing Date	2019-09-27	Docket Number (if applicable)	LGCHEM 3.3F-1654	Art Unit	1729
First Named Inventor	Jae Woong Kim			Examiner Name	A. E. Eggerding		
This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application. Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV							
SUBMISSION REQUIRED UNDER 37 CFR 1.114							
Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).							
☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked. ☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on ☐ Other ☒ Enclosed ☒ Amendment/Reply ☐ Information Disclosure Statement (IDS) ☐ Affidavit(s)/ Declaration(s) ☐ Other							
MISCELLANEOUS							
☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required) ☐ Other							
FEES							
☒ The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed. The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 12-1095							
SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED							
☒ Patent Practitioner Signature Applicant Signature							

Signature of Registered U.S. Patent Practitioner			
Signature	Joseph H. Karlin/	Date (YYYY-MM-DD)	2022-02-18
Name	Joseph H. Karlin	Registration Number	62471

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Certificate of Mailing or Transmission under 37 CFR 1.8

I hereby certify that this correspondence is being:

1. ☐ Deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to:
MS RCE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450
on _____
Date

2. ☐ Facsimile transmitted to the United States Patent and Trademark Office, or
OR

3. ☒ EFS-Web transmitted to the USPTO.

/Joseph H. Karlin/
Signature

February 18, 2022
Date

Joseph H. Karlin
Typed or printed name

(908) 654-5000
Telephone number

Note: Each paper must have its own certificate of mailing or transmission, or this certificate must identify each submitted paper:

Request for Continued Examination (Fillable PDF)

PTO/AIA/22 (10-20)

Approved for use through 11/30/2020. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PETITION FOR EXTENSION OF TIME UNDER 37 CFR 1.136(a)		Docket Number (Optional) LGCHEM 3.3F-1654																															
Application Number 16/498,751-Conf. #4723		Filed September 27, 2019																															
For Secondary Battery And Method For Manufacturing The Same																																	
Art Unit 1729		Examiner A. E. Eggerding																															
This is a request under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above-identified application. The requested extension and fee are as follows (check time period desired and enter the appropriate fee below): <table><thead><tr><th></th><th>Fee</th><th>Small Entity Fee</th><th>Micro Entity Fee</th><th></th></tr></thead><tbody><tr><td>☐ One month (37 CFR 1.17(a)(1))</td><td>\$220</td><td>\$110</td><td>\$55</td><td>\$ _____</td></tr><tr><td>☒ Two months (37 CFR 1.17(a)(2))</td><td>\$640</td><td>\$320</td><td>\$160</td><td>\$ 640.00</td></tr><tr><td>☐ Three months (37 CFR 1.17(a)(3))</td><td>\$1,480</td><td>\$740</td><td>\$370</td><td>\$ _____</td></tr><tr><td>☐ Four months (37 CFR 1.17(a)(4))</td><td>\$2,320</td><td>\$1,160</td><td>\$580</td><td>\$ _____</td></tr><tr><td>☐ Five months (37 CFR 1.17(a)(5))</td><td>\$3,160</td><td>\$1,580</td><td>\$790</td><td>\$ _____</td></tr></tbody></table> ☐ Applicant asserts small entity status. See 37 CFR 1.27. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously. ☐ A check in the amount of the fee is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The Director has already been authorized to charge fees in this application to a Deposit Account. ☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment, to Deposit Account Number 12-1095 . ☒ Payment made via EFS-Web. WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. I am the ☐ applicant. ☒ attorney or agent of record. Registration number 62,471 . ☐ attorney or agent acting under 37 CFR 1.34. Registration number _____ . /Joseph H. Karlin/ Signature February 18, 2022 Date Joseph H. Karlin Typed or printed name (908) 654-5000 Telephone Number NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.					Fee	Small Entity Fee	Micro Entity Fee		☐ One month (37 CFR 1.17(a)(1))	\$220	\$110	\$55	\$ _____	☒ Two months (37 CFR 1.17(a)(2))	\$640	\$320	\$160	\$ 640.00	☐ Three months (37 CFR 1.17(a)(3))	\$1,480	\$740	\$370	\$ _____	☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$1,160	\$580	\$ _____	☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,580	\$790	\$ _____
	Fee	Small Entity Fee	Micro Entity Fee																														
☐ One month (37 CFR 1.17(a)(1))	\$220	\$110	\$55	\$ _____																													
☒ Two months (37 CFR 1.17(a)(2))	\$640	\$320	\$160	\$ 640.00																													
☐ Three months (37 CFR 1.17(a)(3))	\$1,480	\$740	\$370	\$ _____																													
☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$1,160	\$580	\$ _____																													
☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,580	\$790	\$ _____																													

☐ * Total of 1 forms are submitted.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: February 18, 2022

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Electronic Patent Application Fee Transmittal				
Application Number:		16498751		
Filing Date:		27-Sep-2019		
Title of Invention:		Secondary Battery And Method For Manufacturing The Same		
First Named Inventor/Applicant Name:		Jae Woong Kim		
Filer:		Keith McWha/Melinda Cormier		
Attorney Docket Number:		LGCHEM 3.3F-1654		
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 2 months with \$0 paid	1252	1	640	640
Miscellaneous:				
RCE- 1ST REQUEST	1801	1	1360	1360
Total in USD (\$)				2000



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	03/08/2022	EXAMINER	
LGCHEM			EGGERDING, ALIX ECHELMEYER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP				
20 Commerce Drive				
Cranford, NJ 07016			ART UNIT	
			PAPER NUMBER	
			1729	
			NOTIFICATION DATE	
			DELIVERY MODE	
			03/08/2022	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Continued Examination Under 37 CFR 1.114

2. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 2/18/22 has been entered.

3. Claims 1, 2, 7, and 9 are amended. Claims 3, 5, and 11-13 are canceled. Claims 14-23 were previously withdrawn. The rejection of claim 1 over Morishima in view of Martin is withdrawn in light of the amendment; however, a new rejection is made over Sohn in view of Barone and Wolff. Claims 1, 2, 4, and 6-10 are rejected for the reasons provided below.

Claim Rejections - 35 USC § 103

4. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention

and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

1. Claims 1, 2, 4, and 6-10 are rejected under 35 U.S.C. 103 as being unpatentable over Sohn (US 2011/0067227) in view of Barone et al. (US 2013/0252054) and Wolff (US 4,822,377).

Regarding claim 1, Sohn teaches a secondary, or rechargeable, battery (abstract) comprising:

a can member ([0033]);

an electrode, or electrode plate (310), with a coating portion, or active material portion (311), and a non-coating portion, or un-coated portion (312), on which active material is not applied to the electrode collector in a longitudinal direction of the electrode collector (Figures 1 and 6A, [0088]), the electrode being accommodated in the can member ([0085]);

a notching tab part, or first electrode tab group (330), extending from the coating portion (311) in a width direction perpendicular to the longitudinal direction, i.e. in the winding direction, and not coated with active material, the notching tab part having a wound state including two or more overlapping layers, or a plurality of first electrode tabs (331) where the side surfaces are parallel to each other (Figures 5 and 6D, [0083], [0101]).

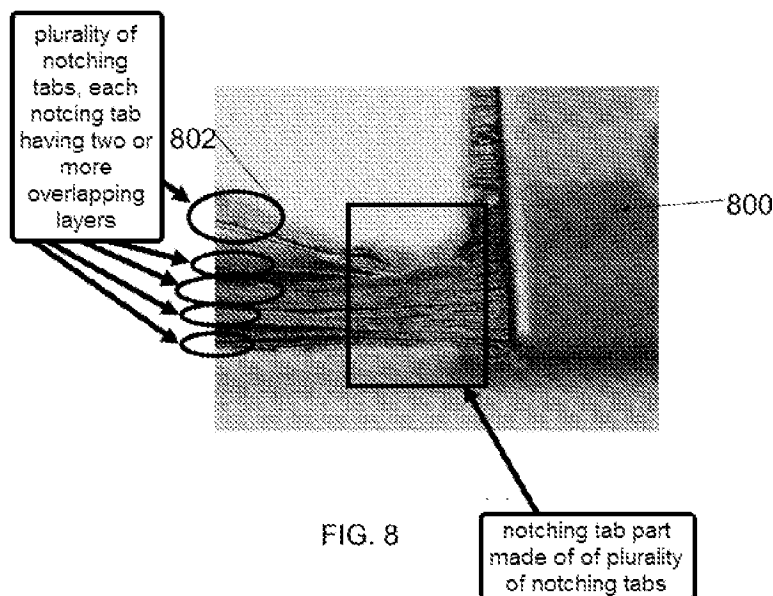
It is seen in Figure 5 of Sohn that the notching tab extends from an outer portion of the end of the electrode; as to the orientation of the can member (top/bottom), that is a relative term, but it is noted that either the positive notching tab group (330) or the negative notching tab group (340) necessarily extends towards a bottom surface of the can while the other extends towards a top surface.

Further regarding claim 1, Sohn teaches the notching tab group (330) includes a plurality of notching tabs (331), but fails to teach specifically the number of overlapping layers.

Barone teaches a secondary battery including an electrode having a notching tab part, or plurality of tabs (602, 604) (Figure 6, [0038]).

Barone further teaches that the number of notching tabs, analogous to the electrode tabs (331) of Sohn, in the notching tab part is preferably one tab per winding in order to reduce impedance and increase heat dissipation ([0030]).

It would have been obvious to the skilled artisan to provide multiple notching tab parts, made of two or more overlapping portions, in the battery of Sohn such as suggested by Barone in order to impedance and increase heat dissipation. The examiner finds that this structure will result in the claimed limitation to the notching tab part:



Further, the skilled artisan will easily understand that each notching tab of Sohn or Barone extends only over a same circumferentially extending angle about a center of the secondary battery.

With further regard to claim 1, Sohn is silent on an insulation member.

Wolff teaches a secondary battery comprising a can member, or cell can (72), accommodating an electrode group (80) having an uncoated portion forming an electrode tab, or positive electrode contact tab (86) (Figure 5).

Wolff further teaches providing an insulation member, or electrical insulator pad (94), which is disposed between the end of the electrode, or the end of the jelly roll electrode unit (80), and the can member, or cover assembly (10) of the can, and wherein the electrode tab (86) is provided on an outer peripheral edge of the insulation member (Figures 5 and 10, column 11 lines 18-20; column 12 lines 29-39).

Wolff teaches that the insulation member (94) is desirable for substantially insulating the can from the electrode roll (column 11 lines 27-31).

It would have been obvious to the skilled artisan at the time of the invention to include a known insulation member in the can member of Sohn, wherein the insulation member is provided between the end of the electrode and the can, and wherein the electrode tabs of Sohn are provided on an outer peripheral edge of the insulation member such as suggested by Wolff in order to substantially insulate the can member from the electrode.

With regard to claim 2, it is seen in Figure 2 of Sohn that the notching tab part, or un-coated portion (312) extends across the entire length of the coating portion in the longitudinal direction.

As for claim 4, Barone teaches that the notching tabs, or electrode tabs, have a width of about 3mm to 8 mm ([0011]). The examiner takes note of the fact that the prior art range completely encompasses the claimed range. Absent any additional and more specific information in the prior art, and absent a showing of criticality of the claimed range relative to the prior art, a prima facie case of obviousness exists. MPEP 2144.05

Regarding claim 6, Barone (Figure 9) and Wolff (Figures 9-11) teaches that a bent part of the notching tabs is bonded to the can member, or cover (910) of Barone/cover assembly (10) of Wolff.

With regard to claim 7, Wolff teaches that the insulator (96) is disposed between the cover assembly, which includes the bent parts, and the insulation member (Figure 10-11, see also column 11 lines 35-39, wherein a hole may be provided in the middle of the insulation member to facilitate welding the electrode tab to the can).

As for claims 8-10, Sohn teaches that the notching tabs as discussed above are electrode tabs (331), which are attached to the non-coating portion by being integral to the non-coating portion ([0082]). There does not appear to be a structural difference between an electrode tab and a notching tab in the currently filed claims. Therefore, a single electrode tab (331) of Sohn in view of Barone and Wolff is considered an electrode tab, while the rest are called notching tabs, and the claimed structure is obvious. The bonding and bent state limitations are addressed above with regard to claim 6.

Response to Arguments

6. Applicant's arguments with respect to claims 1, 2, and 4-10 have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Conclusion

7. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMEYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached 8:00am - 5:00pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

<i>Notice of References Cited</i>	Application/Control No. 16/498,751		Applicant(s)/Patent Under Reexamination Kim et al.	
	Examiner ALIX E EGGERDING		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20130252054-A1	09-2013	BARONE; Michael	H01M50/538	429/94
*	B	US-20110067227-A1	03-2011	Sohn; Young-Bae	H01M4/70	29/623.1
*	C	US-4822377-A	04-1989	Wolff; Merle	H01M50/183	29/422
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
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FOREIGN PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
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NON-PATENT DOCUMENTS		
*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
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	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	1729	
Examiner Name	A. E. Eggerding	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2021-11-02
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Receipt date: 11/02/2021

16/498,751 -- GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (02-18)
Approved for use through 11/30/2020. OMB 0651-0031
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE
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INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
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Application Number	16498751
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First Named Inventor	Jae Woong Kim
Art Unit	1729
Examiner Name	A. E. Eggerding
Attorney Docket Number	LGCHEM 3.3F-1654

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	1	102024936	CN	A	2011-04-20	Samsung Sdi Co., Ltd	U.S. Publication No. 20110067227 is the English language counterpart which was previously cited.	
	2	201450072	CN	U	2010-05-05	Dongguan Ampere Electronics Technology Co., Ltd	U.S. Publication No. 20100310926 is the English language counterpart which was previously cited.	
	3	203250828	CN	U	2013-10-23	A123 Systems, Inc	U.S. Publication No. 20130108903 is the English lanuage counterpart which was previously cited.	

EFS Web 2.1.18

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /A.E.E/

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	1729	
Examiner Name	A. E. Eggerding	
Attorney Docket Number	LGCHEM 3.3F-1654	

	4	204577523	CN	U	2015-08-19	Contemporary Amperex Technology Ltd	Machine generated translation attached.	×
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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Chiinese Search Report for Application No. 201880025247, dated August 24, 2021, 2 pages.	

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EXAMINER SIGNATURE

Examiner Signature	/ALIX E EGGERDING/	Date Considered	03/03/2022
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Electronic Acknowledgement Receipt	
EFS ID:	46136526
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Kevin M. Kocun/Sharon McGee
Filer Authorized By:	Kevin M. Kocun
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	07-JUL-2022
Filing Date:	27-SEP-2019
Time Stamp:	14:57:12
Application Type:	U.S. National Stage under 35 USC 371

Payment information:	
Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 220
RAM confirmation Number	E202277E57370783
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The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:	

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Arguments/Remarks Made in an Amendment	Amendment.pdf	64087	no	8
			d1205a3bcc81d1596e9c3a6363b53e7bc6d188b4		
Warnings:					
Information:					
2	Miscellaneous Incoming Letter	Amendment_Transmittal.pdf	22113	no	1
			8b78dd94050898ab78478864d8ad3faadf0275a4		
Warnings:					
Information:					
3	Extension of Time	One_Month_Request_for_Extension_of_Time_Under_37_CFR_1136a.pdf	23665	no	1
			45e23bcd52a0cf10f90a1e151c53b65cdb8bcb2c		
Warnings:					
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4	Fee Worksheet (SB06)	fee-info.pdf	38300	no	2
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: July 7, 2022

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Dear Sir:

In response to the official action mailed March 8, 2022, applicant submits the following amendments and remarks.

IN THE CLAIMS

1. (Currently Amended) A secondary battery comprising:
a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector, the non-coating portion being adjacent to the coating portion in a longitudinal direction of the electrode collector, the non-coating portion being disposed at a first end of the electrode collector, the electrode being accommodated in the can member;

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part extending from part of the coating portion at a second end of the electrode collector opposite in the longitudinal direction from the first end of the electrode collector, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state; ~~and~~

an electrode tab attached to the non-coating portion, such that when the electrode is in the wound state, the electrode tab extends from a central portion of an end of the electrode; and

an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each notching tab extends towards a bottom surface of the can member only from an outer portion of ~~an~~ the end of the electrode remote from the central portion,

wherein, when the electrode is in the wound state, the plurality of notching tabs are together equally circumferentially distributed about a central axis of the electrode,

wherein, when the electrode is in the wound state, each of the plurality of notching tabs is in a bent state and overlaps the electrode tab, and the electrode tab and the plurality of notching tabs are bonded together to the can member, and

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the

insulation member, and the electrode tab extending through a central opening of the insulation member.

2. (Previously Presented) The secondary battery of claim 1, wherein the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.

3. (Canceled)

4. (Previously Presented) The secondary battery of claim 1, wherein each of the plurality of notching tabs has a width of 4 mm to 5 mm.

5. (Canceled)

6. (Previously Presented) The secondary battery of claim 1, wherein a bent part of each of the plurality of notching tabs is bonded to the can member.

7. (Previously Presented) The secondary battery of claim 6, wherein the insulation member is disposed between the bent parts and the end of the electrode.

8-13. (Cancelled)

14. (Withdrawn) A method for manufacturing a secondary battery, the method comprising:
a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector;

a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction;

a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly;

a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and

an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

15. (Withdrawn) The method of claim 14, wherein the electrode is a negative electrode.

16. (Withdrawn) The method of claim 14, wherein, during the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied are arranged parallel to each other in the width direction of the electrode collector.

17. (Withdrawn) The method of claim 16, wherein, during the notching tab part formation step, the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

18. (Withdrawn) The method of claim 14, wherein, during the notching tab part formation step, notching is performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

19. (Withdrawn) The method of claim 14, wherein, during the notching tab formation step, the notching tab part is cut using a laser.

20. (Withdrawn) The method of claim 14, wherein, during the installation step, the electrode assembly is accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

21. (Withdrawn) The method of claim 20, wherein, during the installation step a bent portion of each of the plurality of notching tabs is welded to the can member.

22. (Withdrawn) The method of claim 20, wherein, during the installation step bent portions of each of the plurality of notching tabs overlap each other and are bonded to the can member together.

23. (Withdrawn) The method of claim 14, wherein, during the winding step, the notching tab part is wound to form two or more overlapping layers.

REMARKS

The present amendments and remarks are being filed in response to the non-final office action dated March 8, 2022. After entry of the foregoing amendments, claims 1, 2, 4, 6, 7, and 14-23 will be pending. Claim 1 is amended. Claims 14-23 have been withdrawn as directed to non-elected species or invention. Claims 3, 5, and 8-13 are canceled. No new matter is added by the present amendments. Applicant respectfully requests that the Examiner reconsider the claims in view of the amendments and remarks herein.

Prior Art Rejections

Claims 1, 2, 4, and 6-10 were rejected under 35 U.S.C. § 103 as allegedly unpatentable over U.S. Patent Application Pub. No. 2011/0067227 to Sohn (“*Sohn*”) in view of U.S. Patent Application Pub. No. 2013/0252054 to Barone et al. (“*Barone*”) and U.S. Patent No. 4,822,377 to Wolff (“*Wolff*”). Applicant submits that the rejections in the Office Action are traversed for the reasons set forth below.

Applicant submits that each and every limitation of claim 1, as amended, is not taught or suggested by any of the references of record, alone or in combination. Claim 1, as amended, recites:

A secondary battery comprising:

a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector, the non-coating portion being adjacent to the coating portion in a longitudinal direction of the electrode collector, the non-coating portion being disposed at a first end of the electrode collector, the electrode being accommodated in the can member;

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part extending from part of the coating portion at a second end of the electrode collector opposite in the longitudinal direction from the first end of the electrode collector, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state;

an electrode tab attached to the non-coating portion, such that when the electrode is in the wound state, the electrode tab extends from a central portion of an end of the electrode; and

an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each notching tab extends towards a bottom surface

of the can member only from an outer portion of the end of the electrode remote from the central portion,

wherein, when the electrode is in the wound state, the plurality of notching tabs are together equally circumferentially distributed about a central axis of the electrode,

wherein, when the electrode is in the wound state, each of the plurality of notching tabs is in a bent state and overlaps the electrode tab, and the electrode tab and the plurality of notching tabs are bonded together to the can member, and

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the insulation member, and the electrode tab extending through a central opening of the insulation member. (emphasis added)

The amendments to claim 1 can find support throughout the specification, and at least in as-filed paragraphs [0019]-[0021], [0072], [0078], [0100], and [0105], original claims 8-10, and figures 1, 8, and 9.

The amendment is directed to the embodiment of figures 8 and 9, in which the notching tabs are bent, overlapped with the electrode tab, and bonded together to the can member. The amendment also recites that the electrode tab extends from the non-coating portion at a first end of the electrode collector, and that the notching tab part extends from part of the coating portion at a second opposite end of the electrode collector. The amendment further recites that the notching tabs are together equally circumferentially distributed about a central axis of the electrode, and the electrode tab extends from a central portion of an end of the electrode.

Applicant respectfully submits that these amended features are together not an obvious variation of the *Barone* reference, which is alleged to teach a multi-layer notching tab part. Since none of the cited references teaches notching tabs equally distributed about a central axis of the electrode and bonded together with an electrode tab to the can member, Applicant respectfully submits the cited references do not the features of claim 1, alone or in combination.

Accordingly, Applicant respectfully submits that each and every limitation of amended claim 1 is not taught or suggested by any of the references of record, alone or in combination. Therefore, Applicant submits that the pending rejection of independent claim 1 under 35 U.S.C. § 103 should be withdrawn. Also, for at least the same reasons, the rejections of dependent claims 2, 4, 6, and 7 should be withdrawn, each of which ultimately depends from independent claim 1.

Application No.: 16/498,751

Docket No.: LGCHEM 3.3F-1654

Conclusion

As it is believed that all of the rejections set forth in the Official Action have been fully met, favorable reconsideration and allowance are earnestly solicited. If, however, for any reason the Examiner does not believe that such action can be taken at this time, it is respectfully requested that the Examiner telephone applicant's attorney at (908) 654-5000 in order to overcome any additional objections which the Examiner might have.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: July 7, 2022

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG,
KRUMHOLZ & MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

AMENDMENT TRANSMITTAL LETTER			Docket No. LGCHEM 3.3F-1654
Application No. 16/498,751-Conf. #4723	Filing Date September 27, 2019	Examiner A. E. Eggerding	Art Unit 1729

Applicant(s): LG Chem, Ltd.

Invention: Secondary Battery And Method For Manufacturing The Same

TO THE COMMISSIONER FOR PATENTS

Transmitted herewith is an amendment in the above-identified application.
The fee has been calculated and is transmitted as shown below.

CLAIMS AS AMENDED					
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate	
Total Claims	15	- 20 =	0	x	
Independent Claims	2	- 3 =	0	x	
Multiple Dependent Claims (check if applicable) ☐					
Other fee (please specify):					
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00

☒ Large Entity

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☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.

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/Joseph H. Karlin/

Joseph H. Karlin (Atty/Agent Reg. No.: 62,471)
LERNER, DAVID, LITTENBERG, KRUMHOLZ & MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000

Dated: July 7, 2022

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).	
Dated: <u>July 7, 2022</u>	Electronic Signature for Joseph H. Karlin: <u>/Joseph H. Karlin/</u>

PTO/AIA/22 (10-20)

Approved for use through 11/30/2020. OMB 0651-0031

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PETITION FOR EXTENSION OF TIME UNDER 37 CFR 1.136(a)		Docket Number (Optional)																															
		LGCHEM 3.3F-1654																															
Application Number		16/498,751-Conf. #4723	Filed																														
For		September 27, 2019																															
Secondary Battery And Method For Manufacturing The Same																																	
Art Unit		1729	Examiner																														
		A. E. Eggerding																															
This is a request under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above-identified application. The requested extension and fee are as follows (check time period desired and enter the appropriate fee below): <table><thead><tr><th></th><th>Fee</th><th>Small Entity Fee</th><th>Micro Entity Fee</th><th></th></tr></thead><tbody><tr><td>☒ One month (37 CFR 1.17(a)(1))</td><td>\$220</td><td>\$110</td><td>\$55</td><td>\$ 220.00</td></tr><tr><td>☐ Two months (37 CFR 1.17(a)(2))</td><td>\$640</td><td>\$320</td><td>\$160</td><td>\$</td></tr><tr><td>☐ Three months (37 CFR 1.17(a)(3))</td><td>\$1,480</td><td>\$740</td><td>\$370</td><td>\$</td></tr><tr><td>☐ Four months (37 CFR 1.17(a)(4))</td><td>\$2,320</td><td>\$1,160</td><td>\$580</td><td>\$</td></tr><tr><td>☐ Five months (37 CFR 1.17(a)(5))</td><td>\$3,160</td><td>\$1,580</td><td>\$790</td><td>\$</td></tr></tbody></table> ☐ Applicant asserts small entity status. See 37 CFR 1.27. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously. ☐ A check in the amount of the fee is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The Director has already been authorized to charge fees in this application to a Deposit Account. ☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment, to Deposit Account Number 12-1095 . ☒ Payment made via EFS-Web.					Fee	Small Entity Fee	Micro Entity Fee		☒ One month (37 CFR 1.17(a)(1))	\$220	\$110	\$55	\$ 220.00	☐ Two months (37 CFR 1.17(a)(2))	\$640	\$320	\$160	\$	☐ Three months (37 CFR 1.17(a)(3))	\$1,480	\$740	\$370	\$	☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$1,160	\$580	\$	☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,580	\$790	\$
	Fee	Small Entity Fee	Micro Entity Fee																														
☒ One month (37 CFR 1.17(a)(1))	\$220	\$110	\$55	\$ 220.00																													
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☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,580	\$790	\$																													
WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. I am the ☐ applicant. ☒ attorney or agent of record. Registration number 62,471 . ☐ attorney or agent acting under 37 CFR 1.34. Registration number . /Joseph H. Karlin/ Signature July 7, 2022 Date Joseph H. Karlin Typed or printed name (908) 654-5000 Telephone Number																																	
NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.																																	

☐ * Total of 1 forms are submitted.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: July 7, 2022

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Electronic Patent Application Fee Transmittal				
Application Number:		16498751		
Filing Date:		27-Sep-2019		
Title of Invention:		Secondary Battery And Method For Manufacturing The Same		
First Named Inventor/Applicant Name:		Jae Woong Kim		
Filer:		Kevin M. Kocun/Sharon McGee		
Attorney Docket Number:		LGCHEM 3.3F-1654		
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	220	220
Miscellaneous:				
Total in USD (\$)				220



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	09/13/2022	EXAMINER	
LGCHEM			EGGERDING, ALIX ECHELMEYER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP			ART UNIT	PAPER NUMBER
20 Commerce Drive			1729	
Cranford, NJ 07016			NOTIFICATION DATE	DELIVERY MODE
			09/13/2022	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):
eOfficeAction@lernerdavid.com

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. This Office Action is in response to the amendment filed 7/7/22. Claims 1, 2, 4, 6, 7, and 14-23 are pending. Claims 14-23 are withdrawn. The rejection of claim 1 over Sohn in view of Barone and Wolff is withdrawn; however, a new rejection is made over Huang and Martin. Claims 1, 2, 4, 6, and 7 are rejected finally for the reasons provided below.

Claim Rejections - 35 USC § 103

3. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 1, 2, 4, 6, and 7 are rejected under 35 U.S.C. 103 as being unpatentable over Sohn (US 2011/0067227) in view of Chun (US 2012/0009450), Martin et al. (US 2013/0108903), and Barone et al. (US 2013/0252054).

Regarding claim 1, Sohn teaches a secondary, or rechargeable, battery (abstract) comprising:

a can member ([0033]);

an electrode, or electrode plate (310), with a coating portion, or active material portion (311), and a non-coating portion, or un-coated portion (312), on which active material is not applied to the electrode collector in a longitudinal direction of the electrode collector (Figures 1 and 6A, [0088]), the electrode being accommodated in the can member ([0085]);

a notching tab part, or first electrode tab group (330), extending from the coating portion (311) in a width direction perpendicular to the longitudinal direction, i.e. in the winding direction, and not coated with active material, the notching tab part having a wound state including two or more overlapping layers, or a plurality of first electrode tabs (331) where the side surfaces are parallel to each other (Figures 5 and 6D, [0083], [0101]).

It is seen in Figure 5 of Sohn that the notching tab extends from an outer portion of the end of the electrode; as to the orientation of the can member (top/bottom), that is a relative term, but it is noted that either the positive notching tab group (330) or the negative notching tab group (340) necessarily extends towards a bottom surface of the can while the other extends towards a top surface.

With further regard to claim 1, Sohn teaches the electrode of claim 1 but fails to teach a non-coating portion being adjacent to the coating portion in a longitudinal direction, disposed at a first end of the electrode collector.

Chun teaches an electrode assembly including an electrode (112) including a non-coating portion, or uncoated region (112a), disposed at a first end of the electrode and having an electrode tab (135a) attached to it, wherein the electrode tab (135a) extends from a central portion of an end of the electrode when the electrode is in a wound state (Figures 3 and 4, [0052])

Chun further teaches that providing an electrode tab (112a) in the center of the wound assembly with an additional electrode tab (135b) at the outer circumference distributes the stresses on the electrode tabs while also maximizing output current to maintain high power ([0054]).

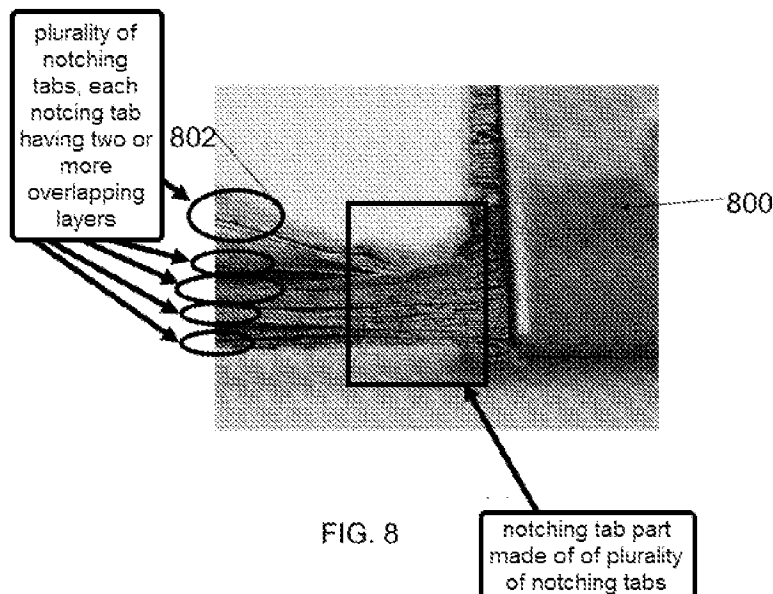
Therefore, it would have been obvious to the skilled artisan to provide a non-coating portion with electrode tab at the center of the wound electrode of Sohn such as suggested by Chun in order to distribute the stresses on the electrode tabs while also maximizing output current to maintain high power.

Further regarding claim 1, Sohn teaches the notching tab group (330) includes a plurality of notching tabs (331), but fails to teach specifically the number of overlapping layers.

Barone teaches a secondary battery including an electrode having a notching tab part, or plurality of tabs (602, 604) (Figure 6, [0038]). The plurality of tabs of Barone are together equally circumferentially distributed about a central axis of the electrode (Figures 6 and 8).

Barone further teaches that the number of notching tabs, analogous to the electrode tabs (331) of Sohn, in the notching tab part is preferably one tab per winding in order to reduce impedance and increase heat dissipation ([0030]).

It would have been obvious to the skilled artisan to provide multiple notching tab parts, made of two or more overlapping portions, in the battery of Sohn such as suggested by Barone in order to impedance and increase heat dissipation. The examiner finds that this structure will result in the claimed limitation to the notching tab part:



Further, the skilled artisan will easily understand that each notching tab of Sohn or Barone extends only over a same circumferentially extending angle about a center of the secondary battery.

With further regard to claim 1, Sohn in view of Chun and Barone is silent on an insulation member.

Martin teaches a secondary battery comprising a can member, or cylindrical tube (23), including an electrode having a center post (1030), analogous to the central tab of Sohn in view of Chun, a plurality of electrode tabs (300/716/1016), analogous to the notching tabs of Sohn in view of Barone, and an insulation member, or insulator portion (24a/24b/34a/34b/50), having a central opening (51) for accommodating a terminal, or central electrode tab/post (Figure 4, [0046]). Martin teaches that the insulator (34a/34b) is provided between the can member, or end cap (28) of the can, and the end of the electrode (33a/33b) (Figure 2A).

Martin teaches that the use of the insulator is desirable to prevent current collecting tabs from contacting a electrode of opposite polarity ([0040]).

Martin further teaches that the electrode tabs (300/716/1016) are in a bent state and overlap the center post (1030) (Figures 5, 7, 8 and 10, [0066], [0068]).

Martin additionally teaches that bending the tabs, analogous to the claimed notching tabs, is desirable to reduce the effective axial stiffness of the tabs, thereby reducing the amount of load transmitted through the tab ([0066]).

It would have been obvious to the skilled artisan at the time of the invention to include an insulation member in the can member of Sohn such as suggested by Martin, wherein the insulation member is provided between the end of the electrode and the can in order to prevent the current collecting tabs from contacting an electrode of opposite polarity.

Additionally, it would have been obvious to the skilled artisan to provide the notching tabs in a bent formation such as suggested by Martin in order to reduce the effective axial stiffness of the tabs, thereby reducing the amount of load transmitted through the tab.

Further regarding claim 1, Sohn in view of Martin fails to teach that the notching tabs are provided on an outer peripheral edge of the insulation member.

Wolff further teaches providing an insulation member, or electrical insulator pad (94), which is disposed between the end of the electrode, or the end of the jelly roll electrode unit (80), and the can member, or cover assembly (10) of the can, and wherein the electrode tab (86) is provided on an outer peripheral edge of the insulation member (Figures 5 and 10, column 11 lines 18-20; column 12 lines 29-39).

Wolff teaches that the insulation member (94) is desirable for substantially insulating the can from the electrode roll (column 11 lines 27-31).

It would have been obvious to the skilled artisan at the time of the invention to include a known insulation member in the can member of Sohn, wherein the insulation member is provided between the

end of the electrode and the can, and wherein the electrode tabs of Sohn are provided on an outer peripheral edge of the insulation member such as suggested by Wolff in order to substantially insulate the can member from the electrode.

With regard to claim 2, it is seen in Figure 2 of Sohn that the notching tab part, or un-coated portion (312) extends across the entire length of the coating portion in the longitudinal direction.

As for claim 4, Barone teaches that the notching tabs, or electrode tabs, have a width of about 3mm to 8 mm ([0011]). The examiner takes note of the fact that the prior art range completely encompasses the claimed range. Absent any additional and more specific information in the prior art, and absent a showing of criticality of the claimed range relative to the prior art, a prima facie case of obviousness exists. MPEP 2144.05

Regarding claim 6, Barone (Figure 9) and Wolff (Figures 9-11) teaches that a bent part of the notching tabs is bonded to the can member, or cover (910) of Barone/cover assembly (10) of Wolff.

With regard to claim 7, Wolff teaches that the insulator (96) is disposed between the cover assembly, which includes the bent parts, and the insulation member (Figure 10-11, see also column 11 lines 35-39, wherein a hole may be provided in the middle of the insulation member to facilitate welding the electrode tab to the can).

Response to Arguments

6. Applicant's arguments, see Remarks, filed 7/7/22, with respect to the rejection of newly amended claim 1 have been fully considered and are persuasive. Therefore, the rejection has been withdrawn. However, upon further consideration, a new ground(s) of rejection is made above.

Conclusion

7. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

8. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMAYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached 8:30am - 4:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

<i>Notice of References Cited</i>	Application/Control No. 16/498,751		Applicant(s)/Patent Under Reexamination Kim et al.	
	Examiner ALIX E EGGERDING		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20140050956-A1	02-2014	HUANG; Wen-Hung	H01M10/052	29/623.5
*	B	US-20130108903-A1	05-2013	Martin; Charles E.	H01M50/172	29/730
*	C	US-20120009450-A1	01-2012	Chun; Kwan-Sic	H01M50/538	429/94
	D					
	E					
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	G					
	H					
	I					
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	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Electronic Acknowledgement Receipt	
EFS ID:	46846719
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	James A. Vezeris/Kathleen Reynolds
Filer Authorized By:	James A. Vezeris
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	19-OCT-2022
Filing Date:	27-SEP-2019
Time Stamp:	09:46:12
Application Type:	U.S. National Stage under 35 USC 371

Payment information:	
Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 260
RAM confirmation Number	E20220I046582545
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The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:	

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM1957_- _JPH11339756A.pdf	496597	no	9
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Warnings:					
Information:					
2	Foreign Reference	LGCHEM1957_- _JPH11339756A_translation.pdf	202856	no	6
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Warnings:					
Information:					
3	Foreign Reference	JP2005276459A.pdf	649856	no	14
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Information:					
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Information:					
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9	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM1957_-_Information_Disclosure_Statement_Fillable_PDF.pdf	1263850	no	4
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Warnings:					
Information:					
11	Fee Worksheet (SB06)	fee-info.pdf	38512	no	2
			c012c428b50db520451596ec8cc4de6e074becb9		
Warnings:					
Information:					
Total Files Size (in bytes):			5807835		

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16498751
	Filing Date		2019-09-27
	First Named Inventor	Jae Woong Kim	
	Art Unit		1729
	Examiner Name	A. E. Eggerding	
	Attorney Docket Number		LGCHEM 3.3F-1654

CERTIFICATION STATEMENT			
Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):			
☒	That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).		
OR			
☐	That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).		
See attached certification statement.			
☒	The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.		
A certification statement is not submitted herewith.			
SIGNATURE			
A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.			
Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2022-10-19
Name/Print	Joseph H. Karlin	Registration Number	62,471
This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.			

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16498751	
	Filing Date		2019-09-27	
	First Named Inventor	Jae Woong Kim		
	Art Unit		1729	
	Examiner Name	A. E. Eggerding		
	Attorney Docket Number		LGCHEM 3.3F-1654	

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	1	20100190056	A	2010-07-29	Turner			
	2	20060127774	A1	2006-06-15	Kim et al.			
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	H11339756	JP	A	1999-12-10	Toyota Central Res & Dev	Machine generated translation attached.	×
	2	2005276459	JP	A	2005-10-06	Sanyo Electric Co	English translation of Abstract only.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		16498751	
	Filing Date		2019-09-27	
	First Named Inventor	Jae Woong Kim		
	Art Unit		1729	
	Examiner Name	A. E. Eggerding		
	Attorney Docket Number		LGCHEM 3.3F-1654	

	3	H0935701	JP	A	1997-02-07	Denso Corp	Machine generated translation attached.	☒
	4	2006128106	JP	A	2006-05-18	Samsung Sdi Co Ltd	U.S. Publication No. 20060127774 is an English language counterpart	☒
	5	2014188501	WO	A1	2014-11-27	Hitachi Automotive Systems, Ltd	Machine generated translation attached.	☒

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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: October 19, 2022

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGCHEM 3.3F-1654
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

**SUBMISSION OF INFORMATION
DISCLOSURE STATEMENT UNDER 37 C.F.R. § 1.97(D)**

Dear Commissioner:

It is respectfully requested that the accompanying Information Disclosure Statement, filed after issuance of a final rejection, be considered pursuant to 37 C.F.R. § 1.97(d). The present submission is accompanied by a certification as required by 37 C.F.R. § 1.97(e). Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material.

The required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: October 19, 2022

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

LD-454\

Electronic Patent Application Fee Transmittal				
Application Number:		16498751		
Filing Date:		27-Sep-2019		
Title of Invention:		Secondary Battery And Method For Manufacturing The Same		
First Named Inventor/Applicant Name:		Jae Woong Kim		
Filer:		James A. Vezeris/Kathleen Reynolds		
Attorney Docket Number:		LGCHEM 3.3F-1654		
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	1806	1	260	260
Total in USD (\$)				260



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	11/21/2022	EXAMINER	
LGCHEM	EGGERDING, ALIX ECHELMEYER			
Lerner, David, Littenberg, Krumholz & Mentlik, LLP			ART UNIT	PAPER NUMBER
20 Commerce Drive			1729	
Cranford, NJ 07016			NOTIFICATION DATE	DELIVERY MODE
			11/21/2022	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 16/498,751	Applicant(s) Kim et al.		
	Examiner ALIX E EGGERDING	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
ALIX E EGGERDING	Primary Examiner	Telephonic
Joseph Karlin	Attorney of Record	

Date of Interview: 15 November 2022

Issues Discussed:

Other

Discussed arguments presented in the agenda. The examiner stated that the arguments appear to be convincing to overcome the current rejection, but further search and consideration are necessary.

Discussed Batson reference in relation to the limitation to the plurality of notching tabs being together equally circumferentially distributed about a central axis of the electrode.

☒ Attachment

/ALIX E EGGERDING/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

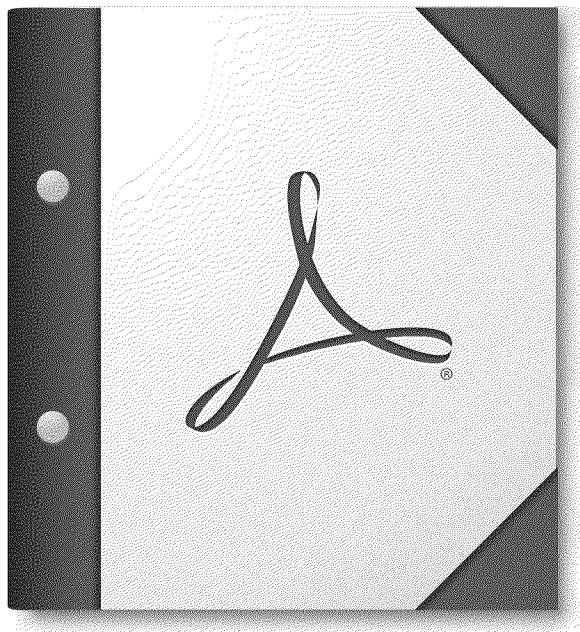
<i>Notice of References Cited</i>	Application/Control No. 16/498,751		Applicant(s)/Patent Under Reexamination Kim et al.	
	Examiner ALIX E EGGERDING		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS						
*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20160351966-A1	12-2016	Batson; David C.	H01M10/052	1/1
	B					
	C					
	D					
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Electronic Acknowledgement Receipt	
EFS ID:	47126305
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	James A. Vezeris/Kathleen Reynolds
Filer Authorized By:	James A. Vezeris
Attorney Docket Number:	LGCHEM 3.3F-1654
Receipt Date:	01-DEC-2022
Filing Date:	27-SEP-2019
Time Stamp:	15:01:01
Application Type:	U.S. National Stage under 35 USC 371

Payment information:					
Submitted with Payment			no		
File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Arguments/Remarks Made in an Amendment	Amendment_After_Final_Action_37_CFR_Section_1.pdf	67394	no	9
			3544256d2c3d7fa4569376e54a86ce63e453fa2a		
Warnings:					

Information:					
2	After Final Consideration Program Request	Certification_and_Request_for_Consideration_Under_the_After_Final_Consideration.pdf	28001	no	1
			ec2f715bf7573c593a1c77a5cc86255ad124e458		
Warnings:					
Information:					
3	Miscellaneous Incoming Letter	Amendment_Transmittal.pdf	25056	no	1
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: December 1, 2022

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

EXPEDITED PROCEDURE

Group Art Unit: 1729

Docket No. LGCHEM 3.3F-1654

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

MS AF

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

AMENDMENT UNDER 37 C.F.R. § 1.116

Dear Commissioner:

In response to the Office Action dated September 13, 2022, finally rejecting claims 1-2, 4, and 6-7, please amend the above-identified U.S. patent application as follows:

IN THE CLAIMS

1. (Previously Presented) A secondary battery comprising:
a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector, the non-coating portion being adjacent to the coating portion in a longitudinal direction of the electrode collector, the non-coating portion being disposed at a first end of the electrode collector, the electrode being accommodated in the can member;

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part extending from part of the coating portion at a second end of the electrode collector opposite in the longitudinal direction from the first end of the electrode collector, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state;

an electrode tab attached to the non-coating portion, such that when the electrode is in the wound state, the electrode tab extends from a central portion of an end of the electrode; and

an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each notching tab extends towards a bottom surface of the can member only from an outer portion of the end of the electrode remote from the central portion,

wherein, when the electrode is in the wound state, the plurality of notching tabs are together equally circumferentially distributed about a central axis of the electrode,

wherein, when the electrode is in the wound state, each of the plurality of notching tabs is in a bent state and overlaps the electrode tab, and the electrode tab and the plurality of notching tabs are bonded together to the can member, and

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the

insulation member, and the electrode tab extending through a central opening of the insulation member.

2. (Previously Presented) The secondary battery of claim 1, wherein the notching tab part continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.

3. (Canceled)

4. (Previously Presented) The secondary battery of claim 1, wherein each of the plurality of notching tabs has a width of 4 mm to 5 mm.

5. (Canceled)

6. (Previously Presented) The secondary battery of claim 1, wherein a bent part of each of the plurality of notching tabs is bonded to the can member.

7. (Previously Presented) The secondary battery of claim 6, wherein the insulation member is disposed between the bent parts and the end of the electrode.

8-13. (Cancelled)

14. (Withdrawn – Currently Amended) A method for manufacturing ~~a~~the secondary battery of claim 1, the method comprising:

a preparation step of preparing ~~an~~the electrode provided with ~~a~~the coating portion coated with ~~an~~the active material on ~~an~~the electrode collector and ~~a~~the non-coating portion on which the active material is not applied to the electrode collector in ~~a~~the longitudinal direction of the electrode collector;

a notching tab part formation step of notching the electrode to form ~~a~~the notching tab part not coated with the active material and extending from the coating portion in ~~a~~the width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction;

a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an electrode assembly;

a notching tab formation step of cutting the notching tab part of the electrode assembly to form ~~a~~the plurality of notching tabs; and

an installation step of accommodating the electrode assembly into ~~a~~the can member and bonding each of the plurality of notching tabs to the can member.

15. (Withdrawn) The method of claim 14, wherein the electrode is a negative electrode.

16. (Withdrawn) The method of claim 14, wherein, during the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied are arranged parallel to each other in the width direction of the electrode collector.

17. (Withdrawn) The method of claim 16, wherein, during the notching tab part formation step, the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

18. (Withdrawn) The method of claim 14, wherein, during the notching tab part formation step, notching is performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

19. (Withdrawn) The method of claim 14, wherein, during the notching tab formation step, the notching tab part is cut using a laser.

20. (Withdrawn) The method of claim 14, wherein, during the installation step, the electrode assembly is accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

21. (Withdrawn) The method of claim 20, wherein, during the installation step a bent portion of each of the plurality of notching tabs is welded to the can member.

22. (Withdrawn) The method of claim 20, wherein, during the installation step bent portions of each of the plurality of notching tabs overlap each other and are bonded to the can member together.

23. (Withdrawn) The method of claim 14, wherein, during the winding step, the notching tab part is wound to form two or more overlapping layers.

REMARKS

The present amendment is responsive to the final office action dated September 23, 2022. This response is enterable after final rejection because it is accompanied by an After-Final Pilot Program 2.0 request form, filed herewith. Claims 1-2, 4, 6-7, and 14-23 are pending. Claim 14 is amended. Claims 14-23 are withdrawn as being directed to a non-elected species or invention. Claims 3, 5, and 8-13 were previously canceled. No new matter is added by the present amendments. Applicant respectfully requests that the Examiner reconsider the claims in view of the amendments and remarks herein.

Interview Summary

Applicants' representative thanks Examiner Eggerding for her time during the telephonic interview conducted on November 15, 2022, with the undersigned. The rejection of claim 1 was discussed and it was agreed that the arguments presented herein would overcome the pending rejection. No agreement was reached regarding patentability.

Prior Art Rejections

Claims 1, 2, 4, 6, and 7 were rejected under 35 U.S.C. § 103 as allegedly unpatentable over U.S. Patent Application Pub. No. 2011/0067227 to Sohn ("*Sohn*"), in view of U.S. Patent Application Pub. No. 2012/0009450 to Chun ("*Chun*"), U.S. Patent Application Pub. No. 2013/0108903 to Martin et al. ("*Martin*"), and U.S. Patent Application Pub. No. 2013/0252054 to Barone et al. ("*Barone*"). Applicant submits that the rejections in the Office Action are traversed for the reasons set forth below.

Applicant submits that each and every limitation of claim 1 is not taught or suggested by any of the references of record, alone or in combination. Claim 1 recites:

A secondary battery comprising:

a can member;

an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector, the non-coating portion being adjacent to the coating portion in a longitudinal direction of the electrode collector, the non-coating portion being disposed at a first end of the electrode collector, the electrode being accommodated in the can member;

a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part extending from part of the coating portion at a second end of the electrode collector opposite in the longitudinal direction from the

first end of the electrode collector, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state;

an electrode tab attached to the non-coating portion, such that when the electrode is in the wound state, the electrode tab extends from a central portion of an end of the electrode; and

an insulation member,

wherein, when the electrode is in the wound state, the notching tab part forms a plurality of notching tabs each formed from a respective portion of the two or more overlapping layers, such that each notching tab has a respective plurality of layer portions each extending only over a same circumferentially extending angle about a center of the secondary battery, and each notching tab extends towards a bottom surface of the can member only from an outer portion of the end of the electrode remote from the central portion,

wherein, when the electrode is in the wound state, the plurality of notching tabs are together equally circumferentially distributed about a central axis of the electrode,

wherein, when the electrode is in the wound state, each of the plurality of notching tabs is in a bent state and overlaps the electrode tab, and the electrode tab and the plurality of notching tabs are bonded together to the can member, and

wherein the insulation member is disposed between the can member and the end of the electrode, the plurality of notching tabs together extending around an outer peripheral edge of the insulation member, and the electrode tab extending through a central opening of the insulation member.

Applicant respectfully submits that some of the limitations of claim 1 are not taught or suggested in the cited references. In particular, Applicant does not believe that any of the five references teaches the feature of claim 1 that recites “when the electrode is in the wound state, the plurality of notching tabs are together equally circumferentially distributed about a central axis of the electrode,” or the feature that recites that “the electrode tab and the plurality of notching tabs are bonded together to the can member.”

Also, the newly cited *Chun* reference is alleged to teach the limitation of claim 1 reciting “the non-coating portion [of the electrode] being adjacent to the coating portion in a longitudinal direction of the electrode collector.” Although *Chun* is alleged to have this teaching, Applicant disagrees with the motivation to modify the primary reference (*Sohn*) to have this feature. The Office Action states that the motivation is to “distribute the stresses on the electrode tabs,” but *Chun* does not say that this is why its battery has the two adjacent portions. Rather, to distribute stresses, *Chun* varies the widths of its electrode tabs. Applicant believes that the alleged motivation to combine these two references to produce the alleged teaching is incorrect.

Furthermore, as mentioned above, it was agreed during the Examiner Interview that the arguments presented herein would overcome the pending rejections of independent claim 1.

Accordingly, Applicant respectfully submits that each and every limitation of amended claim 1 is not taught or suggested by any of the references of record, alone or in combination. Therefore, Applicant submits that the pending rejection of independent claim 1 under 35 U.S.C. § 103 should be withdrawn. Also, for at least the same reasons, the rejections of dependent claims 2, 4, 6, and 7 should be withdrawn, each of which ultimately depends from independent claim 1.

Dependency of Withdrawn Claim 14

The withdrawn formerly independent claim 14 is amended herein to depend from claim 1, thereby including the features of claim 1. Therefore, if claim 1 is determined to be allowable, Applicant respectfully requests that claim 14 be rejoined and allowed by virtue of its dependence from an allowable claim. For at least the same reasons, Applicant requests that if claims 1 and 14 are allowed, dependent claims 15-23 should also be rejoined and allowed by virtue of their ultimate dependence on claim 1 via claim 14.

Conclusion

In view of the above, each of the presently pending claims in this application is believed to be in immediate condition for allowance. Accordingly, the Examiner is respectfully requested to withdraw the outstanding rejection of the claims and to pass this application to issue.

If, however, for any reason the Examiner does not believe that such action can be taken at this time, it is respectfully requested that the Examiner telephone Applicant's attorney at (908) 654-5000 in order to overcome any additional objections which the Examiner might have.

Application No.: 16/498,751

Docket No.: LGCHEM 3.3F-1654

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: December 1, 2022

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/

Joseph H. Karlin

Registration No.: 62,471

LERNER, DAVID, LITTENBERG,

KRUMHOLZ & MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant

CERTIFICATION AND REQUEST FOR CONSIDERATION UNDER THE AFTER FINAL CONSIDERATION PILOT PROGRAM 2.0		
Practitioner Docket No.: LGCHEM 3.3F-1654	Application No.: 16/498,751-Conf. #4723	Filing Date: September 27, 2019
First Named Inventor: Jae Woong Kim	Title: Secondary Battery And Method For Manufacturing The Same	
APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS CONSIDERATION UNDER THE AFTER FINAL CONSIDERATION PILOT PROGRAM 2.0 (AFCP 2.0) OF THE ACCOMPANYING RESPONSE UNDER 37 CFR 1.116.		
1. The above-identified application is (i) an original utility, plant, or design nonprovisional application filed under 35 U.S.C. 111(a) [a continuing application (e.g., a continuation or divisional application) is filed under 35 U.S.C. 111(a) and is eligible under (i)], or (ii) an international application that has entered the national stage in compliance with 35 U.S.C. 371(c). 2. The above-identified application contains an outstanding final rejection. 3. Submitted herewith is a response under 37 CFR 1.116 to the outstanding final rejection. The response includes an amendment to at least one independent claim, and the amendment does not broaden the scope of the independent claim in any aspect. 4. This certification and request for consideration under AFCP 2.0 is the only AFCP 2.0 certification and request filed in response to the outstanding final rejection. 5. Applicant is willing and available to participate in any interview requested by the examiner concerning the present response. 6. This certification and request is being filed electronically using the USPTO's patent electronic filing system. 7. Any fees that would be necessary consistent with current practice concerning responses after final rejection under 37 CFR 1.116, e.g., extension of time fees, are being concurrently filed herewith. [There is no additional fee required to request consideration under AFCP 2.0.] 8. By filing this certification and request, applicant acknowledges the following: • Reissue applications and reexamination proceedings are not eligible to participate in AFCP 2.0. • The examiner will verify that the AFCP 2.0 submission is compliant, i.e., that the requirements of the program have been met (see items 1 to 7 above). For compliant submissions: o The examiner will review the response under 37 CFR 1.116 to determine if additional search and/or consideration (i) is necessitated by the amendment and (ii) could be completed within the time allotted under AFCP 2.0. If additional search and/or consideration is required but cannot be completed within the allotted time, the examiner will process the submission consistent with current practice concerning responses after final rejection under 37 CFR 1.116, e.g., by mailing an advisory action. o If the examiner determines that the amendment does not necessitate additional search and/or consideration, or if the examiner determines that additional search and/or consideration is required and could be completed within the allotted time, then the examiner will consider whether the amendment places the application in condition for allowance (after completing the additional search and/or consideration, if required). If the examiner determines that the amendment does not place the application in condition for allowance, then the examiner will contact the applicant and request an interview. ▪ The interview will be conducted by the examiner, and if the examiner does not have negotiation authority, a primary examiner and/or supervisory patent examiner will also participate. ▪ If the applicant declines the interview, or if the interview cannot be scheduled within ten (10) calendar days from the date that the examiner first contacts the applicant, then the examiner will proceed consistent with current practice concerning responses after final rejection under 37 CFR 1.116.		
Signature /Joseph H. Karlin/	Date December 1, 2022	Practitioner Registration No. 62,471
Name (Print/Typed) Joseph H. Karlin	Telephone Number (908) 654-5000	
Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.		
☐ * Total of 1 forms are submitted.		

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).	
Dated: December 1, 2022	Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

AMENDMENT TRANSMITTAL LETTER			After Final Rejection		
Application No. 16/498,751	Filing Date September 27, 2019	Examiner A. E. Eggerding	Group Art Unit 1729		
Applicant(s): Jae Woong Kim, Hyoung Kwon Kim, and Sang Uck KimLG Chem, Ltd.		Docket No. LGCHEM 3.3F-1654			
Invention: Secondary Battery And Method For Manufacturing The Same					
TO THE COMMISSIONER FOR PATENTS					
Transmitted herewith is an amendment in the above-identified application.					
The fee has been calculated and is transmitted as shown below.					
CLAIMS AS AMENDED					
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate	
Total Claims	15	- 20 =	0	x	
Independent Claims	1	- 3 =	0	x	
Multiple Dependent Claims (check if applicable)					☐
Other fee (please specify):					
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00
☒ Large Entity ☐ Small Entity					
☒ No additional fee is required for this amendment.					
☐ Please charge Deposit Account _____ in the amount _____ . A duplicate copy of this sheet is enclosed.					
☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.					
☐ Payment by credit card. Form PTO-2038 is attached.					
☒ The Commissioner is hereby authorized to charge and credit Deposit _____ 12-1095 as described below.					
☒ Credit any overpayment.					
☒ Charge any additional filing or application processing fees required under 37 CFR 1.16 and 1.17.					
/Joseph H. Karlin/ Joseph H. Karlin Attorney Reg. 62,471			Dated: December 1, 2022		
LERNER, DAVID, LITTENBERG, KRUMHOLZ & MENTLIK, LLP 20 Commerce Dr. Cranford, New Jersey 07016 (908) 654-5000					
I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).					
Dated: December 1, 2022			Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/		

Notice of Allowability	Application No. 16/498,751	Applicant(s) Kim et al.	
	Examiner ALIX E EGGERDING	Art Unit 1729	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. [X] This communication is responsive to the AFCP amendment filed 12/1/22.
[] A declaration(s)/affidavit(s) under 37 CFR 1.130(b) was/were filed on ____.

2. [] An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. [X] The allowed claim(s) is/are 1-2,4,6-7 and 14-23 . As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information , please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. [X] Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) [X] All b) [] Some* c) [] None of the:

1. [X] Certified copies of the priority documents have been received.

2. [] Certified copies of the priority documents have been received in Application No. ____ .

3. [] Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____ .

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. [] CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
[] including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____ .

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. [] DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. [] Notice of References Cited (PTO-892)	5. [] Examiner's Amendment/Comment
2. [X] Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date 10/19/22.	6. [X] Examiner's Statement of Reasons for Allowance
3. [] Examiner's Comment Regarding Requirement for Deposit of Biological Material ____.	7. [X] Other <u>PTO-2323</u> .
4. [] Interview Summary (PTO-413), Paper No./Mail Date. ____.	

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

2. Claim 1 is directed to an allowable product. Pursuant to the procedures set forth in MPEP § 821.04(B), claims 14-23, directed to the process of making or using an allowable product, previously withdrawn from consideration as a result of a restriction requirement, are hereby rejoined and fully examined for patentability under 37 CFR 1.104.

Because all claims previously withdrawn from consideration under 37 CFR 1.142 have been rejoined, **the restriction requirement as set forth in the Office action mailed on 6/3/21 is hereby withdrawn.** In view of the withdrawal of the restriction requirement as to the rejoined inventions, applicant(s) are advised that if any claim presented in a continuation or divisional application is anticipated by, or includes all the limitations of, a claim that is allowable in the present application, such claim may be subject to provisional statutory and/or nonstatutory double patenting rejections over the claims of the instant application. Once the restriction requirement is withdrawn, the provisions of 35 U.S.C. 121 are no longer applicable. See *In re Ziegler*, 443 F.2d 1211, 1215, 170 USPQ 129, 131-32 (CCPA 1971). See also MPEP § 804.01.

Allowable Subject Matter

3. Claims 1-2, 4, 6-7, and 14-23 are allowed.
4. The following is an examiner's statement of reasons for allowance: the prior art fails to teach or fairly suggest the claimed subject matter.

Specifically, as discussed in the interview on 11/15/22 and the remarks filed 12/1/22, the prior art fails to teach or fairly suggest the secondary battery member of claim 1 having the claimed notching tab parts and electrode tab, wherein the notching tab parts are together equally circumferentially distributed about a central axis of the electrode and are in a bent state overlapping the electrode tab.

While Batson (US 2019/0351966) teaches tab parts together equally circumferentially distributed about a central axis of the electrode (see Fig. 12), it is the opinion of the examiner that it would not have been obvious to the skilled artisan to combine Batson with the other references relied upon to reject claim 1 in the Final Rejection of 9/13/22 in such a manner to arrive at the claimed invention in light of the multiple additional limitations of claim 1 which are not taught or suggested by Batson (e.g., the separate notching tabs and electrode tab and the bent state of the notching tabs, and the claimed insulation member), and the multiple references already cited (Sohn, Chun, Martin, and Barone).

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

5. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ALIX ECHELMAYER EGGERDING whose telephone number is (571)272-1101. The examiner can normally be reached 8:30am - 4:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 571-272-1481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ALIX E EGGERDING/
Primary Examiner, Art Unit 1729

OK TO ENTER: /A.E.E/
12/16/2022

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: December 1, 2022
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

EXPEDITED PROCEDURE
Group Art Unit: 1729
Docket No. LGCHEM 3.3F-1654

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 16/498,751

Confirmation No.: 4723

Filed: September 27, 2019

Art Unit: 1729

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: A. E. Eggerding

MS AF
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT UNDER 37 C.F.R. § 1.116

Dear Commissioner:

In response to the Office Action dated September 13, 2022, finally rejecting claims 1-2, 4, and 6-7, please amend the above-identified U.S. patent application as follows:

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	1729	
Examiner Name	A. E. Eggerding	
Attorney Docket Number	LGCHEM 3.3F-1654	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2022-10-19
Name/Print	Joseph H. Karlin	Registration Number	62,471

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Receipt date: 10/19/2022

16/498,751 -- GAU: 1729

Doc code: IDS
Doc description: Information Disclosure Statement (IDS) Filed

PTO/SB/08a (01-22)
Approved for use through 05/31/2024. OMB 0651-0031
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE
Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number	16498751
Filing Date	2019-09-27
First Named Inventor	Jae Woong Kim
Art Unit	1729
Examiner Name	A. E. Eggerding
Attorney Docket Number	LGCHEM 3.3F-1654

U.S.PATENTS

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	1	20100190056	A	2010-07-29	Turner	
	2	20060127774	A1	2006-06-15	Kim et al.	

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FOREIGN PATENT DOCUMENTS

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	1	H11339756	JP	A	1999-12-10	Toyota Central Res & Dev	Machine generated translation attached.	×
	2	2005276459	JP	A	2005-10-06	Sanyo Electric Co	English translation of Abstract only.	☒

EFS Web 2.1.18

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /A.E.E/

Receipt date: 10/19/2022

16/498,751 -- GAU: 1729

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT
(Not for submission under 37 CFR 1.99)

Application Number		16498751
Filing Date		2019-09-27
First Named Inventor	Jae Woong Kim	
Art Unit	1729	
Examiner Name	A. E. Eggerding	
Attorney Docket Number	LGCHEM 3.3F-1654	

3	H0935701	JP	A	1997-02-07	Denso Corp	Machine generated translation attached.	☒
4	2006128106	JP	A	2006-05-18	Samsung Sdi Co Ltd	U.S. Publication No. 20060127774 is an English language counterpart	☒
5	2014188501	WO	A1	2014-11-27	Hitachi Automotive Systems, Ltd	Machine generated translation attached.	☒

If you wish to add additional Foreign Patent Document citation information please click the Add button

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NON-PATENT LITERATURE DOCUMENTS

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1		

If you wish to add additional non-patent literature document citation information please click the Add button

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EXAMINER SIGNATURE

Examiner Signature	/ALIX E EGGERDING/	Date Considered	12/16/2022
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

86765 7590 01/09/2023
LG
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

EXAMINER

EGGERDING, ALIX ECHELMEYER

ART UNIT PAPER NUMBER

1729

DATE MAILED: 01/09/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723

TITLE OF INVENTION: Secondary Battery And Method For Manufacturing The Same

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	04/10/2023

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

86765 7590 01/09/2023
LG
Lerner, David, Littenberg, Krumholz & Mentlik, LLP
20 Commerce Drive
Cranford, NJ 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

	(Typed or printed name)
	(Signature)
	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723

TITLE OF INVENTION: Secondary Battery And Method For Manufacturing The Same

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	04/10/2023

EXAMINER	ART UNIT	CLASS-SUBCLASS
EGGERDING, ALIX ECHELMEYER	1729	429-094000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	--

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.
NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.
NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/498,751	09/27/2019	Jae Woong Kim	LGCHEM 3.3F-1654	4723
86765	7590	01/09/2023	EXAMINER	
LG			EGGERDING, ALIX ECHELMEYER	
Lerner, David, Littenberg, Krumholz & Mentlik, LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
DATE MAILED: 01/09/2023				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)
(Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.** Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b) (2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt	
EFS ID:	47810965
Application Number:	16498751
International Application Number:	
Confirmation Number:	4723
Title of Invention:	Secondary Battery And Method For Manufacturing The Same
First Named Inventor/Applicant Name:	Jae Woong Kim
Customer Number:	86765
Filer:	Keith McWha/Melinda Cormier
Filer Authorized By:	Keith McWha
Attorney Docket Number:	LGES 3.3F-1654
Receipt Date:	07-APR-2023
Filing Date:	27-SEP-2019
Time Stamp:	16:01:46
Application Type:	U.S. National Stage under 35 USC 371

Payment information:	
Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E202347G04048609
Deposit Account	
Authorized User	
The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:	

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	Issue_Fee_Transmittal.pdf	37114	no	1
			38334d26c7d25fbd9a3a01173e7e14749f6081b6		
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	38061	no	2
			12032c71fa8c3c44cd8e71aee21de07cb1d29c2e		
Warnings:					
Information:					
Total Files Size (in bytes):			75175		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PART B–FEE(S) TRANSMITTAL

Complete and send this form, together with the applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

By fax, send to: (571) 273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission
I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

Joseph H. Karlin	(Typed or printed name)
/Joseph H. Karlin/	(Signature)
April 7, 2023	(Date)

APPLICATION NO	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO	CONFIRMATION NO
16/498,751	09/27/2019	Jae Woong Kim	LGES 3.3F-1654	4723

TITLE OF INVENTION: Secondary Battery And Method For Manufacturing The Same

APPLN: TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1,200.00			\$1,200.00	04/09/2023

EXAMINER	ART UNIT	CLASS-SUBCLASS
A. E. Eggerding	1729	429-094000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363) ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-09 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.	1. Lerner David LLP 2. 3.
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE LG Energy Solution, Ltd.

(B) RESIDENCE: (CITY and STATE or COUNTRY) Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment (Please first reapply any previously paid fee shown above):

☒ Electronic Payment via Patent Center or EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 12-1095

5. Change of Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29.	NOTE: Absent a valid Certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.
☐ Applicant asserting small entity status. See 37 CFR 1.27.	NOTE: If the application was previously under micro entity status, checking this box will be taken as a notification of loss of entitlement to micro entity status.
☐ Applicant changing to regular undiscounted fee status.	NOTE: Checking this box will be taken as a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature	/Joseph H. Karlin/	Date	April 7, 2023
Typed or printed name	Joseph H. Karlin	Registration No.	62,471

Electronic Patent Application Fee Transmittal				
Application Number:		16498751		
Filing Date:		27-Sep-2019		
Title of Invention:		Secondary Battery And Method For Manufacturing The Same		
First Named Inventor/Applicant Name:		Jae Woong Kim		
Filer:		Keith McWha/Melinda Cormier		
Attorney Docket Number:		LGES 3.3F-1654		
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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16/498,751	05/16/2023	11652232	LGES 3.3F-1654	4723
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86765	7590	04/26/2023		
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LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO’s Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)
(application filed on or after May 29, 2000)

The Patent Term Adjustment is 101 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional inventors):

Jae Woong Kim, Daejeon, KOREA, REPUBLIC OF;
Hyoung Kwon Kim, Daejeon, KOREA, REPUBLIC OF;
Sang Uck Kim, Daejeon, KOREA, REPUBLIC OF;

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address : COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.
16/498,751	16-MAY-23	11652232

LG Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/498,751		Filing Date 09/27/2019		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$460 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	09/27/2019	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$ 100 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$ 460 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/JAMES W TUNSTALL/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**
If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 16/498,751								
APPLICATION AS FILED - PART I														
		(Column 1)			(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY					
FOR		NUMBER FILED		NUMBER EXTRA		RATE(\$)		FEE(\$)		RATE(\$)		FEE(\$)		
BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A				N/A		300		
SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A		N/A		N/A				N/A		520		
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A				N/A		760		
TOTAL CLAIMS (37 CFR 1.16(j))		20		minus 20 =		*				OR	x	100	=	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))		2		minus 3 =		*					x	460	=	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).										0.00		
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))												0.00		
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL				TOTAL		1580		
APPLICATION AS AMENDED - PART II														
		(Column 1)			(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY				
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)		ADDITIONAL FEE(\$)		RATE(\$)		ADDITIONAL FEE(\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=		OR	x	=			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=		OR	x	=			
	Application Size Fee (37 CFR 1.16(s))								OR					
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								OR					
						TOTAL ADD'L FEE				OR	TOTAL ADD'L FEE			
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)		ADDITIONAL FEE(\$)		RATE(\$)		ADDITIONAL FEE(\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=		OR	x	=			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=		OR	x	=			
	Application Size Fee (37 CFR 1.16(s))								OR					
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								OR					
						TOTAL ADD'L FEE				OR	TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.														

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/498,751		Filing Date 09/27/2019		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$460 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	09/03/2021	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 19	Minus	** 20	= 0	x \$ 100 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/ROLITA M WIMBUSH/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/498,751		Filing Date 09/27/2019		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$460 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	02/18/2022	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 18	Minus	** 20	= 0	x \$ 100 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/BRENDA MURPHY/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/498,751		Filing Date 09/27/2019		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$460 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	07/07/2022	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 15	Minus	** 20	= 0	x \$ 100 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/RENEE D HAWKINS/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/498,751		Filing Date 09/27/2019		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)		(Column 2)					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)	
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		x \$100 =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		x \$460 =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL			
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT	12/01/2022	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 15	Minus	** 20	= 0	x \$ 100 =		0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$ 480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)			
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/PAMELA V THERATT/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**
If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

MULTIPLE DEPENDENT CLAIM FEE CALCULATION SHEET							Application Number		Filing Date		
Substitute for Form PTO-1360 (For use with Form PTO/SB/06)							16498751				
							Applicant(s) Jae Kim				
							* May be used for additional claims or amendments				
CLAIMS		AS FILED		AFTER FIRST AMENDMENT		AFTER SECOND AMENDMENT		*		*	
	Indep	Depend	Indep	Depend	Indep	Depend		Indep	Depend	Indep	Depend
1			1				51				
2				1			52				
3				1			53				
4				1			54				
5				1			55				
6				1			56				
7				1			57				
8				1			58				
9				1			59				
10				1			60				
11			1				61				
12				1			62				
13				1			63				
14				1			64				
15				1			65				
16				1			66				
17				1			67				
18				1			68				
19				1			69				
20				1			70				
21							71				
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42							92				
43							93				
44							94				
45							95				
46							96				
47							97				
48							98				
49							99				
50							100				
Total Indep	0		2		0						
Total Depend	0		18		0						
Total Claims	0		20		0						

Patent Export, 2026-06-01 02:13:24 +0000

Search results for: PNS=(US11652232);

Collections searched: DWPI; ARIPO grant; Argentina UM; Argentina app; Armenia grant; Australia UM; Australia app; Australia grant; Austria UM; Austria app; Austria grant; Belarus UM; Belarus grant; Belgium app; Belgium grant; Brazil UM; Brazil app; Brazil grant; Bulgaria UM; Bulgaria app; Bulgaria grant; Canada app; Canada grant; China, Mainland UM; China, Mainland app; China, Mainland grant; Colombia app; Costa Rica UM; Costa Rica app; Croatia UM; Croatia app; Croatia grant; Cuba app; Cuba grant; Czech Republic UM; Czech Republic app; Czech Republic grant; Czechoslovakia grant; Denmark UM; Denmark app; Denmark grant; Estonia UM; Estonia app; Estonia grant; Eurasian (EAPO) grant; Europe (EP) app; Europe (EP) grant; Finland app; Finland grant; France app; France grant; Georgia grant; German Democratic Republic app; German Democratic Republic grant; Germany UM; Germany app; Germany grant; Greece app; Greece grant; Gulf Cooperation Council grant; Hong Kong UM; Hong Kong app; Hong Kong grant; Hungary grant; Iceland grant; India app; India grant; Indonesia UM; Indonesia app; Ireland UM; Ireland app; Ireland grant; Israel grant; Italy grant; Italy grant; Italy grant; Italy grant; Japan UM; Japan app; Japan grant; Korea UM; Korea app; Korea grant; Latvia grant; Lithuania grant; Luxembourg app; Luxembourg grant; Malaysia app; Malaysia grant; Mexico app; Mexico grant; Moldova UM; Moldova grant; Monaco grant; Mongolia UM; Mongolia grant; Morocco app; Morocco grant; Netherlands app; Netherlands grant; New Zealand grant; Norway app; Norway grant; OAPI grant; PCT (WO) app; Philippines UM; Philippines app; Philippines grant; Poland UM; Poland grant; Portugal UM; Portugal app; Portugal grant; Romania UM; Romania app; Romania granted; Russia UM; Russia app; Russia grant; Saudi Arabia app; Saudi Arabia grant; Serbia UM; Serbia app; Serbia grant; Singapore app; Singapore grant; Slovakia UM; Slovakia app; Slovakia grant; Slovenia UM; Slovenia grant; South Africa app; Spain UM; Spain app; Spain grant; Sweden app; Sweden grant; Switzerland app; Switzerland grant; Taiwan UM; Taiwan appl; Taiwan grant; Thailand UM; Thailand app; Thailand grant; Tunisia app; Turkey UM; Turkey app; UK app; UK grant; USSR grant; Ukraine UM; Ukraine grant; United States app; United States grant; Uruguay app; Vietnam UM; Vietnam app; Vietnam grant; Other

Table of Contents

-
1. US11652232B2 Secondary battery and method for manufacturing the same
-

Family 1/1

1 record(s) per family

Record 1/1 US11652232B2 Secondary battery and method for manufacturing the same

Title: Secondary battery and method for manufacturing the same

Title - DWPI: Secondary battery for use as e.g. AC power source, supplied to e.g. building, has notching tab portion formed and folded if electrode is reeled to be overlapped such that active material is coated from maintenance unit

Priority Number: KR201748645A | WO2018KR4366A | KR201843191A

Priority Number - DWPI: KR201843191A | KR201748645A | KR20178645A | WO2018KR4366A

Priority Date: 2017-04-14 | 2018-04-13 | 2018-04-13

Priority Date - DWPI: 2018-04-13 | 2017-04-14 | 2017-04-14 | 2018-04-13

Application Number: US16498751A

Application Date: 2019-09-27

Publication Date: 2023-05-16

IPC Class Table:

IPC	Section	Class	Subclass	Class Group	Subgroup
H01M0050538	H	H01	H01M	H01M0050	H01M0050538
H01M001004	H	H01	H01M	H01M0010	H01M001004
H01M0050536	H	H01	H01M	H01M0050	H01M0050536

IPC Class Table - DWPI:

IPC - DWPI	Section - DWPI	Class - DWPI	Subclass - DWPI	Class Group - DWPI	Subgroup - DWPI
H01M000226 (IPC 1-7)	H	H01	H01M	H01M0002	H01M000226 (IPC 1-7)
H01M001004 (IPC 1-7)	H	H01	H01M	H01M0010	H01M001004 (IPC 1-7)
H01M00100587 (IPC 1-7)	H	H01	H01M	H01M0010	H01M00100587 (IPC 1-7)
H01M	H	H01	H01M	H01M	H01M
H01M001004	H	H01	H01M	H01M0010	H01M001004

H01M00100587	H	H01	H01M	H01M0010	H01M00100587
H01M001042	H	H01	H01M	H01M0010	H01M001042
H01M000226	H	H01	H01M	H01M0002	H01M000226
H01M000404	H	H01	H01M	H01M0004	H01M000404
H01M000413	H	H01	H01M	H01M0004	H01M000413
H01M0004139	H	H01	H01M	H01M0004	H01M0004139
H01M000470	H	H01	H01M	H01M0004	H01M000470
H01M0050107	H	H01	H01M	H01M0050	H01M0050107
H01M0050528	H	H01	H01M	H01M0050	H01M0050528
H01M0050533	H	H01	H01M	H01M0050	H01M0050533
H01M0050536	H	H01	H01M	H01M0050	H01M0050536
H01M0050538	H	H01	H01M	H01M0050	H01M0050538
H01M0050586	H	H01	H01M	H01M0050	H01M0050586

Assignee/Applicant: LG Energy Solution Ltd. | LG Chem Ltd.,Seoul,KR

Assignee - Current US: LG ENERGY SOLUTION LTD.

JP F Terms:

JP FI Codes:

Item Number: 1

Assignee - Original: LG Energy Solution Ltd. | LG Chem Ltd.

Any CPC Table:

Type	CPC - Current - Invention	CPC - Current - Additional	Version	Office
Current	H01M 10/0422	-	20130101	EP; KR; US
Current	H01M 10/0431	-	20130101	KR; US
Current Current	H01M 10/0587 H01M 10/4235	Y02E 60/10	20130101 20130101	EP; KR EP; KR
Current	H01M 50/538	-	20210101	EP; KR; US
Current	-	H01M 50/536	20210101	EP; US
Current	-	Y02P 70/50	20151101	EP; KR
Original Original	H01M 10/0422 H01M 10/0431	-	20130101 20130101	US US

Original	H01M 50/538	H01M 50/536	20210101	US
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ECLA:

Abstract:

The present invention relates to a secondary battery in which an electrode tab is improved in mechanical strength and a method for manufacturing the same.Also, the secondary battery according to the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector and a notching tab part extending from the coating portion in a width direction of the electrode collector without coating with the active material and overlapping each other to form two or more layers when the electrode is wound. |

La présente invention concerne une batterie secondaire et un procédé de fabrication de batterie secondaire, la résistance mécanique d'une languette d'électrode pouvant être augmentée. De plus, la présente invention est caractérisée en ce qu'elle comprend : une électrode ayant une partie revêtue et une partie non revêtue disposée le long de la direction longitudinale d'un collecteur d'électrode, la partie revêtue ayant un matériau actif appliqué sur le collecteur d'électrode, et la partie non revêtue n'ayant pas de matériau actif appliqué sur le collecteur d'électrode ; et une partie de languette d'encochage s'étendant à partir de la partie revêtue dans la direction transversale du collecteur d'électrode de telle sorte que le matériau actif n'y est pas appliqué, la partie de languette d'encochage étant formée pour chevaucher au moins deux niveaux lorsque l'électrode est enroulée. |

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Language of Publication: EN

INPADOC Legal Status Table:

Gazette Date	Code	INPADOC Legal Status Impact
2023-04-26	STCF	-
Description: INFORMATION ON STATUS: PATENT GRANT PATENTED CASE		
2022-12-07	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL RESPONSE AFTER FINAL ACTION FORWARDED TO EXAMINER		

2022-09-09	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL FINAL REJECTION MAILED		
2022-07-12	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL RESPONSE TO NON-FINAL OFFICE ACTION ENTERED AND FORWARDED TO EXAMINER		
2022-03-04	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL NON FINAL ACTION MAILED		
2021-11-04	AS	-
Description: ASSIGNMENT LG ENERGY SOLUTION, LTD., KOREA, REPUBLIC OF ASSIGNMENT OF ASSIGNORS INTEREST; ASSIGNOR:LG CHEM, LTD.; REEL/FRAME:058295/0068 2021-10-27		
2021-09-20	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL FINAL REJECTION MAILED		
2021-09-08	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL RESPONSE TO NON-FINAL OFFICE ACTION ENTERED AND FORWARDED TO EXAMINER		
2021-06-01	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL NON FINAL ACTION MAILED		
2021-05-16	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL DOCKETED NEW CASE - READY FOR EXAMINATION		
2020-06-22	STPP	-
Description: INFORMATION ON STATUS: PATENT APPLICATION AND GRANTING PROCEDURE IN GENERAL APPLICATION DISPATCHED FROM PREEXAM, NOT YET DOCKETED		
2019-09-30	AS	-
Description: ASSIGNMENT LG CHEM, LTD., KOREA, REPUBLIC OF ASSIGNMENT OF ASSIGNORS INTEREST; ASSIGNORS:KIM, JAE WOONG; KIM, HYOUNG KWON; KIM, SANG UCK; REEL/FRAME:050557/0127 2019-09-25		

2019-09-27	FEPP	-
Description: FEE PAYMENT PROCEDURE ENTITY STATUS SET TO UNDISCOUNTED (ORIGINAL EVENT CODE: BIG.); ENTITY STATUS OF PATENT OWNER: LARGE ENTITY		

Post-Issuance (US):

Reassignment (US) Table:

Assignee	Assignor	Date Signed	Reel/Frame	Date
LG ENERGY SOLUTION LTD.,SEOUL,KR	LG CHEM, LTD.	2021-10-27	058295/0068	2021-11-04
Conveyance: ASSIGNMENT OF ASSIGNOR'S INTEREST				
Corresponent: DENTONS US LLP 1900 K STREET NW WASHINGTON, DC 20006				
LG CHEM LTD.,SEOUL,KR	KIM, JAE WOONG	2019-09-25	050557/0127	2019-09-30
	KIM, HYOUNG KWON	2019-09-25	-	-
	KIM, SANG UCK	2019-09-25	-	-
Conveyance: ASSIGNMENT OF ASSIGNOR'S INTEREST				
Corresponent: LERNER, DAVID, LITTENBERG, KRUMHOLZ & MENTLIK, LLP 600 SOUTH AVENUE WEST WESTFIELD, NJ 07090				

Maintenance Status (US):

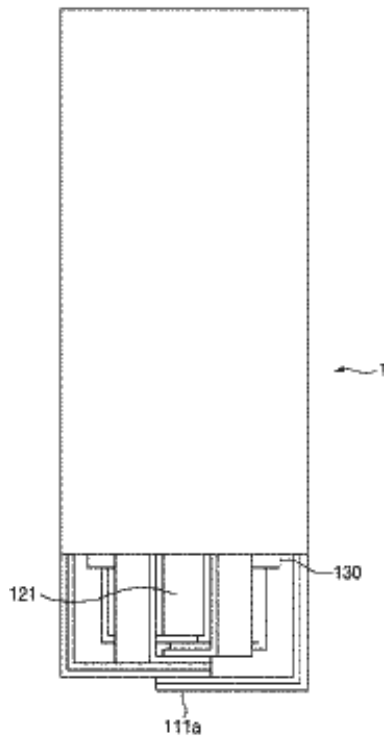
Litigation (US):

Opposition (EP):

License (EP):

EPO Procedural Status:

Front Page Drawing:





United States Patent and Trademark Office

Office of the Commissioner for Patents

Secondary Battery And Method For Manufacturing The Same

PATENT #

11652232

APPLICATION #

16498751

FILING DATE

09/27/2019

ISSUE DATE

05/16/2023

Payment Window Status

						FEE CODE	AMOUNT
WINDOW		STATUS		FEES		1551	\$2,150.00
3.5 Year		Open		Due		Total	\$2,150.00
Window	First Day to Pay	Surcharge Starts	Last Day to Pay	Status	Fees	Fee information is available at uspto.gov	
3.5 Year	05/16/2026	11/17/2026	05/17/2027	Open	Due		
7.5 Year	05/16/2030	11/19/2030	05/16/2031	Not Open	Not Due		
11.5 Year	05/16/2034	11/17/2034	05/16/2035	Not Open	Not Due		

Patent Holder Information

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Entity Status

UNDISCOUNTED

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